Image Based Biometry

By Peter Peer,

Žiga Emeršič; *Blaž Meden & Matej Vitek & Peter Rot & Tim Oblak & Ajda Lampe & Nicolas Larue & Darian Tomašević & ? ☺*

+ thanks to Walter Scheirer & Vitomir Štruc

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Biometry?!?

biometry

/bʌɪˈɒmɪtri/

noun

noun: **biometry**; noun: **biometrics**

the application of statistical an

Translate biometry to Slove

1. biometrija

Use over time for: biometry

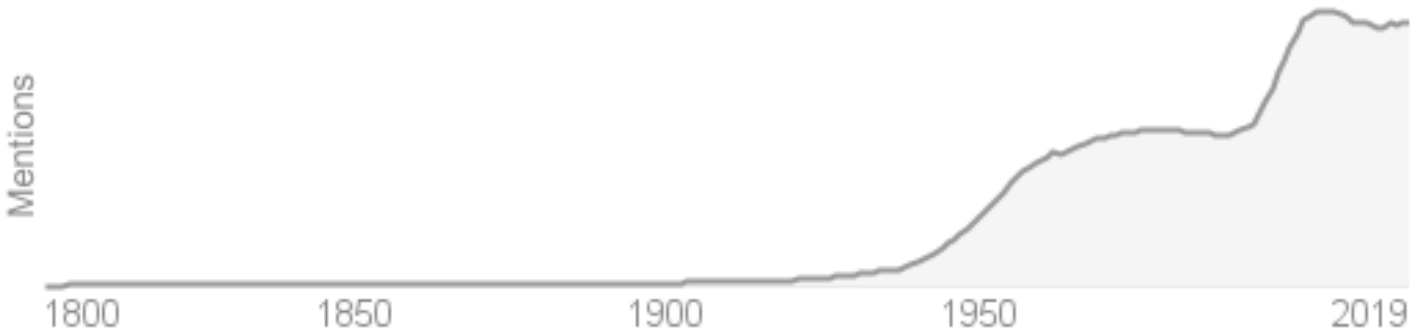
noun: **biometrics**

the application of statistical analysis to biological data.

Translate biometrics to Slovenian

1. biometrija

Use over time for: biometrics



What Is This Course All About?

- Biometrics:
  "the measurement of the properties of
  living beings"
- Greek: βίος = "life", μέτρον = "measurement"
- Or maybe it's:
  "the use of physical or behavioral properties of human beings for automatic identity recognition"
- Image-based! Computer vision! Programming skills!

Course Info/Overview

The course relies mostly on computer vision, as most biometrics technologies are based on it. Students interested in cutting edge technology, much of which is still in a research stage, are the intended target for the course. The main content (will evolve due to developments in the field):

1. Biometry basics
2. Biometrical modalities
3. Structure of a typical biometric system
4. Recognition/verification/identification
5. Metrics
6. Conditions for correct comparisons of the systems (databases, frameworks)
7. Performance and usefulness of the systems
8. Computer vision as the foundation of the biometric systems
9. Fingerprint
10. Iris
11. Face
12. Gait
13. Ear
14. Multi-biometric systems / multi-modality / fusions
15. Key problems of modalities/systems (research challenges)

The lectures introduce the approaches and explain their operation. At tutorial the knowledge is applied to practical problems in Matlab and open source tools.

Groups ARIS

Computer Vision Laboratory

Research activity

FRASCATI classification

CODE	SCIENCE	FIELD	SUBFIELD
2.07.07	Engineering sciences and technologies	Computer science and informatics	Intelligent systems - software

Keywords

Computer vision, image processing, machine vision, robot vision, multimedia, web applications

Evaluation (rules)

Evaluation of bibliographic research performance indicators according to ARIS methodology

Points	A''	A'	A1/2	CI10	CImax	h10	A1	A3
4,164.06	196.37	1,207	1,872.89	2,441	201	27	13.27	0.26

Data for the last 5 years (citations for the last 10 years) on September 24, 2024; A3 for period 2018-2022

Citations → Citations for bibliographic records in COBIB.SI that are linked to records in citation databases

Scopus

DATABASE	LINKED RECORDS	CITATIONS	PURE CITATIONS	AVERAGE PURE CITATIONS
WoS	130	1,768	1,593	12.25
Scopus	238	3,791	3,409	14.32

1. PERNUŠ, Martin, BHATNAGAR, Mansi, SAMAD, Badr, SINGH, Divyanshu, PEER, Peter, ŠTRUC, Vitomir, DOBRIŠEK, Simon. ChildNet : structural kinship face synthesis model with appearance control mechanisms. *IEEE access*. 2023, vol. 11, str. 49971-49991, ilustr. ISSN 2169-3536. https://ieeexplore.ieee.org/document/10126110, DOI: 10.1109/ACCESS.2023.3276877. [COBISS.SI-ID 152348163], [JCR, SNIP, WoS, Scopus]
kategorija: 1A2 (Z, A1/2); uvrstitev: SCIE, Scopus, MBP (COMPENDEX, DOAJ, INSPEC); tip dela je verificiral OSICT
točke: 12.93, št. avtorjev: 7

2. ROT, Peter, GRM, Klemen, PEER, Peter, ŠTRUC, Vitomir. PrivacyProber : assessment and setection of soft-biometric privacy-enhancing techniques. *IEEE transactions on dependable and secure computing*. 2023, vol. , iss. , str. 1-18, ilustr. ISSN 1941-0018. https://ieeexplore.ieee.org/abstract/document/10264192, DOI: 10.1109/TDSC.2023.3319500. [COBISS.SI-ID 166631427], [JCR, SNIP]
kategorija: 1A1 (Z, A', A1/2); uvrstitev: SCIE, Scopus, MBP (COMPENDEX, INSPEC); tip dela še ni verificiran
točke: 36.06, št. avtorjev: 4

3. VITEK, Matej, PEER, Peter, ŠTRUC, Vitomir, et al. Exploring bias in sclera segmentation models : a group evaluation approach. *IEEE transactions on information forensics and security*. [Print ed.]. 2023, vol. 18, str. 190-205, ilustr. ISSN 1556-6013. https://ieeexplore.ieee.org/document/9926136, DOI: 10.1109/TIFS.2022.3216468. [COBISS.SI-ID 127112963], [JCR, SNIP, WoS do 5. 9. 2023: št. citatov (TC): 1, čistih citatov (CI): 1, čistih citatov na avtorja (CIAu): 0.07, Scopus do 24. 9. 2023: št. citatov (TC): 3, čistih citatov (CI): 2, čistih citatov na avtorja (CIAu): 0.14]
kategorija: 1A1 (Z, A', A1/2); uvrstitev: SCIE, Scopus, MBP (COMPENDEX, INSPEC, METADEX); tip dela je verificiral OSICT
točke: 9.88, št. avtorjev: 29

4. HROVATIČ, Anja, PEER, Peter, ŠTRUC, Vitomir, EMERŠIČ, Žiga. Efficient ear alignment using a two-stack hourglass network. *IET biometrics*. [Print ed.]. Mar. 2023, vol. 12, iss. 2, str. 77-90, ilustr. ISSN 2047-4938. https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/bme2.12109, DOI: 10.1049/bme2.12109. [COBISS.SI-ID 145179139], [JCR, SNIP, WoS, Scopus do 18. 8. 2023: št. citatov (TC): 1, čistih citatov (CI): 1, čistih citatov na avtorja (CIAu): 0.25]
kategorija: 1A4 (Z); uvrstitev: SCIE, Scopus, MBP (COMPENDEX, DOAJ, INSPEC); tip dela je verificiral OSICT
točke: 14.76, št. avtorjev: 4

5. KUMAR, Sumit, KUMAR SINGH, Satish, PEER, Peter. Occluded thermal face recognition using BoCNN and radial derivative Gaussian feature descriptor. *Image and vision computing*. [Print ed.]. Apr. 2023, vol. 132, str. 1-16, ilustr. ISSN 0262-8856. https://www.sciencedirect.com/science/article/abs/pii/S0262885623000203?via%3Dihub, DOI: 10.1016/j.imavis.2023.104646. [COBISS.SI-ID 140724739], [JCR, SNIP, WoS do 24. 7. 2023: št. citatov (TC): 1, čistih citatov (CI): 1, čistih citatov na avtorja (CIAu): 0.33, Scopus do 27. 7. 2023: št. citatov (TC): 1, čistih citatov (CI): 1, čistih citatov na avtorja (CIAu): 0.33]
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točke: 38.31, št. avtorjev: 3

6. MEDEN, Blaž, GONZALEZ-HERNANDEZ, Manfred, PEER, Peter, ŠTRUC, Vitomir. Face deidentification with controllable privacy protection. *Image and vision computing*. [Print ed.]. Jun. 2023, vol. 134, [article no.] 104678, str. 1-19, ilustr. ISSN 0262-8856. https://www.sciencedirect.com/science/article/pii/S0262885623000525, https://repozitorij.uni-lj.si/IzpisGradiva.php?id=146103, DOI: 10.1016/j.imavis.2023.104678. [COBISS.SI-ID 150487811], [JCR, SNIP, WoS, Scopus]
kategorija: 1A1 (Z, A', A1/2); uvrstitev: SCIE, Scopus, MBP (COMPENDEX, INSPEC); tip dela je verificiral OSICT
točke: 28.73, št. avtorjev: 4

7. VITEK, Matej, BIZJAK, Matic, PEER, Peter, ŠTRUC, Vitomir. IPAD : Iterative Pruning with Activation Deviation for sclera biometrics. *Mağalatć ğamatć al-maläik Saud : ùlm al-îhasib wa al-malumat*. (Print ed.). Sep. 2023, vol. 35, iss. 8, [article no.] 101630, str. 1-21, ilustr. ISSN 1319-1578. https://www.sciencedirect.com/science/article/pii/S1319157823001842?via%3Dihub, DOI: 10.1016/j.jksuci.2023.101630. [COBISS.SI-ID 157745667], [JCR, SNIP, Scopus]
kategorija: 1A1 (Z, A', A1/2); uvrstitev: SCIE, Scopus, MBP (DOAJ, INSPEC); tip dela je verificiral OSICT
točke: 28.47, št. avtorjev: 4

8. FAIZAN ANSARI, Mohd, KASPROWSKI, Pawel, PEER, Peter. Person-specific gaze estimation from low-quality webcam images. *Sensors*. Apr. 2023, vol. 23, iss. 8, str. 1-17, ilustr. ISSN 1424-8220. https://www.mdpi.com/1424-8220/23/8/4138, DOI: 10.3390/s23084138. [COBISS.SI-ID 150104067], [JCR, SNIP, WoS do 19. 7. 2023: št. citatov (TC): 1, čistih citatov (CI): 1, čistih citatov na avtorja (CIAu): 0.33, Scopus do 29. 7. 2023: št. citatov (TC): 1, čistih citatov (CI): 1, čistih citatov na avtorja (CIAu): 0.33]
kategorija: 1A2 (Z, A1/2); uvrstitev: SCIE, Scopus, MBP (ASFA, CAB, COMPENDEX, DOAJ, INSPEC, MEDLINE, METADEX, PUBMED); tip dela je verificiral OSICT
točke: 32.08, št. avtorjev: 3

9. OBLAK, Tim, HARAKSIM, Rudolf, BESLAY, Laurent, PEER, Peter. Probabilistic fingermark quality assessment with quality region localisation. *Sensors*. Apr. 2023, vol. 23, iss. 8, str. 1-22, ilustr. ISSN 1424-8220. https://www.mdpi.com/1424-8220/23/8/4006, DOI: 10.3390/s23084006. [COBISS.SI-ID 149322755], [JCR, SNIP, WoS, Scopus]
kategorija: 1A2 (Z, A1/2); uvrstitev: SCIE, Scopus, MBP (ASFA, CAB, COMPENDEX, DOAJ, INSPEC, MEDLINE, METADEX, PUBMED); tip dela je verificiral OSICT
točke: 24.06, št. avtorjev: 4

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computer vision biometrics

FOLLOW

TITLE	CITED BY	YEAR
Generating bimodal privacy-preserving data for face recognition D Tomašević, F Boutros, N Damer, P Peer, V Štruc Engineering Applications of Artificial Intelligence 133, 108495		2024
DiCTI: Diffusion-based Clothing Designer via Text-guided Input A Lampe, J Stopar, DK Jain, S Omachi, P Peer, V Štruc 2024 IEEE 18th International Conference on Automatic Face and Gesture …		2024
ASPECD: Adaptable Soft-Biometric Privacy-Enhancement Using Centroid Decoding for Face Verification P Rot, P Terhörst, P Peer, V Štruc 2024 IEEE 18th International Conference on Automatic Face and Gesture …		2024
Preliminary Study on Detection of Breasts MP Vrešnjak, A Perušić, Ž Emeršič, P Peer, B Batagelj 2024 47th MIPRO ICT and Electronics Convention (MIPRO), 2079-2083		2024
Integrating AI into VET: Insights from AIM@ VET's First Training Activity Ž Emeršič, P Peer, G Hrastnik, NM Peer, JM Bey, M Meizoso-García, ... IntechOpen		2024
Recent Advances in Synthesis and Interaction of Speech, Text, and Vision L Orynbay, B Razakhova, P Peer, B Meden, Ž Emeršič Electronics 13 (9), 1726	1	2024
Enhancing Gender Privacy with Photo-Realistic Fusion of Disentangled Spatial Segments P Rot, J Križaj, P Peer, V Štruc ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and …		2024
AI-KD: Towards Alignment Invariant Face Image Quality Assessment Using Knowledge Distillation Ž Babnik, F Boutros, N Damer, P Peer, V Štrue 2024 12th International Workshop on Biometrics and Forensics (IWBF), 1-6		2024
eDifFIQA: Towards Efficient Face Image Quality Assessment Based On Denoising Diffusion Probabilistic Models Ž Babnik, P Peer, V Štruc IEEE Transactions on Biometrics, Behavior, and Identity Science	1	2024
W-TDL: Window-Based Temporal Deepfake Localization L Dragar, P Rot, P Peer, V Štruc, B Batagelj		2024
Semiotics of visualisation in comparative literature: the case of the NEWW women writers database A Vaupotič, N Bovcon Traveling concepts		2024

Cited by VIEW ALL

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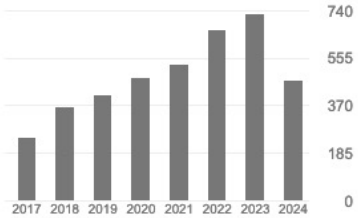
i10-index	124	74



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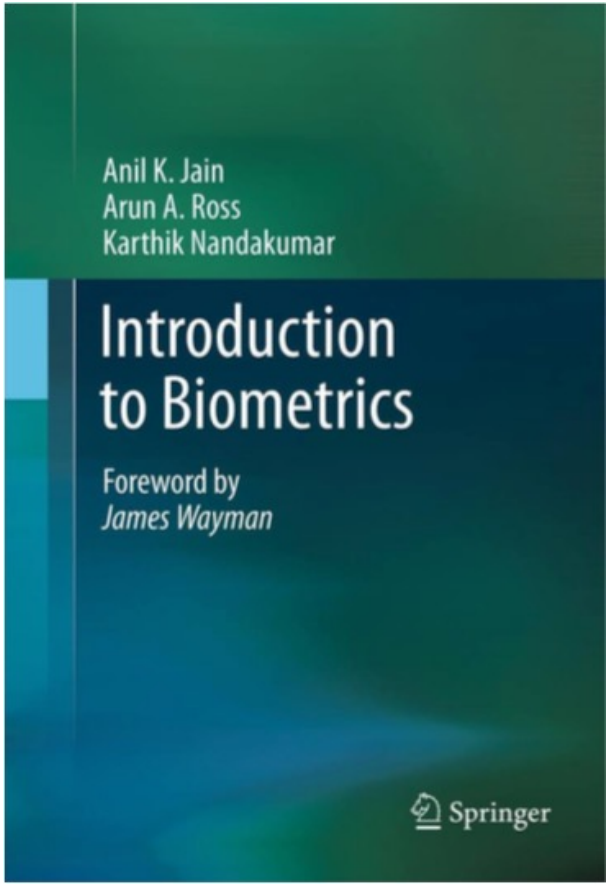
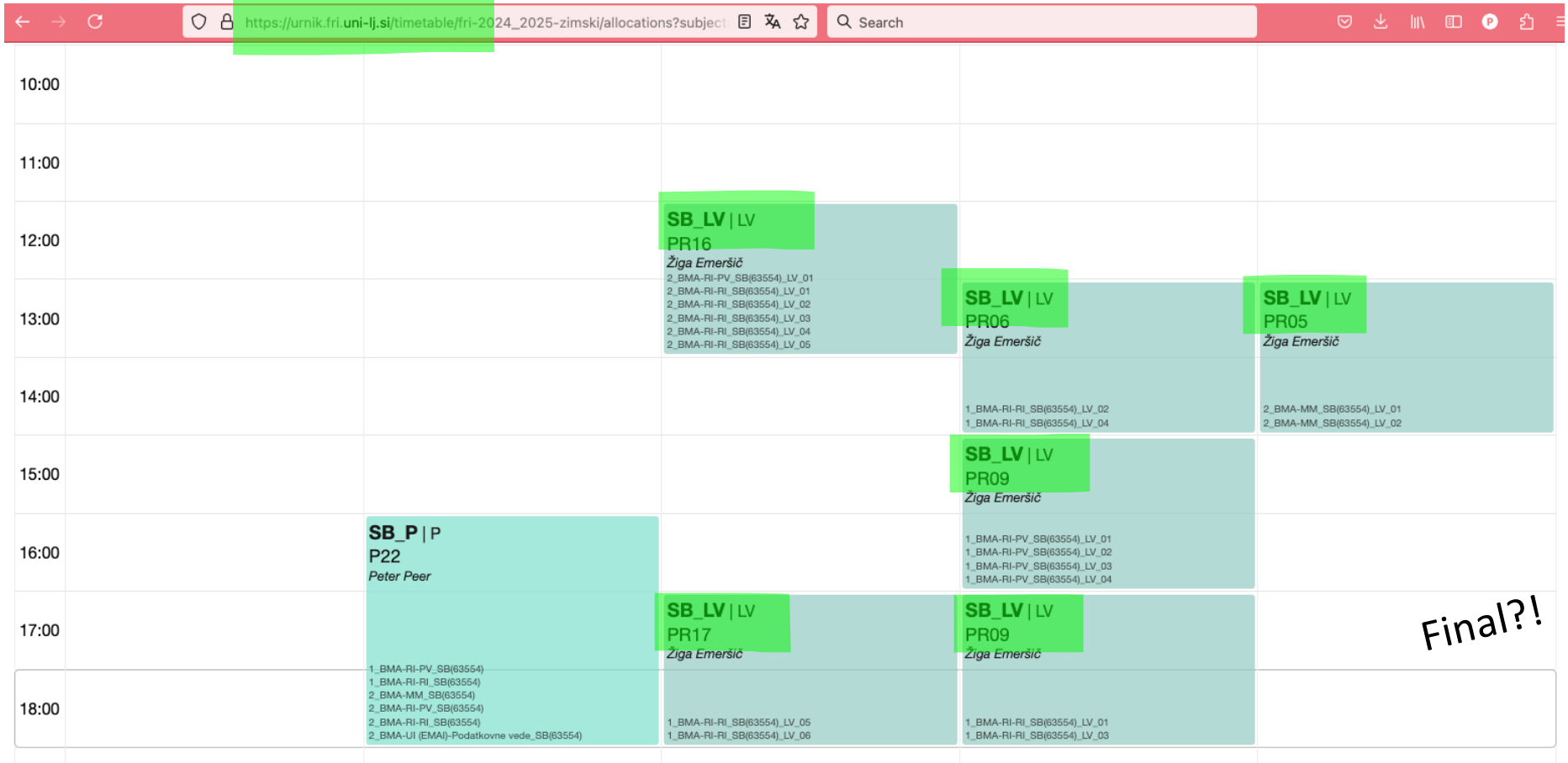


Literature ?!?

+ some other books &
a bunch of scientific articles

Tutorials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1_BMA-RI-RI_SB(63554)_LV_06
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Final?!

Tutorials

- Hands on practice!
- 2 tracks: X assignments // Y assignments + 1 project
- (Really) short reports with IMRAD structure (with proper professional and scientific terminology)
- 2/3 of contact hours in the form of labs,
  1/3 in the form of a seminar
- What is a seminar? Panels/discussions based on articles, presentations of your work etc.

Predavanja Lectures	Seminar Seminar	Vaje Tutorial	Klinične vaje Laboratory work	Druge oblike študija Field work	Samost. delo Individ. work	ECTS
45	10	20	/	/	105	6

10-12h/week

Grading

Weight (in %)	**Assessment:**
	Type (examination, oral, coursework, project):
67%	**Continuing** (assignments/project, presentations)
33%	**Final** (written or oral exam)
	Grading: 6-10 pass, 1-5 fail.

Motivation … Through Pictures

India from 2009 …
Collected Information

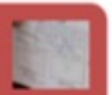
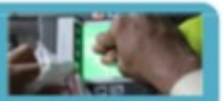
KYR Fields – Name, Address, Gender, DOB 

Photo & Address Verification

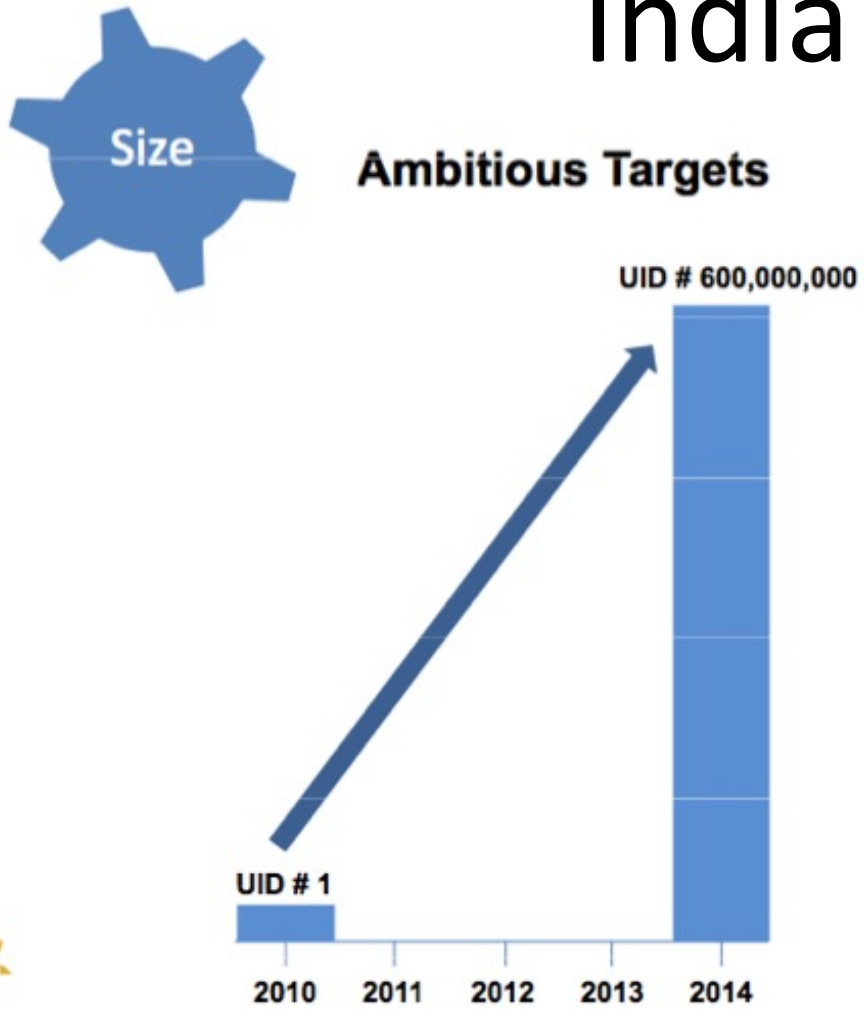
Photo

10-fingerprints on Slap scanner 

Iris Scan

आधार

India … Goals



- July 2015:
  879M enrollments
- Now?

- Problems:
  legislation, privacy,
  security, ethics

Where Should We Start?

With perhaps some basic terms and definitions?

Understanding the Term Biometrics

- "the use of physical or behavioral properties of human beings for automatic identity recognition"

- Something characteristic **only** to me
  (e.g. my face, my handwriting, my voice)
- **Not** something I know (e.g. a password or PIN)
- **Not** something I have
  (e.g. a card, key or authentication token)

Understanding the Term Biometrics

- "the use of physical or behavioral properties of human beings for automatic identity recognition"

- We need a **living subject**

Understanding the Term Biometrics

- "the use of physical or behavioral properties of human beings for automatic identity recognition"

- Why do we want computers to do this?
- **High throughput // Repeatability // Predictability**

Biometric Modality

- A **single** physical or behavioral property we use for biometric recognition

- Face
- Fingerprint
- Iris

Biometric Recognition Types

- Positive Recognition
- A sample represents a subject **known** to the system (i.e., already registered)

- Negative Recognition
- A sample represents a subject **unknown** to the system (i.e., not yet registered)

Authentication Types

- Pair matching: Do these two images match?
- Verification: Does this sample of subject *X* match the one in our system? (1:1)
- Identification: Does this person exist in our system? (1:n)
- Negative authentication: I'm not the subject *X* / I'm not a member of the group **X**
- Deduplication: Do we have to enroll you?

Probe & Gallery

- Probe: A sample presented to a biometric system
- Gallery: A collection of enrolled templates

"Who is this?"

"Halle Barry"

Enrollment Pipeline

- New user comes in

- Acquisition $\Longrightarrow$
  Feature extraction $\Longrightarrow$
  Template generation

Recognition Pipeline

- We need to make a decision

- Acquisition $\Longrightarrow$
  Feature extraction $\Longrightarrow$
  Matching $\Longrightarrow$
  Labeling

Enrollment – User's Perspective

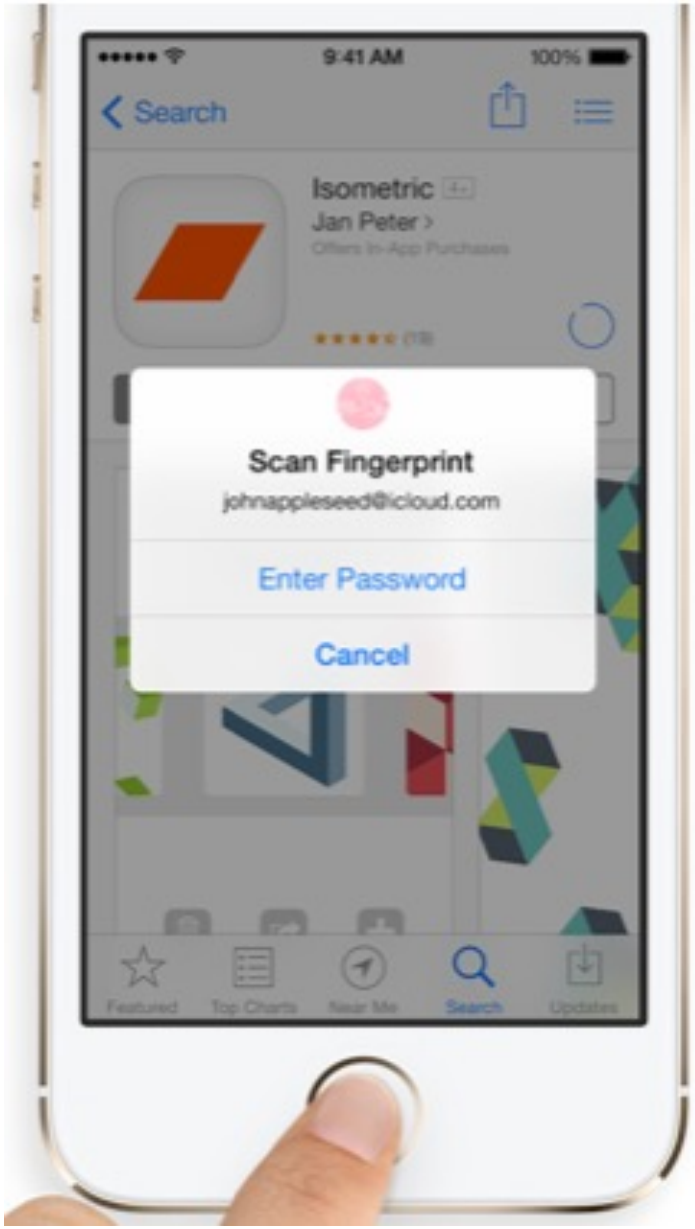
Multiple measurements

- Quality control
- Select best samples
- Merge samples

Enrollment – Security Engineer's Perspective

- Is meta-data associated with the record?
- Are multiple templates stored for a user?
- Operator supervision?
- How is the template stored?
  - Encryption
  - Template protection

Authentication – User's Perspective



Authentication – Security Engineer's Perspective

- How is the stored template retrieved?
  - On device
  - Local server
  - Cloud
- Operational matching threshold?
- How many attempts?
- Speed vs. security tradeoff?
- Auditing?

Error Statistics

- **False Match (Type I Error):** an impostor sample matched a reference template

Error Statistics

- **False Non-Match (Type II Error):** a genuine sample did not match a reference template

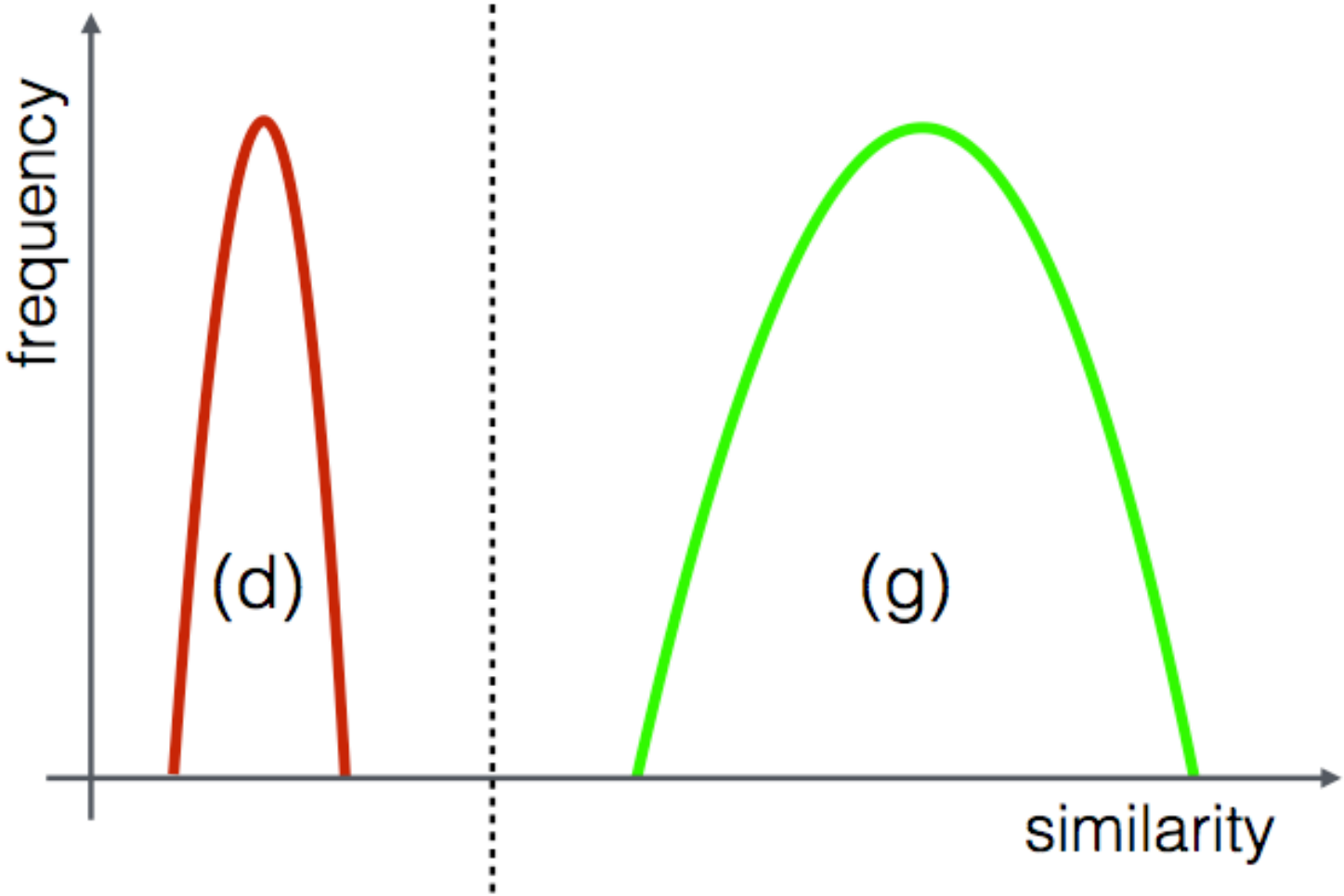
Decisions

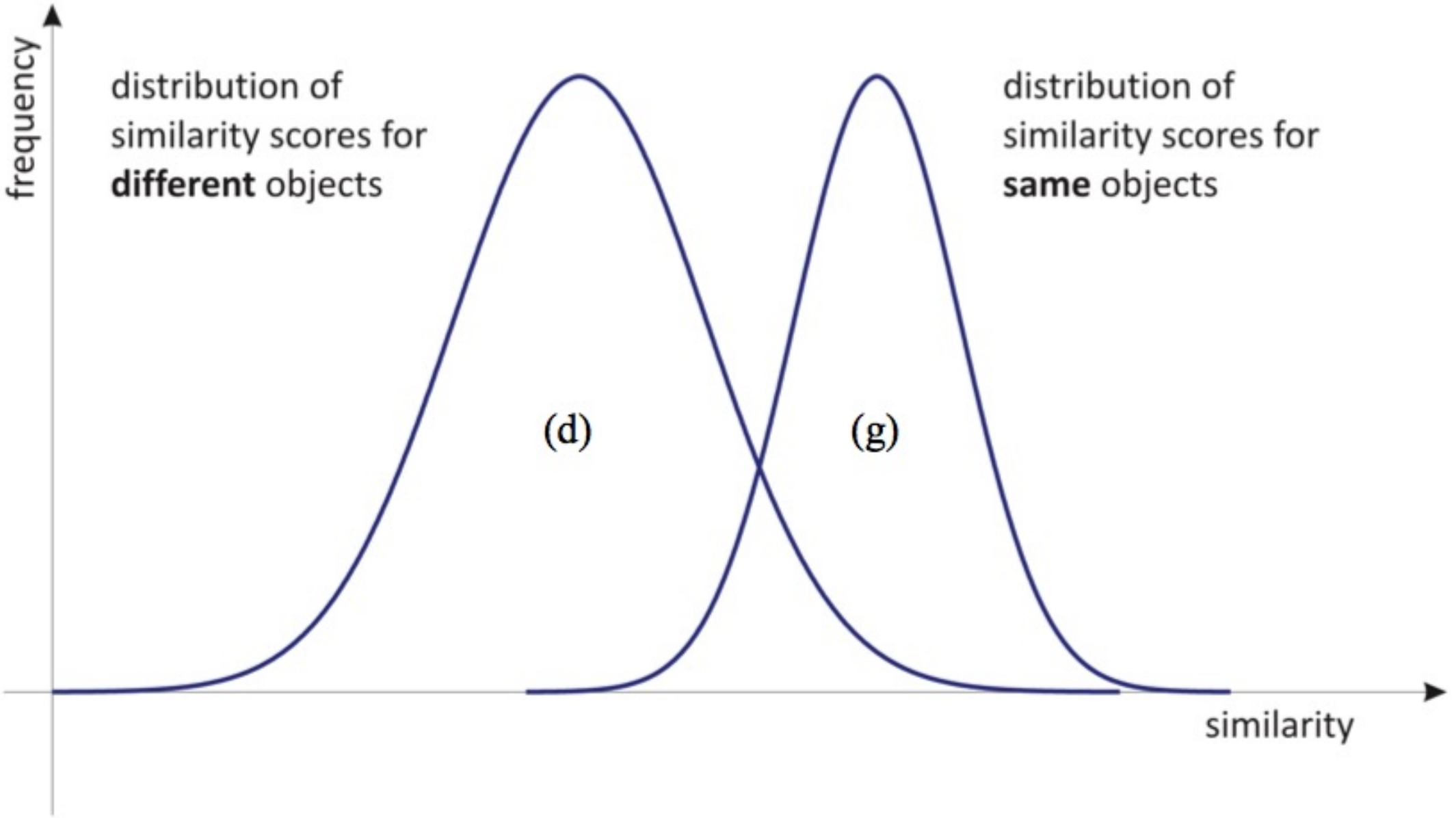
- **Similarity** score: higher is better
- **Distance** score: lower is better (0 for self match)
- Decision threshold: τ

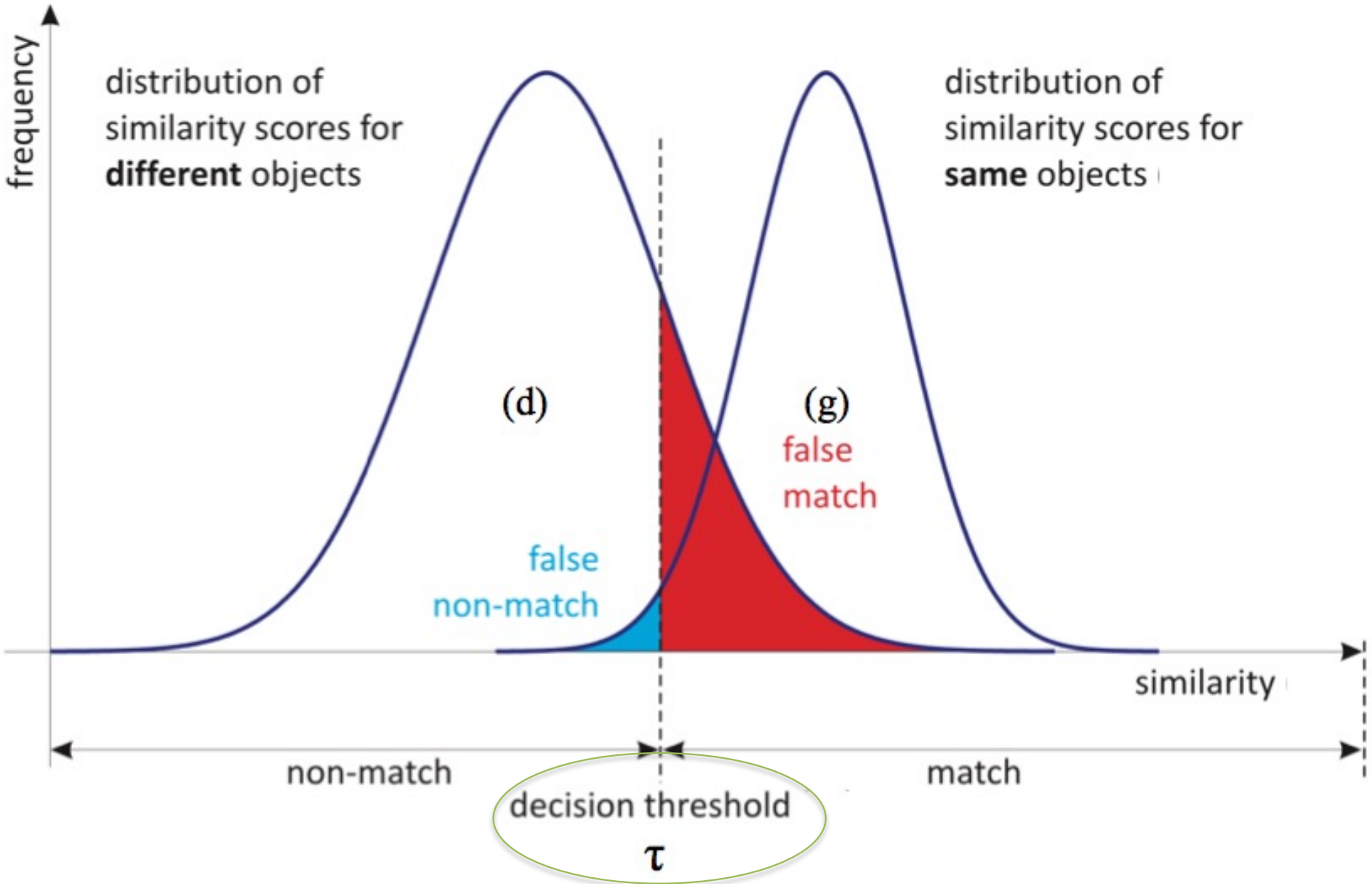
- Similarity Score Match: $s_y > \tau$
- Distance Score Match: $s_y < \tau$

Score Distributions

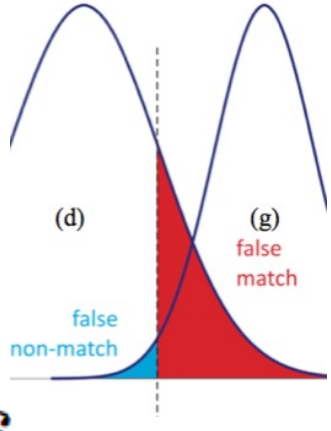
Ideal world: different (d) and genuine (g) score distributions well-separated




frequency
distribution of similarity scores for **different** objects
distribution of similarity scores for **same** objects
(d)
(g)
similarity


frequency
distribution of similarity scores for **different** objects
distribution of similarity scores for **same** objects
(d)
(g)
false match
false non-match
similarity
non-match
match
decision threshold
τ

False Non-Match Rate



- If we know $\mathbf{g}$:

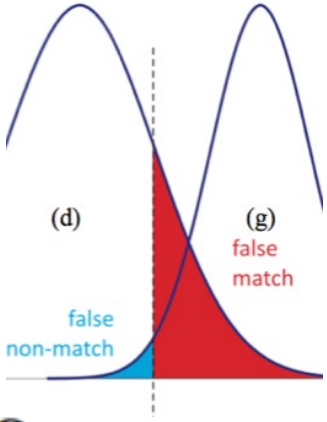
$$g_{FNM}(\tau) = \int_{-\infty}^{\tau} g(s)ds$$

- But typically we don't, thus:

$$\hat{g}_{FNM}(\tau) = FNMR(\tau) = \frac{\text{Number of false non-matches for } \tau}{\text{Number of all genuine comparisons}}$$

- Calculate with a **benchmark data set**

False Match Rate



- If we know $\mathbf{d}$:

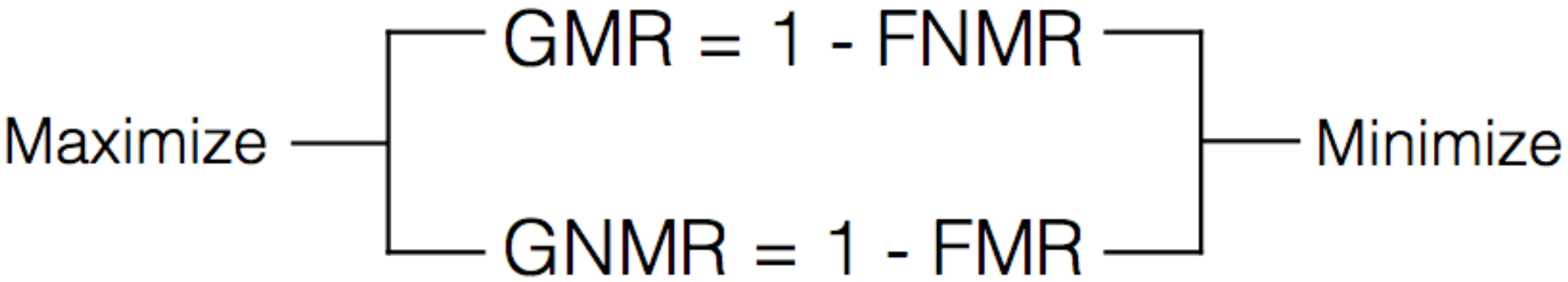
$$\mathrm{d}_{\mathrm{FM}}(\tau) = \int_{\tau}^{\infty} \mathrm{d}(s)ds$$

- But typically we don´t, thus:

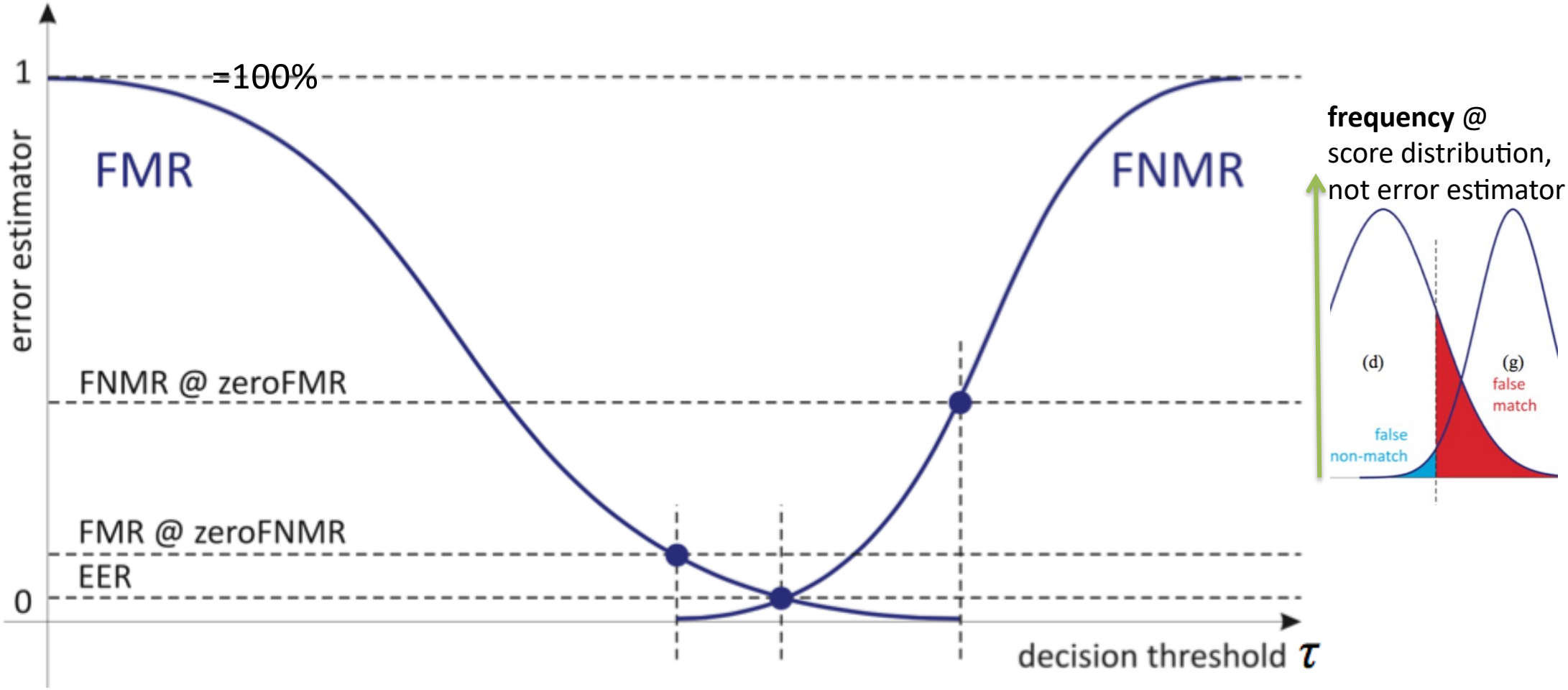
$$\hat{\mathrm{d}}_{\mathrm{FM}}(\tau) = \mathrm{FMR}(\tau) = \frac{\text{Number of false matches for } \tau}{\text{Number of all impostor comparisons}}$$

- Calculate with a **benchmark data set**

How to Measure **Correct** Decisions?

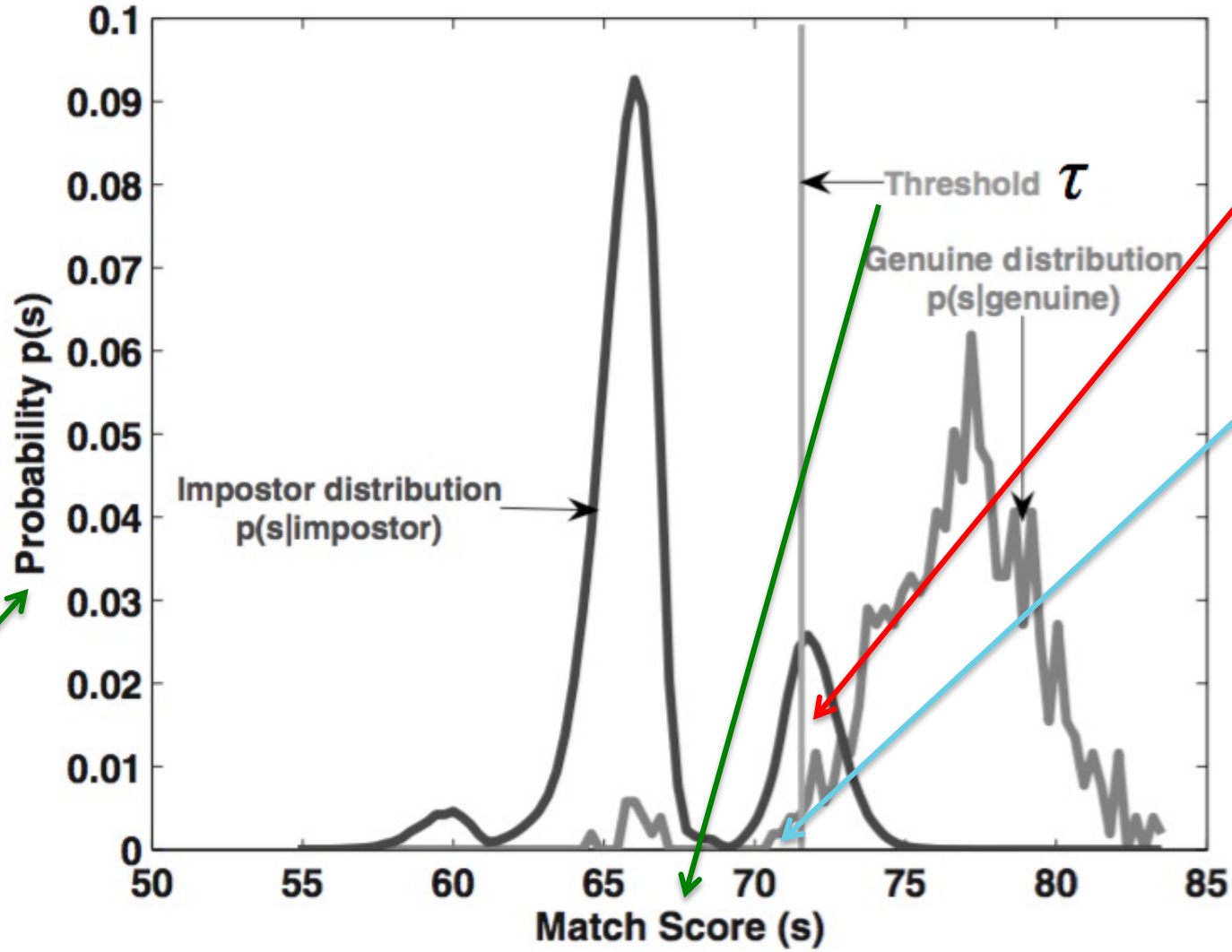


Error Estimators as a Function of τ



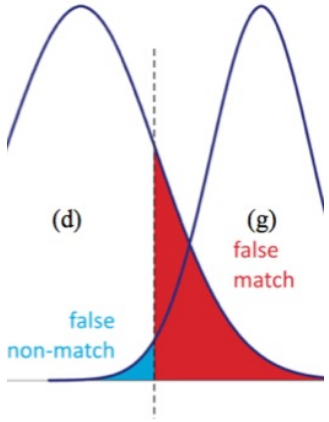
frequency @
score distribution,
not error estimator

BTW: Score Distribution – Real World Case



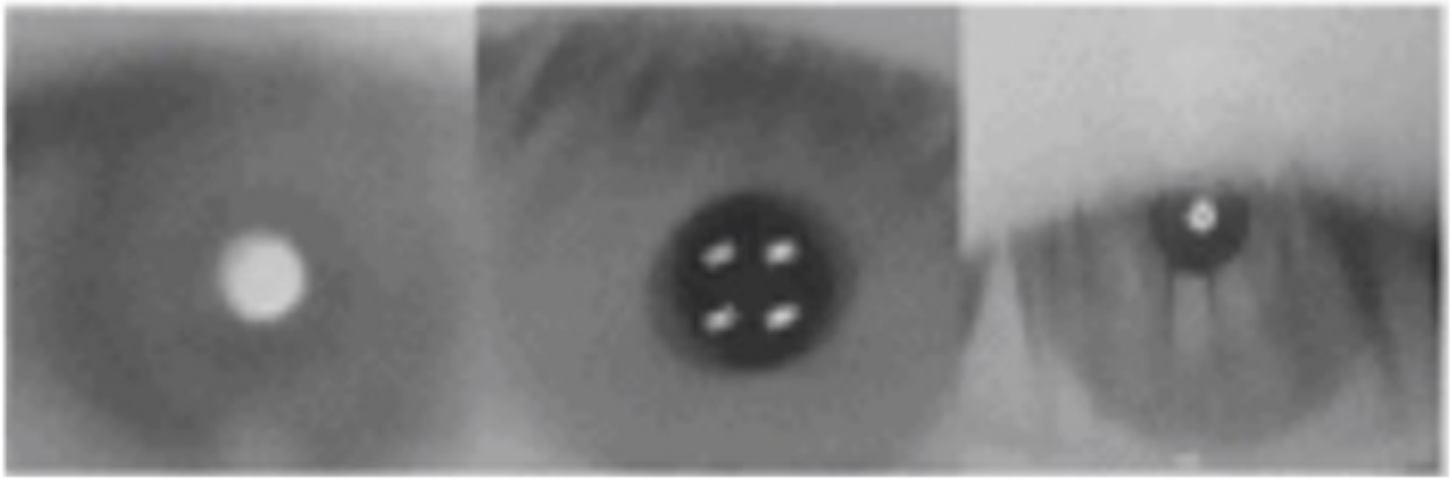
Calculated from the frequency

The FAR and the FRR cannot be reduced *simultaneously* by adjusting the threshold!



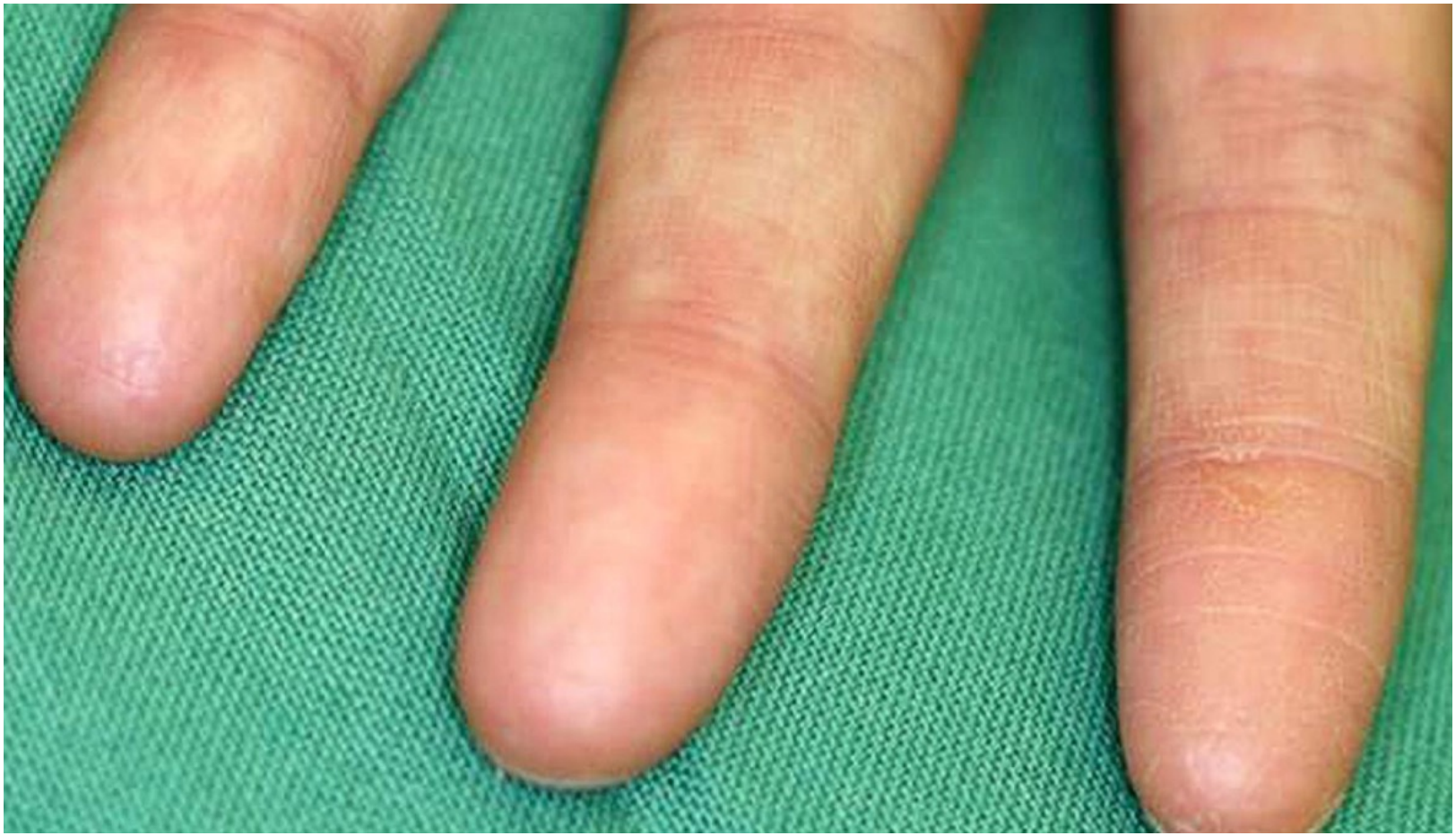
Failure to Acquire

Falsely rejected biometric samples
(due to problems at **acquisition**)



Failure to Enroll

Falsely rejected biometric samples (due to problems at **enrollment**)



Misunderstandings in Biometry

- **Uniqueness** -> the physical trait itself may not be unique
  - *Harper*'s headline, 1910: "Only once during the existence of our solar system will two human beings be born with similar finger markings"
  - *Scientific American*, 1911: "Two like fingerprints would be found only once every 1048 years"
  - Uniqueness of biometric modalities has not been scientifically clearly established
- **Permanence** -> also not an established scientific fact

Facts

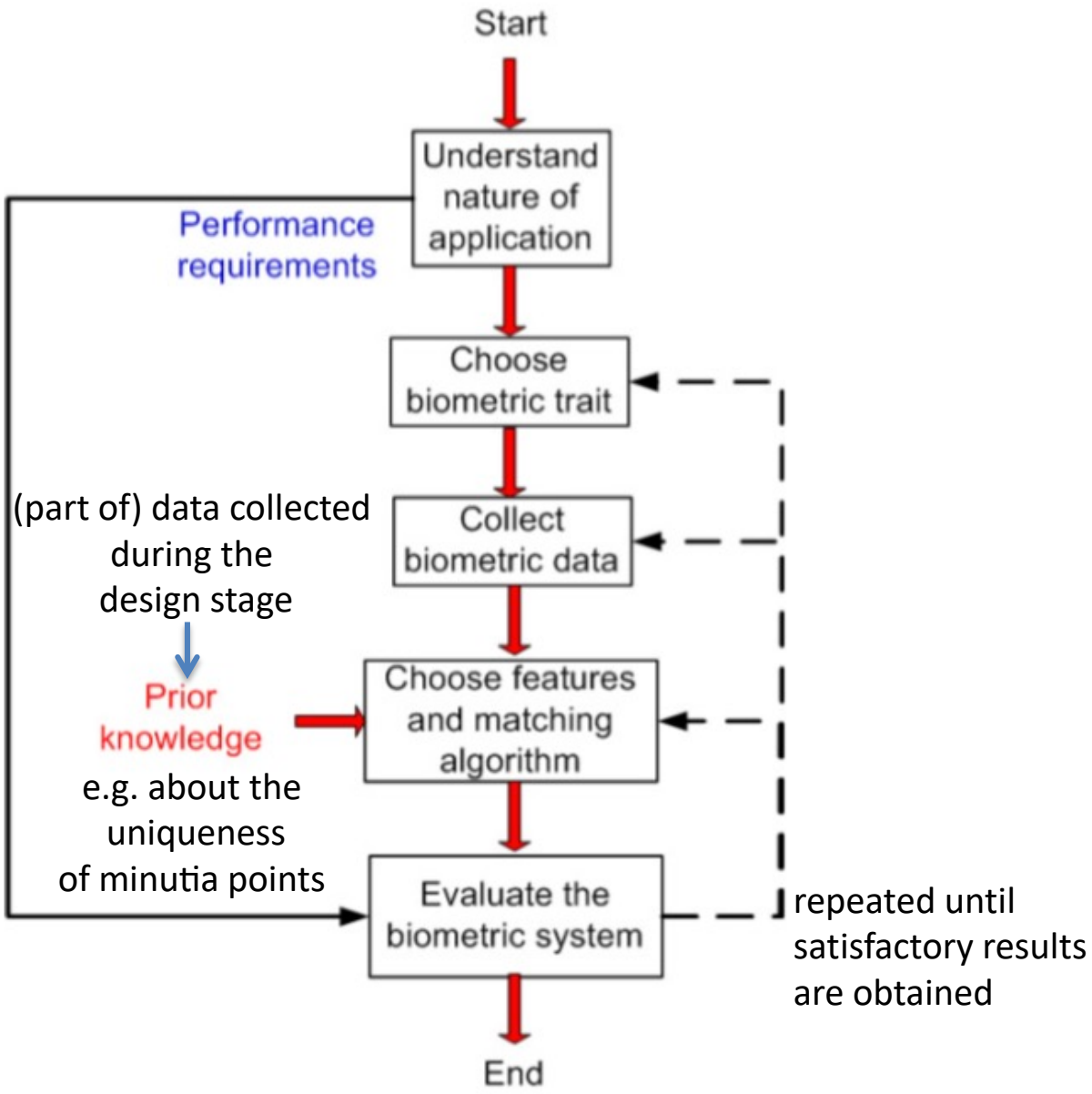
- Fact 1: biometric systems rely only on the **digital** measurements
- Fact 2: sensing introduces **variations** in the samples of the same biometric trait of a user obtained over a period of time
- Fact 3: feature sets obtained from different samples of the same biometric trait of a user are **seldom** identical

  => we look for **close** and not perfect match

Intra-User Variations

The same person?

The **Design Cycle** of a Biometric System



*general pipeline

Applications

FORENSICS	GOVERNMENT	COMMERCIAL
Corpse identification	National ID card	ATM
Criminal investigation	Driver's license; voter registration	Access control; computer login
Parenthood determination	Welfare disbursement	Mobile phone
Missing children	Border crossing	E-commerce; Internet; banking; smart card

Schipol Airport (Amsterdam)

Employs **iris** scan smart cards to speed up the immigration procedure

Passengers who are **voluntarily** enrolled in this scheme insert their **smart card** at the gate and peek into a camera

Ben Gurion International Airport (Tel Aviv)

Outbound passengers are required to provide **fingerprints** and **facial** images, which are stored in a smart card that is issued to each passenger

These smart cards are then used to **track** the passengers as they pass through different locations at the airport

Dubai airport will scan your face as you walk through a video tunnel …

US-VISIT Program

Visitor and **I**mmigration **S**tatus **I**ndicator **T**echnology

Utilizes all ten fingerprints and face to validate the travel documents of visitors to the US

Ten fingerprints are matched against a dynamic watch list containing **up to a few million records** in less than 10 sec

Example of negative recognition, where the purpose is to find if the visitor has multiple aliases

If not on a **watch-list**, the visitor is admitted into the US and the person's fingerprints are enrolled into the database for future matching

Unique Identity (UID) Card Project in India

Facilitate efficient delivery of various welfare schemes

Ten fingers, two irises, and face

An example of de-duplication (negative recognition), where a fusion of 10 fingers and 2 irises is determining if the same person is trying to acquire two different 12-digit identification numbers

Disney World in Orlando

Prevent ticket fraud –
prevents more than one person using the same ticket

System works in all weather conditions because it uses a rugged fingerprint sensor that is able to capture good quality fingerprint images even in adverse imaging conditions

ATMs in Japan

Palm-vein authentication systems

Using a near infrared lighting source

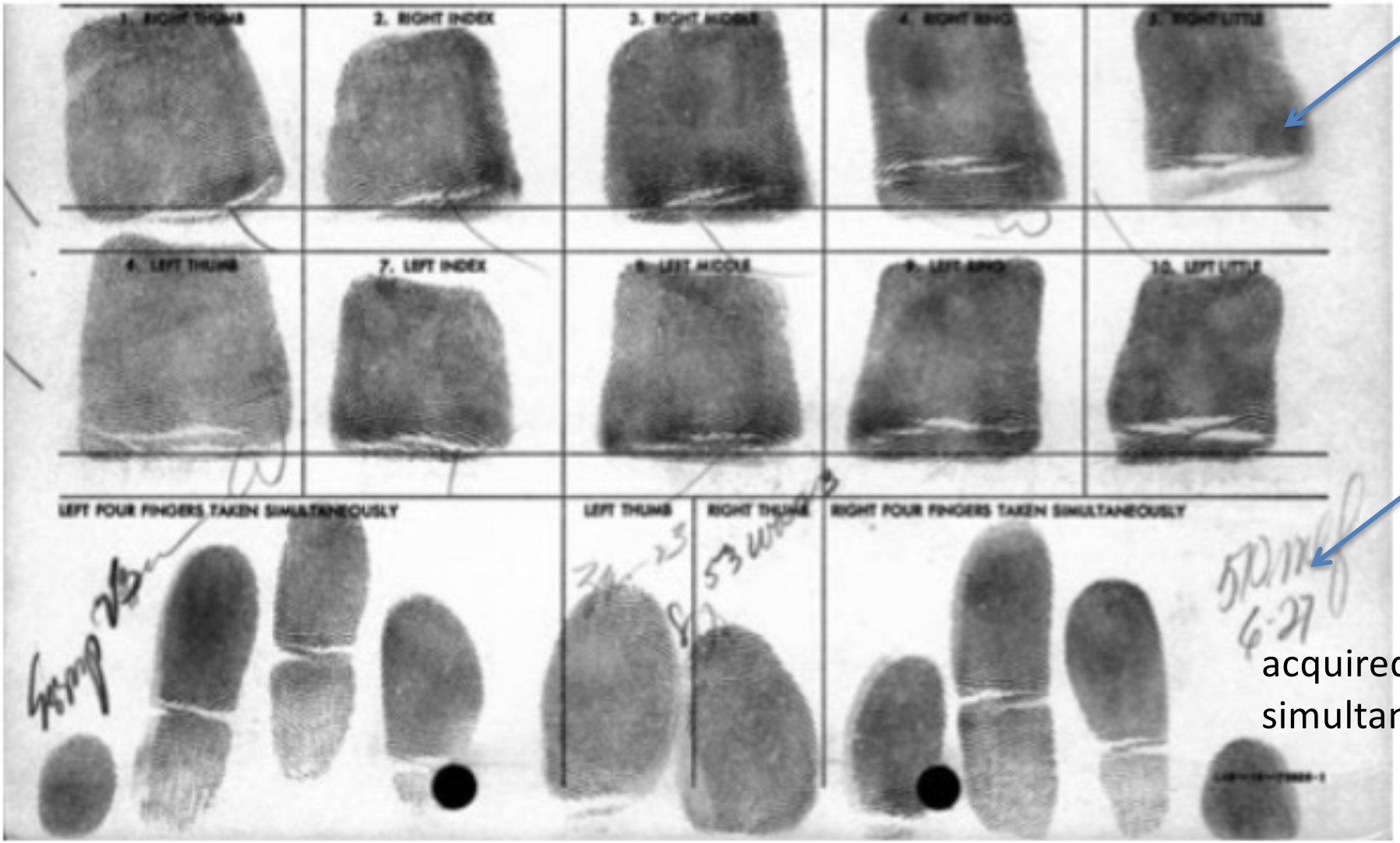
Importance of Protocols

- Needed for **sound comparisons** of techniques / **repeatability**
- Define procedure, metrics, databases, frameworks/toolboxes
- Example:
  - AWE db & toolbox
  - partition the data into a *development set* that contains 60% percent of all images and a *test set* that contains the remaining 40%
  - 5-fold cross validation: means and standard deviations
  - bootstrapping on test set: means and standard deviations
  - files containing lists for the experiments are distributed with the dataset
  - metrics: ROC & ERR (VER), CMC & Rank-1 (ID)

FINGERPRINT

- The ridges increase the friction with the contact surface – grasping made easier
- A typical young male has, on an average, 20.7 ridges per centimeter, while a female has 23.4 ridges per centimeter
- Is claimed to be unique
- Became synonym for biometric recognition in the minds of the general public

A tenprint card

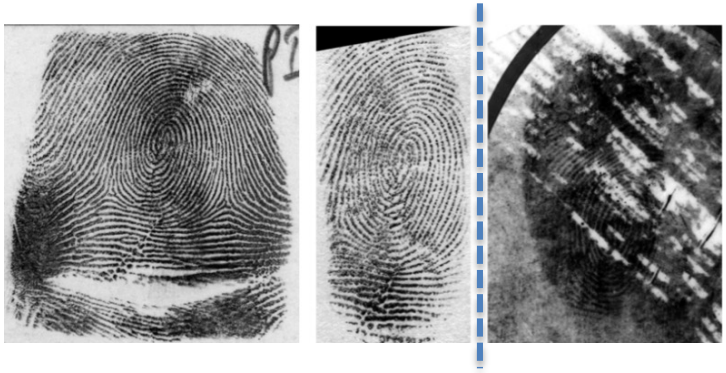


rolled

plain

acquired simultaneously

A latent fingerprint / **fingermark**!



AFIS

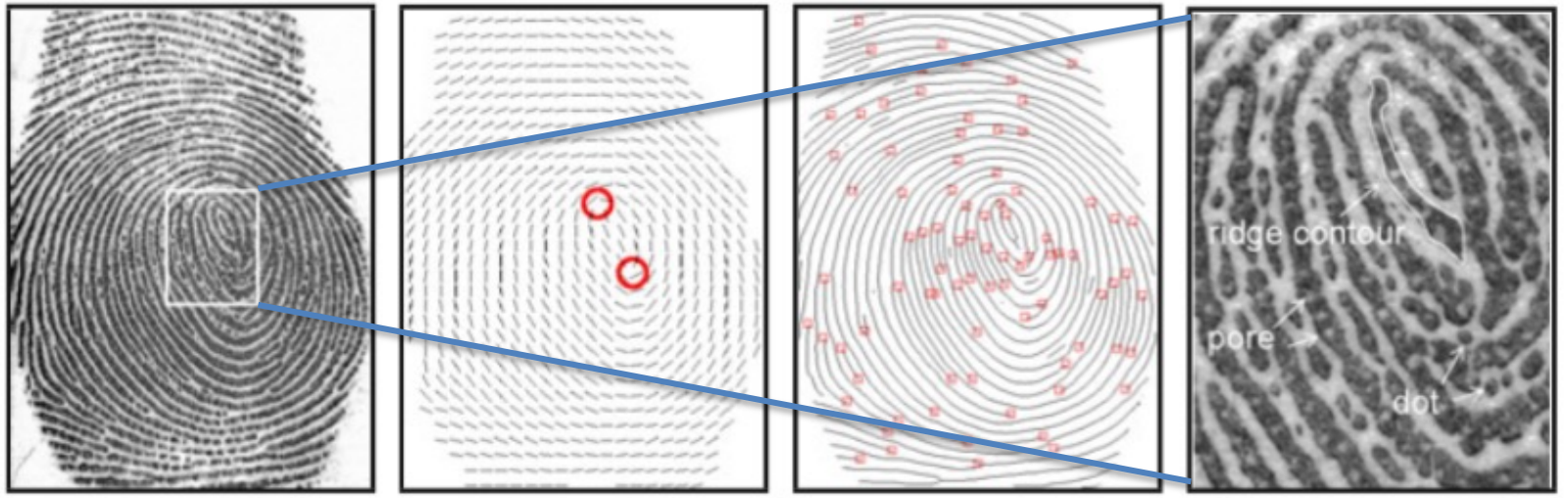
Michigan State Police

3.2 million tenprint cards

Performs 700,000+ searches each year

Features

- Features used have a **physical interpretation**
- Three levels of features (L1-3)



L1: orientation field or ridge flow and singular points

L2: ridge skeleton and minutiae

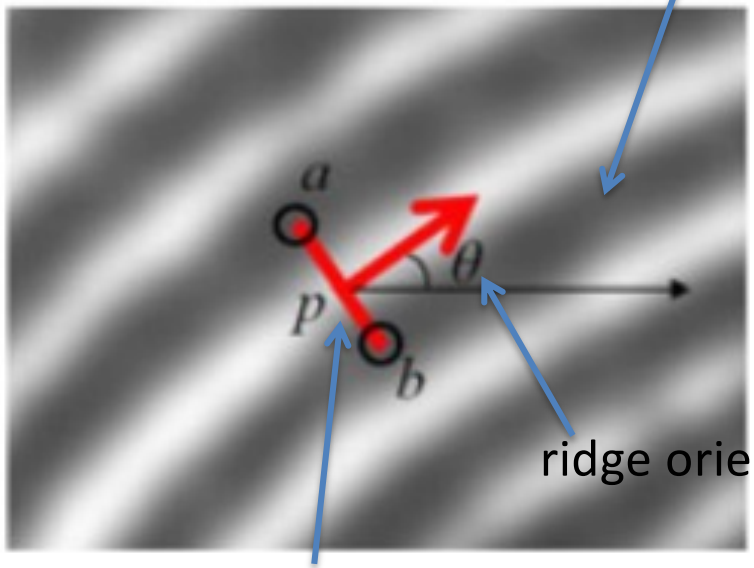
L3: ridge contour, pore, dot

L1 Details

- Oriented texture pattern – its global shape and structure can be defined by the **orientation** and **frequency** of its ridges
- Only the ridge flow and ridge frequency are observed; the exact location and dimensional details of ridges are **ignored**
- Low-resolution image sensors capable of scanning 250 ppi can be used

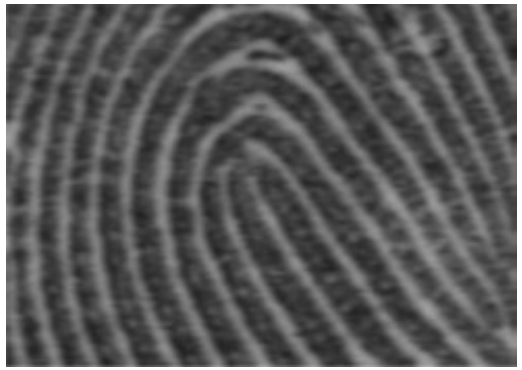
L1 Details

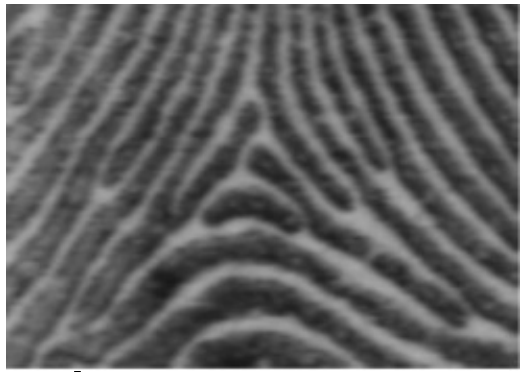
- Coarse features @ pixel *p*
- Ridge orientation information is viewed as being more important



Singular Points

- Salient locations where the ridge orientations change abruptly
- Two basic types: **loop** and **delta**
- Loop can be used as a landmark point to **align** the fingerprint



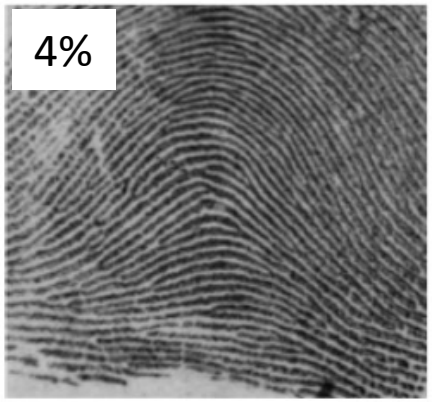


- **Core** point corresponds to the north most loop-type singular point
- No singular points => core usually refers to the point of maximum ridge curvature

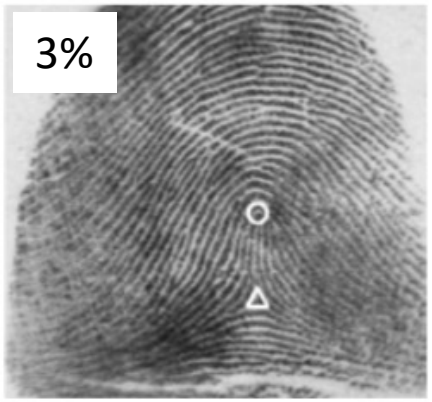
Singular Points

- Singular points can be viewed as an **abstract** representation of the orientation map =>
  - Orientation map can be roughly predicted based on the number and location of singular points
- Even better abstract representation is the **pattern type (fingerprint class)** =>
  - Deduced based on the **number** of loops and deltas and the spatial relationship between them

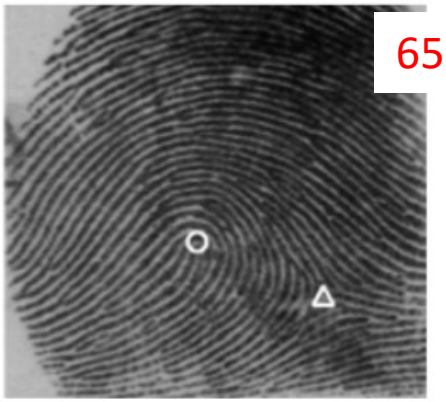
Pattern Types



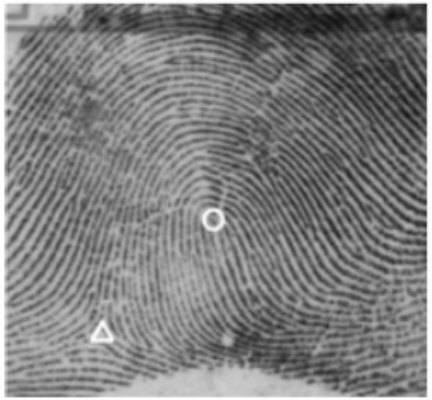
Plain arch



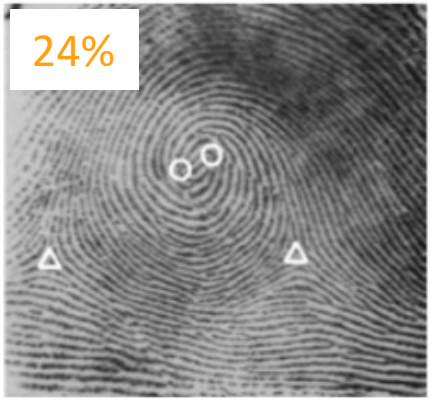
Tented arch



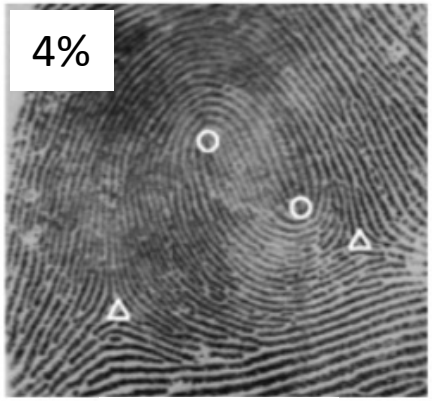
Left loop



Right loop



Whorl



Twin loop

@Tutorials:
PCASYS – Pattern Classification System

Classifies fingerprint patterns into **six classes**:

- **A** = Arch
- **L** = Left Loop
- **R** = Right Loop
- **T** = Tented Arch
- **W** = Whorl
- ***S** = Scar*

```
pcasys pcasys.prc
```

Key features:

1. PCA-based algorithm:

- Uses **Principal Component Analysis (PCA)** to reduce image data to key features

2. Neural network classifier:

- Classifies fingerprints into one of the six classes based on reduced data

*Part of NBIS (NIST Biometric Image Software)

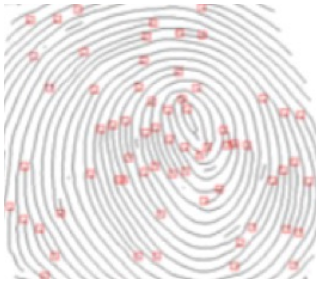
How PCASYS Works?

Process:

1. Image preprocessing:
    - Enhances and normalizes fingerprint images
2. Feature extraction with PCA:
    - Extracts key features for classification
3. Classification:
    - The neural network assigns the fingerprint to one of the six classes

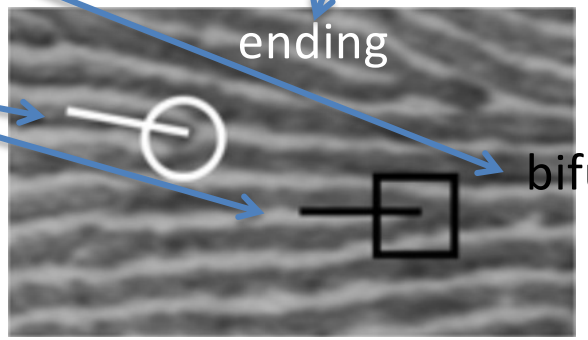
Use case:

Ideal for large-scale fingerprint systems, streamlining indexing and retrieval by classifying fingerprints into predefined categories



L2 Details

- Represented as a ridge skeleton image
- **Locations** (a) where a ridge emerges or ends, splits or merges with another ridge are termed as ridge characteristics or **minutiae**
- Minutia generally has two other properties: **direction** (b) and **type** (c)
- Easily observed at resolution of 500 ppi



JK Re: [Assignment 1] - png2wsq
od JANEZ KUHAR - nedelja, 13. oktober 2024, 23.05

Thank you!

I've since noticed something else regarding **pcasys** that might be helpful to others:

> **5.1.1.1. Segmentor** [/NBIS/Main/pcasys/src/lib/pca/sgmnt.c; sgmnt()]
> The segmentor routine performs the first stage of processing needed by the classifier. It reads the input fingerprint image. The image must be an 8-bit grayscale raster of width ***at least 512 pixels and height at least 480 pixels*** (these dimensions can be changed in the file include/pca.h), and scanned at about 19.69 pixels per millimeter (500 pixels per inch).
> *[Source: documentation]*

We have to pick a dataset with fingerprint images larger than or equal to **512x480** pixels in order to run **pcasys** to detect fingerprint types. I've been getting the following error when running **pcasys** on 300x300 pixel images:

```
$ pcasys pcasys.prs
initialization

101_1.wsq:
 segment
ERROR: sgmnt: original-raster width, 300, is < sw
```

AFAIK, there is only one free dataset (among those provided in the assignment instructions) that contains fingerprint files larger than 512x480 pixels (and is thus suitable for **pcasys**):

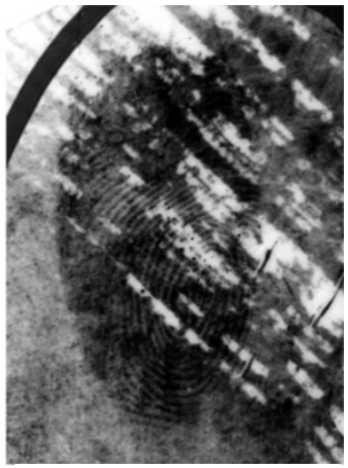
1. FVC2004 DB1 B
   Size: 10 fingers x 8 impressions
   Format: TIFF, 500dpi, 640x480px

Minutiae & AFIS

- Number varies a lot; according to the acquisition method, sensor etc.
- Minutiae are abstract representation of the ridge skeleton
- Capture much of the **discriminative** information
- Are storage efficient
- Reasonably robust to various sources of degradation

Rolled – 136

Plain – 56



Fingermark – 18

@Tutorials: MINDTCT – Fingerprint Minutiae Detector

Functionality:

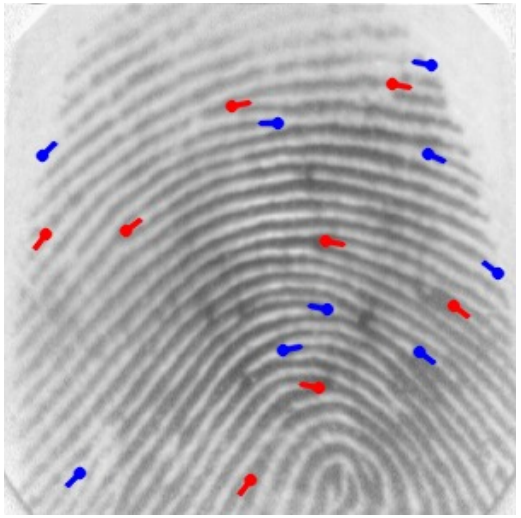
- Automatically detects and records **minutiae points** (ridge endings and bifurcations) in fingerprint images
- The system also includes minutiae quality assessment based on local image conditions. It assigns a quality score to each detected minutia based on image clarity
- Outputs a standardized minutiae data format for use in matching algorithms (like **Bozorth3)**

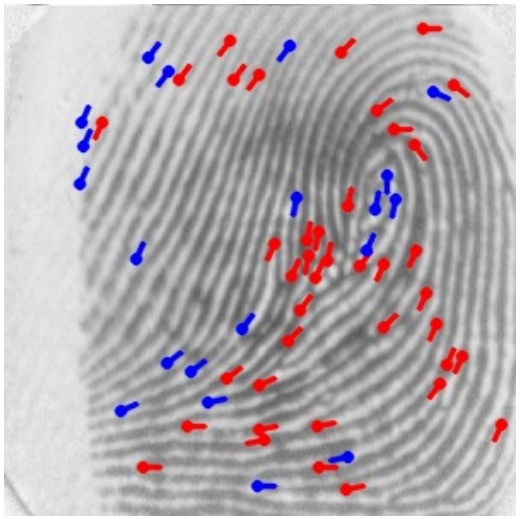
```
mindtct img.png out/img
```

*Part of NBIS (NIST Biometric Image Software)

How MINDTCT Works?

- **Fingerprint image analysis**:
  - Analyzes the fingerprint image's ridges and valleys
  - Detects **minutiae types**: ridge endings & bifurcations
- **Minutiae filtering**:
  - Removes bad minutiae caused by noise or poor image quality
- **Quality scoring**:
  - Each minutia is assigned a quality score reflecting its reliability
- **Output**:
  - Multiple files of minutiae data (positions, orientations, type, quality)





MINDTCT Outputs

Outputs we are using:

.xyt (Minutiae Data)

- **Content**: Minutiae detection results (x, y coordinates, theta, quality)
- **Use**: Format used by **bozorth3** matching algorithm

.min (Minutiae Details)

- **Content**: Detailed minutiae attributes (x, y, theta, quality, **type**)
- **Details**: Format includes multiple fields separated by : for a more detailed minutiae information

Other outputs:

.dm (Direction Map)

- Content: Grid of ridge flow directions in 8x8 pixel neighborhoods
- Details: 16 quantized direction units, each representing 11.25° increments. Value -1 indicates undetermined ridge flow

.hcm (High-Curvature Map)

- Content: Bi-level map marking regions of high ridge curvature (e.g., core/delta)
- Details: 1 = high-curvature, 0 = low curvature

.lcm (Low-Contrast Map)

- Content: Bi-level map identifying regions with low image contrast
- Details: 1 = low-contrast, typically background, 0 = sufficient contrast

.lfm (Low-Flow Map)

- Content: Bi-level map marking regions with indeterminable ridge flow
- Details: 1 = no dominant directional frequency found, 0 = ridge flow determined

.qm (Quality Map)

- Content: Map representing varying levels of image quality
- Details: Values from 0 (lowest quality) to 4 (highest quality) in 8x8 pixel neighborhoods

@Tutorials: Bozorth3 – Fingerprint Matcher Overview

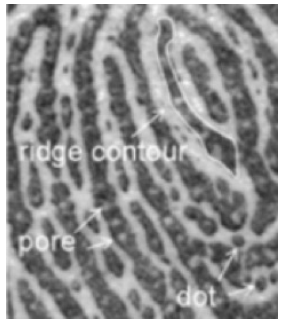
- Enhanced from "bozorth98" by Allan S. Bozorth, FBI
- **Key features**:
  - **Rotation & translation invariant**: Matches fingerprints despite variations in position/orientation
  - **Pairwise minutiae comparison**: Analyzes minutiae point relationships for similarity
  - **Match score**: Outputs a score indicating the likelihood of fingerprints being from the same finger

```
bozorth img1.xyt img2.xyt
```

*Part of NBIS (NIST Biometric Image Software)

How Bozorth3 Works?

- **Intra-fingerprint minutiae comparison**:
  - Records spatial relationships between pairs of minutiae
  - Data is sorted and truncated by distance
- **Inter-fingerprint comparison**:
  - Creates a compatibility table based on distance and angle tolerances
- **Match score calculation**:
  - Builds clusters of compatible minutiae pairs
  - Higher score = higher likelihood of matching fingerprints

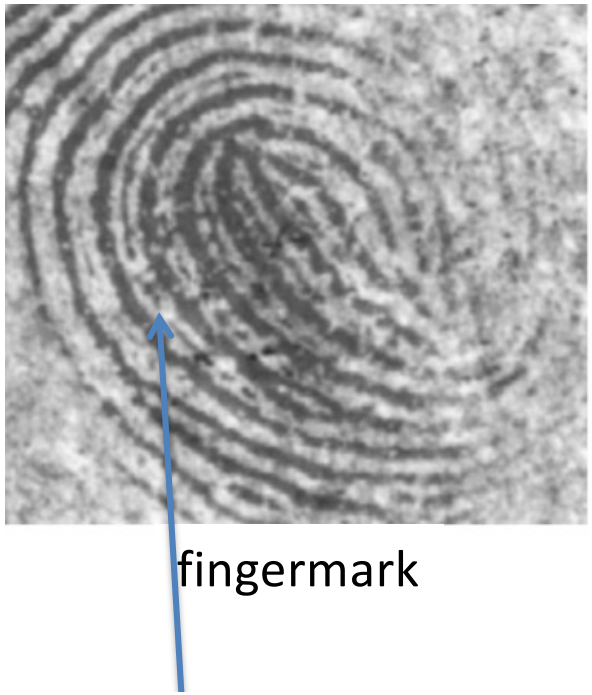


L3 Details

- Ridges are no longer viewed as being simple, one-pixel wide skeletal images
- Using the inner holes (**sweat pores**) and outer **contours** (edges)
- **Incipient** ridges are immature ridges, thinner than **mature** ridges and contain no sweat pores
- **Dot** is a very short ridge containing only a single ridge unit

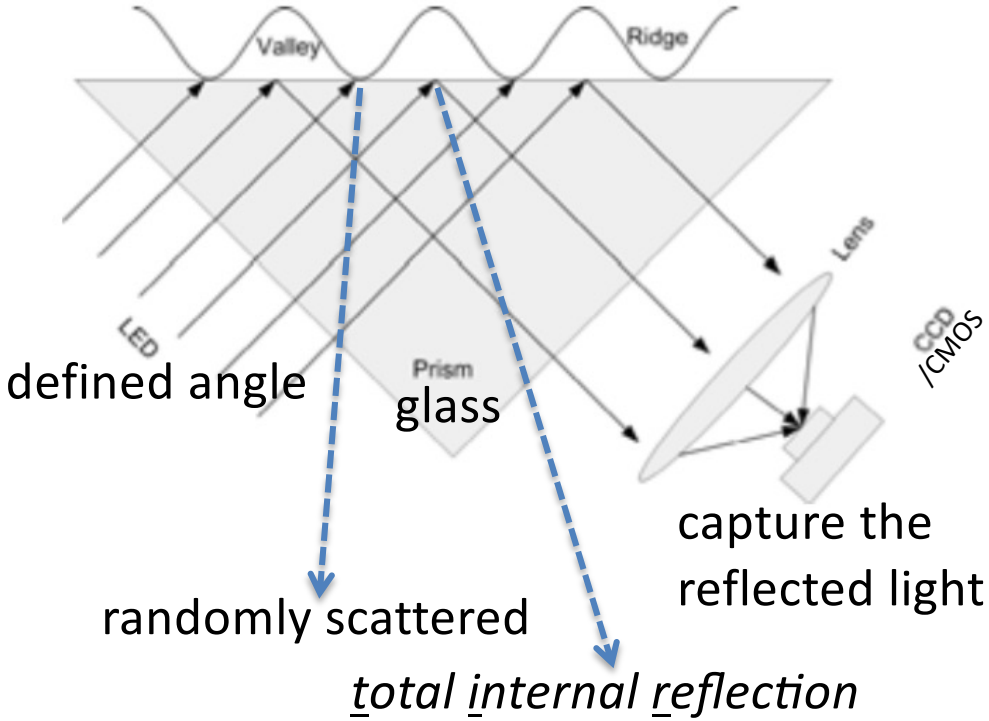
L3 Details

- 1000 ppi scanning capability is needed to capture L3 details
- importance in matching **fingermarks** – contain much fewer minutiae than rolled or plain ones

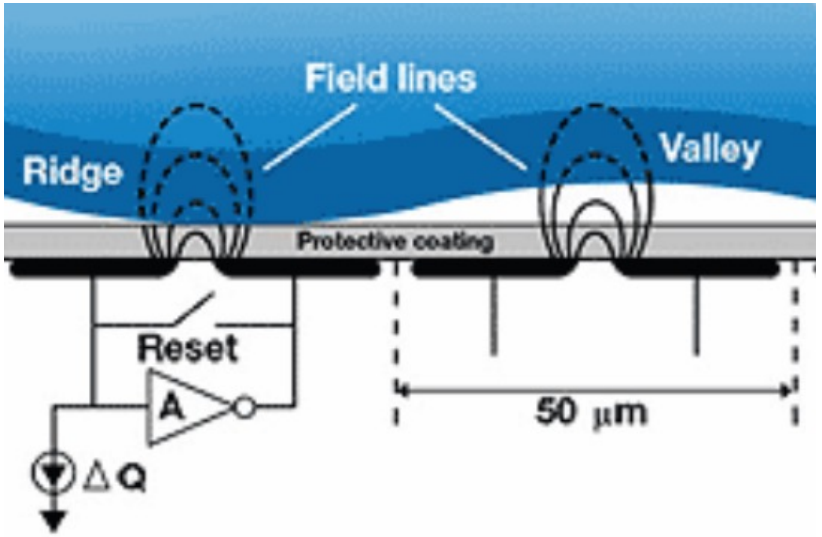


Live-Scan Sensors

- Optical FTIR-based sensor:
  - Difficult to have this arrangement in a compact form; focal length of small lenses can be very large
  - Image distortions are possible when the reflected light is not focused properly



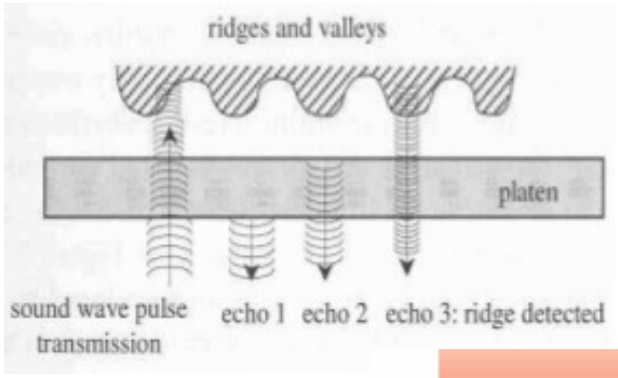
Live-Scan Sensors

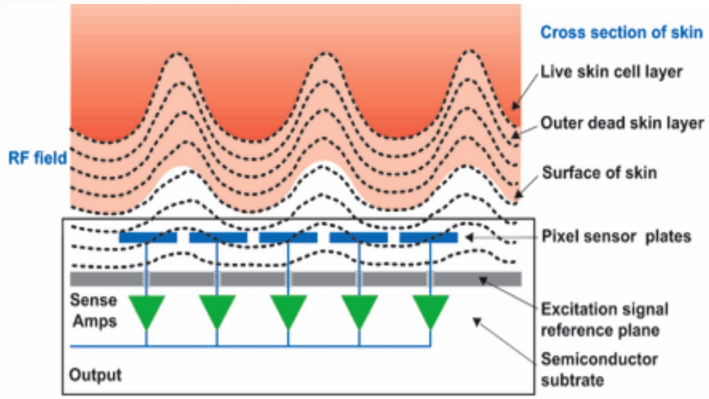


- Capacitance-based sensor:
  - More **commonly used**
  - Very small in size
  - An array of electrodes
  - Fingerprint skin acts as the other electrode, forming a miniature capacitor
  - Magnitude of electrical charges depends on the distance between the fingerprint surface and the capacitance plates
  - Capacitance due to the ridges is higher than those formed by valleys
  - Proper grounding needed due to tip electrostatic discharges

Live-Scan Sensors

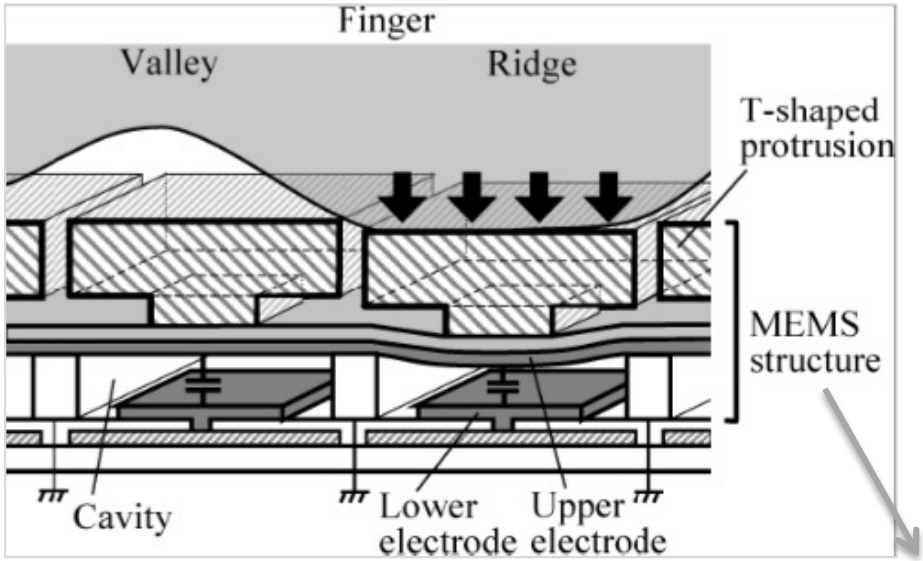
- Ultrasound reflection-based sensor:
  - Sends acoustic signals toward the tip and captures the echo signal
  - Computes the **range image**
  - Images the **sub**surface of the fingerprint =>
  - Resilient to dirt and oil accumulations
  - Expensive





Live-Scan Sensors

- Piezoelectric effect-based sensor:
  - **Pressure**-sensitive sensor
  - Produces an electrical signal when a **mechanical** stress is applied
  - Generates a small amount of **current** => piezoelectric effect
  - Does not capture the structure accurately because of its low sensitivity



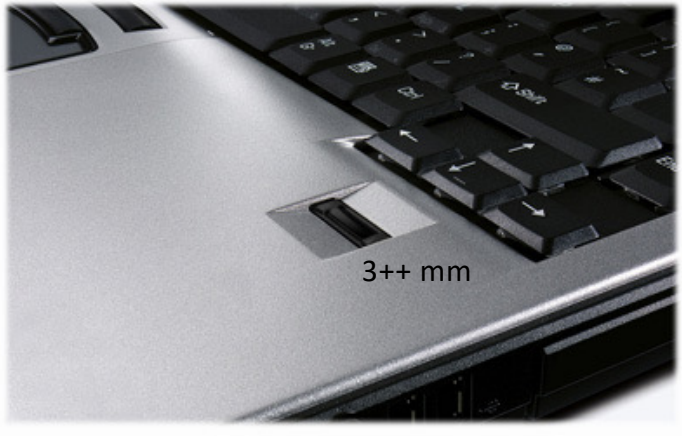
mechanical devices such as sensors, embedded in semiconductor chips

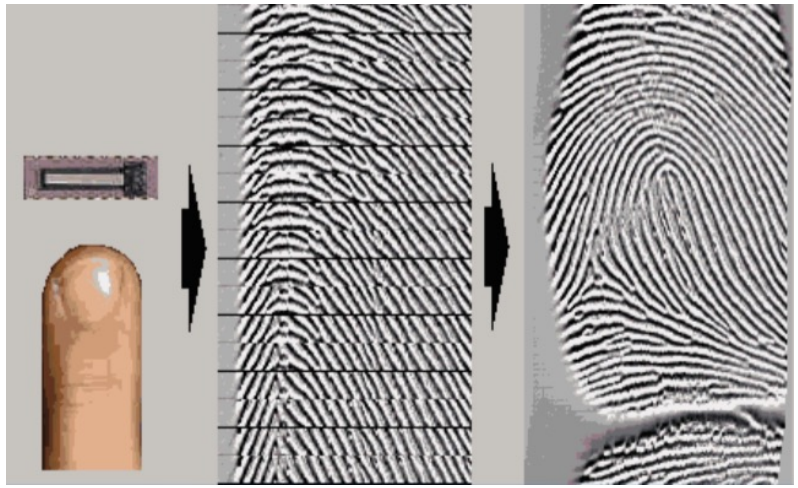
Live-Scan Sensors

- Temperature differential-based sensor:
  - Made of pyro-electric material that generates a current based on temperature differentials (lasts approx. 100 ms)
  - Sensors are typically maintained at a high temperature by electrically heating them up
  - Only four-bit grayscale from sensor (Atmel, 2005) ?!

Swipe Sensors

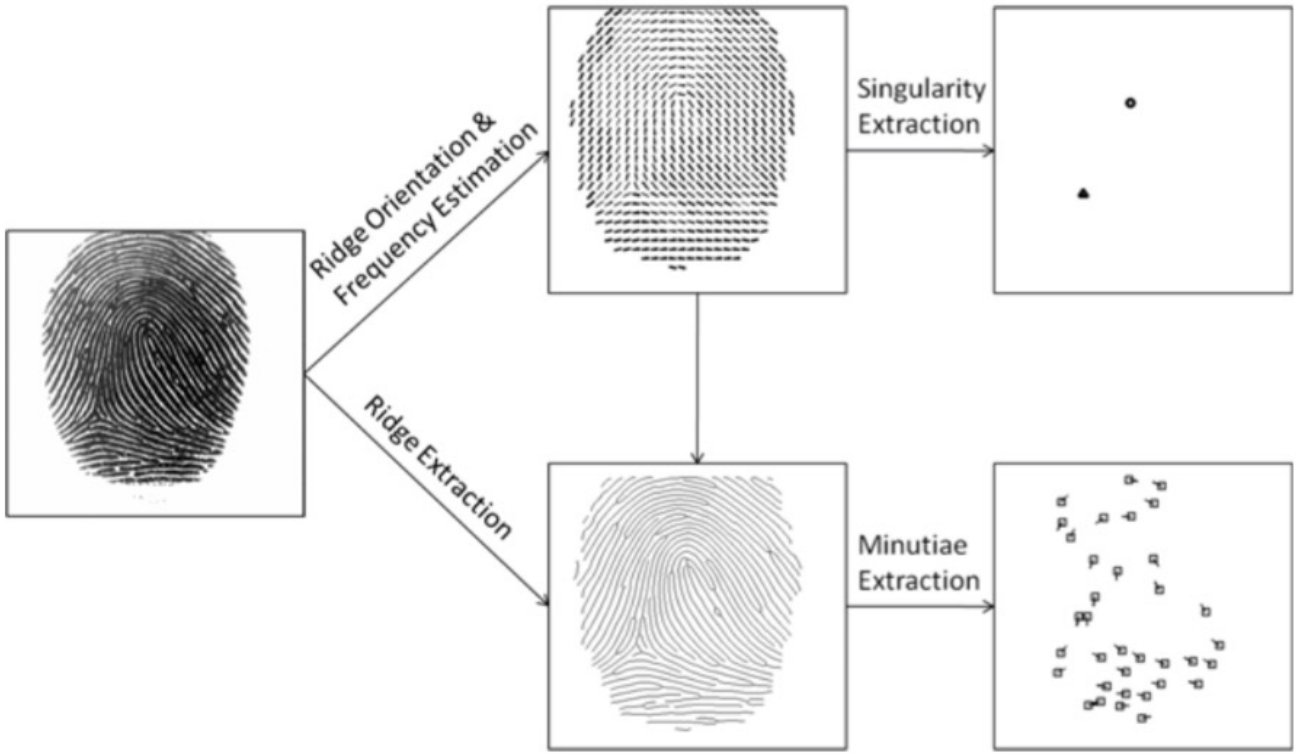
- Software stitching process
- May produce **artifacts**
  => unacceptable in forensic quality fingerprints





Feature Extraction

Commercial systems => L1&2 features



Feature Extraction

Ridge orientation and frequency estimation

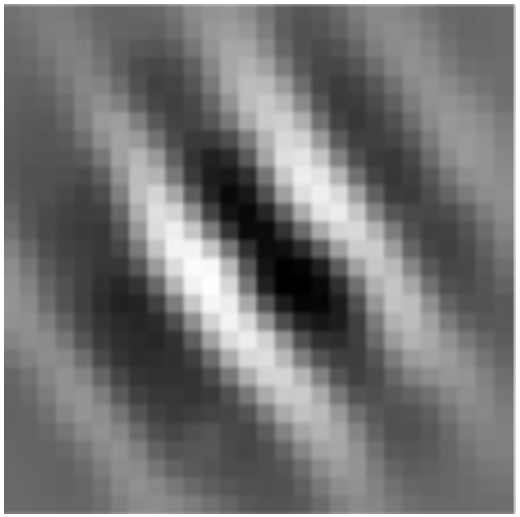
- Approximate ridge pattern in a local area by a cosine wave (defined by amplitude, **frequency** & **orientation** – parameters)
- Compute its 2D Fourier transform (using FFT)
- The wave parameters can be obtained by detecting the maximum value of the magnitude spectrum

- But …

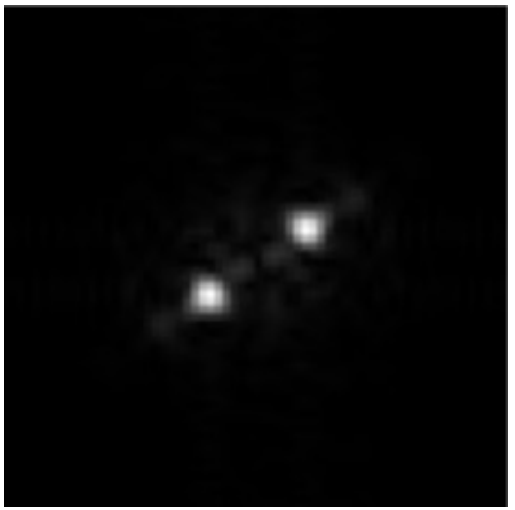
Feature Extraction

Ridge orientation and frequency estimation

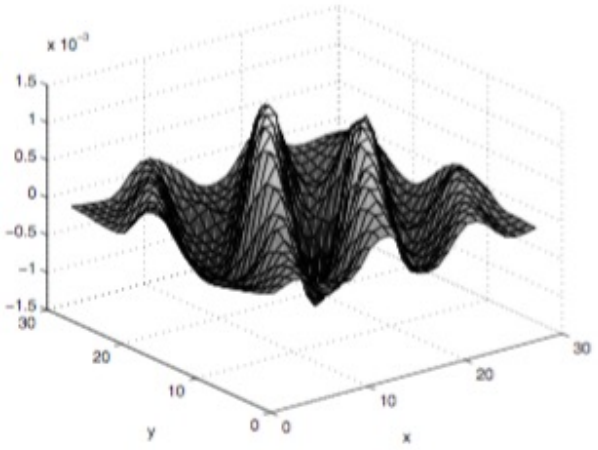
- But#1 … ridge pattern is not exactly a cosine wave
- FT gives a pair of blurred impulses
- **Magnitude spectrum** is first **smoothed** using a low pass filter and then the maximum value is detected => better parameters obtained



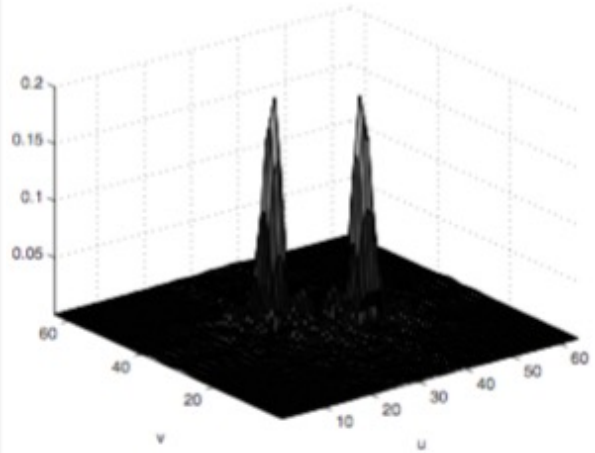
Local ridge



Magnitude spectrum



3D visualization

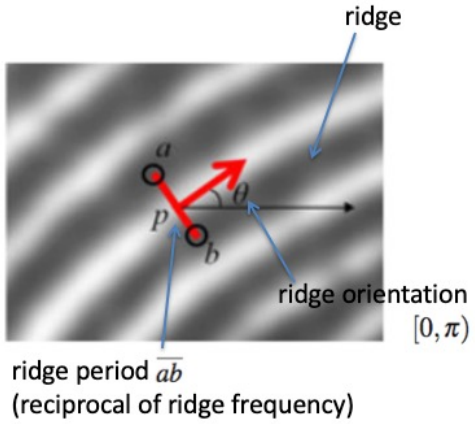


3D visualization

Feature Extraction

Ridge orientation and frequency estimation

- But#2 … problem of **creases**? => smoothing
- Ridge orientation of fingerprints in a local area is defined in the range [0,π) =>
  θ = (θ + π) => average between 1° and 179°
  should be 0° rather than 90°
- Smooth using a vector field
  $V = (V_x, V_y) = (\cos 2\theta, \sin 2\theta)$



input

initial (noisy) orientation field

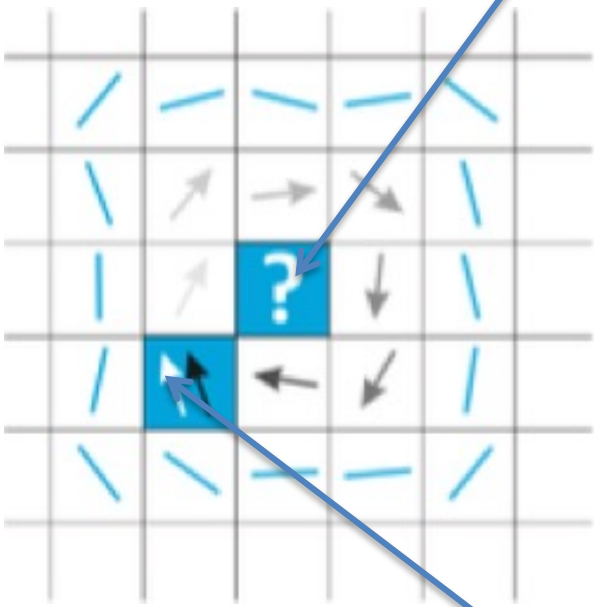
smoothed orientation field

Feature Extraction

Singularity extraction

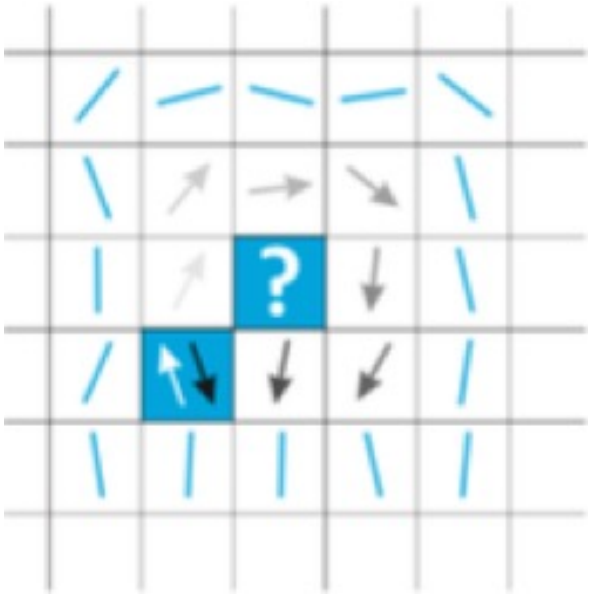
- Poincaré index method
- = cumulative change of orientations along a closed path in an orientation field
- Generally evaluated using the eight neighbors of a pixel

Calculating the Poincaré index >>>

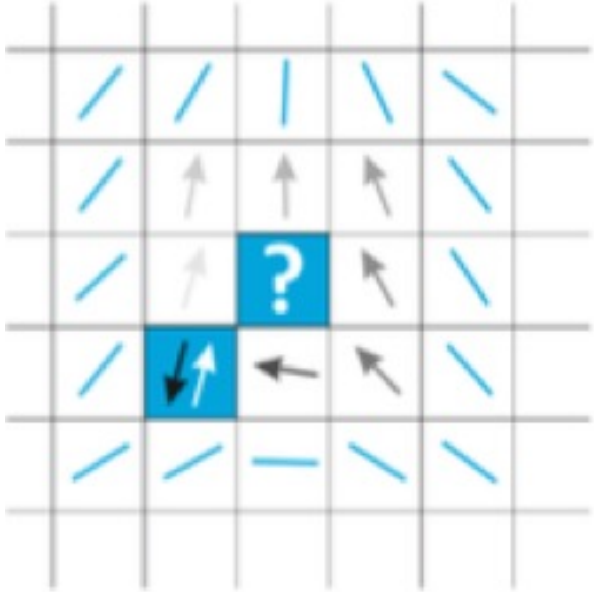


p = 360° (whorl)

(white and black arrow are in the same direction)



p = 180° (loop)



p = -180° (delta)

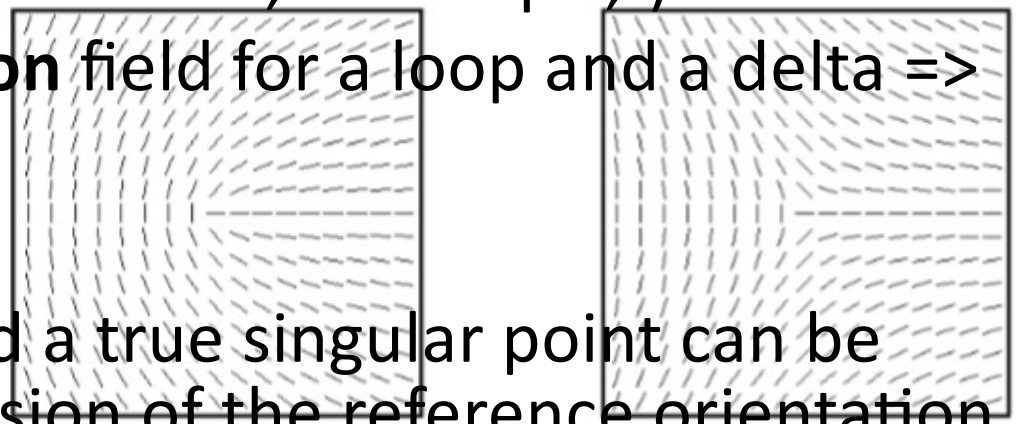
1. Select a random orientation for **one** example element (<u>white</u>) of the directional image
2. For the remaining elements select the **closest** orientation to the neighboring element
3. Travel along the closed curve and sum up the rotations

if 0° => no singular point

Feature Extraction

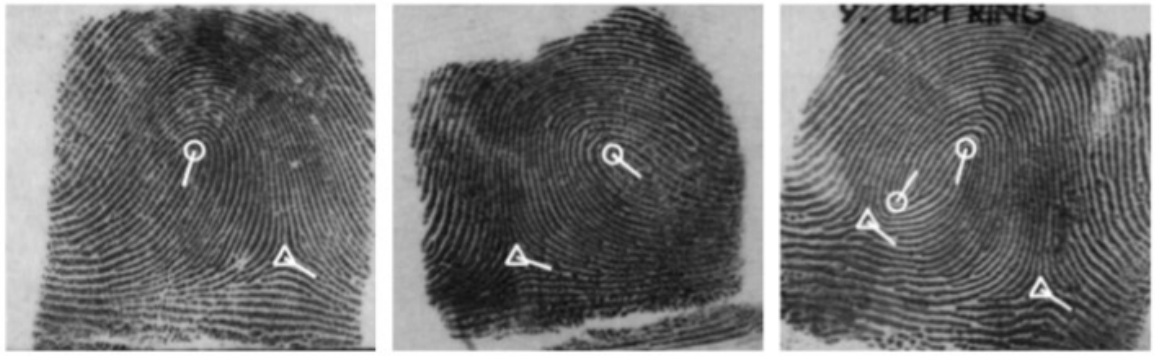
Direction of singularity

- Not absolutely needed for classification, but helps ;-)
- Define the **reference orientation** field for a loop and a delta =>



- The orientation field (α) around a true singular point can be **approximated** by a rotated version of the reference orientation field
- Calculate α based on the difference between the local orientation field around a singularity and the reference orientation field

Remember classes?



- Left loop, right loop, twin loop; plain arch, tented arch, whorl

- It requires the fingerprint to be **complete** (i.e. a good quality rolled fingerprint)
- **Fingermark** obtained from crime scenes is typically considered as **unknown** and fingermarks are often searched against all types of fingerprints in the database

Feature Extraction

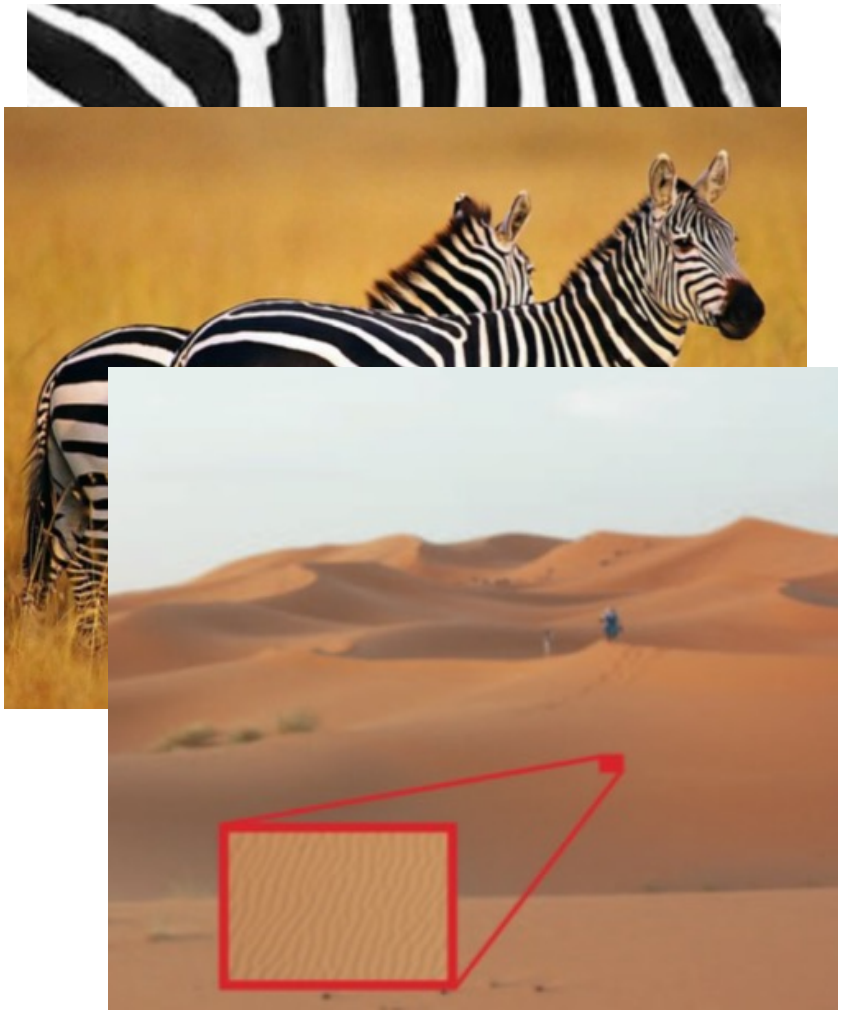
Ridge extraction (to get minutiae ;-)

[PDF] A threshold selection method from gray-level histograms
N **Otsu** - IEEE transactions on systems, man, and cybernetics, 1979 - cw.fel.cvut.cz
A nonparametric and unsupervised method of automa-tic threshold selection for picture segmentation is presented. An optimal threshold is selected by the discriminant criterion,
☆ Save 99 Cite Cited by 50678 Related articles All 29 versions »
segmentation is presented. An optimal threshold is selected by the discriminant criterion, r
☆ Save 99 Cite Cited by 47424 Related articles All 34 versions »

- Enhanced image is converted to a **binary** image by using either a global threshold (Otsu thresholding) or thresholds that are locally computed
- A morphological operation **thinning** is used to reduce the ridge width to one pixel

Feature Extraction

What is this?

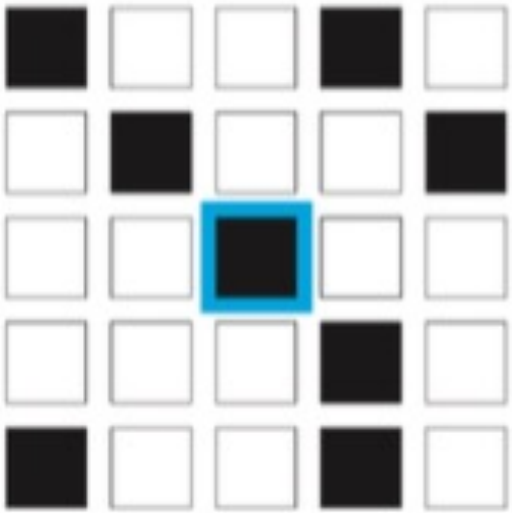


Minutiae extraction

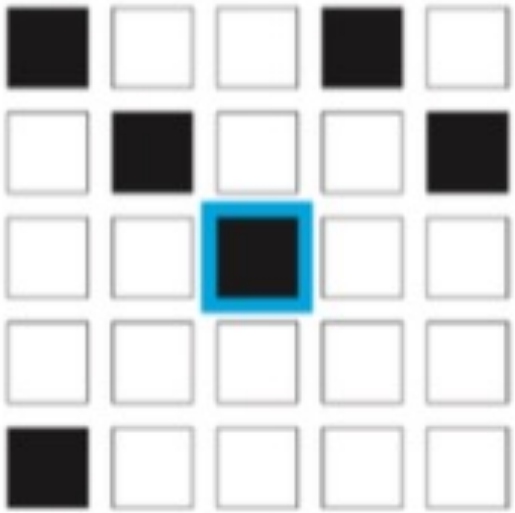
- Ridge pixels with 3 ridge pixel neighbors are identified as ridge **bifurcations**
- Those with only one ridge pixel neighbor are identified as ridge **endings**

Feature Extraction

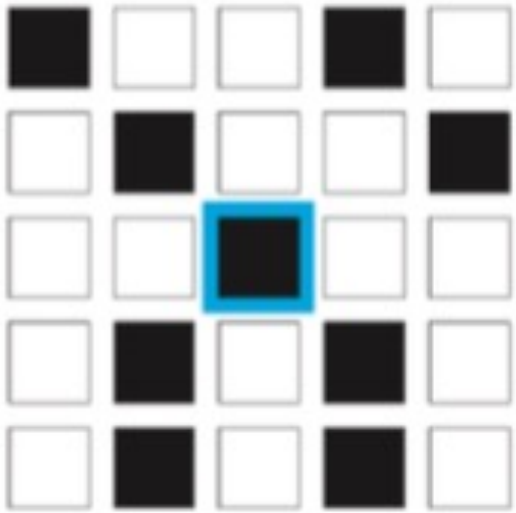
Minutiae extraction



#ridge_neighbors=2 =>
ridge (no minutiae)



=1 => ridge ending

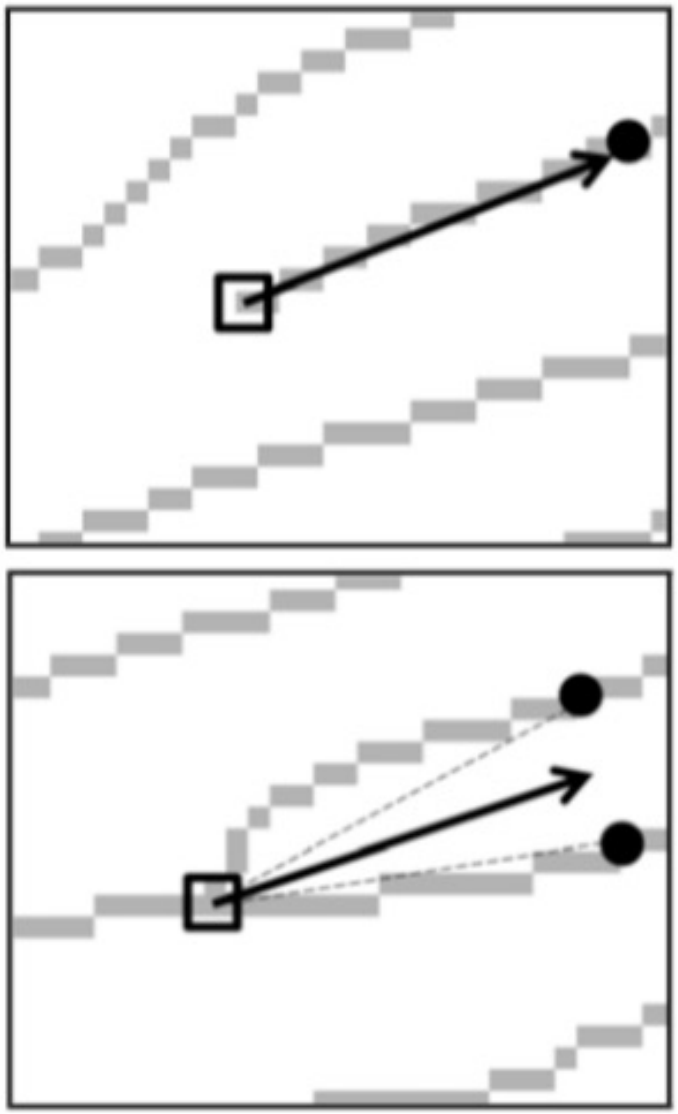


=3 => bifurcation

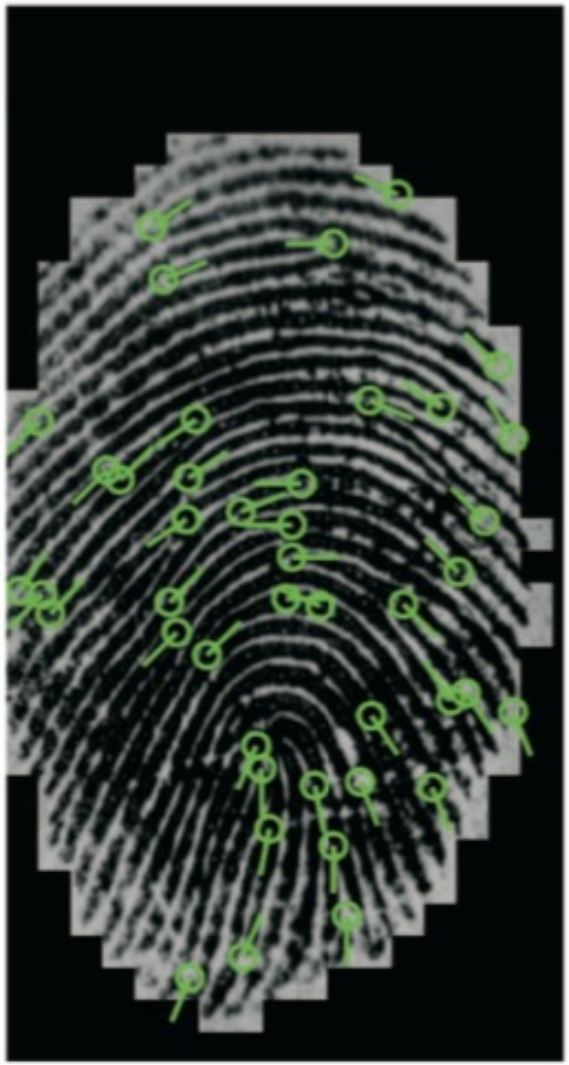
Feature Extraction

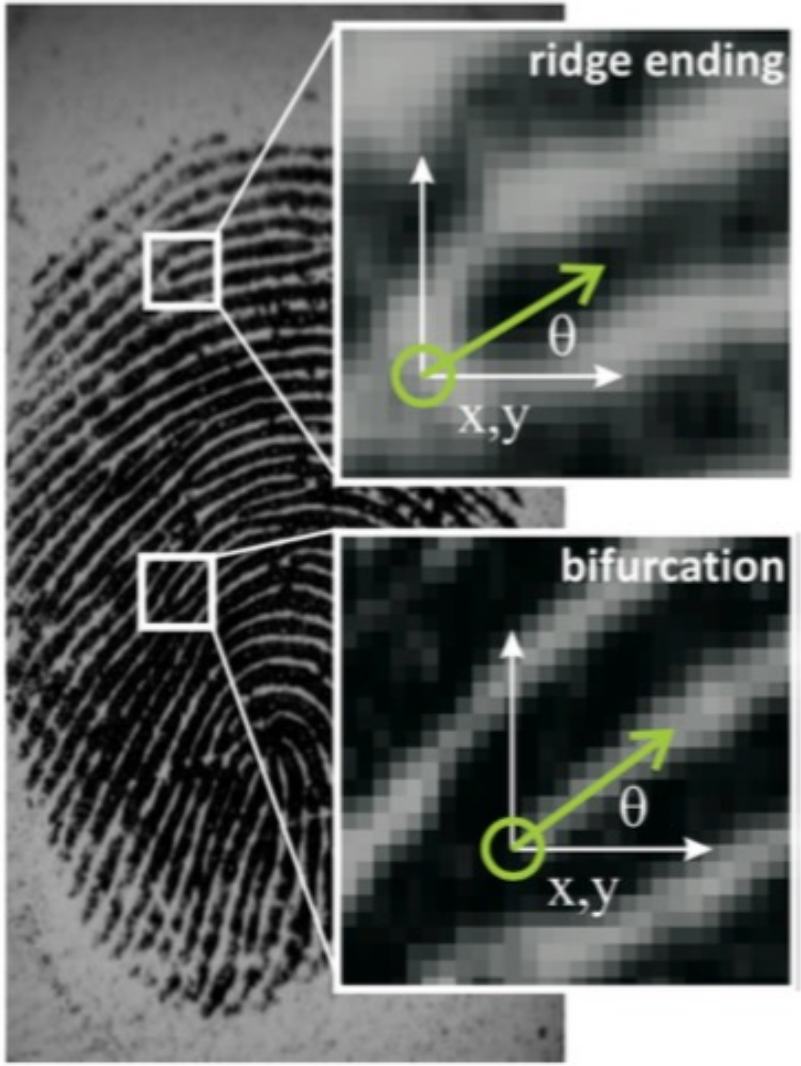
Minutiae **direction**

- Ending => **trace** the associated ridge to a fixed distance (say 10 pixels)
- Bifurcation => obtain three points by tracing the ridges to a fixed distance & calculate mean of the two directions, which difference is the **smallest** among the three ridges

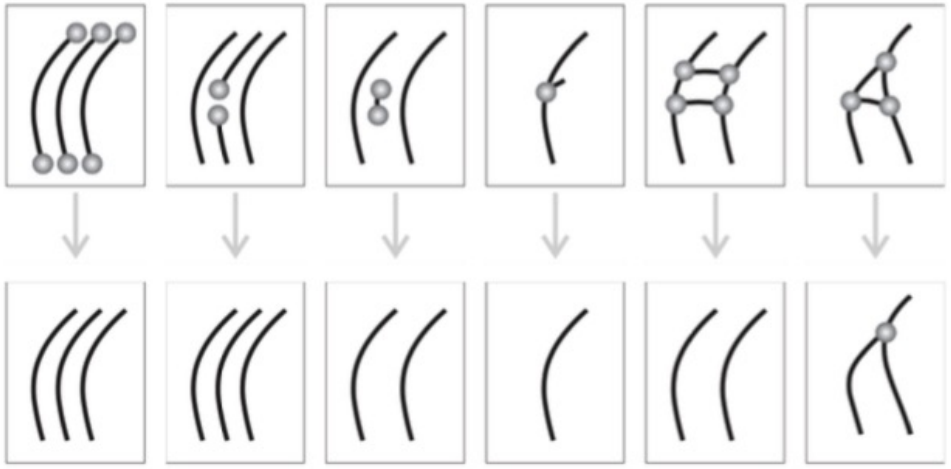


Importance of standards:
Minutiae directions compliant to ISO/IEC 19794-2:2011





Feature Extraction



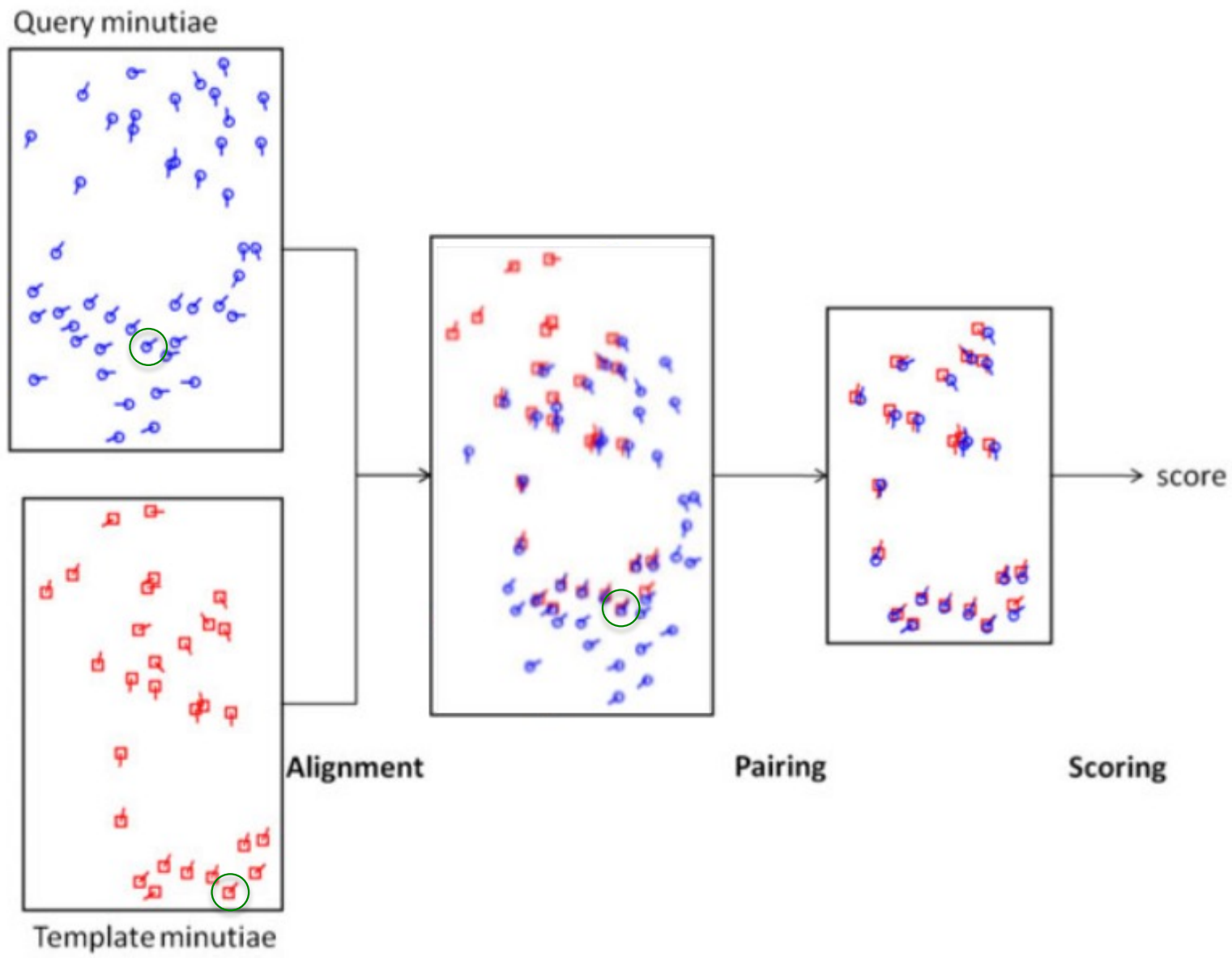
Minutiae **filtering**

- Removal of spurious minutiae
- Heuristic methods:
  remove minutiae located at boundaries,
  that are close in location and almost opposite in direction,
  too many minutiae in a small neighborhood
- Duality of minutiae: simultaneous processing of positive and negative images

Alignment … Matching

1. Alignment: determine the geometric transformation between the template & query minutiae sets so that they are in **the same coordinate system**

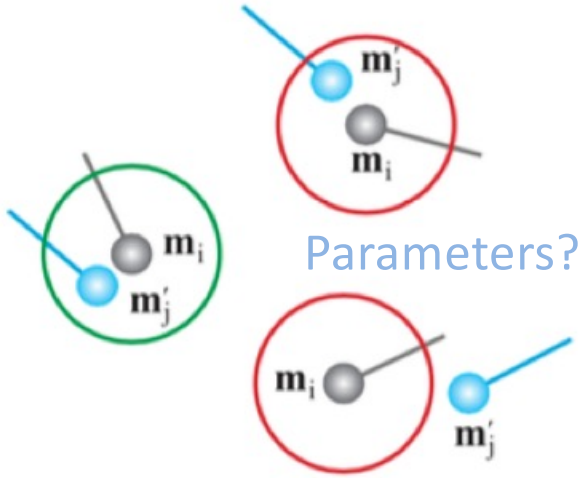
2. Correspondence: form **pairs** of corresponding minutiae
3. Score generation: compute the **match score** based on the corresponding minutiae points


Query minutiae
Template minutiae
Alignment
Pairing
Scoring
score

Alignment … Matching

Alignment/registration

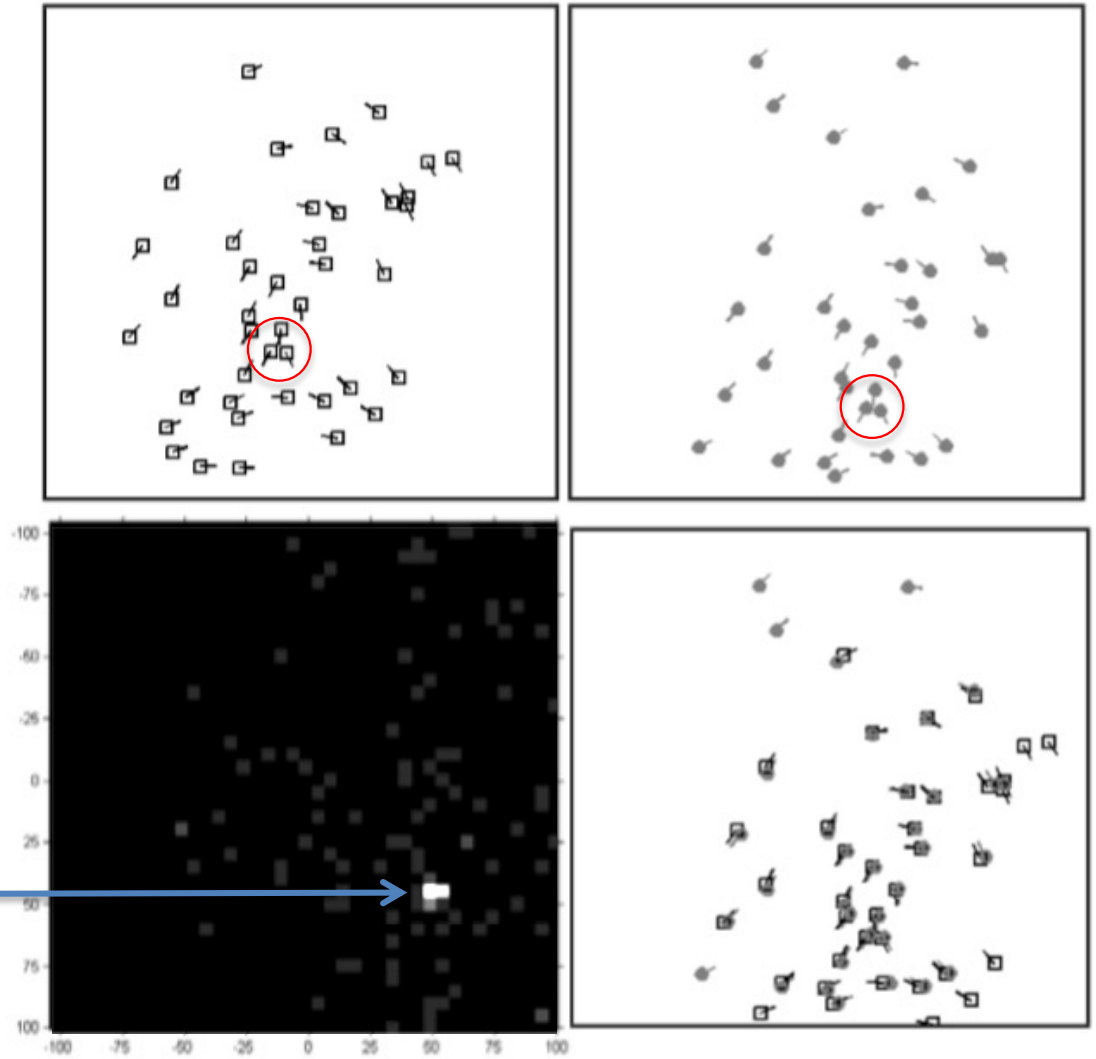
- Transforms one image in such a way that it is geometrically aligned with the other
- Rigid transformation is normally sufficient:
  linear mapping +
  compensation for non-linear deformations (e.g. use of tolerance “box”)



- Use **generalized Hough transform** for estimating the spatial transformation between two point sets =>

input : Two minutiae sets $\{x_i^T, y_i^T, \theta_i^T\}_{i=1}^{M}$ and $\{x_j^Q, y_j^Q, \theta_j^Q\}_{j=1}^{N}$
output: Transformation parameters
Initialize accumulator array A to 0
for $i = 1, 2, \cdots, M$ **do**
    for $j = 1, 2, \cdots, N$ **do**
        $\Delta\theta = \theta_i^T - \theta_j^Q$
        $\Delta x = x_i^T - x_j^Q cos(\Delta\theta) - y_j^Q sin(\Delta\theta)$
        $\Delta y = y_i^T + x_j^Q sin(\Delta\theta) - y_j^Q cos(\Delta\theta)$
        $A[\Delta\theta][\Delta x][\Delta y] = A[\Delta\theta][\Delta x][\Delta y] + 1$
    end
end
return location of peak in A

Simplified example for visualization purposes when only translation is in question (2D case) =>



receives the most votes

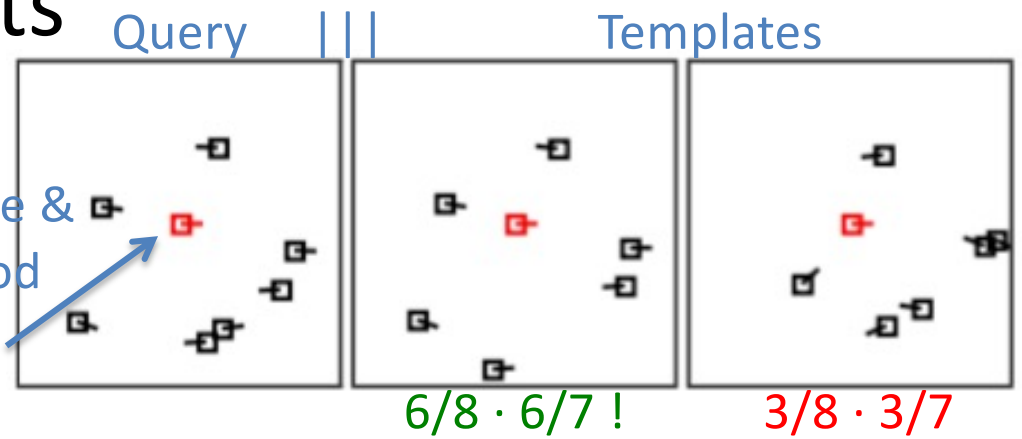
Alignment … Matching

Useful modifications for HT

- Add fourth dimension to accumulator array for **scale**
- Simultaneous weighted incrementation of location in array and its neighbors
- Parallelization of HT is easy
- Coarse-to-fine approaches

Alignment … Matching

Alignment … if the overlapping area between two fingerprints contains very **few** minutiae (fingermarks?!!)



1. Define minutia descriptor
2. Pairing of neighboring minutiae in it is established (similar as on the next slide)
3. Compute the product of the percentages of matched minutiae

Matching

Pairing minutiae

- Minutia *a* is in correspondence with *b* if their distance is within a predefined **distance** threshold (e.g. 15 pixels)
- And the angle between their directions is within another predefined **angle** threshold (e.g. 20 degrees)
- *a* is allowed to match to **at most one** *b* and vice versa (use of two flag arrays) =>

input : Two minutiae sets $\{x_i^T, y_i^T, \theta_i^T\}_{i=1}^{M}$ and $\{x_j^Q, y_j^Q, \theta_j^Q\}_{j=1}^{N}$;
Transformation parameters $(\Delta\theta, \Delta x, \Delta y)$
output: List of paired minutiae
Initialize: set flag arrays f^T, f^Q, and *count* as 0; *list* as empty
for $i = 1, 2, \cdots, M$ **do**
  for $j = 1, 2, \cdots, N$ **do**
    if $f^T[i] == 0$ & $f^Q[j] == 0$ & *distance between minutiae i and* $j < t_d$
    & *rotation between them* $< t_r$ **then**
      $f^T[i] = 1$
      $f^Q[j] = 1$
      $count = count + 1$
      $list[count] = \{i, j\}$
    end
  end
end
return *list*

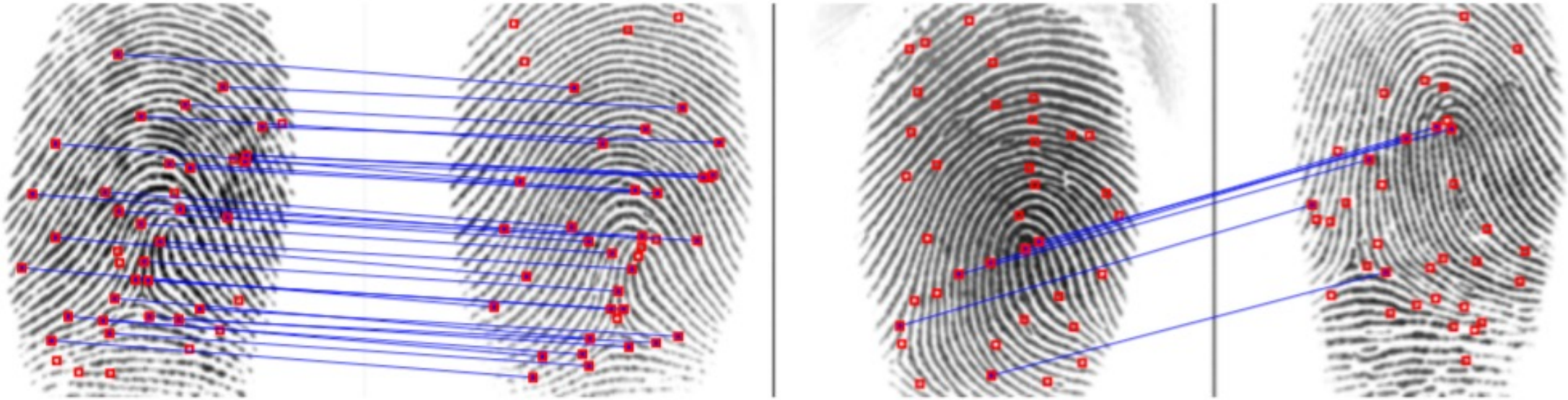
Matching

Match score

- Number of paired minutiae
- Percentage of matched minutiae in the overlap area
- Number of paired minutiae divided by average number of minutiae in both maps

- Then compare to a predefined **threshold** to classify the two fingerprints as a **genuine** or an **impostor** match (two-class classification)

Neurotechnology VeriFinger SDK 4.2, commercial matcher



31 matched minutiae

6 matched minutiae

match score is computed as some function of the number of matched minutiae and *some other parameters that are proprietary* to the commercial matcher

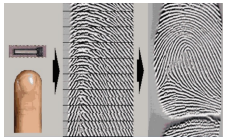
match score = 614

match score = 7

More reading material?

Bule, Peer, Fingerprint matching, ROSUS 2012.
Pohar, Bule, Peer, Ridge-based fingerprint matching, ROSUS 2013.

Performance evaluation

Evaluation	Data	Best reported accuracies
NIST FpVTE 2003 (MST)	10,000 plain fingerprints	FNMR = 0.6% at FMR = 0.1%
FVC2006 	140 fingers, 12 images per finger Electric field sensor (250 ppi)	FNMR = 15% at FMR = 0.1%
	Optical sensor (569 ppi)	FNMR = 0.02% at FMR = 0.1%
	Sweep sensor (500 ppi)	FNMR = 3% at FMR = 0.1%
NIST ELFT 2008 (Phase II)	835 latent prints, 100,000 rolled fingerprints	FNMR = 8% at FMR = 1%

Note: variability due to data characteristics (e.g. number of subjects, demographic distribution), sensors, environments (including indoor/outdoor, temperature, humidity), user interactions with the sensor, specific application (i.e. Disney World is using low FNMR)

=> direct comparison possible only on the same DB!!!

Image Quality

- Quality factors: image **resolution**, finger **area**, **clarity** of ridge pattern

- 500 ppi (L2) => distance between adjacent ridges is approximately 9 pixels
- Civilian applications => sensors with <500 ppi are often used to reduce the cost

Image Quality – Area

- The finger has to be rolled on the sensor surface to obtain the full fingerprint
- In consumer electronic products => swipe sensors to further reduce the cost

Image Quality – Clarity

- If finger is dry, the ridges may have many breaks
- If moist, adjacent ridges may be joined

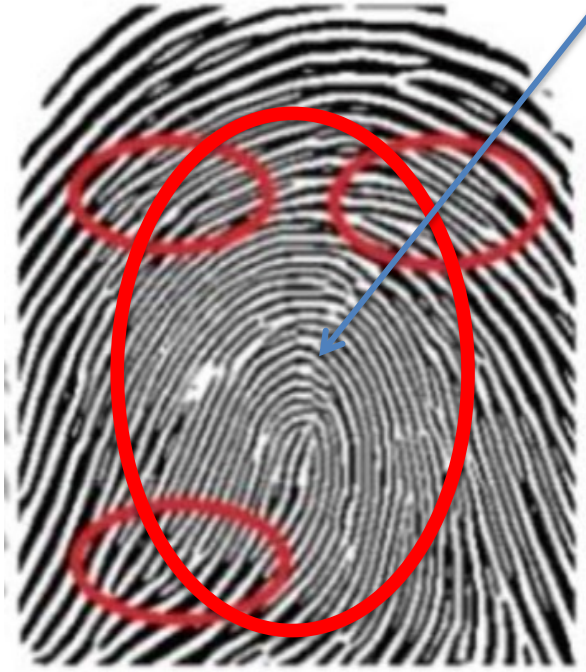
dry finger

wet finger

many creases

How to Assess the Quality?

Is this one good or not?



Where do we have a problem?

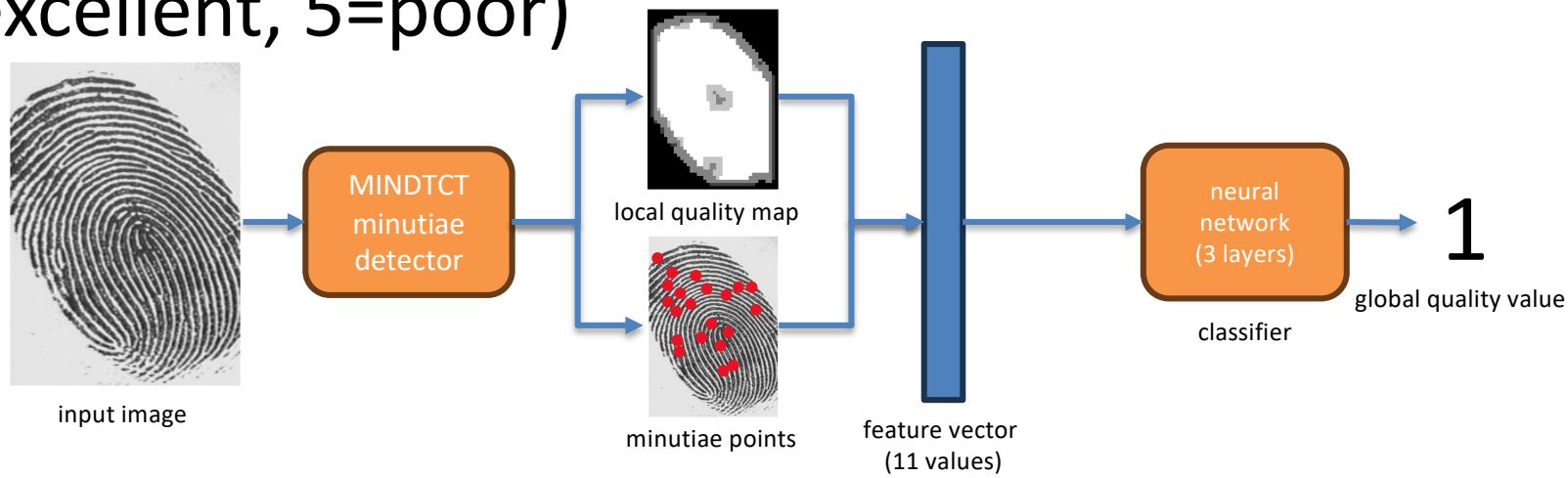
Image Quality Assessment

With NFIQ 2

NIST Fingerprint Image Quality

History – NFIQ

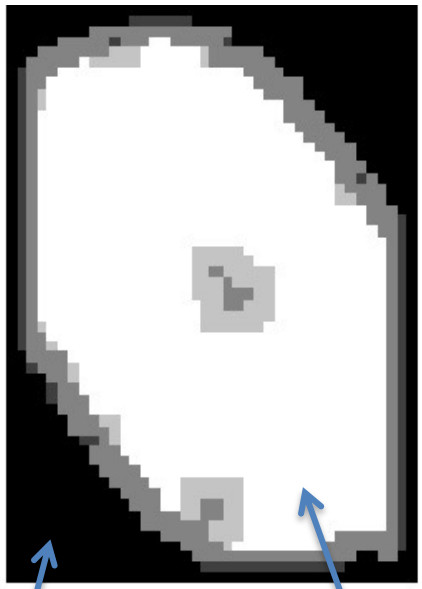
- NIST Fingerprint Image Quality (2004)
  - Quality value predicts recognition performance
  - Open-source
- 5 quality values for the whole input (1=excellent, 5=poor)



How to Evaluate the Local Quality?

quality map
class 1 (excellent)

(a)



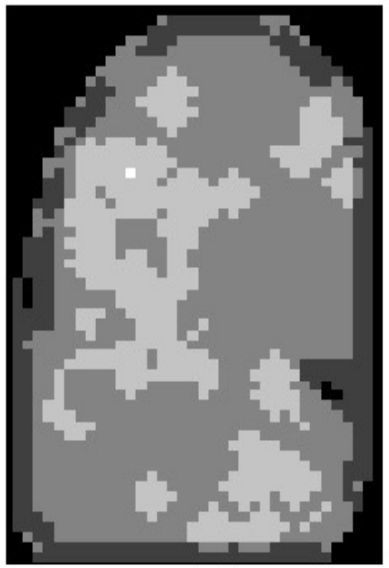
(b)

background –
block quality 0

block quality 4
(best)

(a)

class 4



(b)

=> Registration with only class 1 and 2 quality is acceptable

	NAME	DESCRIPTION
1	**foreground**	number of blocks that are quality 1 or better; i.e. foreground = $\sum_{i=1}^{4} U_i$ where U_i is number of blocks with quality i
2	**total #of minutia**	number of total minutiae found in the fingerprint
3	**min05**	number of minutiae that have quality 0.5 or better
4	**min06**	number of minutiae that have quality 0.6 or better
5	**min075**	number of minutiae that have quality 0.75 or better
6	**min08**	number of minutiae that have quality 0.8 or better
7	**min09**	number of minutiae that have quality 0.9 or better
8	**quality zone 1**	percentage of the foreground blocks of quality map with quality =1
9	**quality zone 2**	percentage of the foreground blocks of quality map with quality =2
10	**quality zone 3**	percentage of the foreground blocks of quality map with quality =3
11	**quality zone 4**	percentage of the foreground blocks of quality map with quality =4

Table 3. Feature vector description

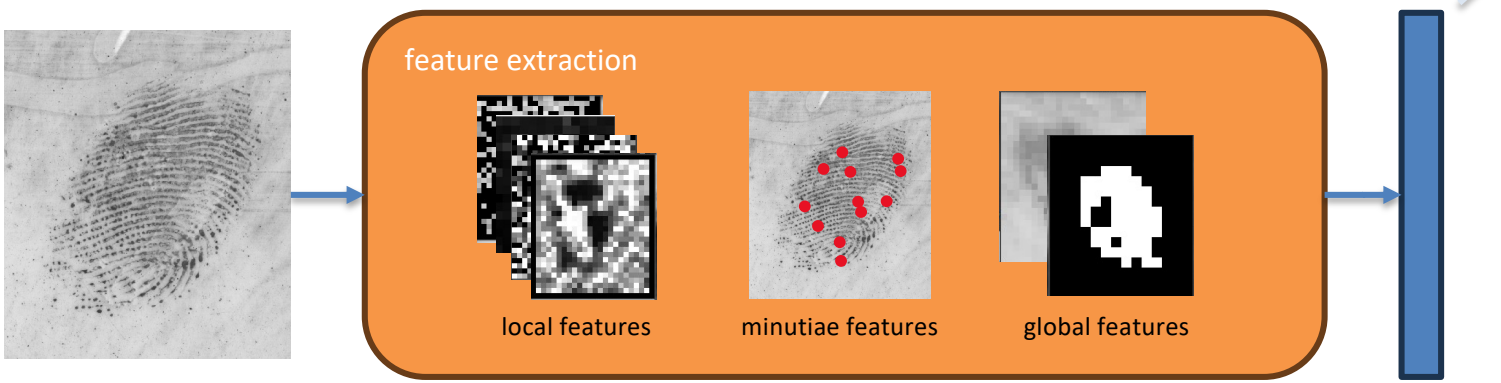
Development of NFIQ 2

- NIST starts development of NFIQ 2 (2010)
  - Developed in collaboration with industry and stakeholders
- Addresses disadvantages of original NFIQ
  - No single failure point (more feature extractors)
  - Increased robustness (random forest classifier)
  - Expanded quality range
- Reference implementation of an ISO standard (ISO/IEC 29794-4)
- Use-cases: IDs, Schengen Information System, criminal AFIS, airport control, etc.
- Still open-source!

Feature Extraction

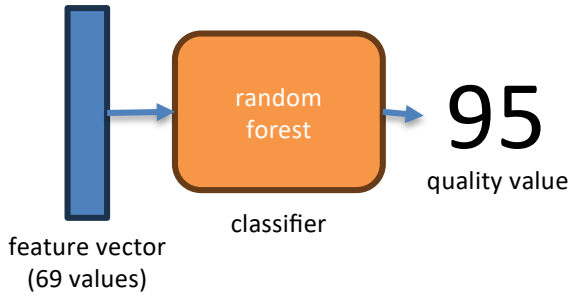
- Quality components (defined in ISO/IEC 29794-4)
  - Local features (sliding window)
  - Global features
  - Minutiae points (FingerJetFX detector)
- Fixed length feature vector => histograms!

$$\mathbf{Q}_{\text{NFIQ 2}} = \Big(Q^{\mu}_{\text{FDA}}, Q^{\mu}_{\text{LCS}}, Q^{\mu}_{\text{OCL}}, Q^{\mu}_{\text{OFL}}, Q^{\mu}_{\text{RVU}}, Q^{\sigma}_{\text{FDA}}, Q^{\sigma}_{\text{LCS}}, Q^{\sigma}_{\text{OCL}}, Q^{\sigma}_{\text{OFL}}, Q^{\sigma}_{\text{RVU}}, \mathbf{Q}_{\text{FDA}}, \mathbf{Q}_{\text{LCS}}, \mathbf{Q}_{\text{OCL}}, \mathbf{Q}_{\text{OFL}}, \mathbf{Q}_{\text{RVU}}, Q_{\text{MU}}, Q_{\text{MMB}}, Q^{rel}_{\text{COH}}, Q^{sum}_{\text{COH}}, Q^{\mu}_{\text{AREA}}, Q^{cnt}_{\text{MIN}}, Q^{com}_{\text{MIN}}, Q^{mu}_{\text{MIN}}, Q^{ocl}_{\text{MIN}}\Big).$$



Random Forest Classifier

- Random forest with 100 binary classifiers
- Trained on 2 classes:
  - Class 0 – only excellent fingerprints (NFIQ = 1)
  - Class 1 – only poor fingerprints (NFIQ = 5)
- Quality score = number of trees predicting Class 0
- Range of quality [0 - 100]



97 77 57 37 17 0

NFIQ 2 Versions

2.0 (2016, initial release)

- Written in C++
- Uses open-source

2.1 (June 2021)

- Added command line interface
- Output formatting options
- Enforced 500 PPI for input images
- Pre-compiled for multiple platforms
- Bugfixes

2.2 (November 2021)

- Pre-processing bugfixes, improvements
- Official report published by NIST (IR 8382)
- Support for Raspberry Pi OS

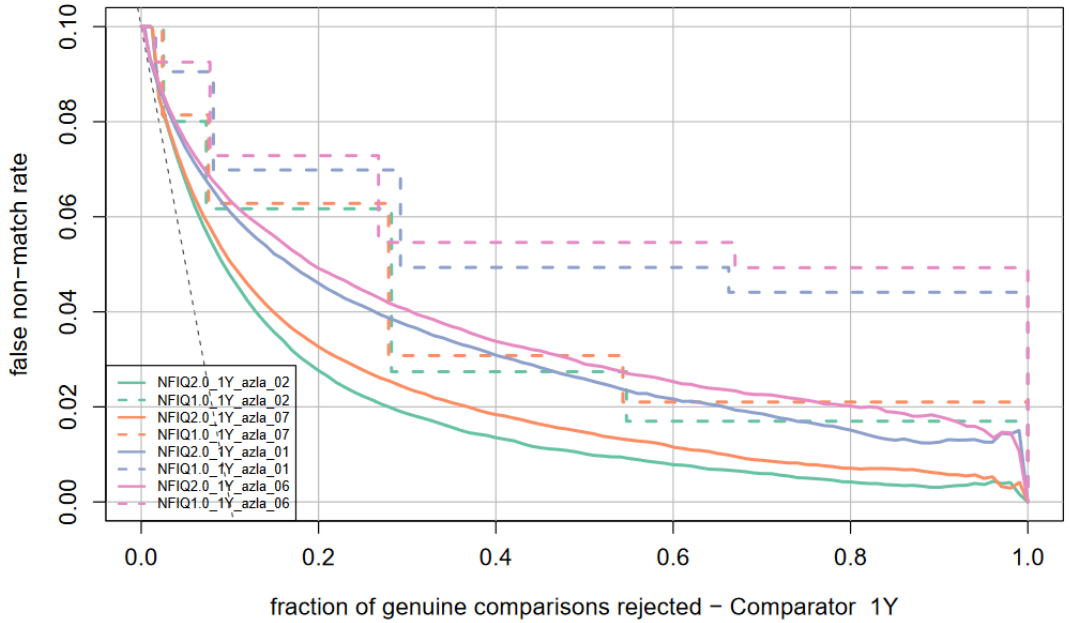
2.3 (October 2024, latest)

- Feature extraction bugfixes
- Model retraining
- Faster, smaller model
- Updated dependencies
- Better alignment with ISO/IEC 29794-4

Performance

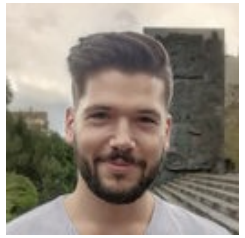
- Performs best on contact-based optical sensor images
- Questionable performance on other sensors
  - Contactless
  - Forensic fingermarks

- Significant improvement over NFIQ 1
- Considered SOTA and industry proven to this day



NIST website for NFIQ 2 (https://www.nist.gov/services-resources/software/nfiq-2) + ISO/IEC 29794-4:2024 (Finger image data)

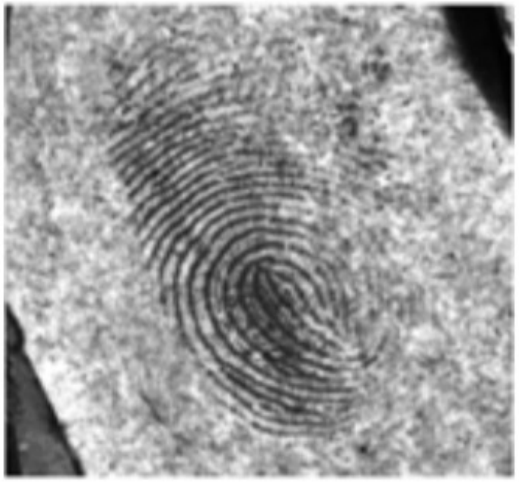
Open Research Questions



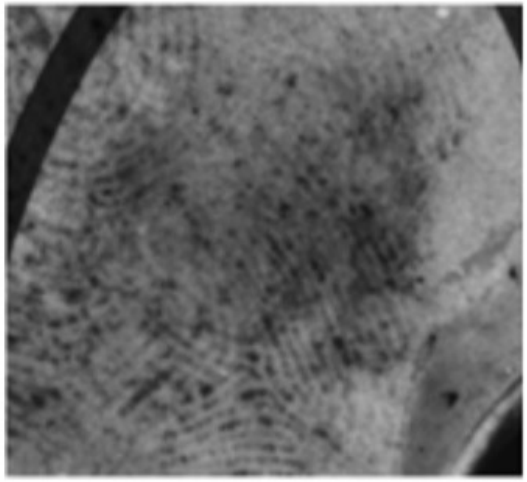
Fingermarks/latents matching

- Poor image quality, small finger area, large(r) nonlinear distortion of most fingermarks

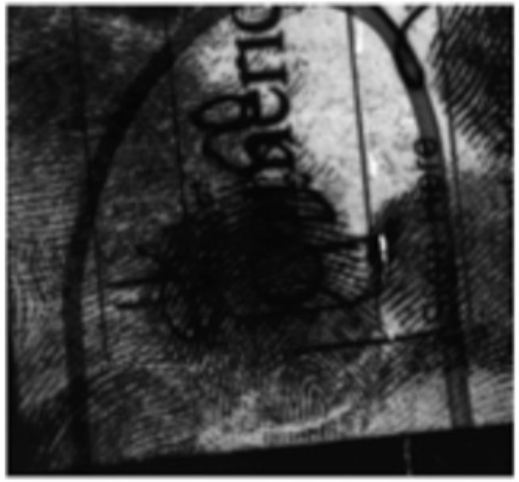
partial print



unclear structure



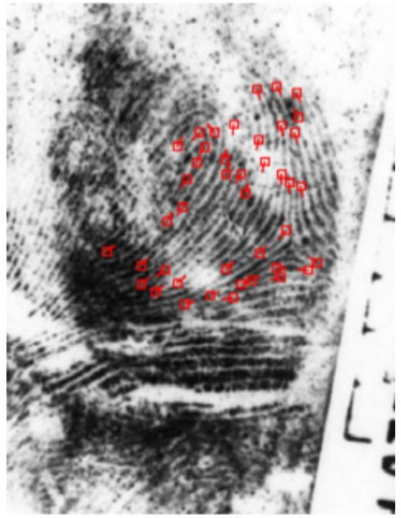
overlapped prints

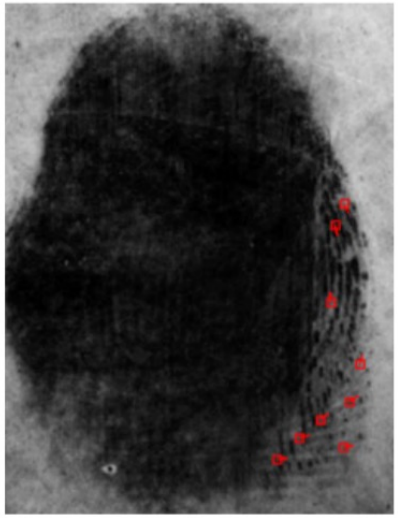


overlapped with complex background

Open Research Questions

NIST SD27 latents DB:
5..82 minutiae, avg=**21**





Fingermark matching

- State-of-the-art feature extractors do not work well => semi-manual extraction
- Limited number of minutiae => L3 features
- Main challenge is how to reliably encode and match L3 features in poor quality fingermarks

Open Research Questions

Fingermark matching

- **Daubert standard** for source of forensic evidence (**admissibility of scientific testimony**):
  1. underlying scientific basis should be accepted widely
  2. its **error rate** should be known **=>**
- Problem: estimate involves human factors (fingermark development, minutia extraction,…) for which the error rate is **not** known!
- Solution: keep improving the performance of automated fingerprint systems and ultimately replace human experts with automated systems ☺

Open Research Questions

Anti-spoofing / attack detection

Reconstruction of a fingerprint from features

Acquisition of 3D fingerprints in a touchless mode

Persistence and uniqueness over time

Age 34

Age 40

Age 42

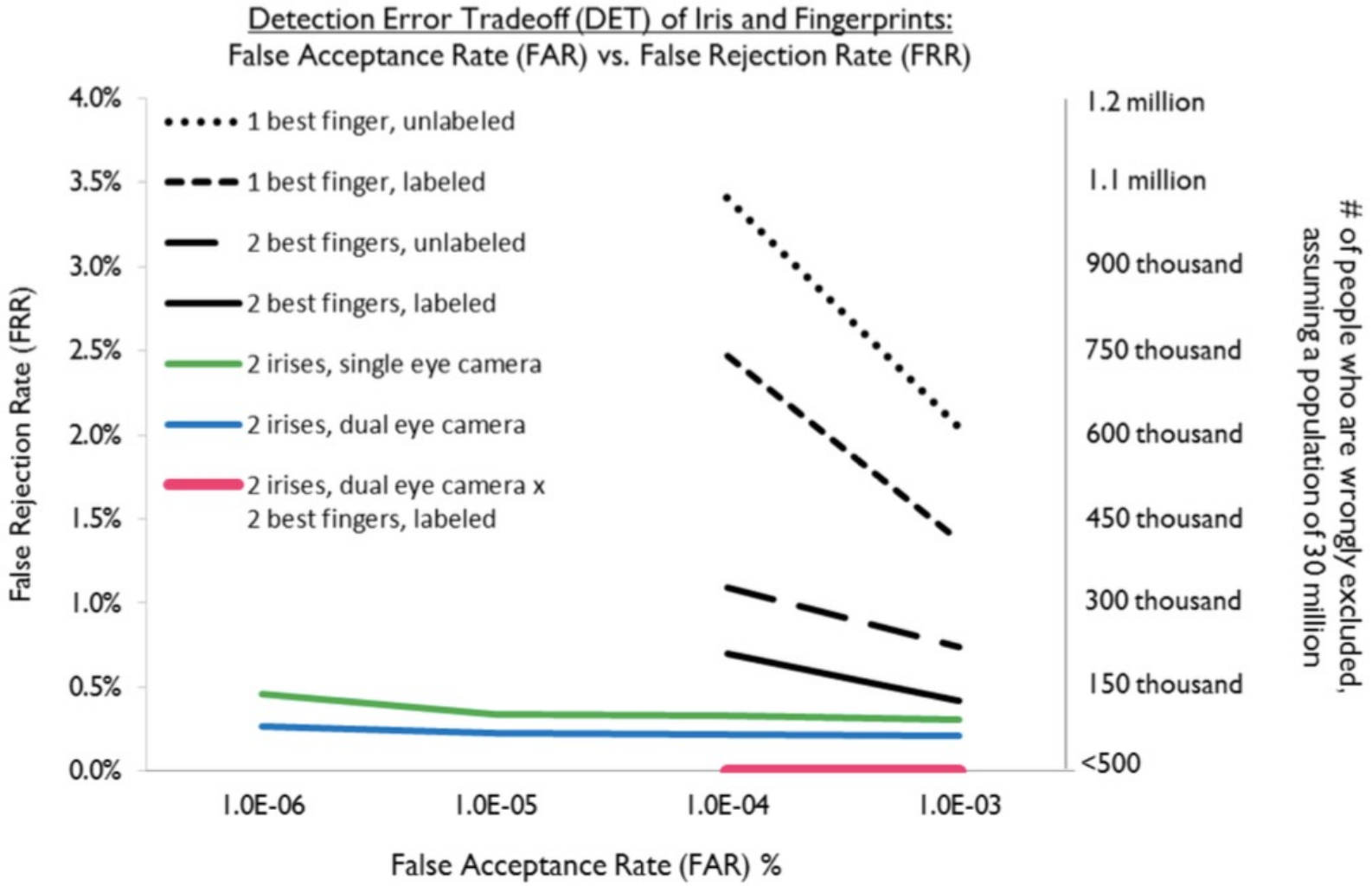
Bigger time interval between prints => <u>decreased</u> genuine match score

As subject's age increases /or/ image quality decreases => <u>decreased</u> GMS

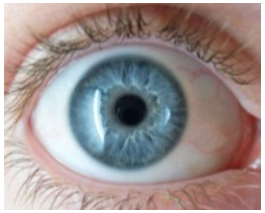
Probability of true acceptance at operational FAR settings <u>remains</u> close to 1

But with poor quality the uncertainty(!) becomes large

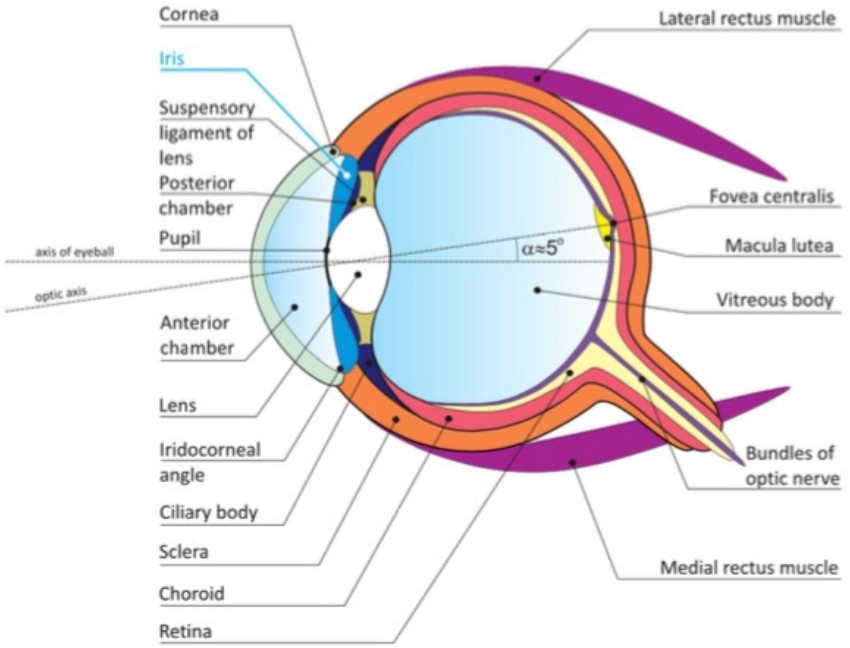
Motivation for the Next Modality ☺

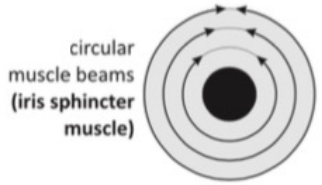


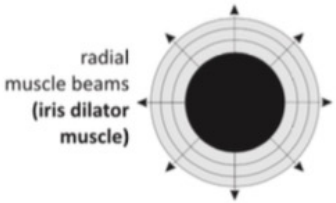
IRIS

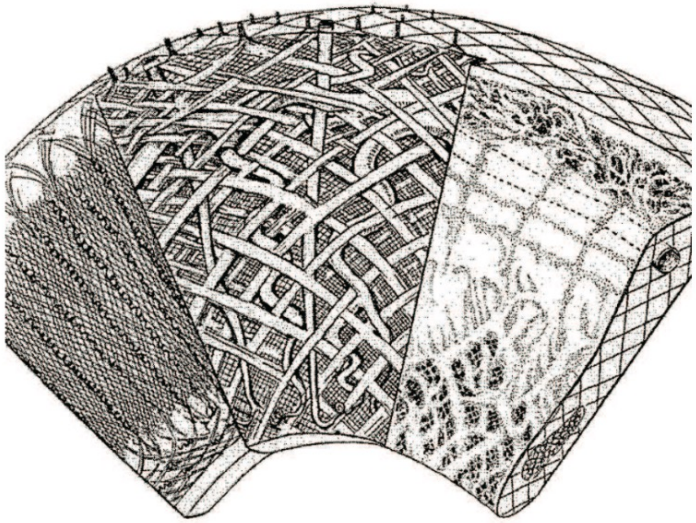


- The **texture** of an iris is the source of an **iris code**
- Iris codes are **unique** to an iris and stable across a range of conditions
- Iris structure: complicated 3D meshwork of muscle beams, blood and nerves

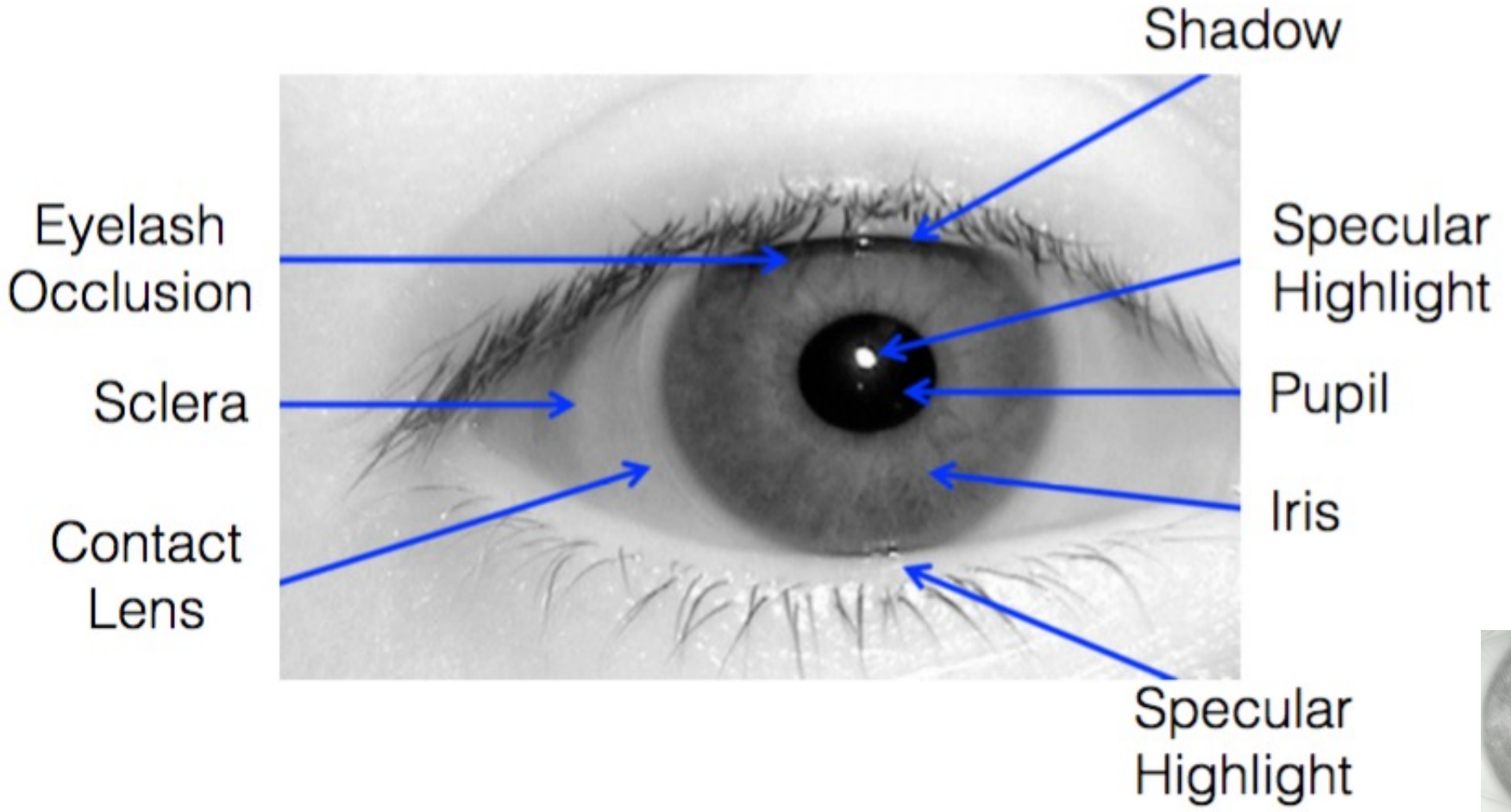








Typical (IR) Image for Iris Recognition



Do it Yourself: Building a Low-Cost Iris Recognition System at Home Using Off-The-Shelf Components

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Abstract

Among the different biometric traits that can be used for person recognition, the human iris is generally considered to be among the most accurate. However, despite a plethora of desirable characteristics, iris recognition is not widely as widely used as competing biometric modalities likely due to the high cost of existing commercial iris-recognition systems. In this paper we contribute towards the availability of low-cost iris recognition systems and present a prototype system built using off-the-shelf components. We describe the prototype device, the pipeline used for iris recognition, evaluate the performance of our solution on a small in-house dataset and discuss directions for future work. The current version of our prototype includes complete hardware and software implementations and has a combined bill-of-materials of 110 €.

1 Introduction

The human iris is often described as one of the most suitable traits for biometric authentication due to several desirable characteristics. The iris has a complex texture that

Figure 1: Illustration of our iris-recognition prototype. Th
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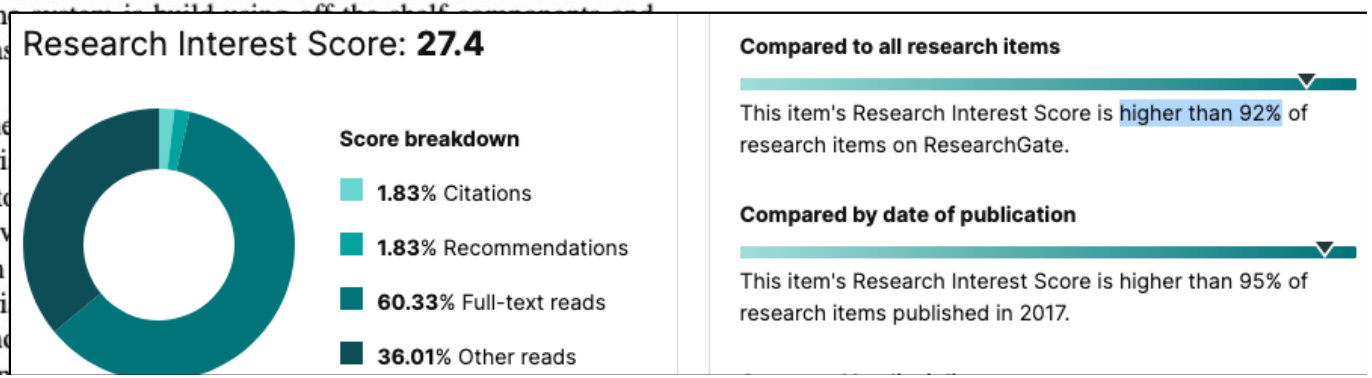
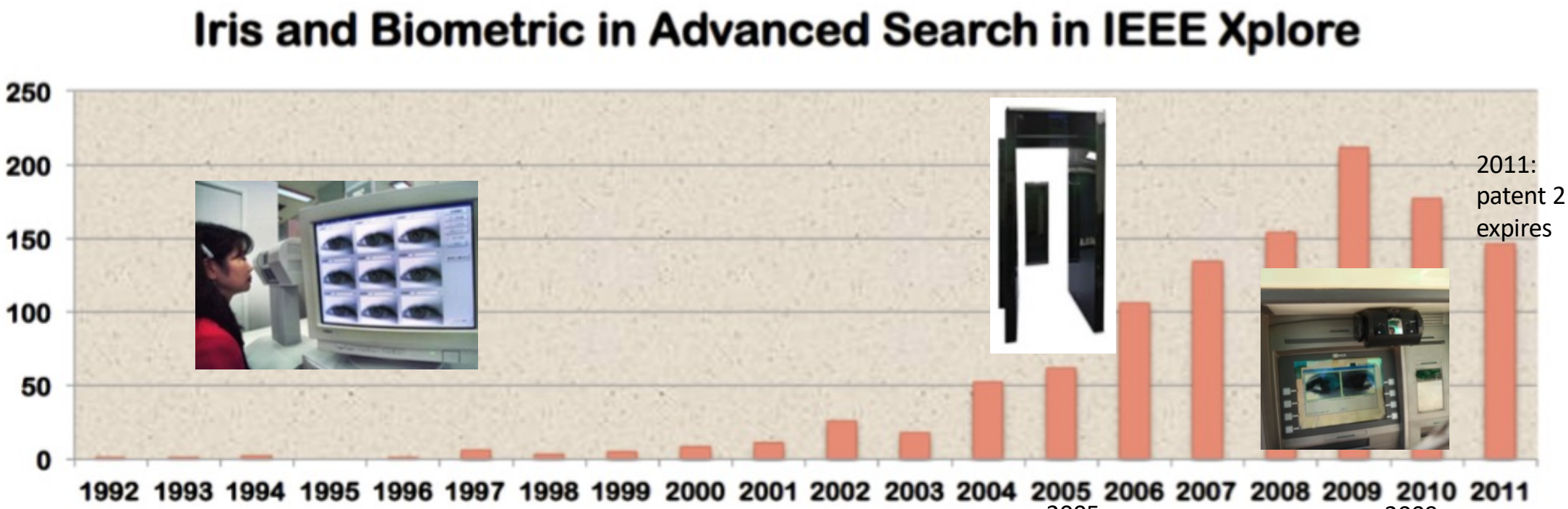
and



A Brief Historical Timeline



1987:
Flom & Safir patent

1994:
Daugman patent

1995:
IriScan, Oki commercial systems

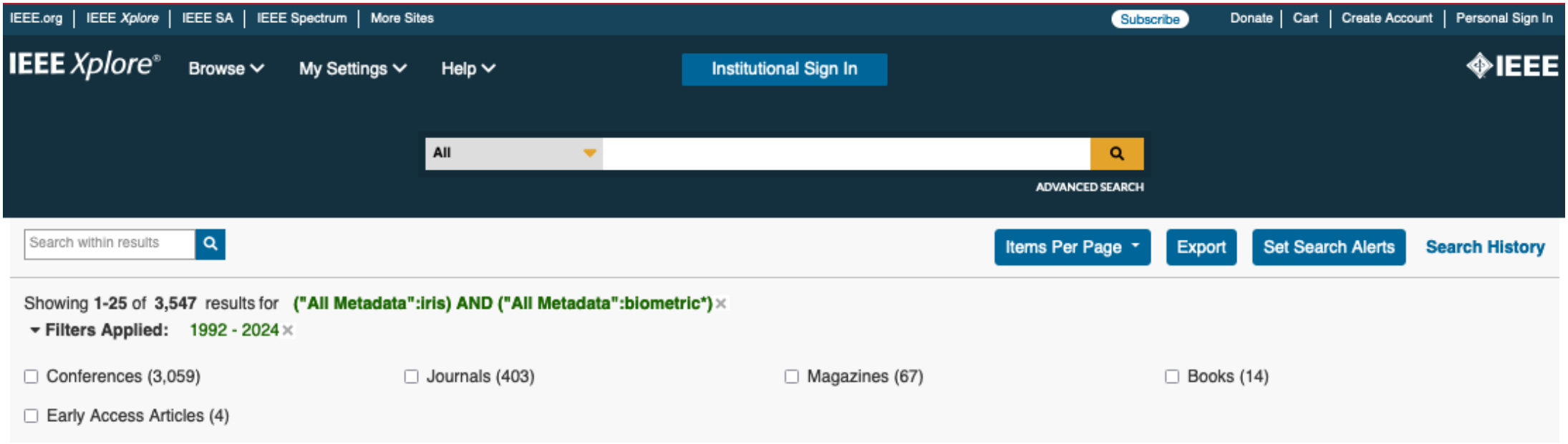
2003:
UAE border control system

2005:
"Iris on the move" from Sarnoff & patent 1 expires

2005, 2006:
NIST Iris Challenge Evaluations

2009:
Cairo-Amman Bank single-factor authentication at ATMs

2011:
patent 2 expires


IEEE.org | IEEE Xplore | IEEE SA | IEEE Spectrum | More Sites
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My Settings
Help
Institutional Sign In
IEEE
All
ADVANCED SEARCH
Search within results
Items Per Page
Export
Set Search Alerts
Search History
Showing 1-25 of 3,547 results for ("All Metadata":iris) AND ("All Metadata":biometric*)
Filters Applied: 1992 - 2024
Conferences (3,059)
Journals (403)
Magazines (67)
Books (14)
Early Access Articles (4)

What is Iris Biometrics Today?

- A particular reference implementation developed by Daugman
- An industry with successful applications and strong projected growth!
- An active, fast-growing research area with some important open questions and new topics opening up!

Do you recognize the image?

1984: titled Afghan Girl

The First World's Third World *Mona Lisa*

Cover is one of the most famous of the National Geographic

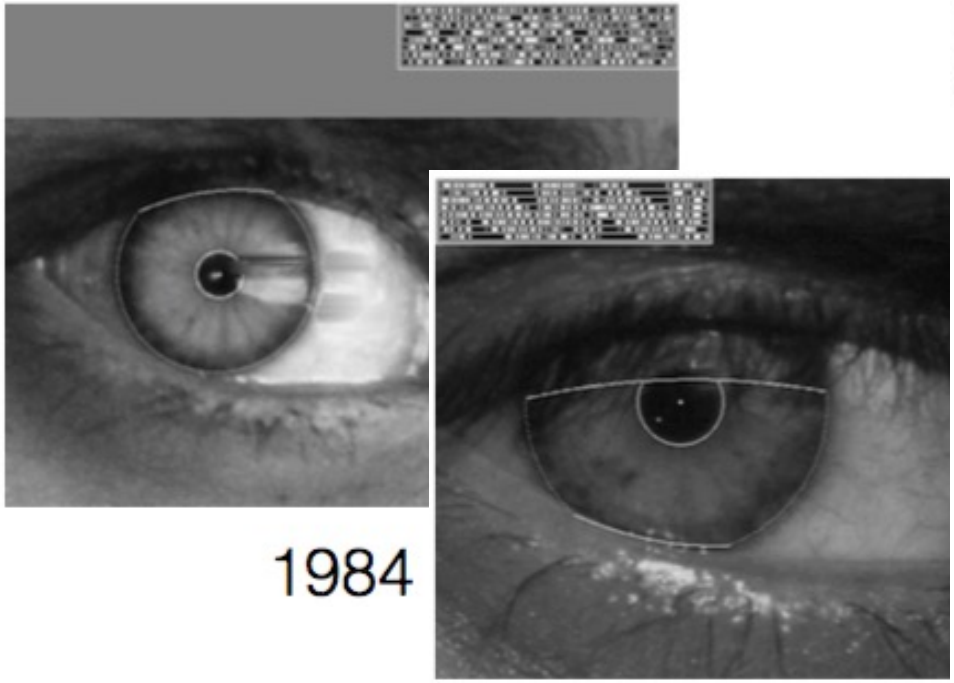
Unsuccessful attempts to locate her afterwards

2002: Sharbat Gula

Confirmed by Daugman using iris recognition

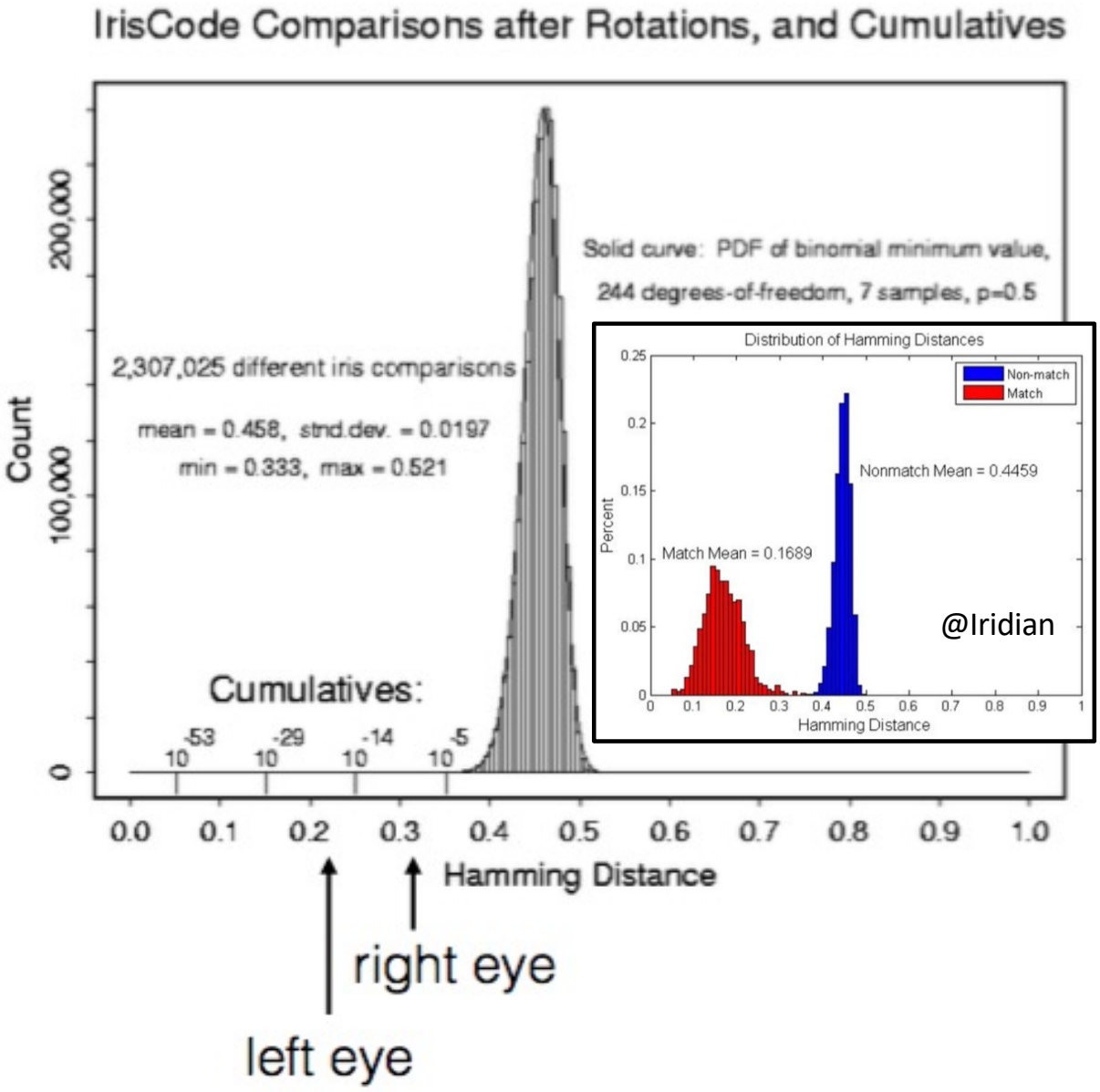
(McCurry)

Calculation?

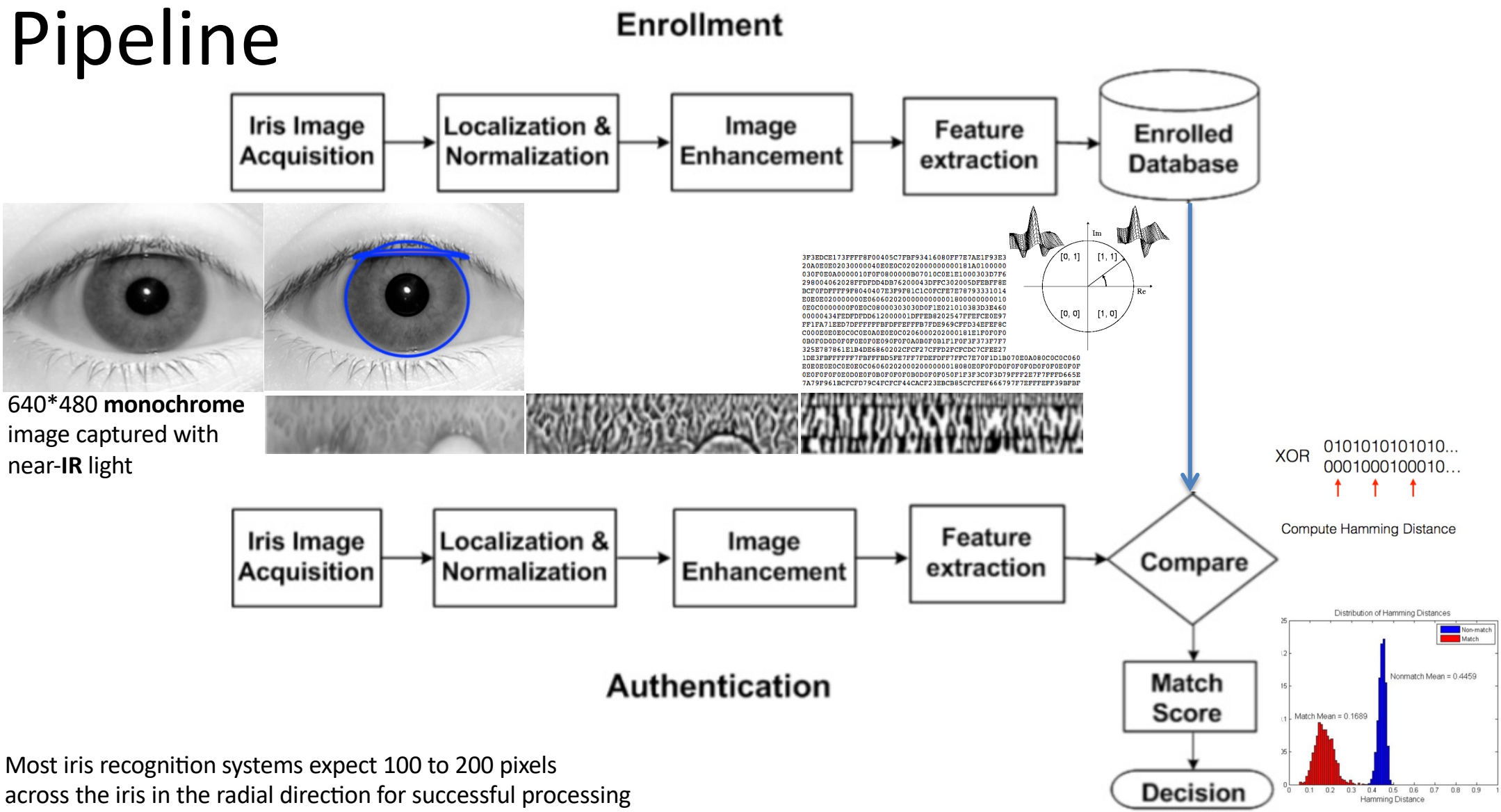


1984

2002



Pipeline



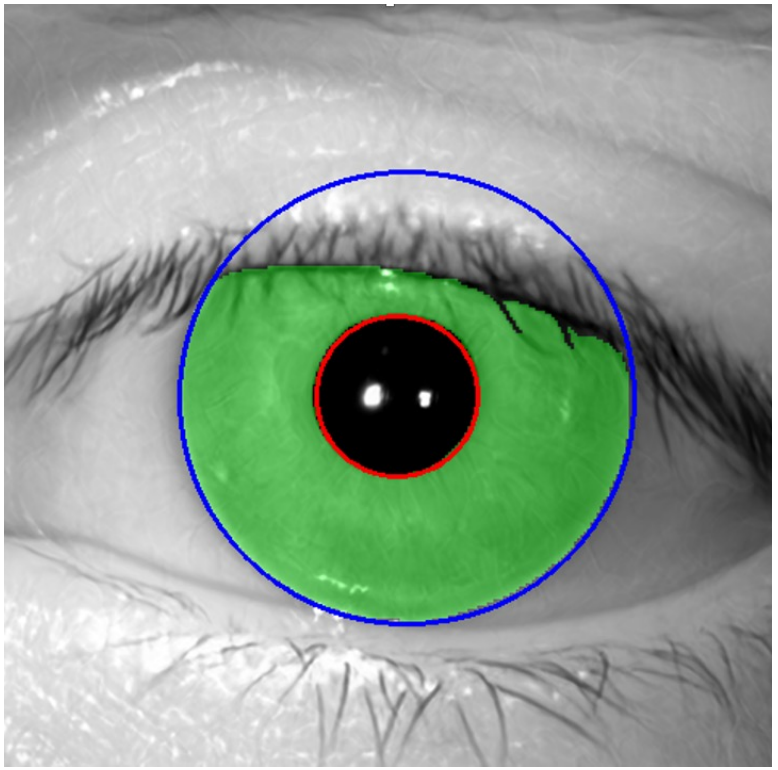
Most iris recognition systems expect 100 to 200 pixels across the iris in the radial direction for successful processing

@Tutorials:
Notre Dame's CVRL Iris Project

Focus on HDBIF

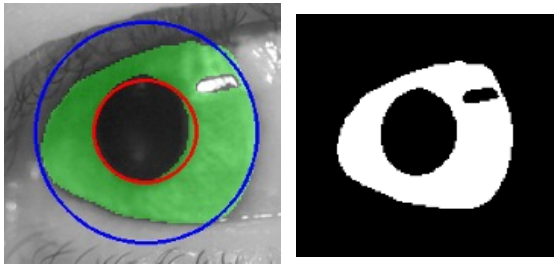
Open Source Iris Recognition Project

- **Project root:** https://github.com/CVRL/OpenSourceIrisRecognition
- **HDBIF subfolder**: `/methods/HDBIF/Python for Iris Recognition`
- Focus on **H**uman-**D**riven **B**inary **I**mage **F**eatures (HDBIF)
- Core **modules** for iris segmentation, encoding, and matching; simple all-vs-all comparison already included
- **Python-based** with traditional Conda installation (libraries: OpenCV, NumPy, and PyTorch**)**



Key Functions (in hdbif.py)

- **Segmentation:**
  - `irisRec.segment_and_circApprox(im)`
  - Extracts iris/pupil masks and circular boundaries
  - Converts images to ISO aspect ratio (640×480) for compliance



- **Polar transformation:**
  - `irisRec.cartToPol_torch(im, mask, pupil_xyr, iris_xyr)`
  - Converts Cartesian coordinates to polar, enabling radial encoding
- **Encoding:**
  - `irisRec.extractCode(im_polar)`

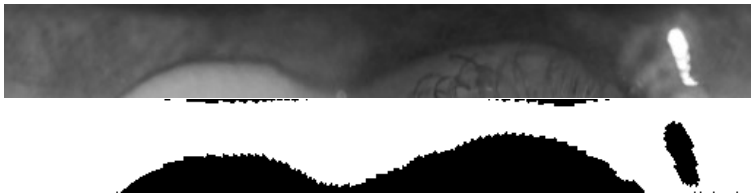


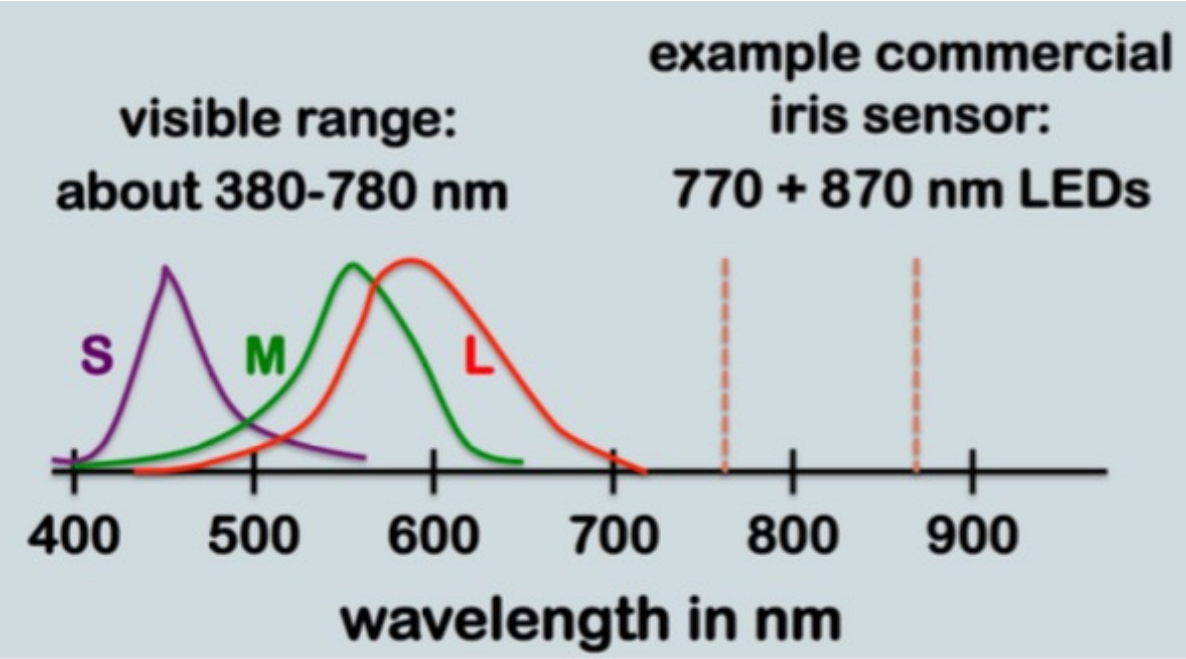
Image Processing and Matching

- **Image processing:**
  - Saves processed results in folders `dataProcessed` and `templates`
  - Files saved: segmentation mask, polar-transformed images, and encoded template
  - [`segmentVis` visualizes segmentation boundaries]
- **Matching:**
  - `irisRec.matchCodesEfficient`
  - Inputs: computed codes and masks in polar format
  - Matching method:
    - All-vs-all comparison
    - Scores and mutual rotations printed for each match
    - Enables scalable large-database matching

Acquisition

The system typically captures a **series** of images of the eye … and based on a **quality** evaluation scheme retains only few (or one) images that are deemed to have sufficient iris texture information for further processing

Why using near-IR spectrum?

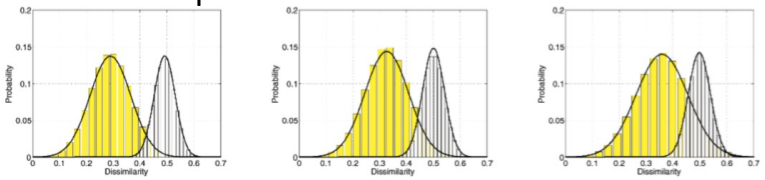


Panasonic BM-ET 330

Oki IrisPass

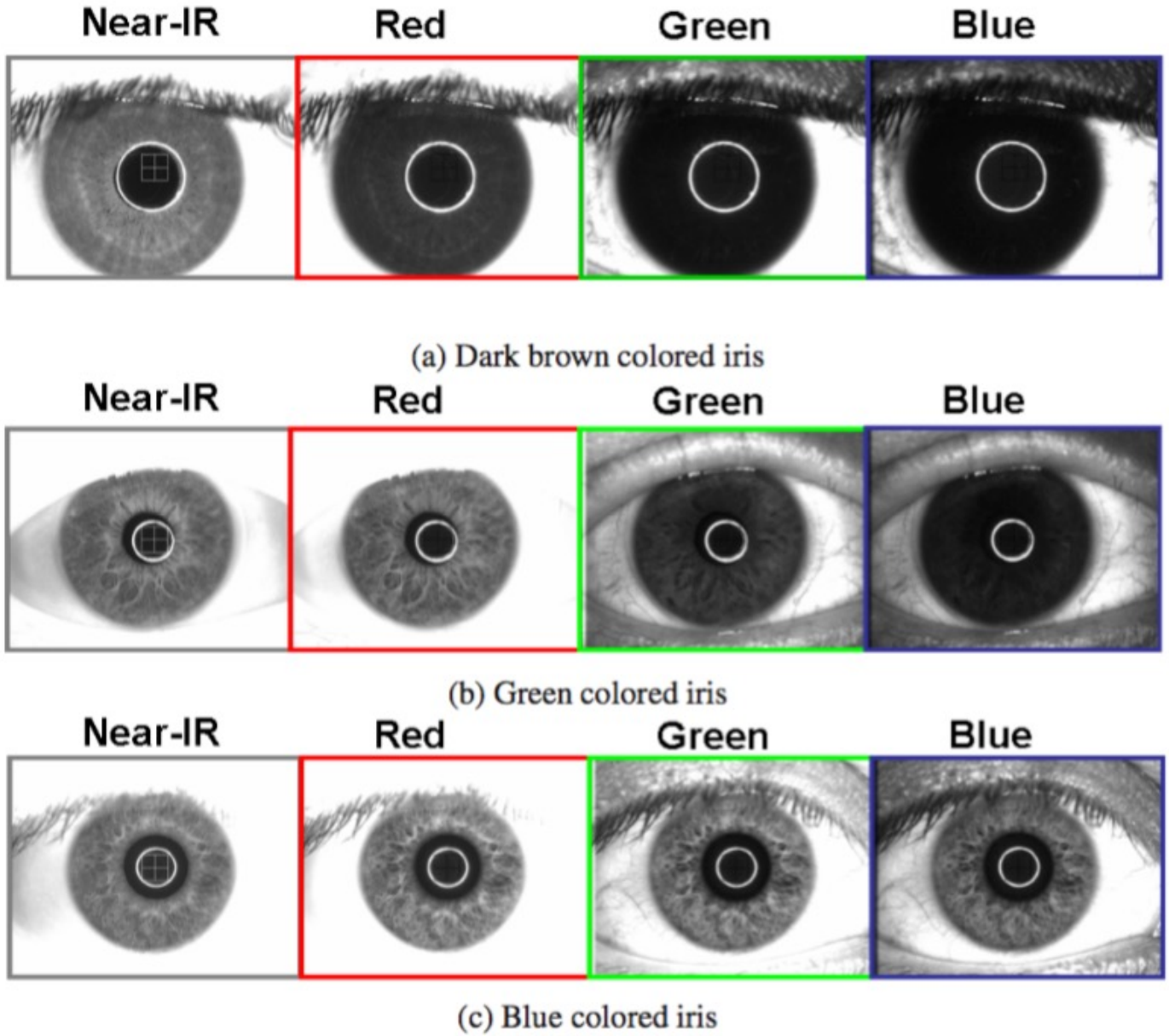
Retica MobileEyes

… in visible spectrum =>

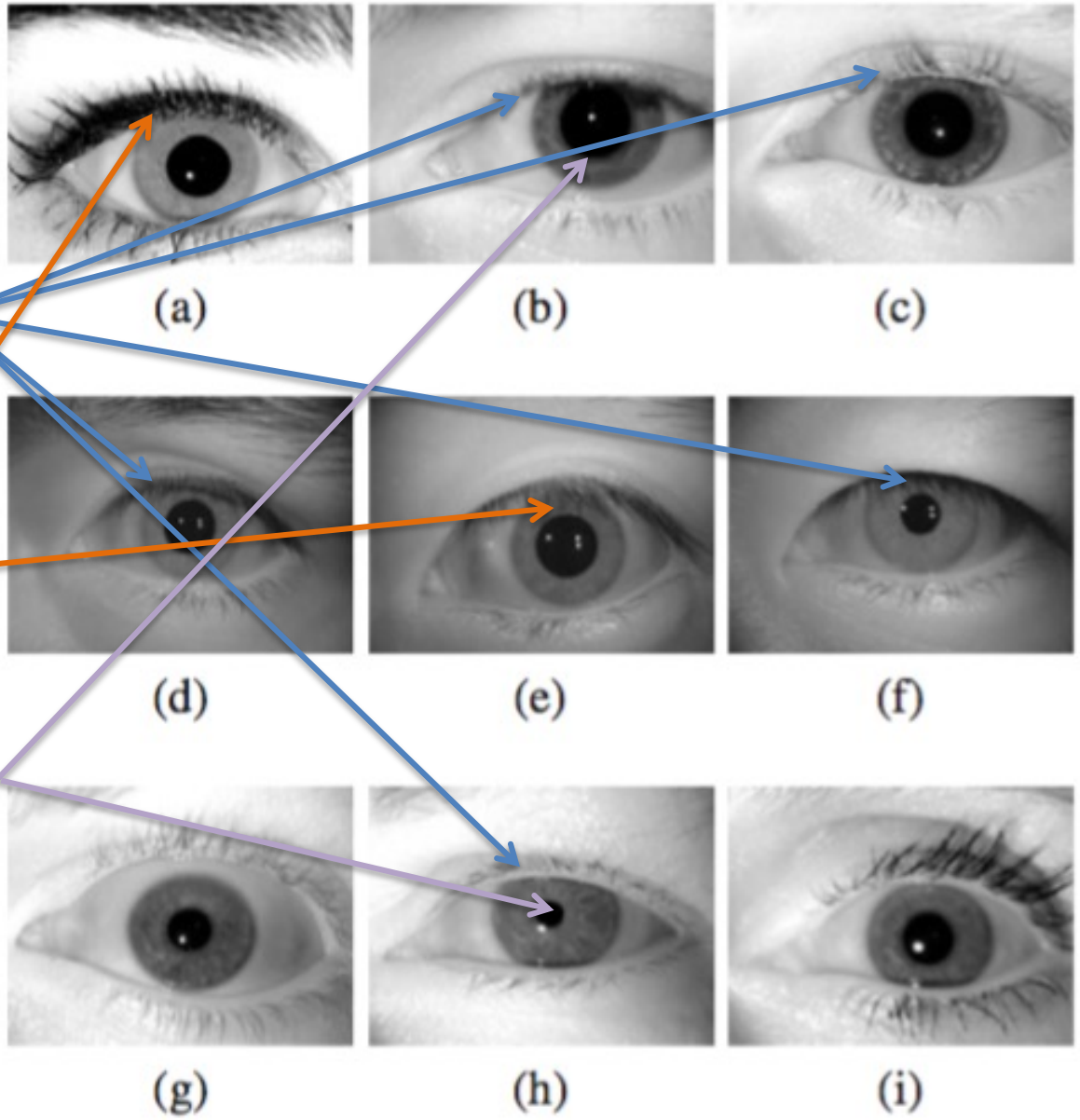


Poor ⟶ Very Poor

Because using near-IR illumination allows usable texture imaging even for **dark** irises



Negative impacts?



partially closed eyelids

intruding eyelashes

extremely dilated or constricted pupil

harsh or non-uniform illumination

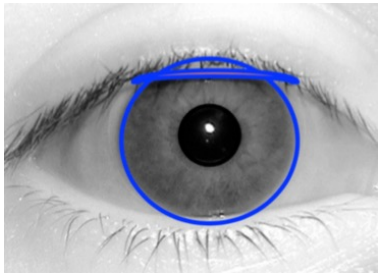
low resolution

ICE db
(LG EOU 2200 cam)

CASIA v3 db
(OKI IRISPASS device)

MBGC db
(LG EOU 2200 cam)

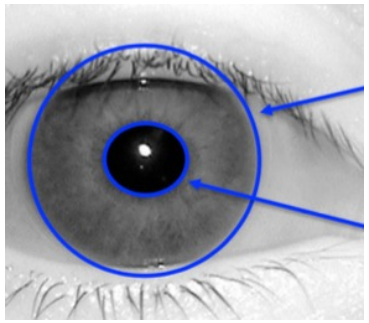
Segmentation



- To determine pixels in the image that correspond to the iris region
- Challenges:
  - contains numerous “edge” like features that are randomly distributed (no go for appearance-based schemes)
  - boundaries (especially the limbus boundary) may not be very sharp (over-segmentation or under-segmentation problem)
  - irregularly shaped due to eyelids
  - detecting (and masking) eyelids, eyelashes, reflections

Segmentation Assumptions

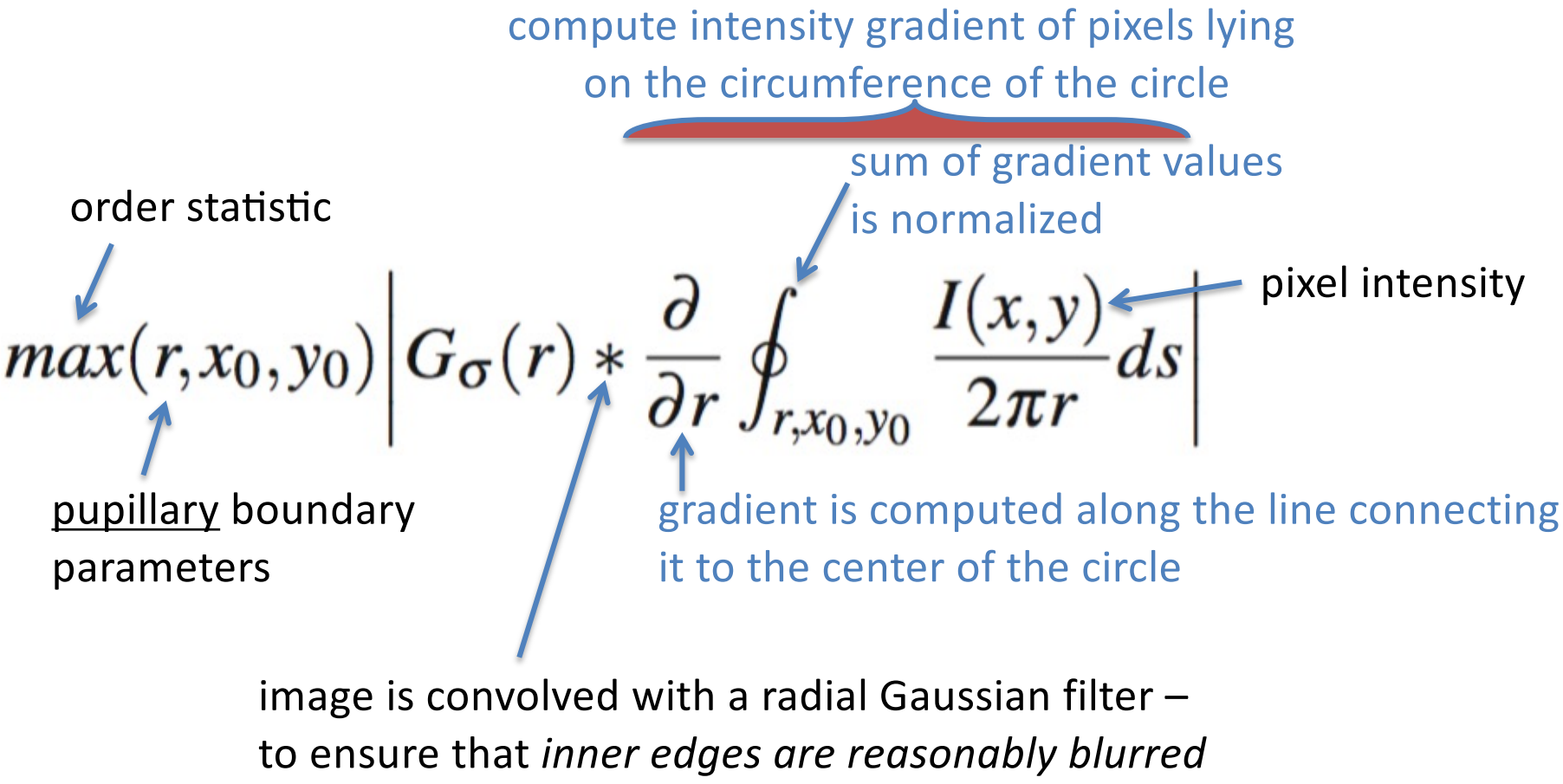
(for Integro-Differential Operator)



- The contrast in image intensity between the pupil and the iris offers a good cue for identifying the **pupillary** boundary (cataract problem?)
- The contrast between the iris and the sclera can be used to detect the **limbus** boundary
- Both these boundaries can be approximated(!) using **circles**
- The magnitude of the edge pixels contributing to these boundaries is **stronger** than those pertaining to other circular(!) contours in the image

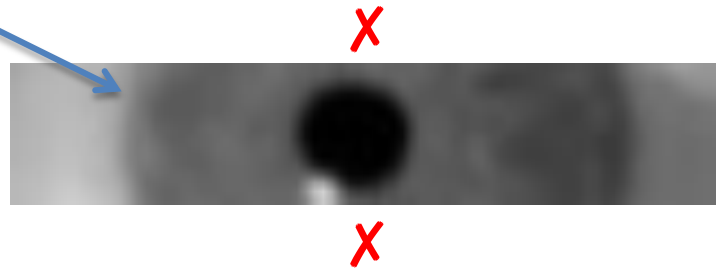
Daugman's Integro-Differential Operator

(or Circular Edge Detector)



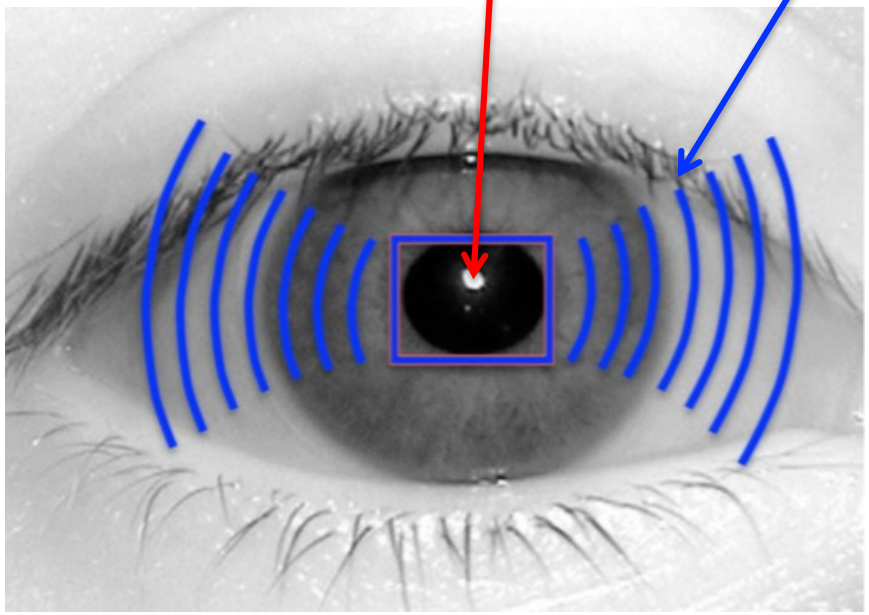
What About the Limbus Boundary?

- Arc of integration is constrained to the near-**vertical** pixels on the circumference of the circle
- Necessary since the contour of the limbus may be interrupted by the upper and lower eyelids – the effect of those pixels has to be minimized



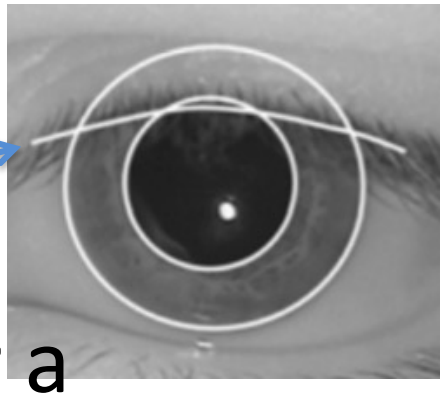
How to Limit the Search Space?

The bounding box of the largest dark region might be a range to search for x_0, y_0 with r in a feasible range



Note:
Typically the iris is **not** concentric with the pupil

Postprocessing



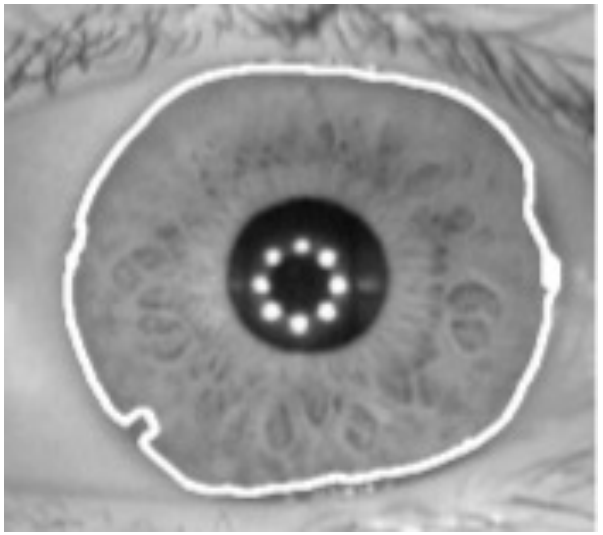
- Eyelid can be detected by searching for a **parabolic edge** within the region defined by the outer circle – can use **spline**-fitting procedure
- Detect eye-lashes infringing into the iris texture by searching for strong near-**vertical edges** in the segmented iris

$$\mathbf{G}_x = \begin{bmatrix} +1 & 0 & -1 \\ +2 & 0 & -2 \\ +1 & 0 & -1 \end{bmatrix} * \mathbf{A}$$

$$\mathbf{G}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} * \mathbf{A}$$

But Boundaries are Actually Non-Circular !?

- Fitting ellipses
- Active contours[1] – simultaneously demarcates the iris from the sclera as well as the eyelids/eyelashes
- Fitting circles and adjusting using active contours[2]

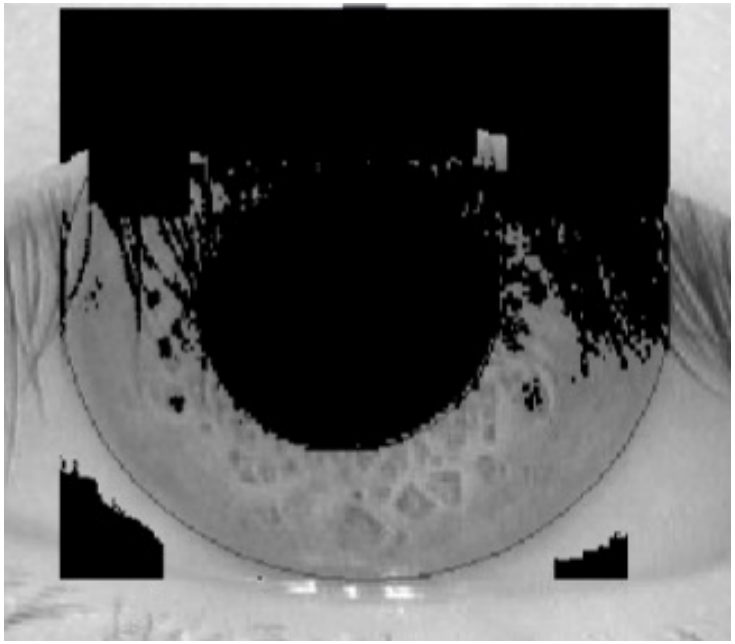


[1]Shah and Ross, IEEE TIFS, 2009 & our primary literature
[2]Park and Savvides, Encyclopedia of Biometrics, 2009

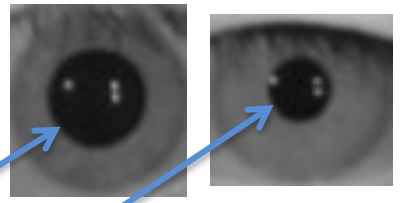
Noise Mask

Records the locations of undesired iris occlusions:

- eyelids,
- eyelashes,
- shadows,
- specular reflections

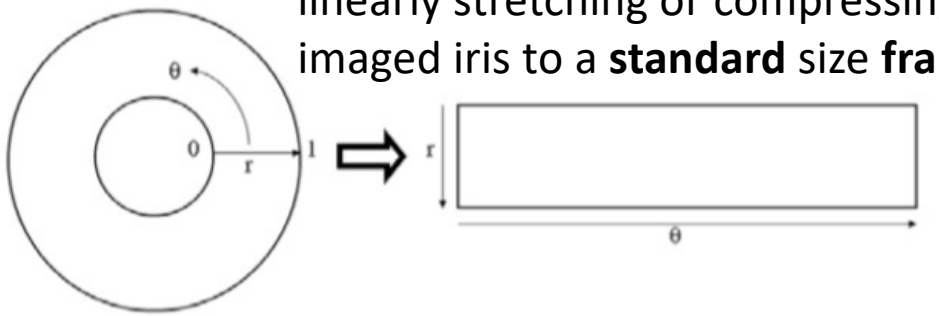


Normalization



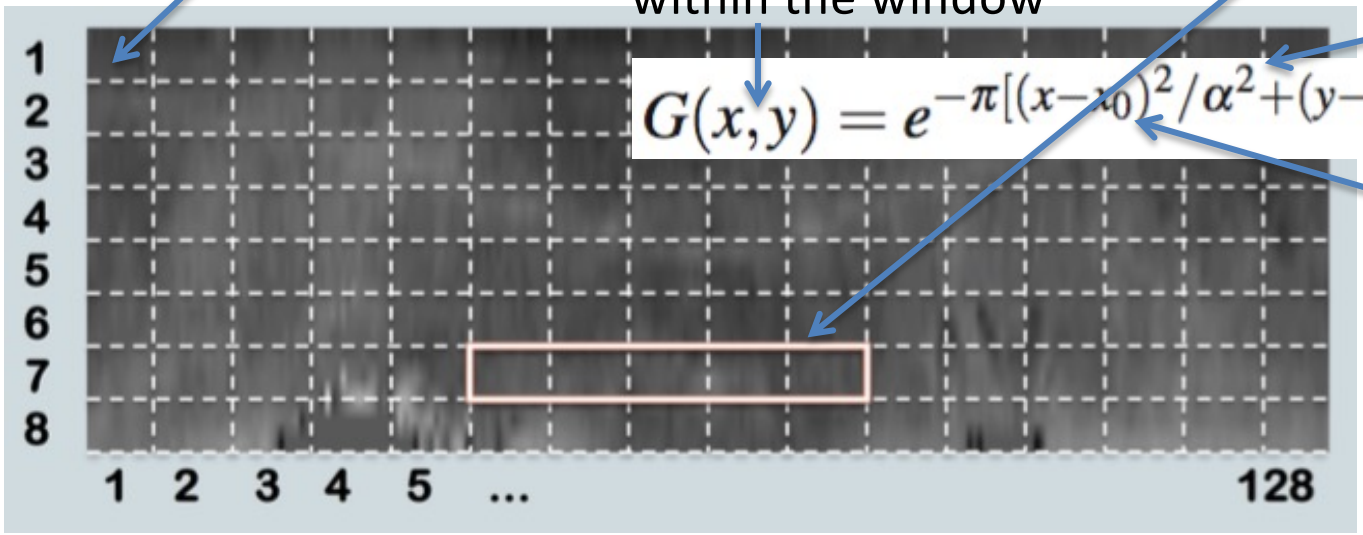
- Addresses the problem of variations in pupil size across multiple images, subjects, sensors, imaging distances
  (main concern: dilation <-> contraction)
- Daugman's rubber sheet model

linearly stretching or compressing the imaged iris to a **standard** size **frame**



Iris Encoding => Iris Code

- Extracting a numerical feature set
- Typical approach: apply a 2D Gabor filter at *M*N* grid **points** on the image



convolution

within the window

effective width & length

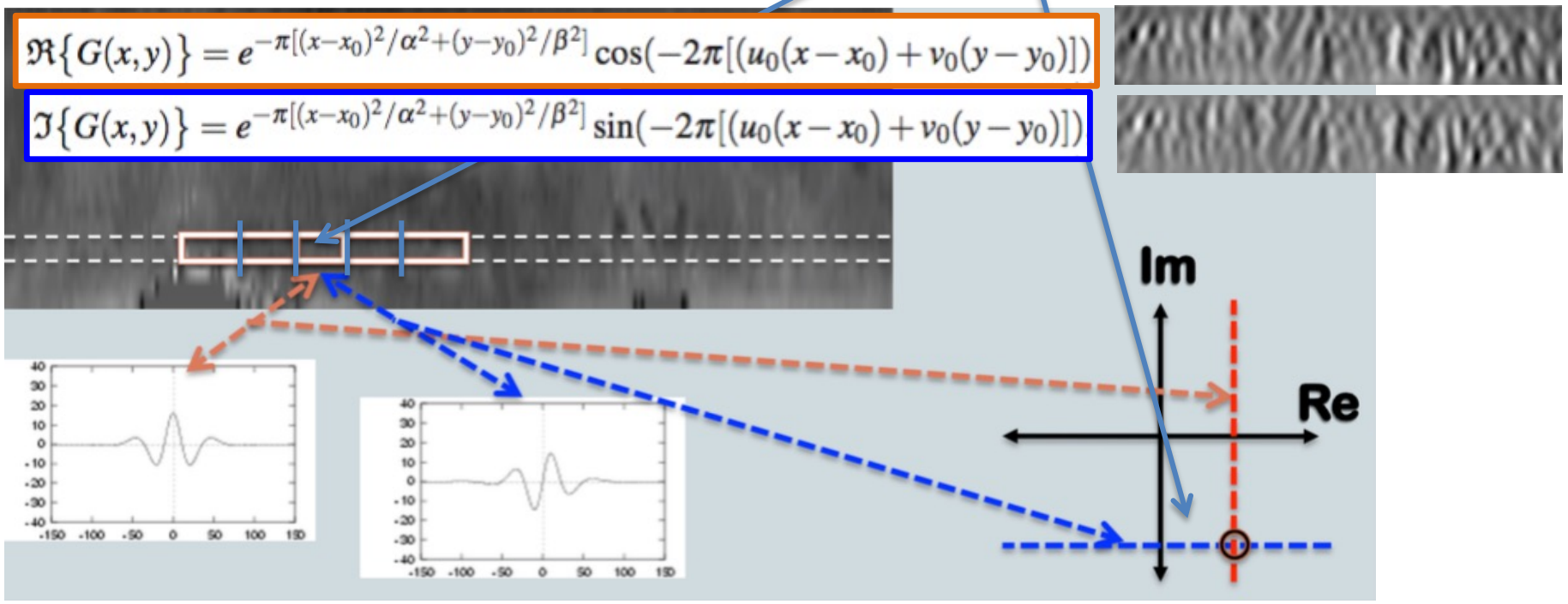
$$G(x,y) = e^{-\pi[(x-x_0)^2/\alpha^2+(y-y_0)^2/\beta^2]}e^{-2\pi i[(u_0(x-x_0)+v_0(y-y_0)]}$$

center position

give the wave direction with a spatial frequency $\omega_0 = \sqrt{u_0^2+v_0^2}$

Iris Encoding => Iris Code

Applying the Gabor filter at a **location** in the iris image results in a **point in a complex plane**



How to obtain a point in a complex plane?

- Demodulate Gabor output to compress the data
- By quantizing the **phase** information into four different levels, one for each quadrant of the complex plane

demodulation and phase quantization process:

normalized image in polar coordinates => filter:

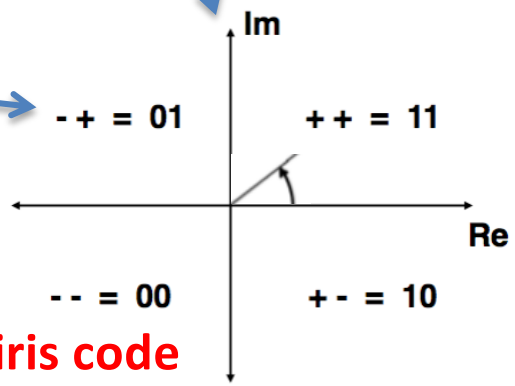
$$H(r,\theta) = e^{-i\omega(\theta-\theta_0)} e^{-(r-r_0)^2/\alpha^2} e^{-i(\theta-\theta_0)^2/\beta^2}$$

center frequency

$$h_{Re,Im} = sign_{Re,Im} \int_\rho \int_\phi I(\rho,\phi) e^{-i\omega(\theta_0-\phi)} e^{-(r_0-\rho)^2/\alpha^2} e^{-(\theta_0-\phi)^2/\beta^2} \rho d\rho d\phi$$

normalized iris

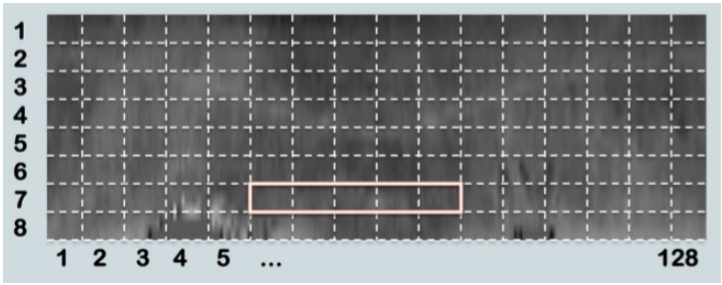
complex value, which real and imaginary components are dependent on (only) the **sign** of the integral

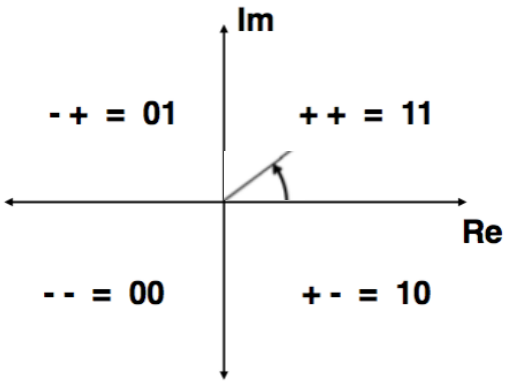


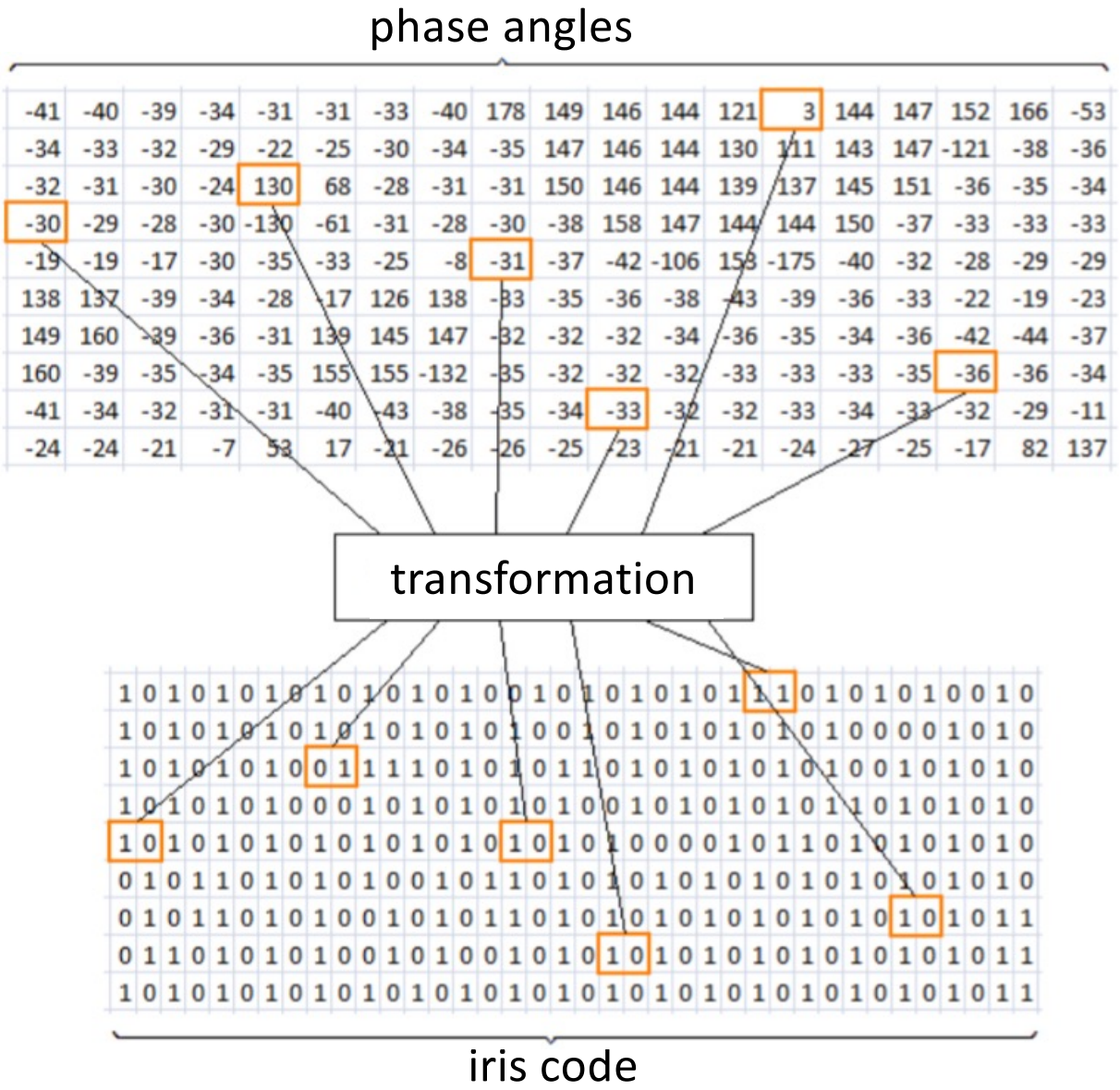
the phase of the complex value makes **two bits in the iris code**

How Big is the Iris Code?

- The standard commercial iris code is **2048** bits (plus 2048-bit mask)
- 2048 / 2 = 1024 sample points in the iris
- 1024 = 8 radii * 128 samples is a plausible sampling configuration




Im
- + = 01
+ + = 11
Re
- - = 00
+ - = 10

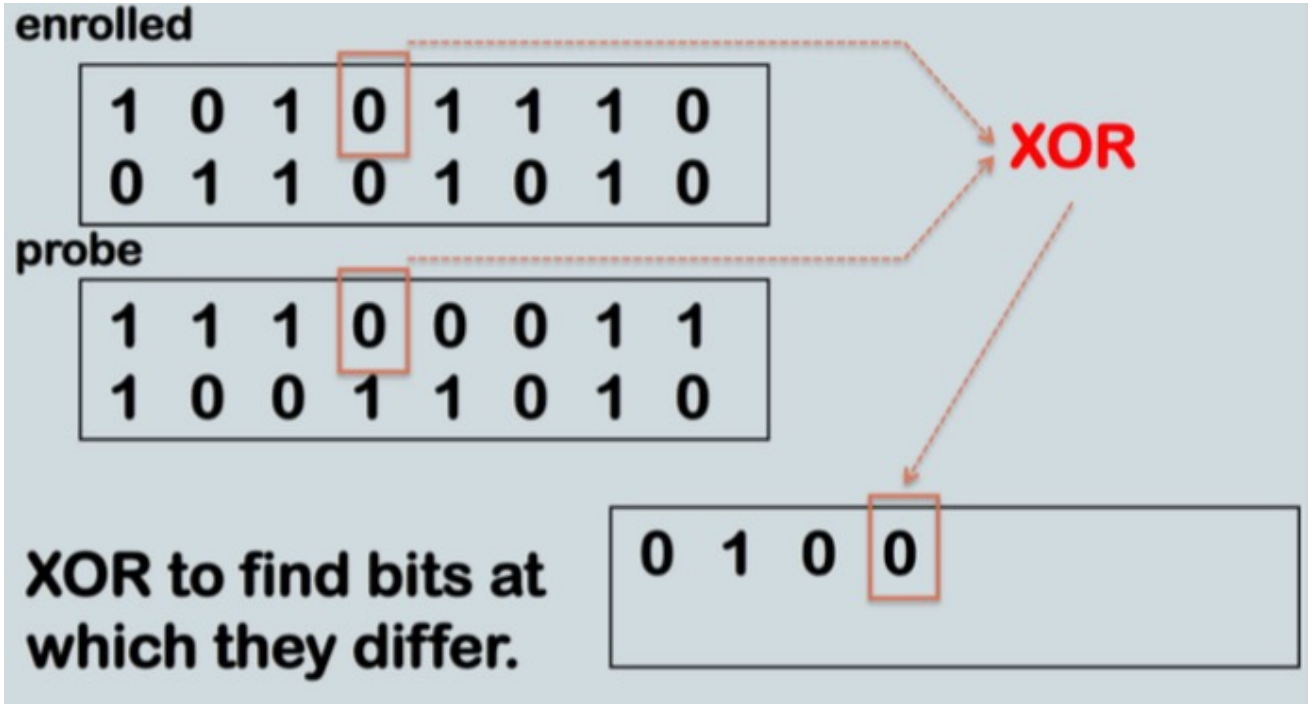

phase angles
-41 -40 -39 -34 -31 -31 -33 -40 178 149 146 144 121 3 144 147 152 166 -53
-34 -33 -32 -29 -22 -25 -30 -34 -35 147 146 144 130 111 143 147 -121 -38 -36
-32 -31 -30 -24 130 68 -28 -31 -31 150 146 144 139 137 145 151 -36 -35 -34
-30 -29 -28 -30 -130 -61 -31 -28 -30 -38 158 147 144 144 150 -37 -33 -33 -33
-19 -19 -17 -30 -35 -33 -25 -8 -31 -37 -42 -106 153 -175 -40 -32 -28 -29 -29
138 137 -39 -34 -28 -17 126 138 -33 -35 -36 -38 -43 -39 -36 -33 -22 -19 -23
149 160 -39 -36 -31 139 145 147 -32 -32 -32 -34 -36 -35 -34 -36 -42 -44 -37
160 -39 -35 -34 -35 155 155 -132 -35 -32 -32 -32 -33 -33 -33 -35 -36 -36 -34
-41 -34 -32 -31 -31 -40 -43 -38 -35 -34 -33 -32 -32 -33 -34 -33 -32 -29 -11
-24 -24 -21 -7 53 17 -21 -26 -26 -25 -23 -21 -21 -24 -27 -25 -17 82 137
transformation
1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 1 0 1 0 1 0 1 0 1 1 1 0 1 0 1 0 1 0 0 1 0
1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 0 0 0 0 1 0 1 0
1 0 1 0 1 0 1 0 0 1 1 1 1 0 1 0 1 0 1 1 0 1 0 1 0 1 0 1 0 1 0 0 1 0 1 0 1 0
1 0 1 0 1 0 1 0 0 0 1 0 1 0 1 0 1 0 1 0 0 1 0 1 0 1 0 1 0 1 1 0 1 0 1 0 1 0
1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 0 0 1 0 1 1 0 1 0 1 0 1 0 1 0
0 1 0 1 1 0 1 0 1 0 1 0 0 1 0 1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
0 1 0 1 1 0 1 0 1 0 0 1 0 1 0 1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 1
0 1 1 0 1 0 1 0 1 0 0 1 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 1
1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 1
iris code

Why Gabor Filters?

- Possible relevance to human vision
- Daugman: “Different members of this optimal filter family are an excellent description of the 2D neural receptive fields found in the visual cortex.”
- Good theoretical properties

Matching

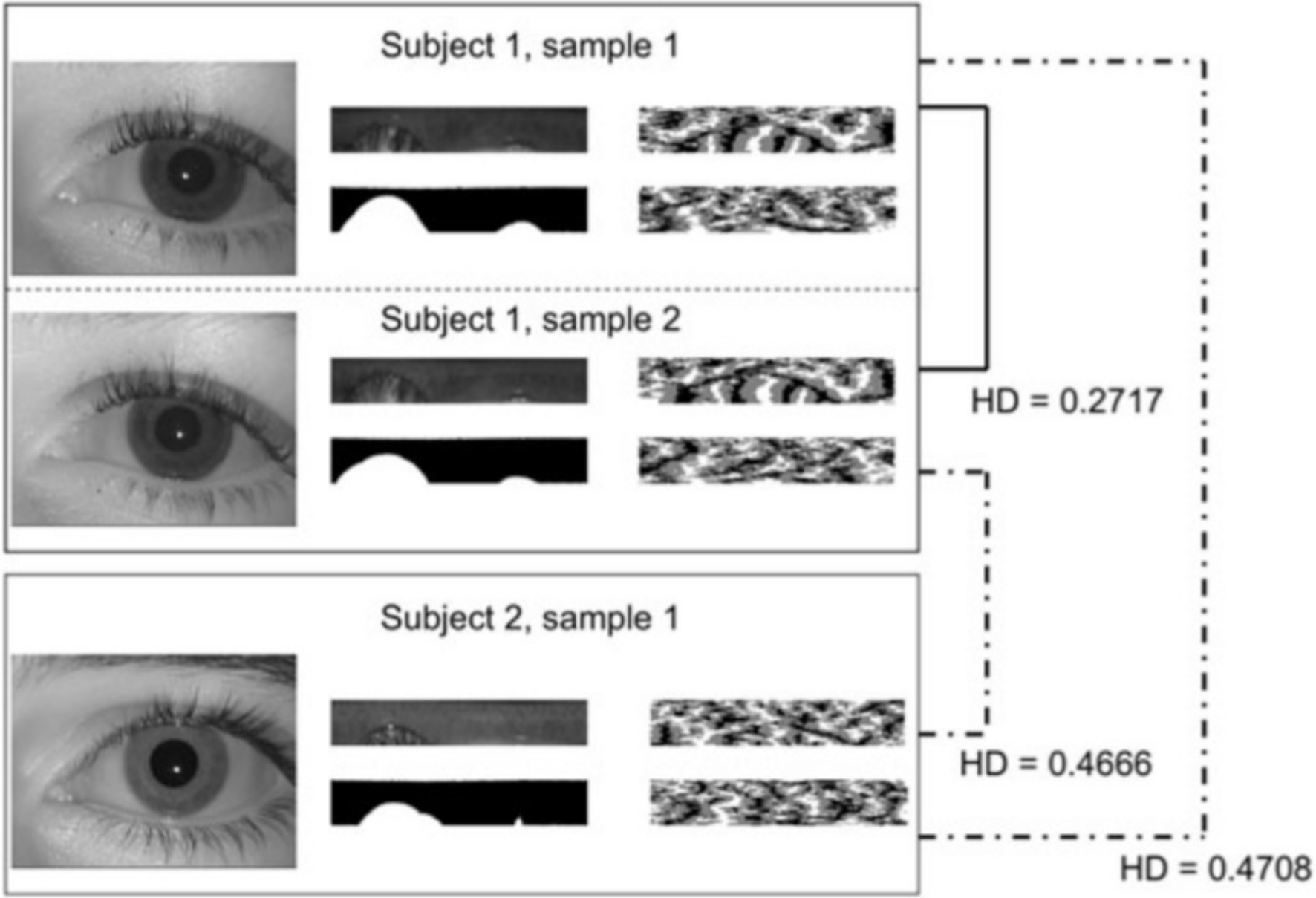
Metric: Hamming distance



Matching

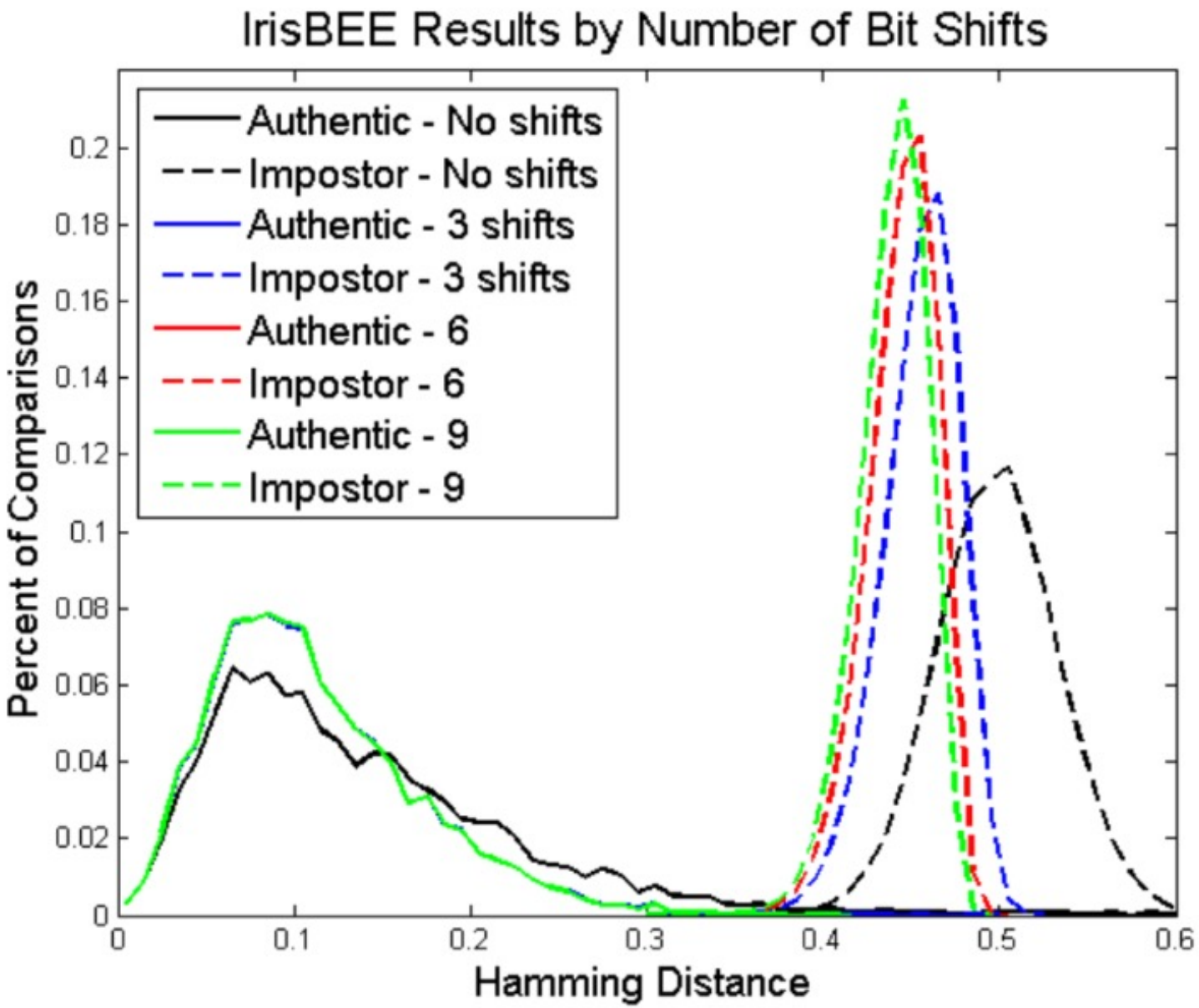
- Normalized Hamming distance – fraction of bits that differ, using **masks** (1 = iris, 0 = occlusion)
- Min. value of 0.0 means **no** difference
- Max. value of 1.0 means **totally** different
- 0.5 means **random** agreement
- The denominator helps in normalizing the total number of bits that disagree in the interval [0,1]

$$\frac{\left| \left(\text{Mask}_1 \text{ AND } \text{Mask}_2\right) \text{ AND } \left(\text{Code}_1 \text{ XOR } \text{Code}_2\right) \right|}{\left| \text{Mask}_1 \text{ AND } \text{Mask}_2 \right|}$$


Subject 1, sample 1
Subject 1, sample 2
Subject 2, sample 1
HD = 0.2717
HD = 0.4666
HD = 0.4708

Matching & Rotation of the Iris?

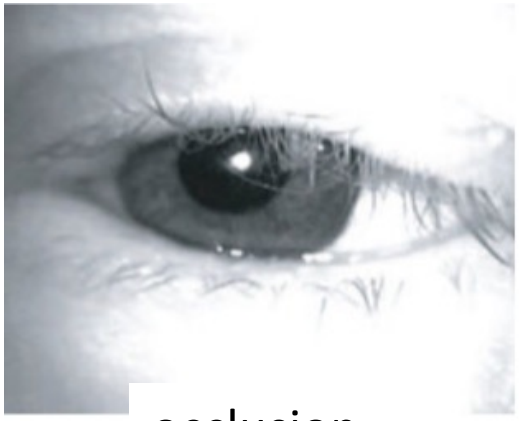
- To account for varying head tilt we **rotate** codes
- Metric is computed for a range of circular **shifts** and the minimum is kept
- This corrects the authentic distribution – but **increases** time for matching & **changes** the impostor distribution =>


IrisBEE Results by Number of Bit Shifts
Authentic - No shifts
Impostor - No shifts
Authentic - 3 shifts
Impostor - 3 shifts
Authentic - 6
Impostor - 6
Authentic - 9
Impostor - 9
Percent of Comparisons
Hamming Distance
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0
0.1
0.2
0.3
0.4
0.5
0.6

Iris Quality Assessment

Factors influencing the quality:

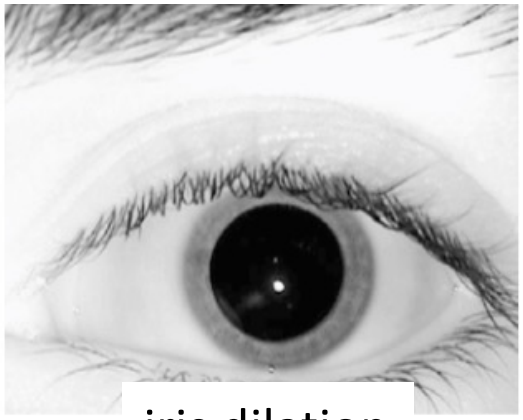
- *Occlusion*
- *Defocus / out-of-focus*
- *Motion blur*
- Non-uniform illumination
- Low resolution / large imaging distance
- *Pupil dilation*
- Off-angled imaging
- Presence of accessories; such as fake or printed contact *lenses*



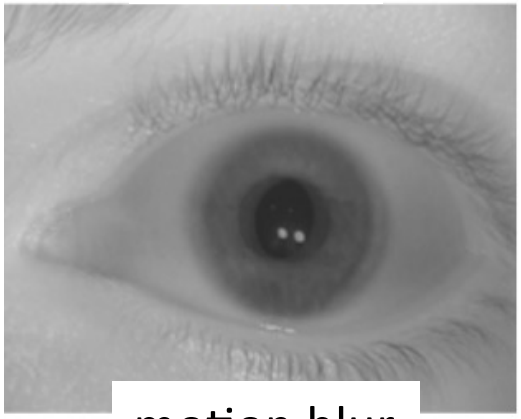
occlusion

defocus

low resolution sensor

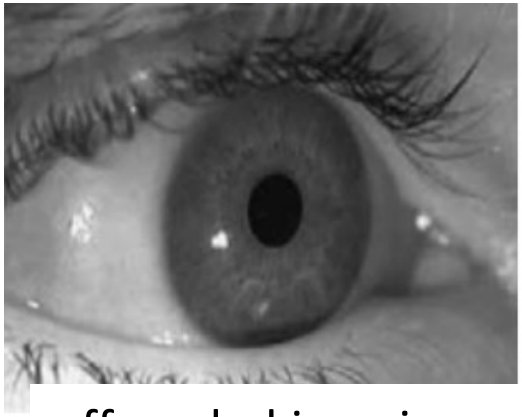


iris dilation

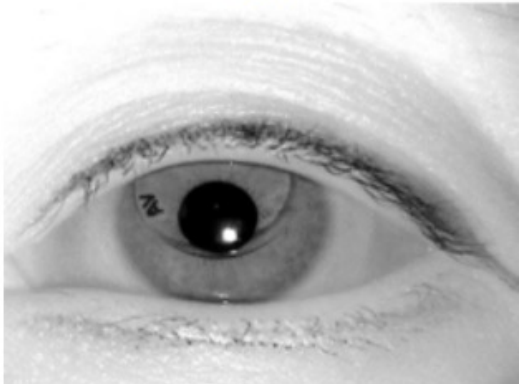


motion blur

non-uniform illumination



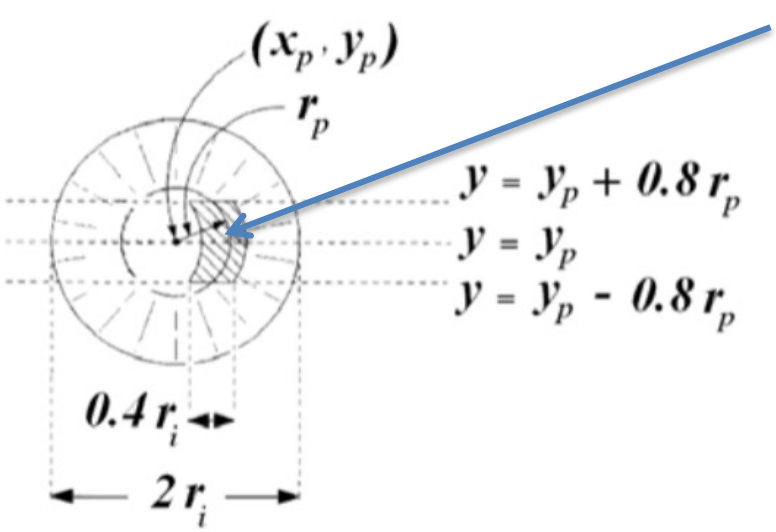
off-angled imaging



presence of a (printed!?) contact lens

Sharpness/Focus Evaluation

Sharpness is essentially computed as the normalized magnitude of the intensity **gradient near the pupillary boundary**



ROI (Region of Interest):

$$(y_p - 0.8r_p) < y < (y_p + 0.8r_p)$$

$$-\sqrt{r_p^2 - (y-y_p)^2} + 0.1r_i < x - x_p < -\sqrt{r_p^2 - (y-y_p)^2} + 0.2r_i$$

Sharpness/Focus Evaluation

1. From ROI compute median values:
   - of the pixels falling in the pupil region, M_p
   - of the pixels falling in the iris region, M_i
2. For all the pixels in ROI with intensities in $[M_p..M_i]$ calculate absolute values of their horizontal gradients
3. Calculate the average of the top 20 values = S
4. Calculate sharpness: $S / (M_i - M_p)$
5. If >0.5 => considered to be well focused

Sharpness/Focus Evaluation Using 2D Fourier Transform

- Quantifying the energy of **high** spatial frequencies over the **entire** image

image

$$F(\mu,\nu) = \frac{1}{(2\pi)^2}\int\int I(x,y)\exp(-i(\mu x+\nu y))dxdy$$

frequencies

- Defocused image is related to the FT of the corresponding in-focus image by the following model:

$$D_\sigma(\mu,\nu) = \exp\left(-\frac{\mu^2+\nu^2}{\sigma^2}\right)F(\mu,\nu)$$

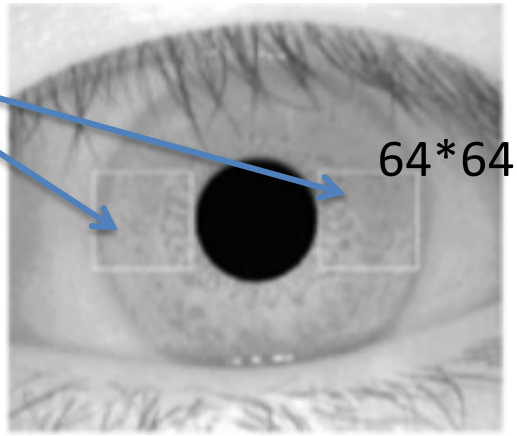
blur parameter

Sharpness/Focus Evaluation Using 2D Fourier Transform

- Defocusing primarily **reduces** the **highest** frequencies in the image, while the lower frequency components are virtually unaffected

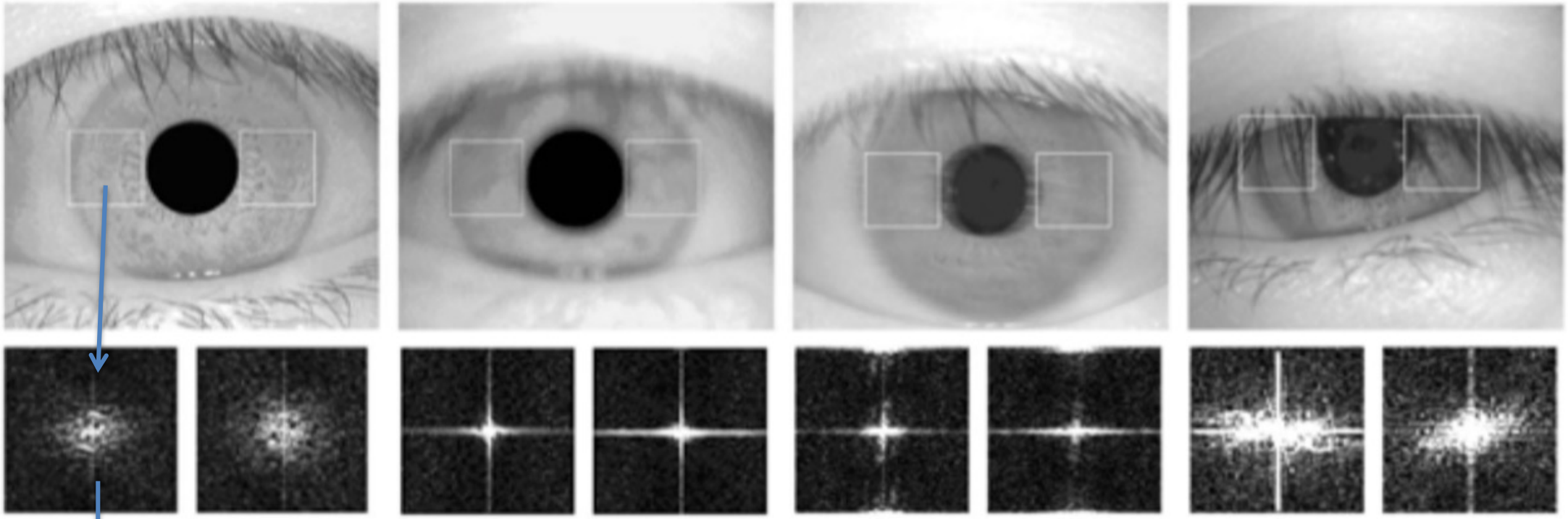
- How to then identify a defocused image?
- **Measure its total power in the Fourier domain at higher spatial frequencies** (as they are most reduced by defocus)

Analyzing the Fourier Spectra of Local Iris Regions



Applicable to:

- out-of-focus blur
  - Fourier spectrum is supposed to be largely dominated by the **low** frequency components
- motion blur
  - lacks **middle** and **high** frequency components and has frequency distribution similar to that of an out-of-focus image
- occlusion due to the eyelashes, eyelids
  - contains significant **middle** and **high** frequency components



Compute quality descriptor values: $D = \left[(F_1 + F_2 + F_3); \frac{F_2}{F_1 + F_3}\right]$

power of the low, middle, and high frequency components

$$F_i = \int\int_{\Omega = \{(u,v)|f_1^i < \sqrt{u^2+v^2} <= f_2^i\}} |F(u,v)| \, dudv \quad i = 1,2,3$$

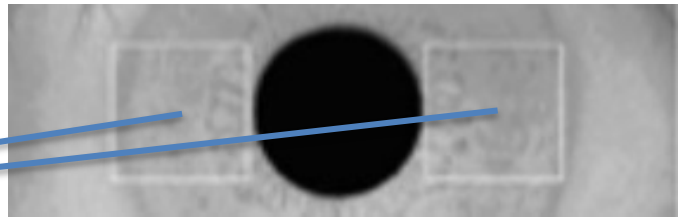
radial frequency pair that form the extrema of the corresponding frequency components

2D Fourier spectrum of the iris region

$$D = \left[(F_1 + F_2 + F_3); \frac{F_2}{F_1 + F_3} \right]$$

Two discriminating frequency features:

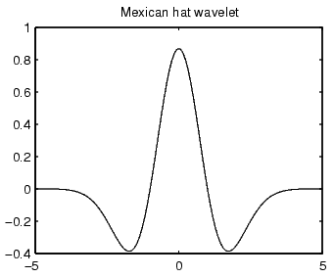
- **Total** spectrum power of the iris region that can effectively discriminate severely occluded iris images from high quality images
- Ratio of the **middle** frequency power to the other frequency powers – for a clearly focused image this ratio is high when compared to out-of-focus or motion blurred images



Mean of the resulting two local quality descriptors is regarded as an appropriate quality measure of the iris image

Measuring the energy from 2D **wavelets** at local concentric **bands** of segmented iris

- Local quality measure is used as a **weighting** scheme in the matching process
- Method uses **higher** weights for **inner** regions of the iris – these are more stable compared to the outer regions that are more prone to occlusions



Continuous Wavelet Transform
(convolution with a **series** of wavelet functions)

Mexican hat wavelet (band pass filter for <u>edge</u> detection at scale *s*)

$$w(s,x_0,y_0) = \frac{1}{\sqrt{s}} \int\int_{R^2} I(x,y)\phi\left(\frac{x-x_0}{s}, \frac{y-y_0}{s}\right)dxdy$$

scale & translation

image

Measuring the energy from 2D **wavelets** at local concentric bands of segmented iris

capture various features at multiple scales *s*:

$$w^{mul}(s_1, s_2, s_3) = w(s_1) \times w(s_2) \times w(s_3)$$

+iris is partitioned into multiple concentric bands of fixed width

energy of the *t*-th band

$$E_t = \frac{1}{N_t} \sum_{i=1}^{i=N_t} | w_{t,i}^{mul} |^2$$

total number of wavelet coefficients

(high energy value suggests a good quality image)

quality index of the entire iris

$$Q = \frac{1}{T} \sum_{t=1}^{T} (m_t \times \log(E_t))$$

total number of bands

weight

mean radius of the *t*-th band with respect to l_c

center of the pupil

$$m_t = \exp\{- \| l_t - l_c \|^2 / (2q)\}$$

Measuring the energy from 2D **wavelets** at local concentric bands of segmented iris

Possibility to incorporate local (band) quality measures into the matching scheme =>

Weighted Hamming distance

$$HD_w = \frac{1}{B} \frac{\sum_{i=1}^{B} \sqrt{E_{g(i)}^X \times E_{g(i)}^Y} \times (X_i \otimes Y_i)}{\sum_{i=1}^{B} \sqrt{E_{g(i)}^X \times E_{g(i)}^Y}}$$

iris codes

total number of bits

local quality measures of the *g(i)*-th band

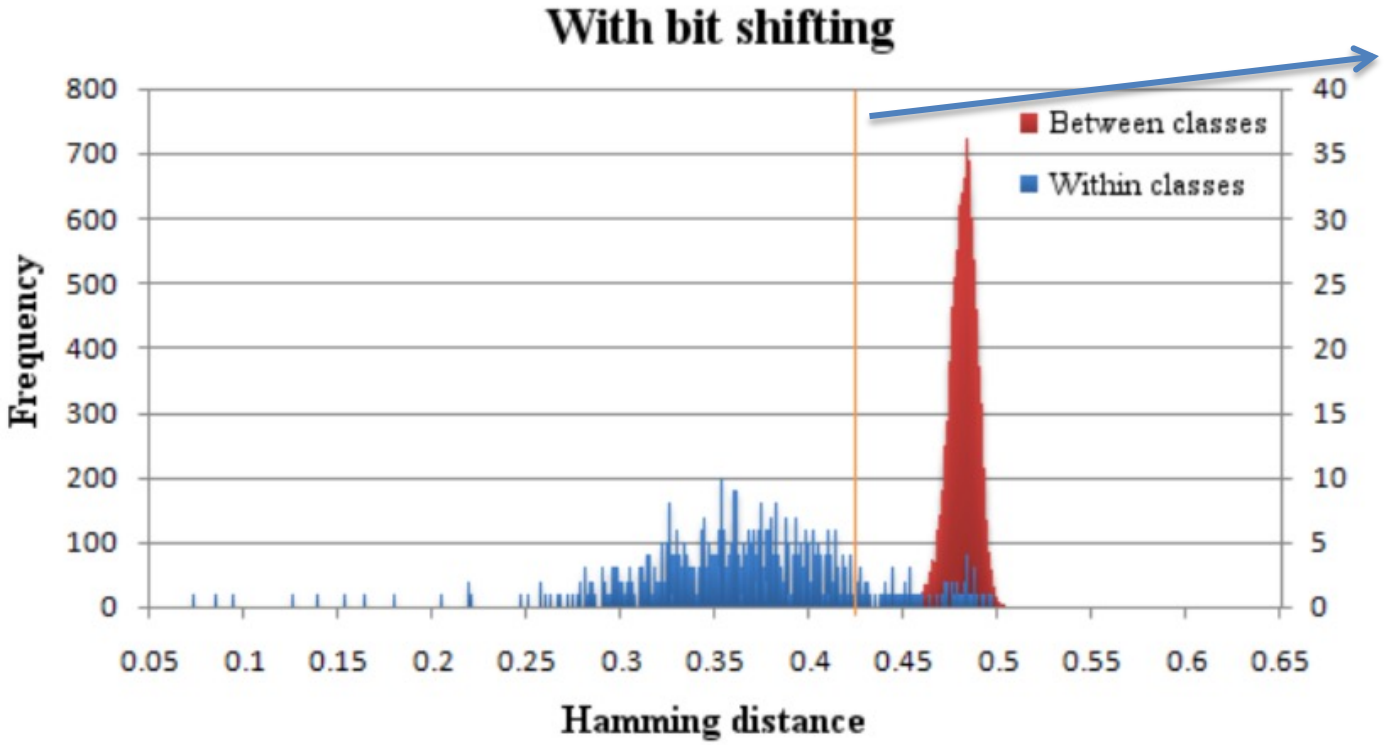
index of the band that contains the *i*-th bit

Performance Evaluation

- Iris is substantially diverse across the population
- Even monozygotic twins exhibit structural differences
- Daugman's experiment: 632,500 iris images (316,250 persons, 152 nationalities) suggest the possibility of a decision policy that could yield **zero** error rates
- This rate is predicated on the **quality** of the iris image, which must be strictly monitored to ensure reasonable textural clarity
- 2006 NIST Iris Challenge Evaluation:
  - involving a broad range of image quality
  - FNMR=1.1 … 1.4 % @ FMR=0.1 %

Playing Around with Our Prototype

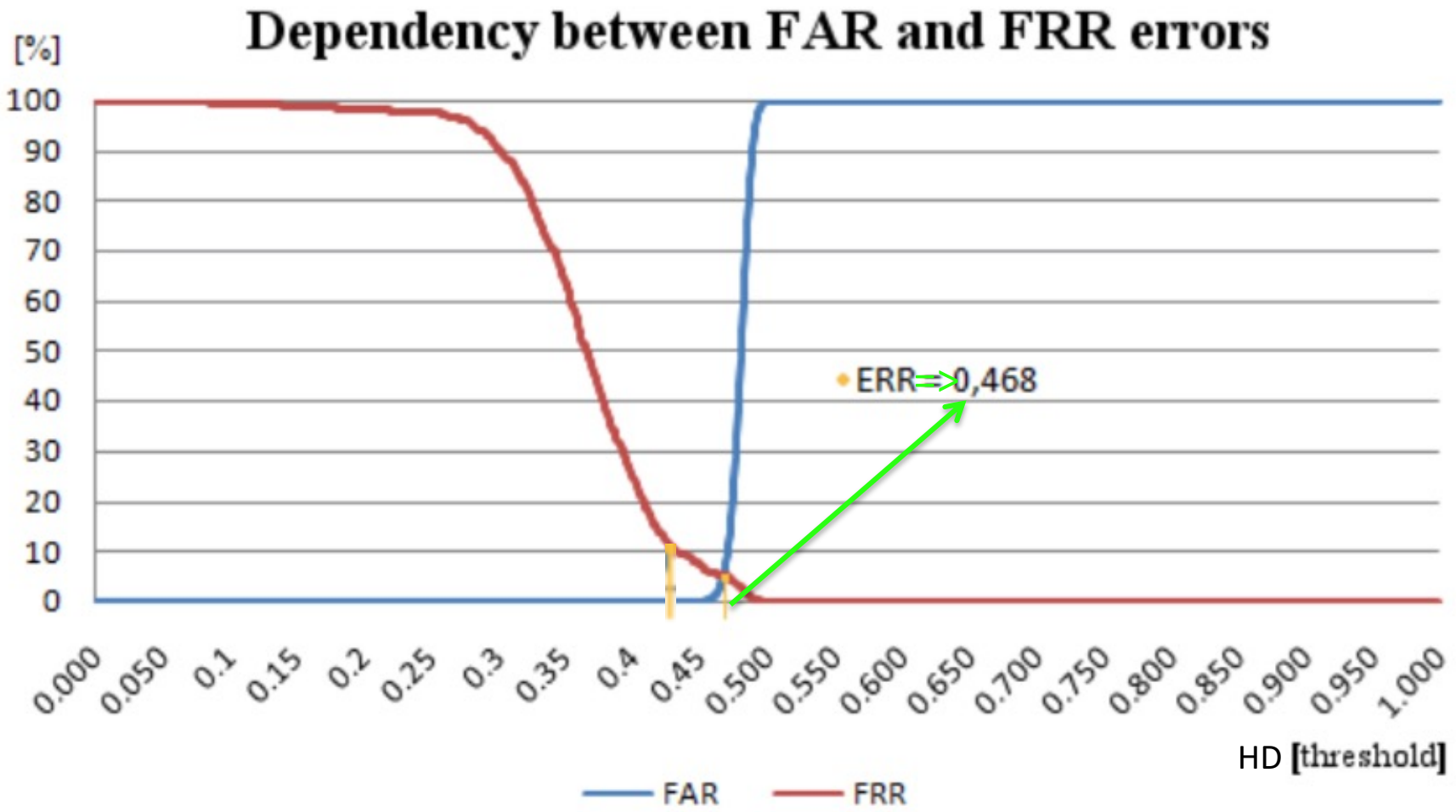
Histogram of comparisons *within classes* and *between classes* on CASIA v1 db:



HD = 0.427 =>

FRR = 11.58 % @ FAR = 0 %

Playing Around with Our Prototype



2006 NIST Iris Challenge Evaluation Details

FRR @ FAR = 0.1 %, where three FRR estimations are reported;
due to 30 different partitions of the iris database (ICE db) for test purposes

Group	FRR [%]		
	Minimum	*Median*	*Maximum*
Sagem-Iridian	0.473	1.22	2.31
Cambridge	1.06	1.93	3.29
Iritech	0.993	2.06	3.84

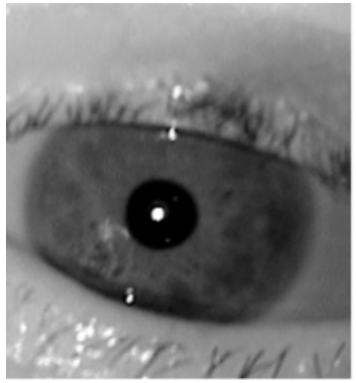
Just to contrast our prototype results (unfortunately on CASIA db, thus not directly comparable):
FRR = 7.7 % (@ FAR = 0.1 %)

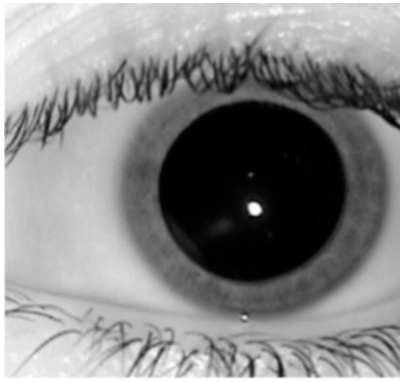
Pupil Dilation Influence

Why it varies?

- Change in ambient lighting
- Emotional state
- Health
- Medications
- *Age*
- Decision making

$$\text{Pupil Dilation Ratio} = \frac{\text{Pupil Radius}}{\text{Iris Radius}}$$

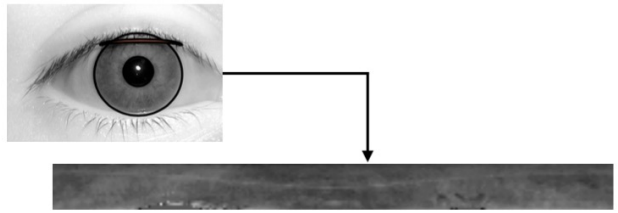




Statements

- “Although the iris stretches and contracts ... Such distortions in the texture can readily be reversed mathematically ... to extract and encode an iris signature that remains ***the same*** **over a wide range** of dilations.”, J. Daugman, U.S. Patent #5291560, 1994
- “Even though the visible portion of the iris changes as a function of pupil dilation, this **does not adversely affect** authentication.”, Alfred C. Weaver, How Things Work: Biometric Authentication, IEEE Computer, Feb. 2006

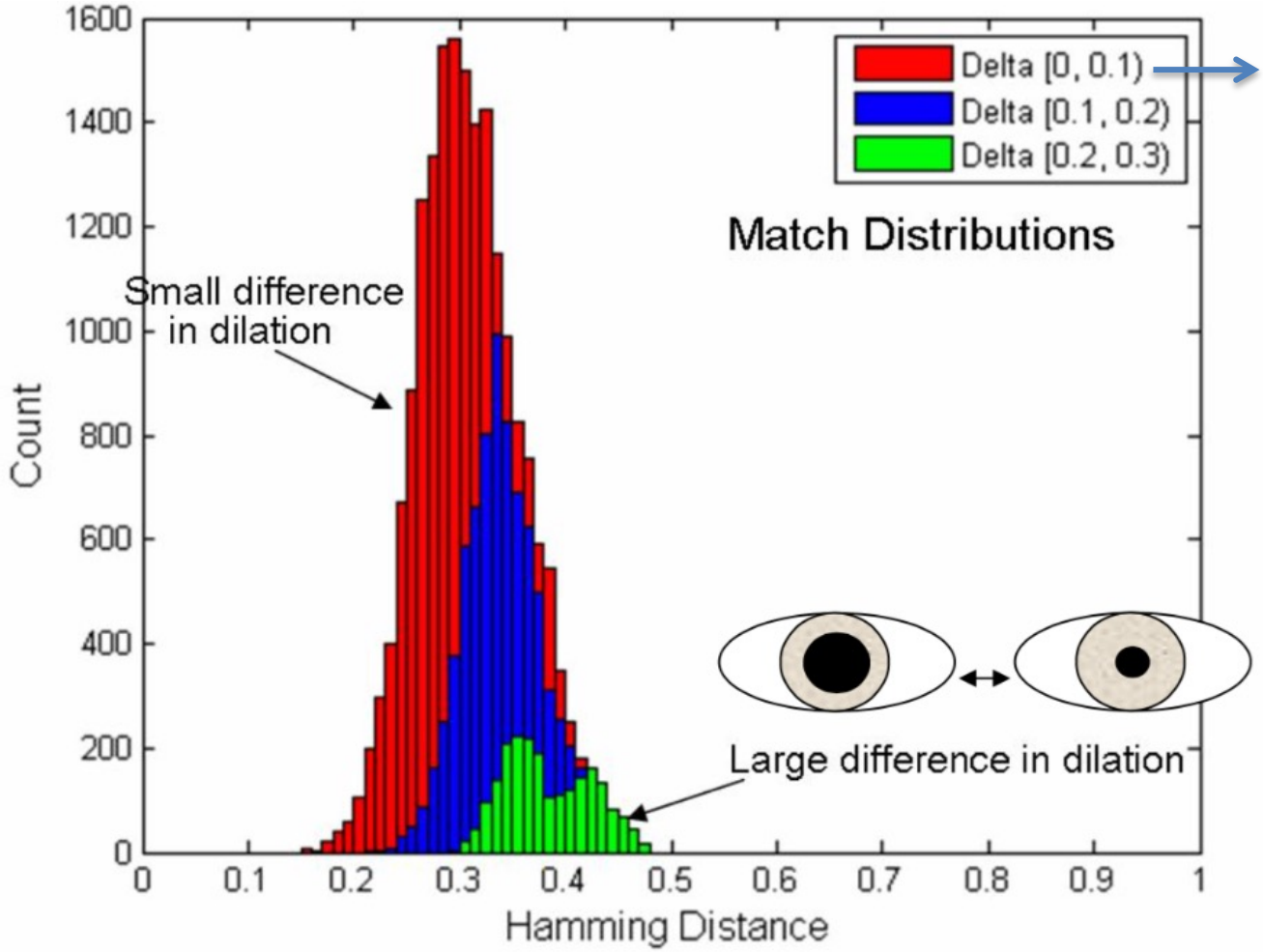
Experiment



- Fact: Dilation info is available at the segmentation stage, but current systems **discard** this info in going to the normalized iris image

- Data: 18 subjects, 632 iris images acquired via LG 2200, 28% of images taken with lights off, to induce normal dilation
- Method(ology):
  - Difference in dilation ratio for each pair of images matched
  - Modified ICE baseline software to compute and match iris codes
  - **Assumption:** authentic and impostor distributions are conditioned by dilation

Match Distributions



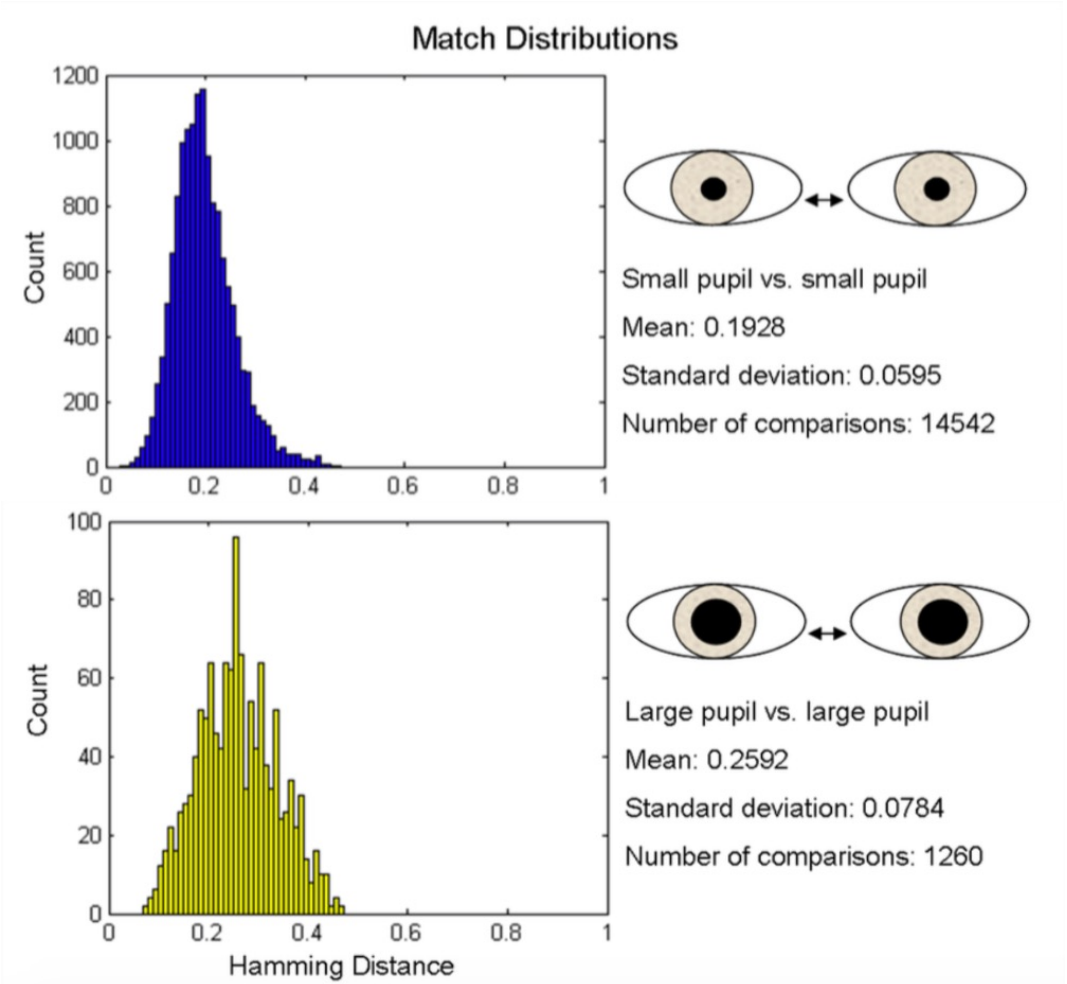
Difference in dilation ratio

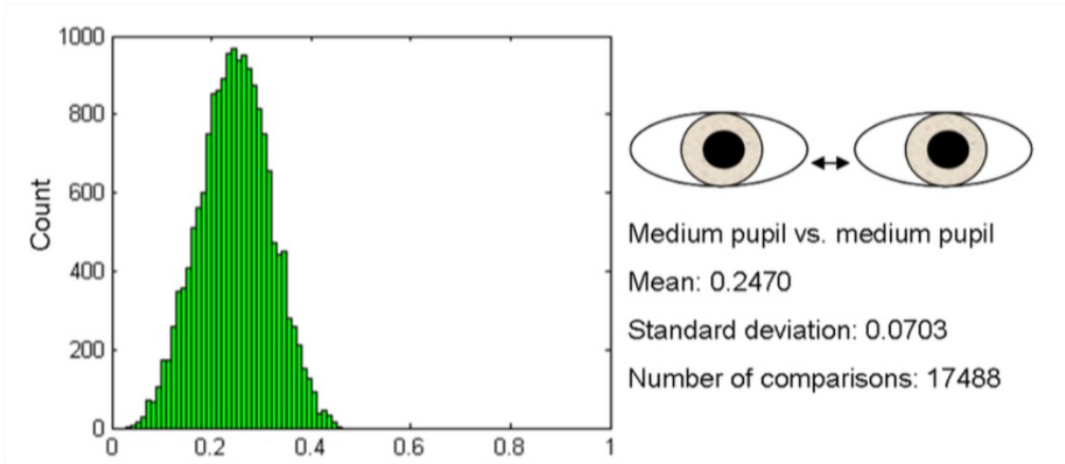
Increased difference in dilation **shifts** the match distribution

+No change in impostor distribution

Match Distribution Means

(Using **Similar** Dilation Ratio)

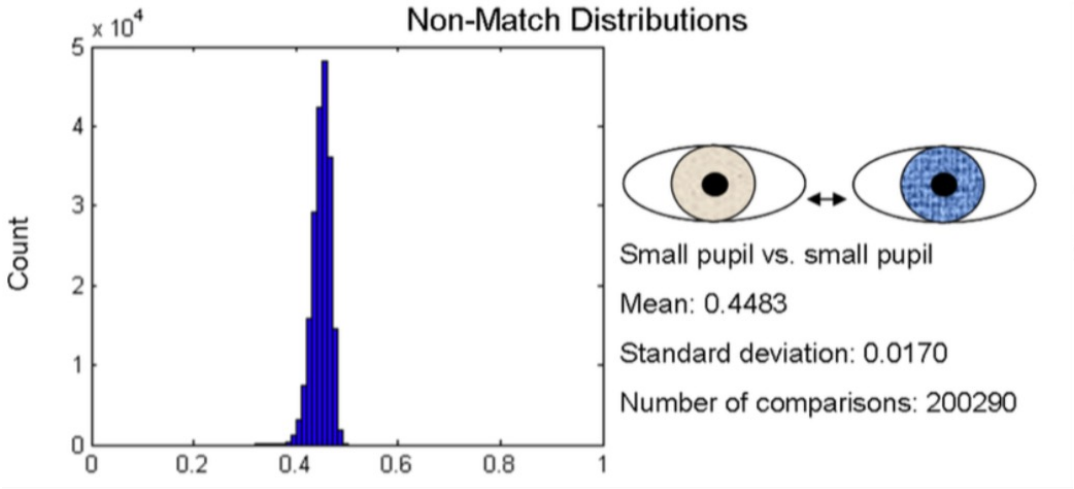


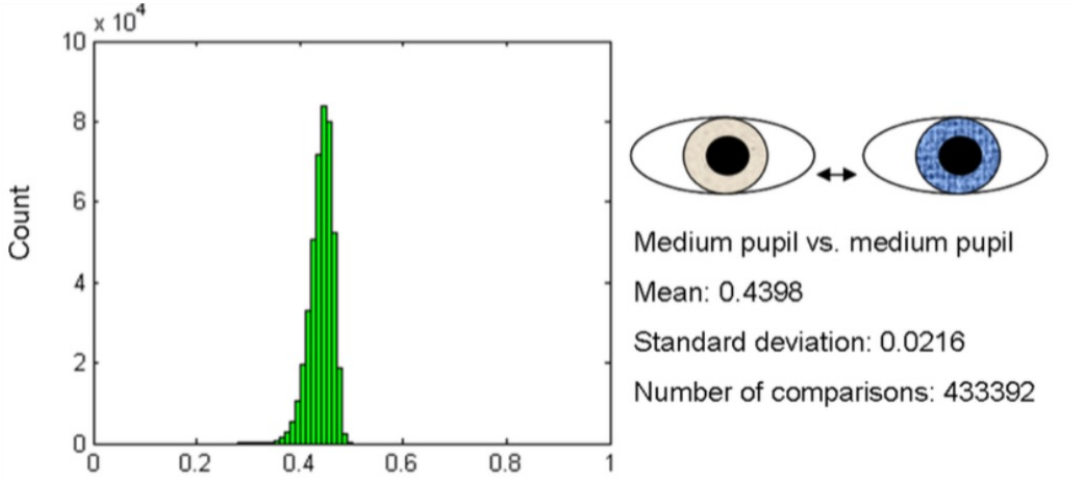


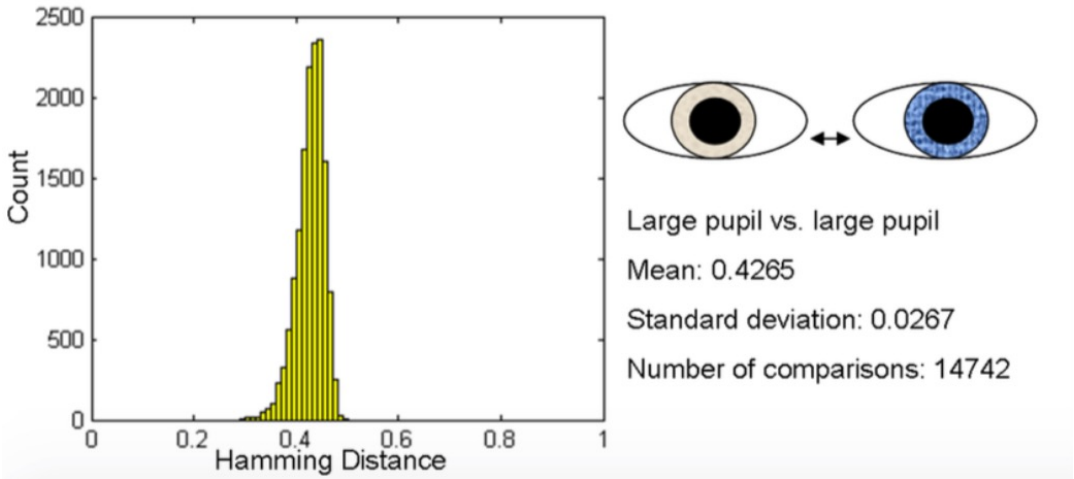
Larger pupil dilation **increases** the mean of the match distribution

Non-Match Distribution Means

(Using **Similar** Dilation Ratio)







Larger pupil dilation **somewhat decreases** the mean of the non-match distribution

This is somewhat surprising; but likely not a problem in typical practice

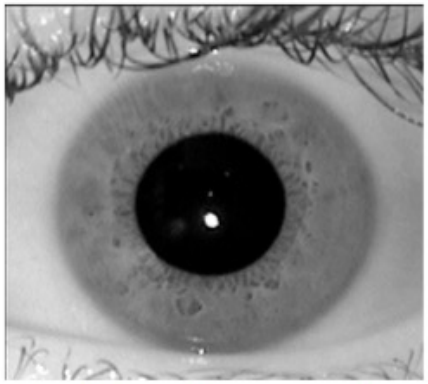
Dilation Influence – Conclusion

- When the difference in pupil dilation between two images is **large**, the **FNMR increases**

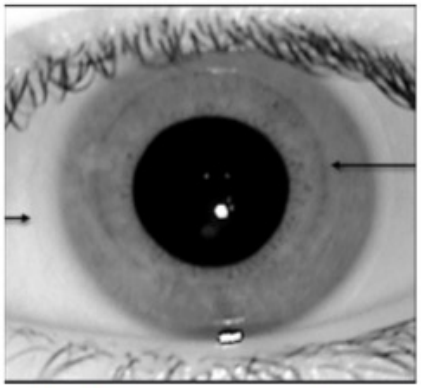
- NIST's IREX 1 report found similar results using different data
- The draft ISO standard for iris data format has a field for the pupil dilation ratio -> evidence that the role of dilation is (now) acknowledged

Influence of Lenses

- Myth: Wikipedia, “Iris Recognition”, Oct. 2010: “Iris recognition efficacy is rarely impeded by glasses or contact lenses.”
- Current page no longer contains this language, but instead says: “Iris recognition works with clear contact lenses ...”
- Let’s dig into this claim >>>



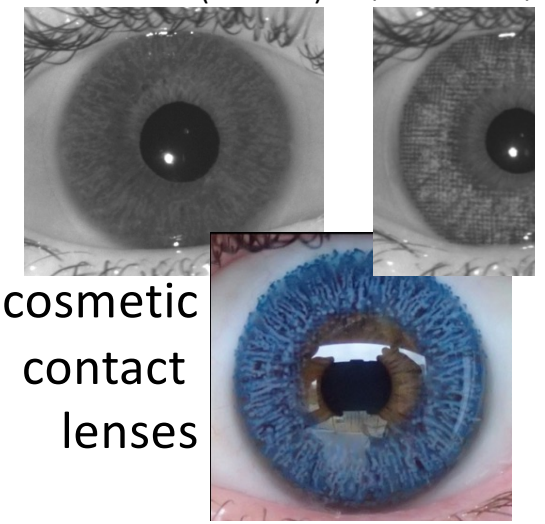
iris **without** contact lens



iris **with** contact lens

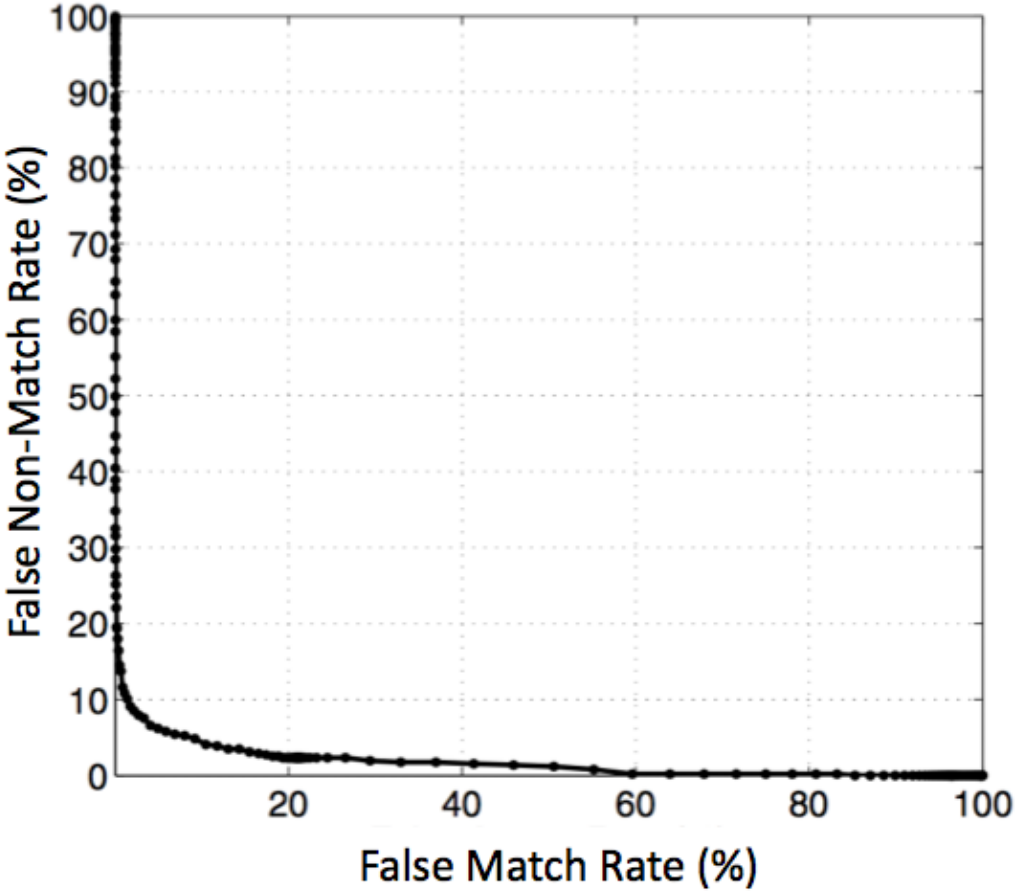
clear contact lenses

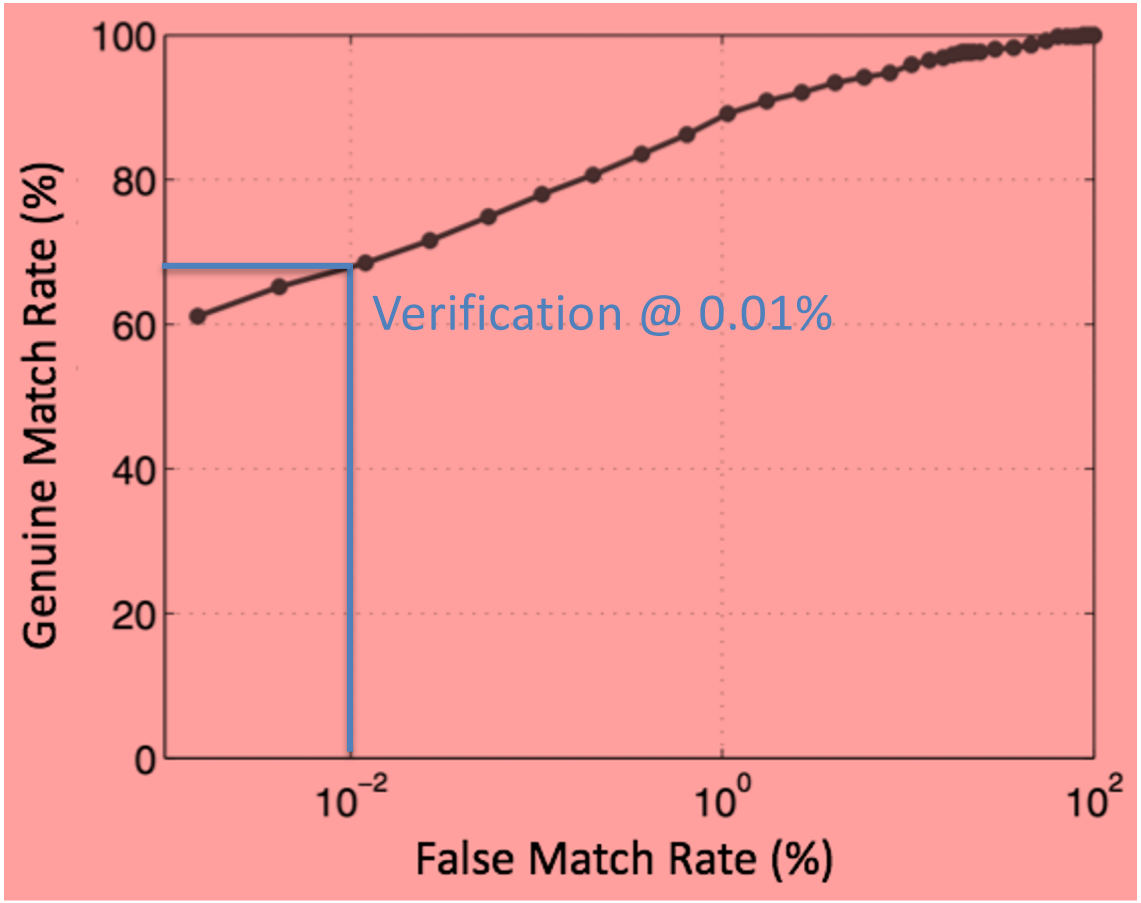
(AcuVue) (FreshLook)



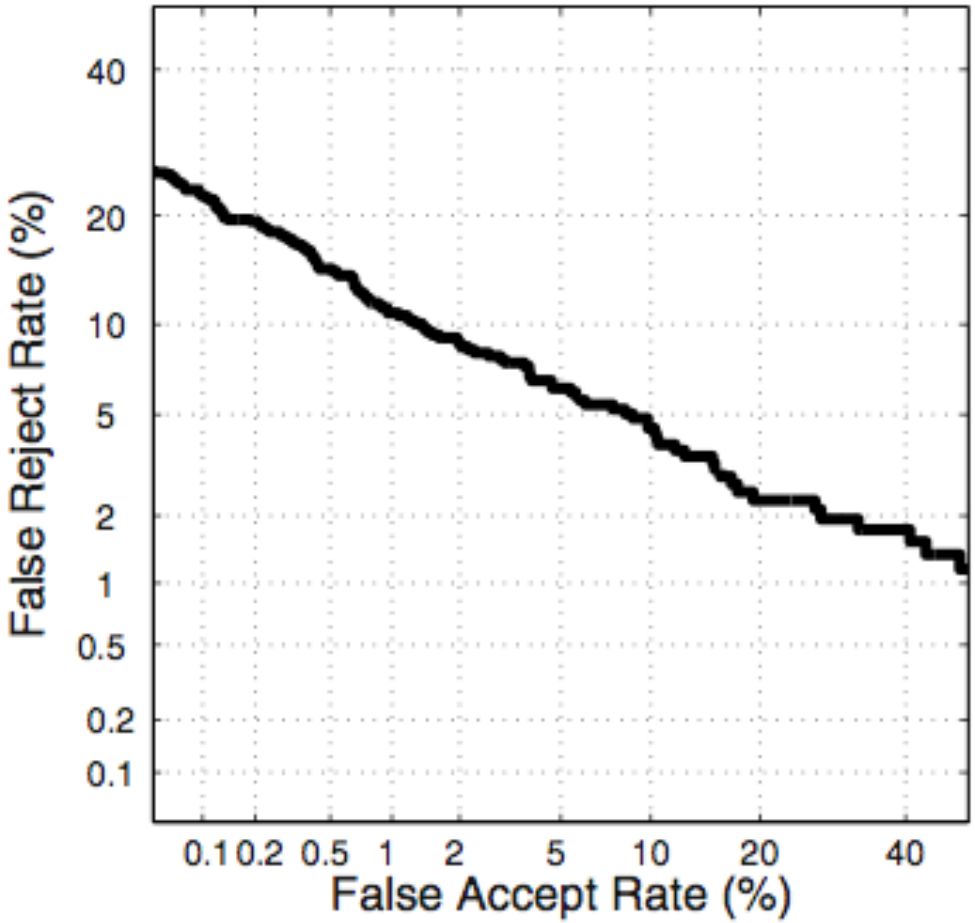
cosmetic contact lenses

Intermezzo: **R**eceiver **O**perating **C**haracteristic Curve (ROC)





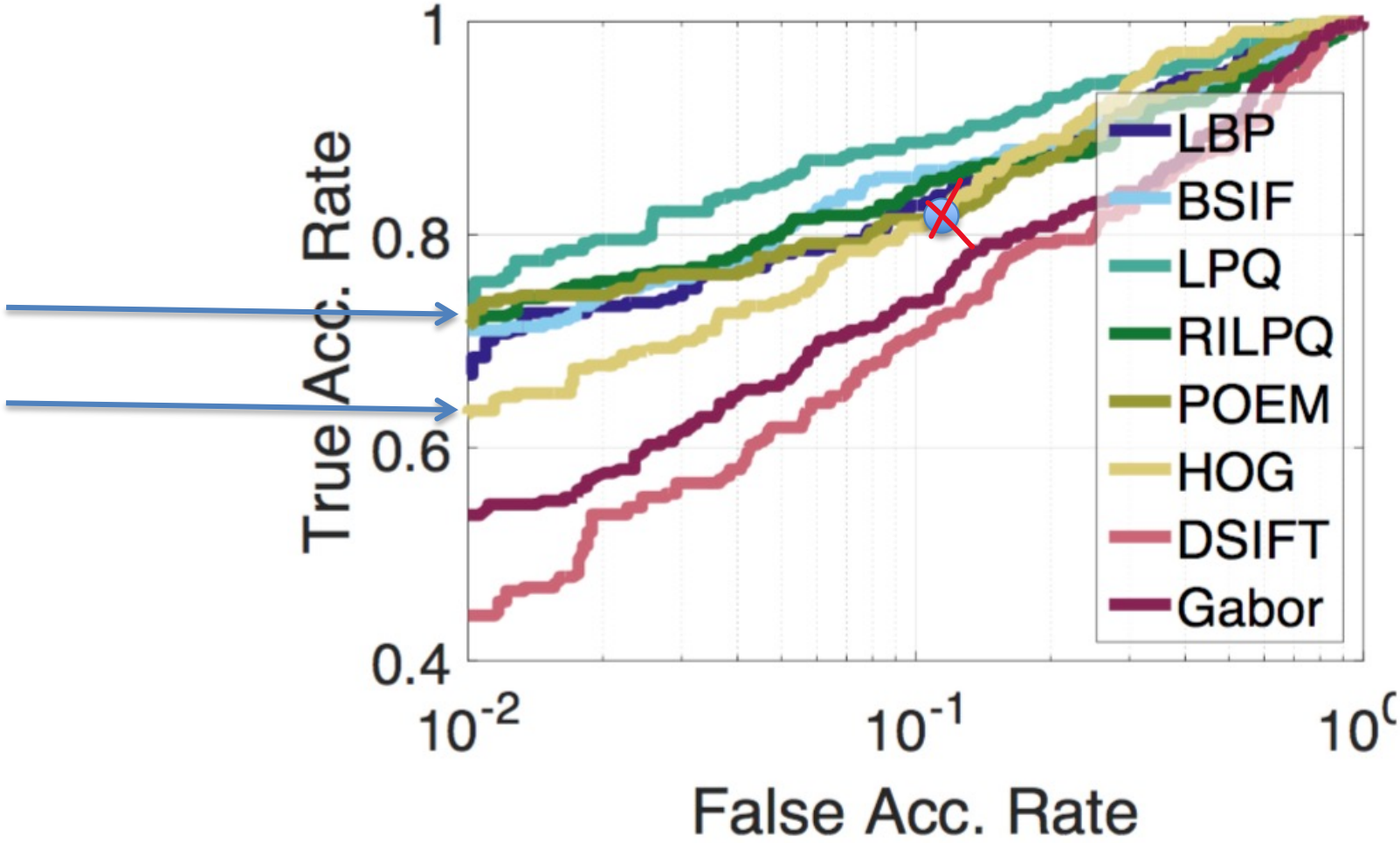
Detection **E**rror **T**radeoff Curve (DET)



ROC reveals …

- The best way to compare the performance of two biometric systems is to examine their ROC curves
- If the FRR of one biometric system (A) is consistently lower than the FRR of the other system (B) for corresponding values of FAR, one can conclude that the matching performance of biometric system A is better than that of B

Area **U**nder the ROC **C**urve (AUC)



0.9 - 1.0 = excellent
0.8 - 0.9 = good
0.7 - 0.8 = fair
0.6 - 0.7 = poor
0.5 - 0.6 = fail

use when you can't pick a threshold

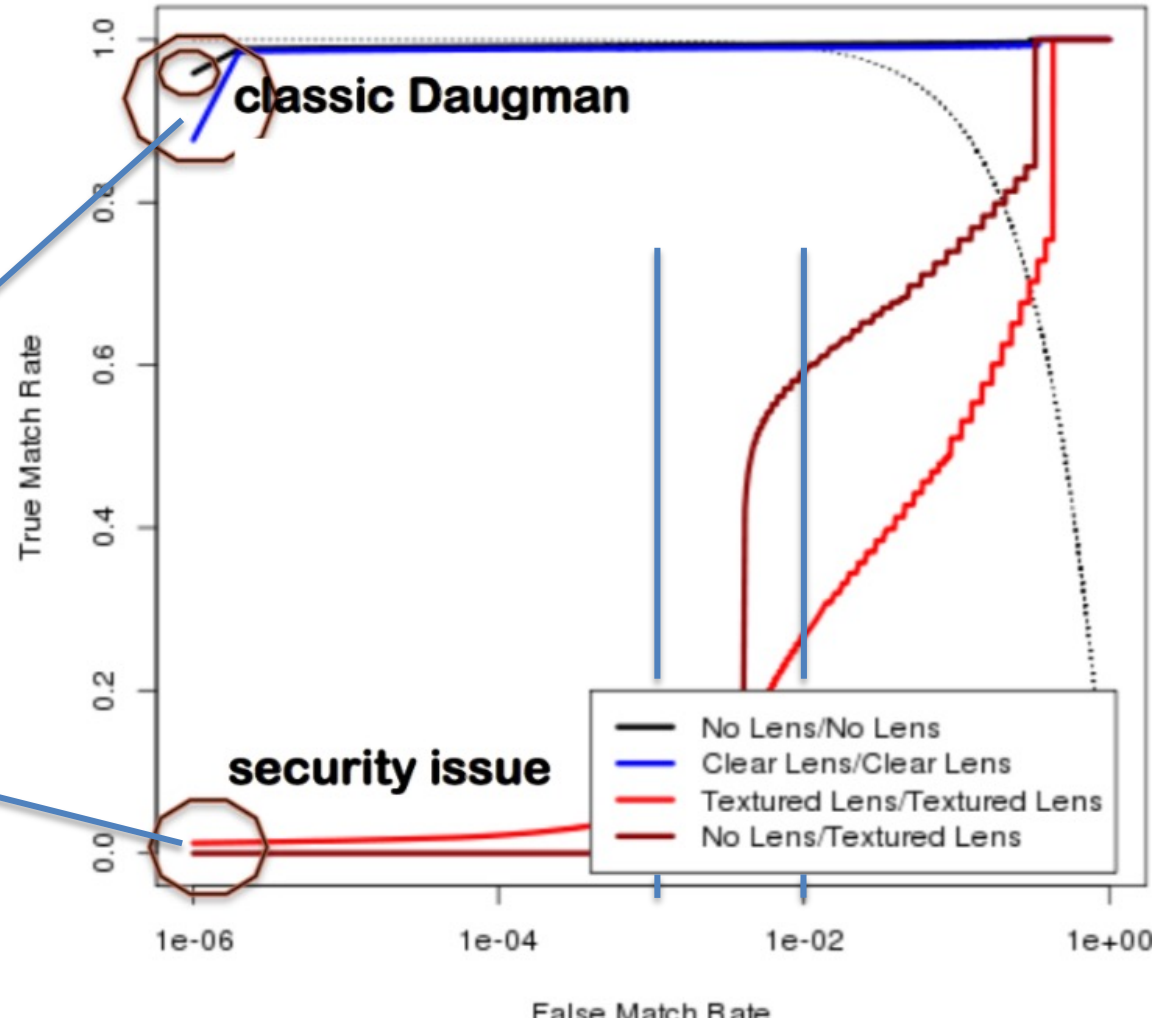
ROC Analysis for Lenses

NDCLD12 database

VeriEye 4.1 matcher

Clear lenses increase the FNMR slightly;
a minor social impact issue

Textured lenses are a security issue;
automatic detection needed



Variation within One Brand

For just the Freshlook line
there are 4 types
with 3 to 12 colors each

So how many different types & colors of textured lenses are out there?

Malicious Use of Contact Lenses

- Evade detection: intentionally create a false non-match (**easy** to do)
- Create a synthetic identity (**maybe**)
- Impersonate a target identity (maybe, but even **harder**)

Detecting Contact Lenses

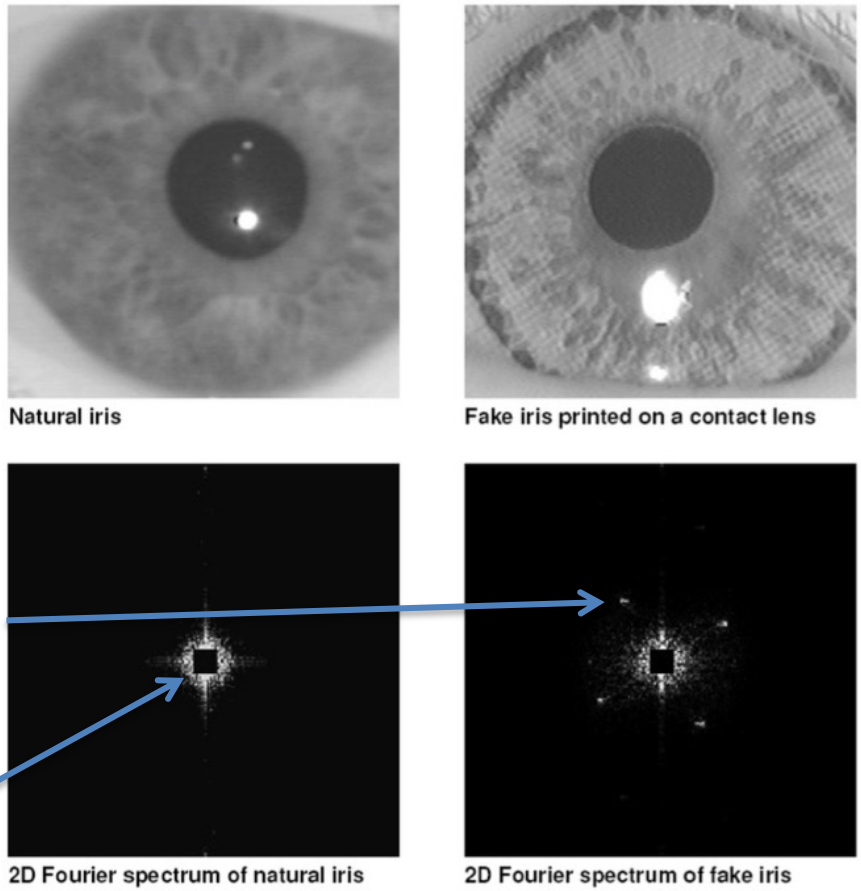
Daugman's frequency analysis approach:

- Fourier analysis for lens detection
- A **printed** iris typically exhibits some artifacts that can be detected by analyzing the 2D Fourier spectrum of the iris image

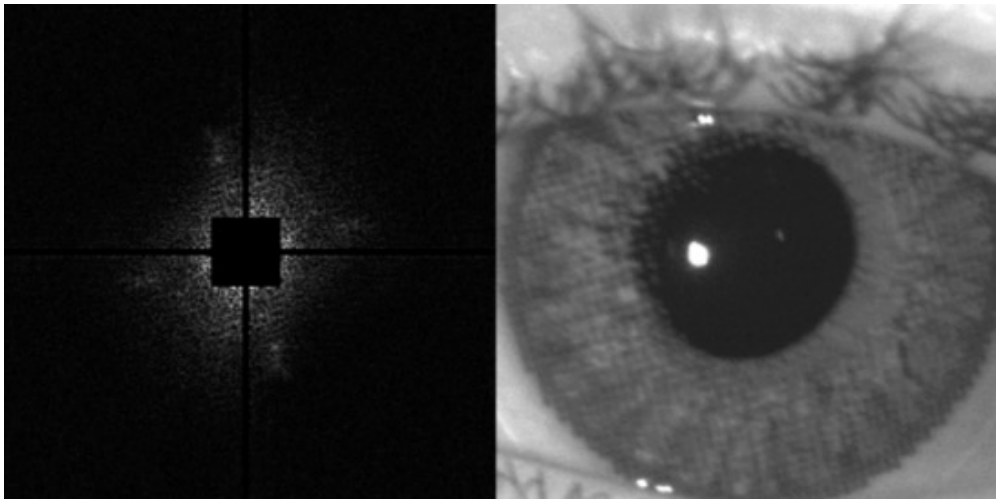
dot matrix printing process generates 4 points of spurious energy

the central square of spectrum has been blanked out to prevent its domination

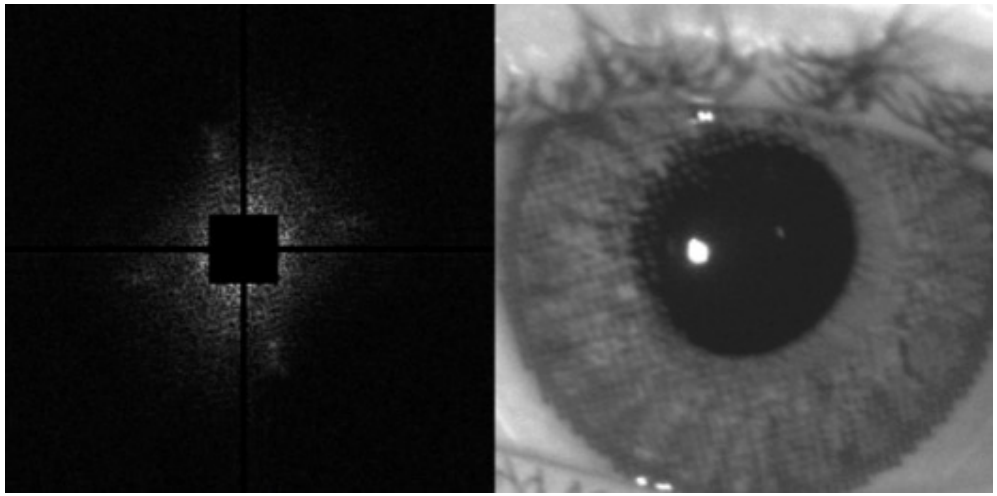
iris spoof detection



Various Producers

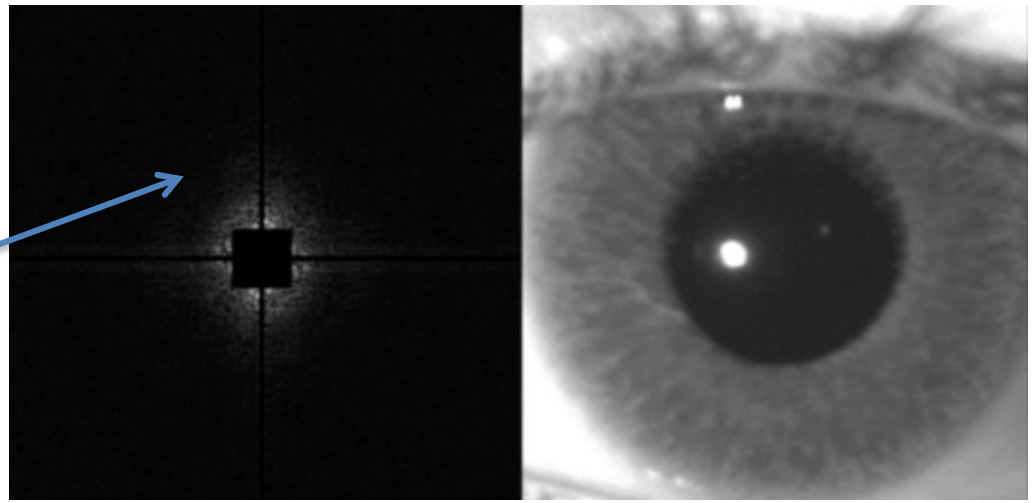


CooperVision



FreshLook

?!



AcuVue

Detecting Contact Lenses

Other approaches:

- Texture analysis (*features*: iris edge sharpness, iris-texton feature, co-occurrence matrix feature,… + *classifiers*)

person 1, lens type 1 person 2, lens type 2 person 3, lens type 3	person 1, lens type 2 person 2, lens type 3 person 3, lens type 1	person 1, lens type 3 person 2, lens type 1 person 3, lens type 2
person 1, lens type 1 person 1, lens type 2 person 1, lens type 3	person 2, lens type 2 person 2, lens type 3 person 2, lens type 1	person 3, lens type 3 person 3, lens type 1 person 3, lens type 2
person 1, lens type 1 person 2, lens type 1 person 3, lens type 1	person 1, lens type 2 person 2, lens type 2 person 3, lens type 2	person 1, lens type 3 person 2, lens type 3 person 3, lens type 3

Standard Machine Learning Approach

Person Disjoint

Lens Type Disjoint

Person-disjoint 10-fold cross-validation gives estimated performance of 100% correct classification using:

- LBP
- Variety of supervised classifiers ?!

Classifier	All
Naïve Bayes	98.0
Bagging	100
LogitBoost	100
JRip	100
J48	100
Random Forest	100

Wei, ICPR'08

- Printed iris pattern does not undergo any distortions when the pupil changes in size

Lens Detection – Conclusions

- Detection of textured lenses seen in training data is a solved problem
- Detection of **new textured (Open Set Recognition)** lenses is not a solved problem

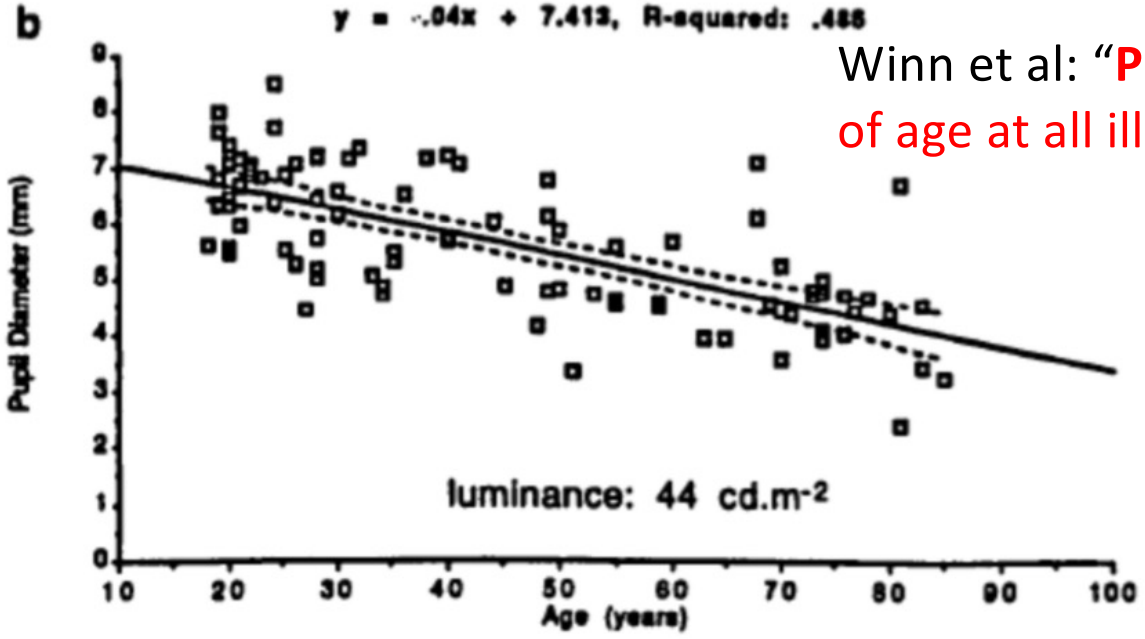
Influence of Aging

ISO/IEC 19795-1:2006 “Information technology - Biometric performance testing and reporting”, Section 6.4.6.:

- “Longer time intervals generally make it more difficult to match samples to templates due to the phenomenon known as ‘template aging’.”
- “This refers to the increase in error rates caused by time-related changes in the biometric pattern, its presentation and the sensor.”

What does medicine tell us?

From medical literature we know that the **iris changes functionally** with increased age



Winn et al: "**Pupil size** decreased linearly as a function of age at all illuminance levels. …"

Smaller average pupil size with age

=>

Greater time lapse translates into greater average difference in dilation

=>

Assumption: Greater difference in dilation degrades iris match scores

Flom and Safir's Patent

- 1987 patent allows for the possibility that **re-enrollment** might be needed to maintain performance =>

- "... the significant features of the iris remain extremely stable and do not change over a period of **many** years."
- "Even features which do develop over time... usually develop rather **slowly**, so that an updated iris image will permit ID for a substantial period ..."

Daugman's Patent

"The iris of every human eye has a unique texture of high complexity, which proves to be essentially immutable over a person's life"

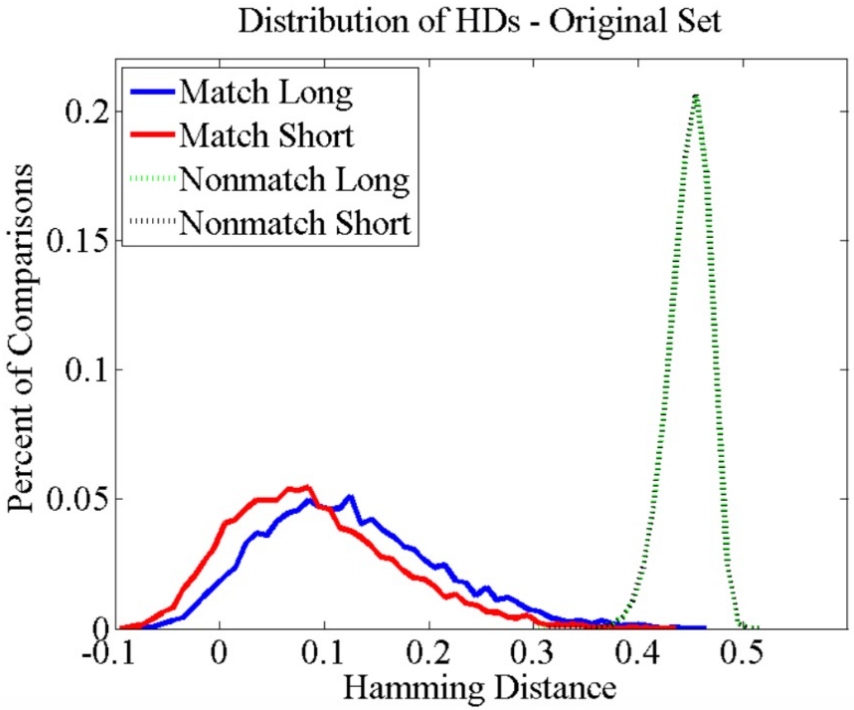
Which narrative won?

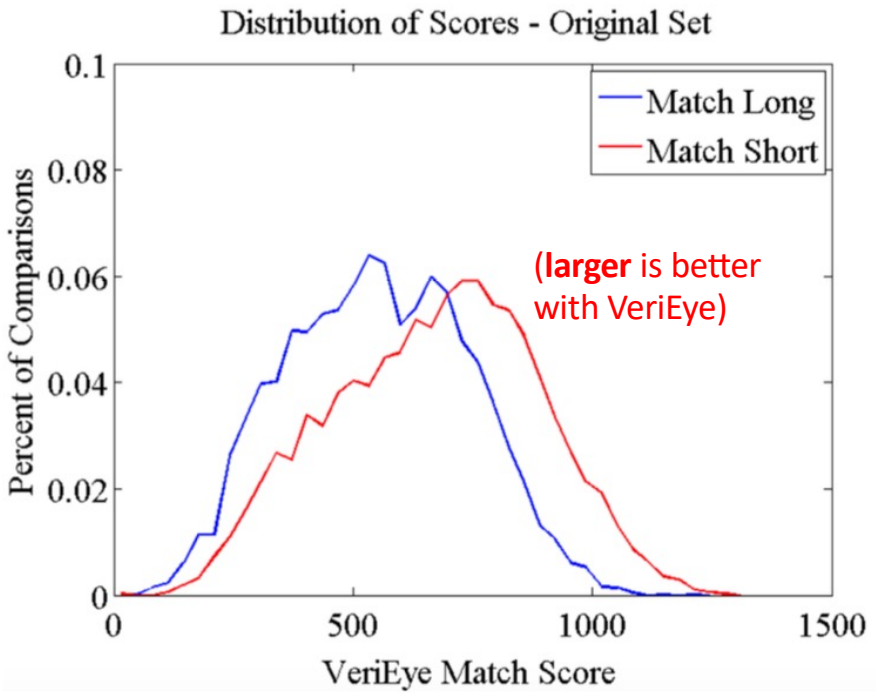
A single enrollment can last a lifetime ;-)

… but =>

Experiments

- 43 subjects, 86 irises
- LG 4000 iris sensor
- Two year time lapse
- 1830 images
- 9k short-term matches, 5 to 51 days
- 10k long-term matches, about 2 years
- Two matchers: IrisBEE, VeriEye



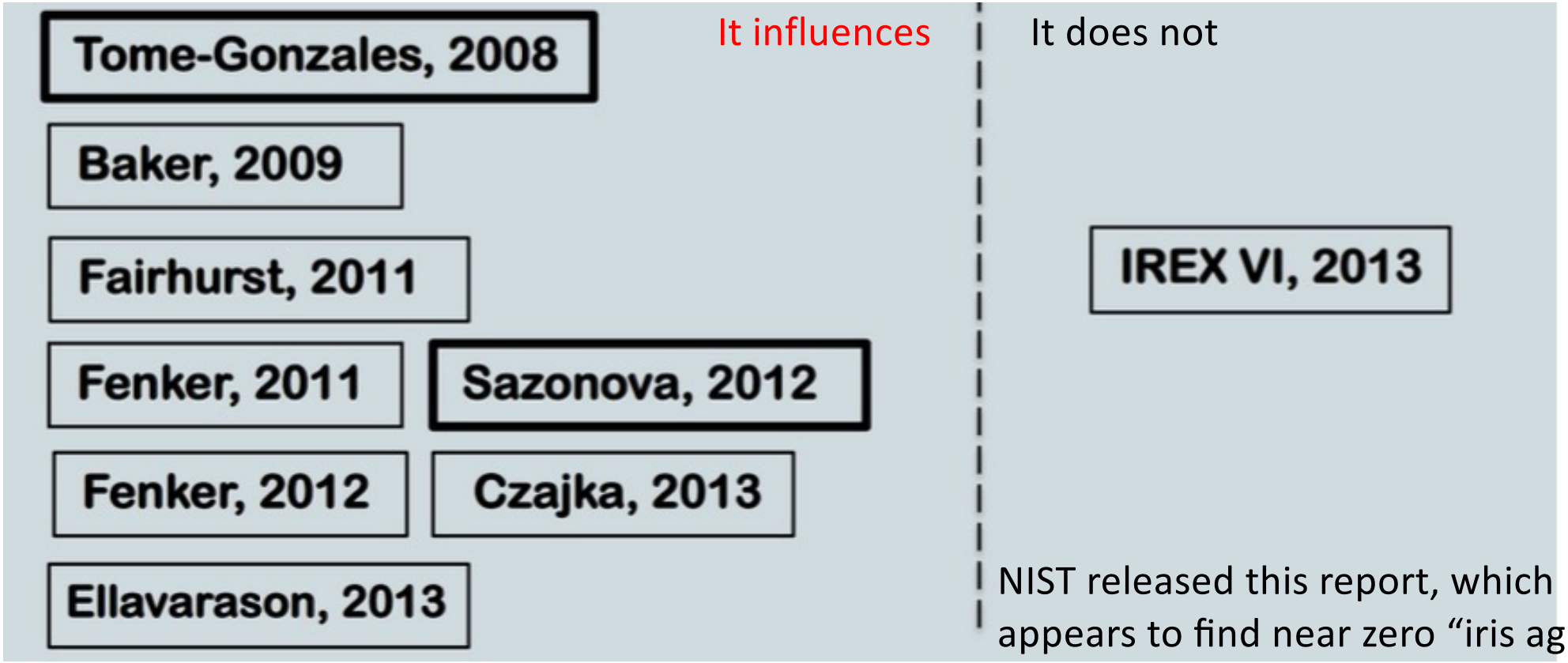


IrisBEE authentic distribution is clearly degraded, impostor distribution seems unchanged

VeriEye authentic distribution is also clearly degraded

- 84 of 86 had an increased IrisBEE mean HD with longer time lapse
- 82 of 86 had a decreased VeriEye mean score
- 2 years time-lapse:
  - IrisBEE: 157 % to 305 % increase in FRR at thresholds of 0.28 to 0.34
  - VeriEye: 239 % to 370 % increase in FRR at thresholds of 60 to 120

Iris Template Aging is Controversial



NIST released this report, which appears to find near zero “iris aging” on a large “operational” dataset (with some “strange” particularities)

Open Research Questions?

Acquisition:

- How well can iris recognition be performed with little or no explicit user **cooperation**? *[*on-the-move; the iris is a slightly oscillating object within a moving object (the eye-ball), which is located within yet another moving object (the head) on top of the moving body
  + long focal length needed
  + activating the NIR illuminant ☺ *]*
- How well can iris recognition be performed using **visible** light or cross-wavelength images?

Open Research Questions?

Masking:

- How well can iris texture **occlusions** by specularities, eyelashes, shadows, etc. be found?

Iris code:

- Is there a better way to **compute** an “iris code” from the iris texture? Perhaps higher-res?
- Is there simply a **different** way that could be combined with the iris code?

Open Research Questions?

Quality:

- While initial research has shown the benefits of incorporating iris quality, its **assessment** and **use** in a real-time environment is still an open challenge

Dilation:

- What is the **best way** to deal with the effects of difference in pupil dilation between images?
  - Control acquisition?
  - Enroll multiple samples?
  - Better model of dilation? // Something else?!

Open Research Questions?

Lenses:

- Can the presence of contact lenses be automatically **detected**?
- Can the artifacts created by clear contact lenses be **reversed** – or at least **masked** out?

Aging:

- Does template aging **occur**?
- Does the rate of template aging vary with **demographic** factors?
- Can **aging-resistant** algorithms be designed?

General:

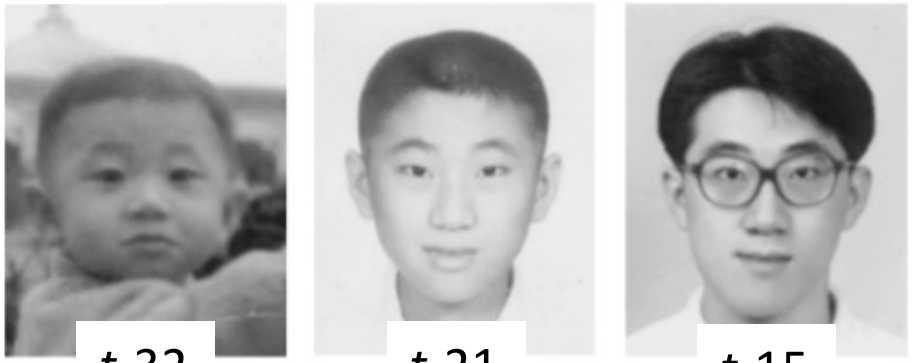
- Although researchers have used match-score distributions and IrisCode statistics to infer the iris biometric's degrees of freedom, no one has yet directly used the iris's biological underpinnings to **ascertain its individuality**
  => This interesting problem has implications for using iris recognition in a court of law in accordance with Daubert's admissibility criteria and Federal Rules of Evidence (remember: as with fingerprints)

MODALITY #3: FACE

- Human face is a specific modality – why?
- Not just that it enables recognition, but gives other attributes like **gender**, **age**, **ethnicity**, and **emotional** state (i.e. facial expression) of a person (soft modalities)
- The face is considered to be the most commonly used biometric trait **by humans**

Many Variations?

perfection ;-) pose illumination expression



t-32 *t*-21 *t*-15

glasses sunglasses cap scarf

\+
facial hair,
make-up,
…

Not Enough **Inter-Class** Variations?

Twins

Family

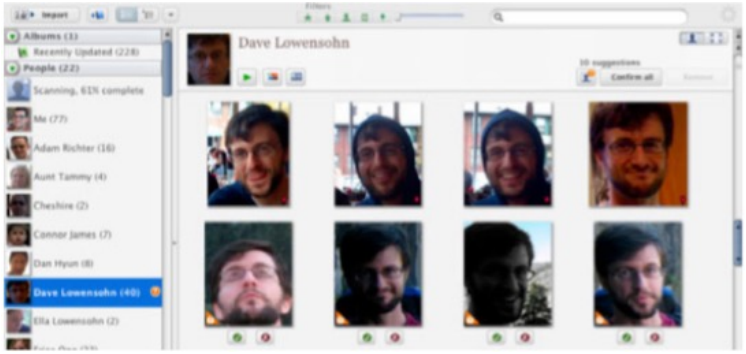
Advantages

1. Face can be captured at a longer stand-off distance using non-contact sensors => surveillance applications
2. Conveys also emotions, gender, age, ethnicity => interactive human-computer interfaces
3. There are large legacy face databases (e.g. U.S. driver's license repositories cover over 95% of the adult population), which enable **large scale** analysis of the face modality in terms of individuality or scalability
4. People are generally more willing to **share** their face images in the public domain (e.g. Facebook, Instagram)

Applications

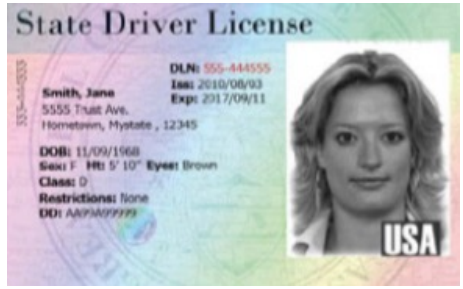
- law enforcement
- civilian identification
- surveillance systems
- entertainment/amusement systems

Australia's SmartGate system:
faster immigration clearance



Picasa,…: automated face tagging

Kinect:
personalizing XBOX



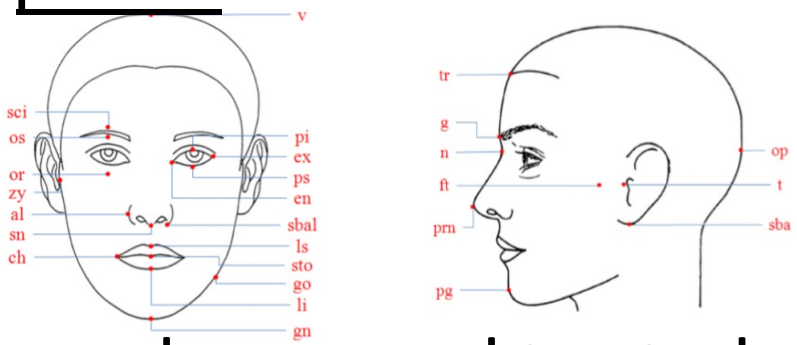
Morpho's driver license solution:
prevents a single person
obtaining multiple licenses

Humans & Face Recognition (*Psychology*)

- Fact: humans find it difficult to detect or recognize faces that are **inverted**
- *Prosopagnosia* – a disorder in which an individual loses the ability to recognize only faces
- Fact: humans perceive the face based on certain higher-level characteristics

Facial Features

- **Anthropometric** studies have attempted to characterize the dimensions of the face based on a set of **anatomically** meaningful **landmark** or **fiducial** points



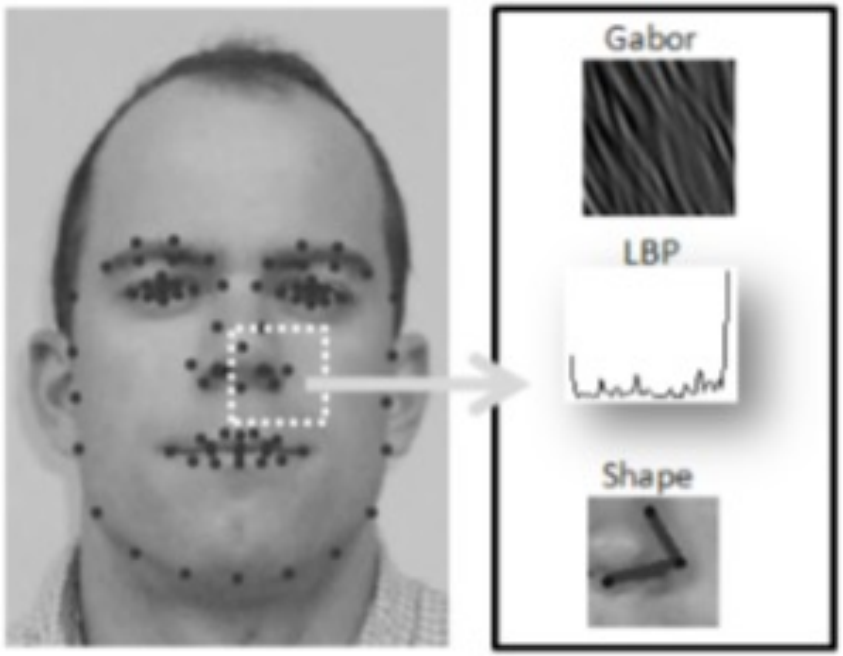
- Anthropometric measurements have been used to study the **growth patterns** in humans as well as understand characteristics of the face as it pertains to **gender** and **ethnicity**
- … but lack of **distinctiveness** for automated FR!

3 Levels of Features (again ;-)

- L1: gross facial characteristics like **geometry** of the face and global skin **color**
  - a short round face and an elongated thin face
  - faces exhibiting predominantly male and female characteristics
  - faces from different races
  - extracted even from low resolution face images (< 30 inter-pupillary distance (IPD))

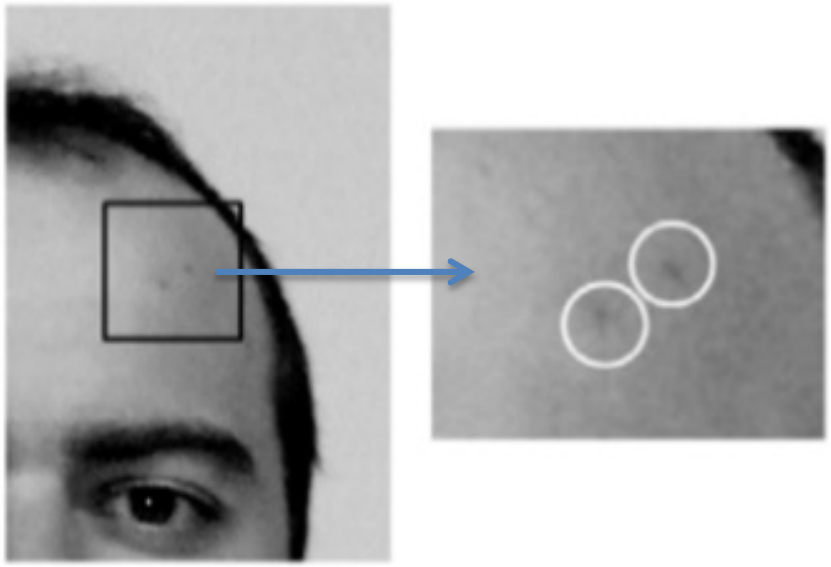
3 Levels of Features

- L2: localized face information such as the structure of the face **components** (e.g. eyes), the **relationship** between facial components, **precise shape** of the face
  - accurate face recognition (!)
  - local regions characteristics can be represented using **geometric** or **texture** descriptors
    - 30 to 75 IPD



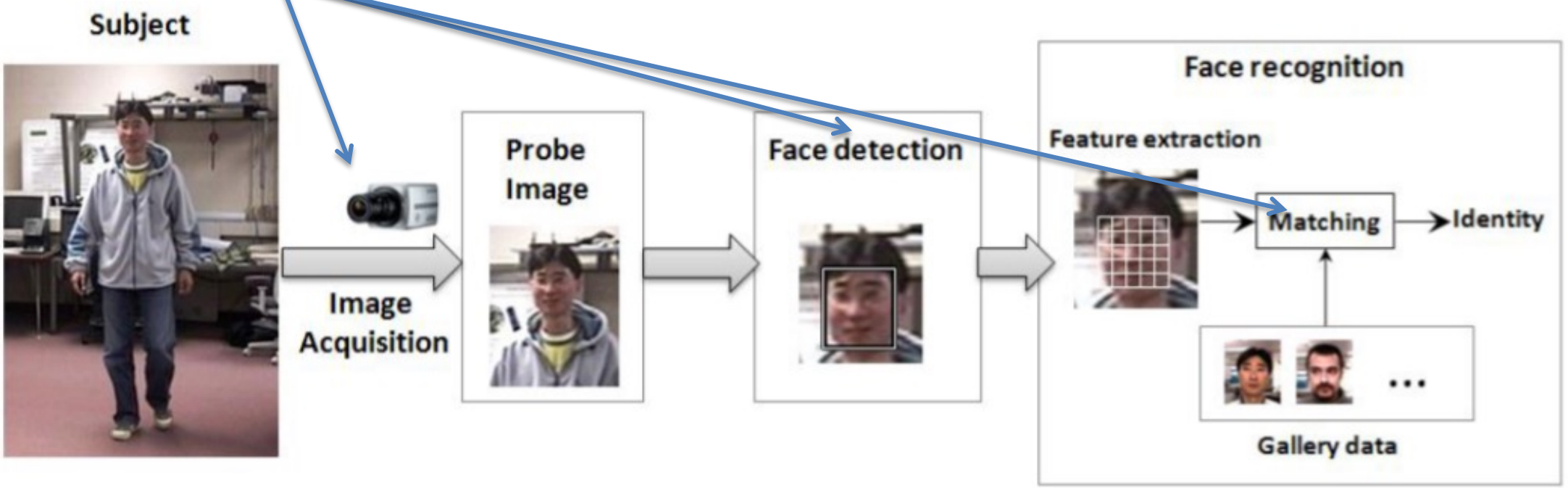
3 Levels of Features

- L3: unstructured, **micro level** features on the face, which includes scars, freckles, skin discoloration, and moles
  - discrimination of identical twins (?!)

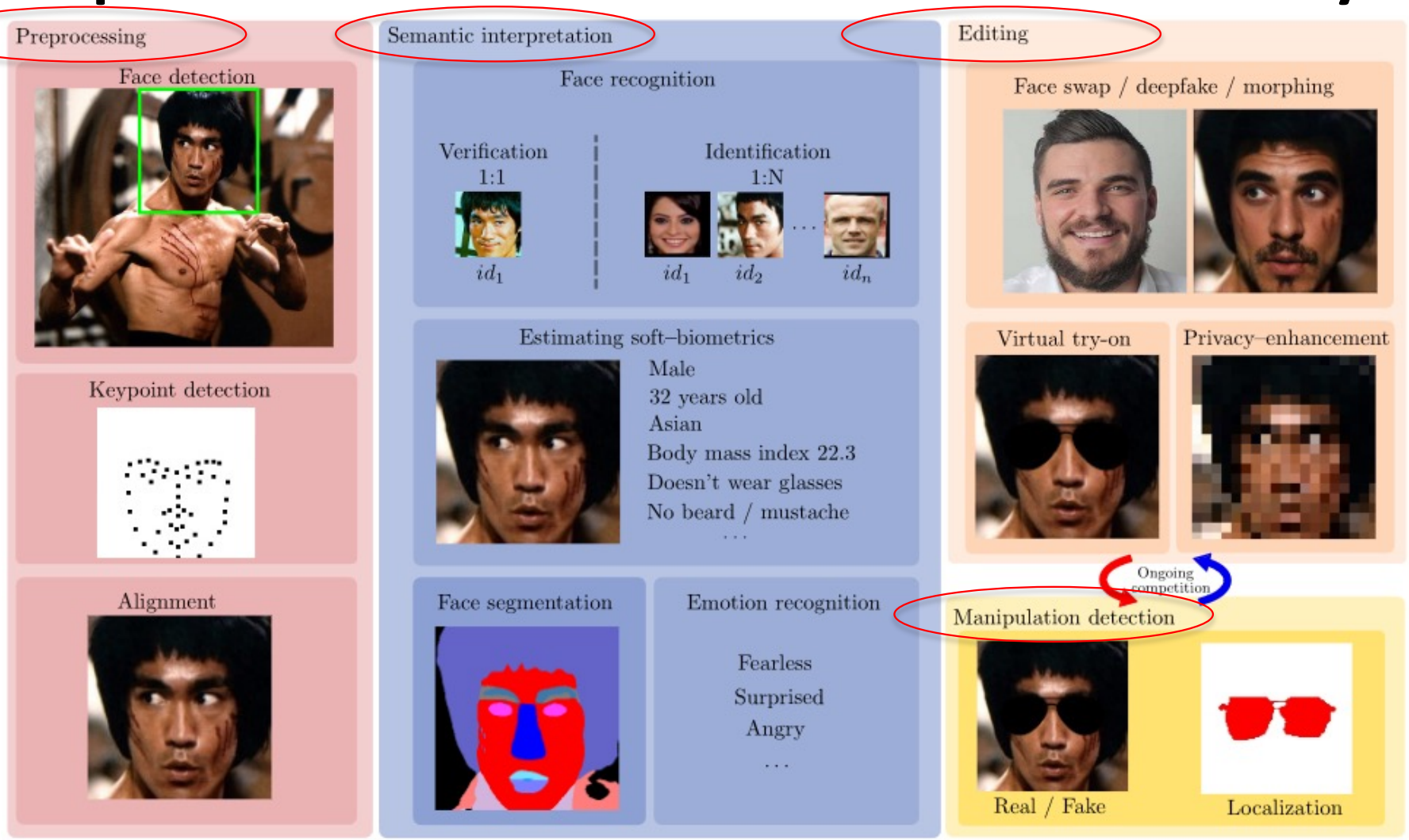


Basic FR Pipeline

- Most of the automated FR systems make use of **2D** images acquired in the **visible** spectrum
- Most commercial FR engines first detect the **two** eyes
- Three modules



Comparison to Broader Face Analytics



From 1973 to the Present

Figure 3-3

Typical sequence of the analysis steps.

(a) top of head
(b) cheeks and sides of face
(c) nose, mouth, and chin
(d) chin contour
(e) face-side lines
(f) nose lines
(g) eyes
(h) face axis

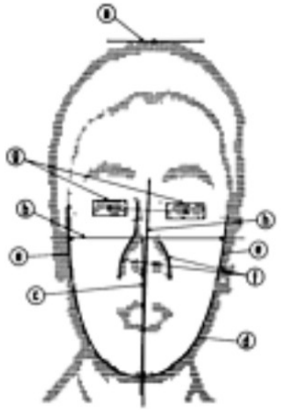
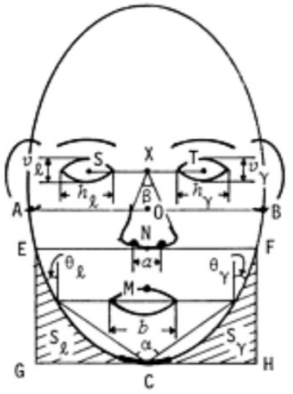


Figure 3-4

Result of feature extraction.

(a) Original photograph (b) Printout of the digital gray-level picture (c) Binary picture

Figure 3-2 Picture input and line extraction. The dark horizontal line in the upper part is due to the burn in the CRT surface of the FSS used for digitization.



S_ℓ S_Y : hatched areas

x_1 = AB/OC
x_2 = ST/AB
x_3 = NC/OC
x_4 = {curvature of the top of the chin
x_5 = $(\Delta EGC+\Delta FCH)/(S_\ell+S_Y)$
x_6 = α
x_7 = $\frac{1}{2}(\theta_\ell+\theta_Y)$
x_8 = β
x_9 = XN/XC
x_{10} = NM/XC
x_{11} = $(h_\ell+h_Y)/ST$
x_{12} = $(v_\ell+v_Y)/ST$
x_{13} = a/ST
x_{14} = b/ST
x_{15} = ST/XN
x_{16} = $(h_\ell+h_Y)/(v_\ell+v_Y)$

Figure 4-12 Facial parameters.

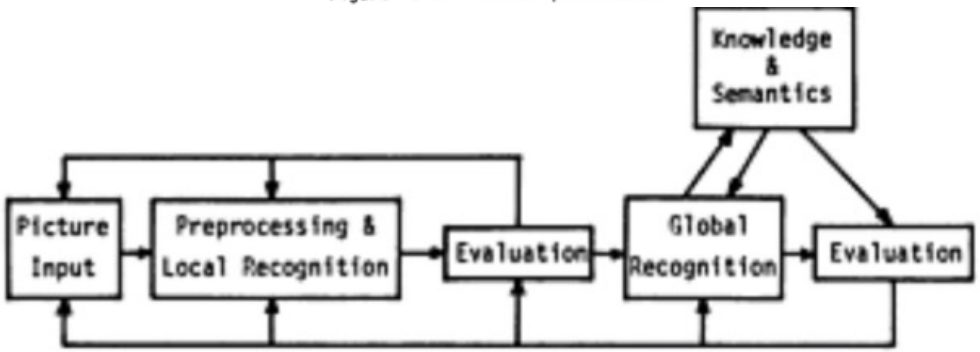


Figure 2-1 Advanced picture processing scheme.

Image Credit: T. Kanade 1973

Acquisition → Normalization → Alignment → Features → Recognition

Acquisition

2D sensors >>>

Multiple 2D cameras

Pan-Tilt-Zoom camera

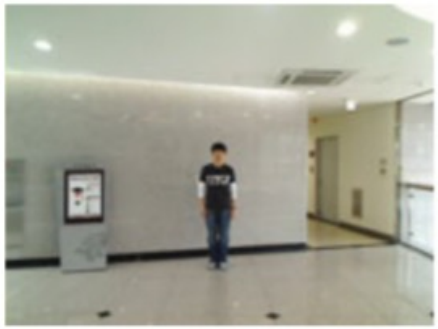
Long range infrared camera

Why such sensors?

To address:

- spatial resolution
- variations in illumination
- self-occlusion (frontal, profile)
- the pose variation problem

640 × 480 >>>



1m

IPD=35

3m

IPD=12

5m

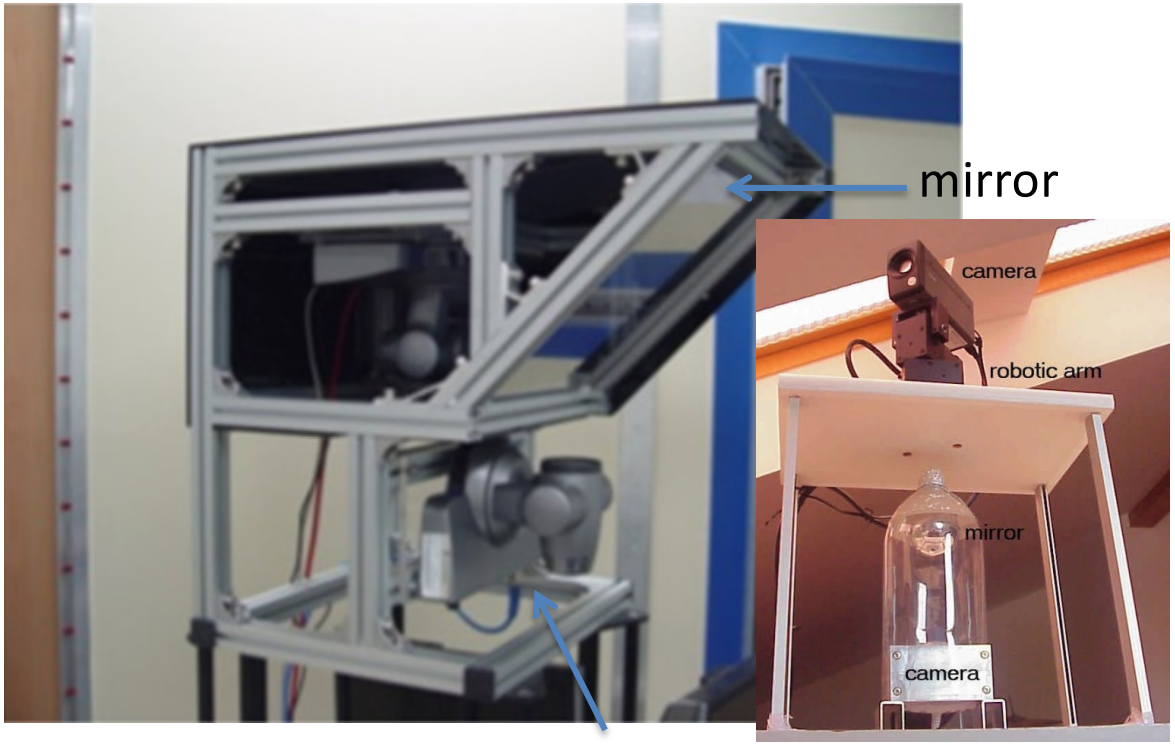
IPD=7

VISIBLE
950 nm
1050 nm
1150 nm
1250 nm
1350 nm
1450 nm
1550 nm

Surveillance System Setup

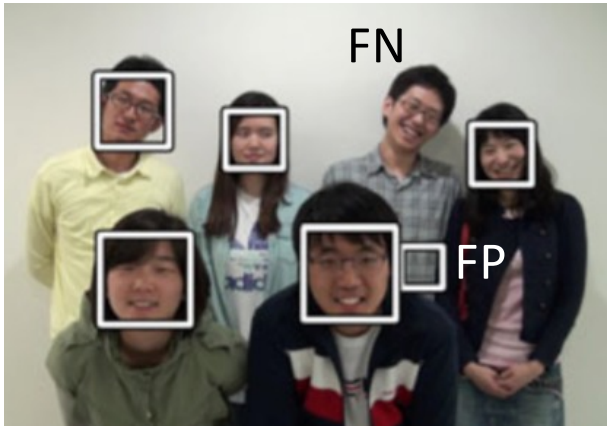
Detect (and track) human

Turn the camera and zoom in on the face



Face Detection

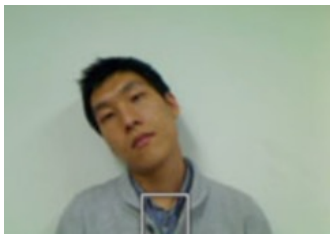
- State-of-the-art face detection methods are typically based on extracting **local texture features**
- Next approach follows the seminal work done by **Viola and Jones**



… but is not perfect =>

cluttered background

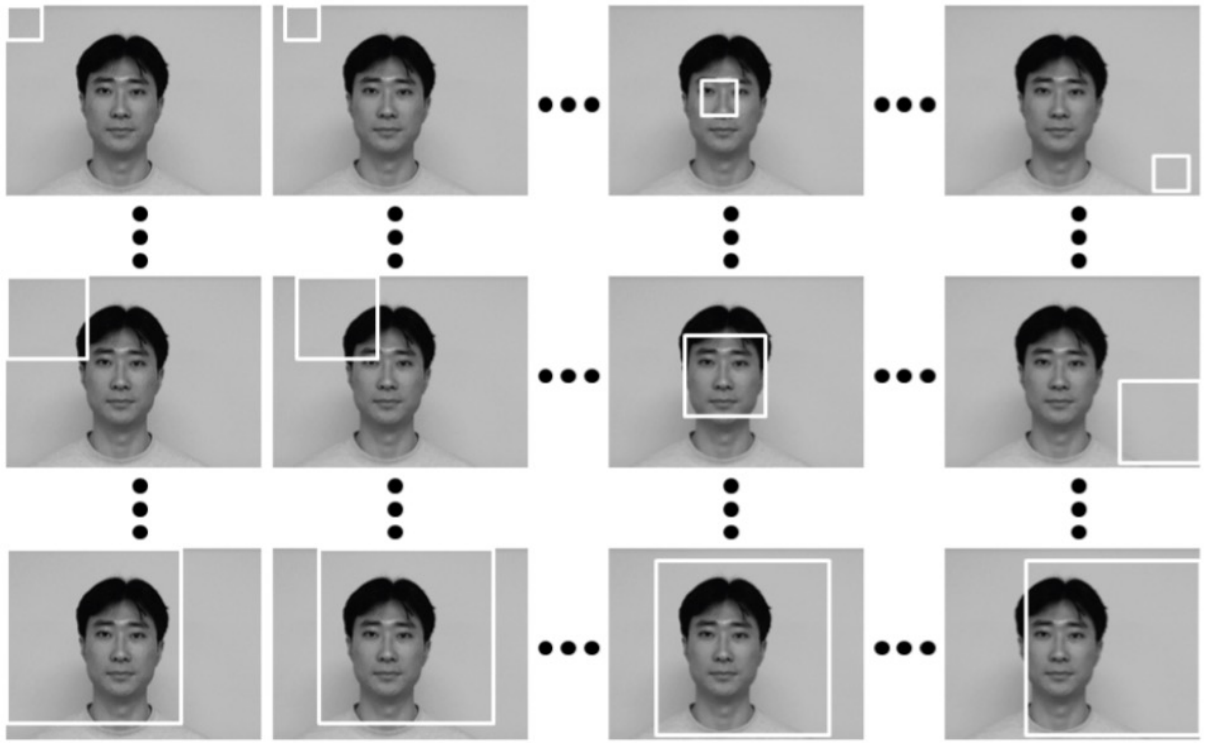
tilted face



inverted face

Viola-Jones Face Detector

Scans through the input image with detection **windows** of different sizes and decides whether each window contains a face or not



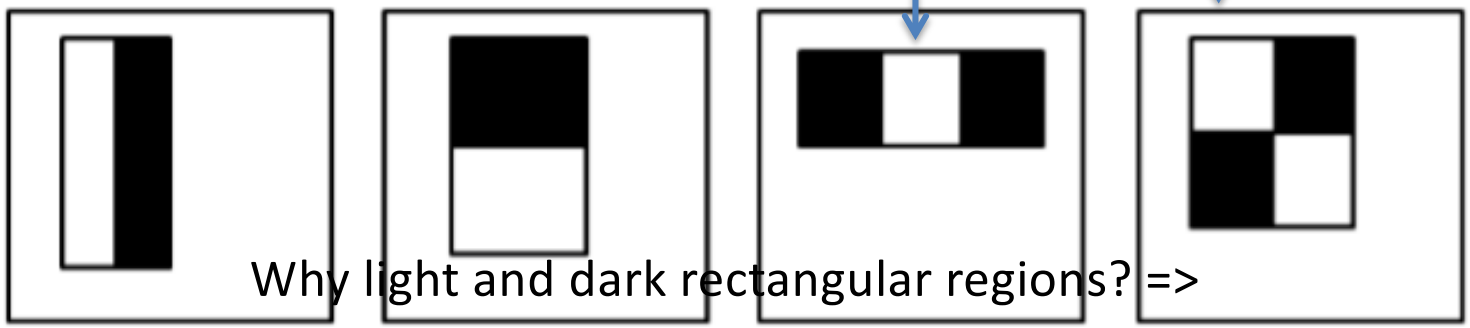
Viola-Jones Face Detector & **Filters**

In each window the existence of a face candidate is decided by applying a **classifier** to simple **local features** derived using **rectangular filters** (Haar-like)

… i.e. **two**-rectangle, **three**-rectangle and **four**-rectangle filters >>>

detection window

When the **combination** of filter responses (features) in a certain window exceeds a threshold, a face is said to have been detected

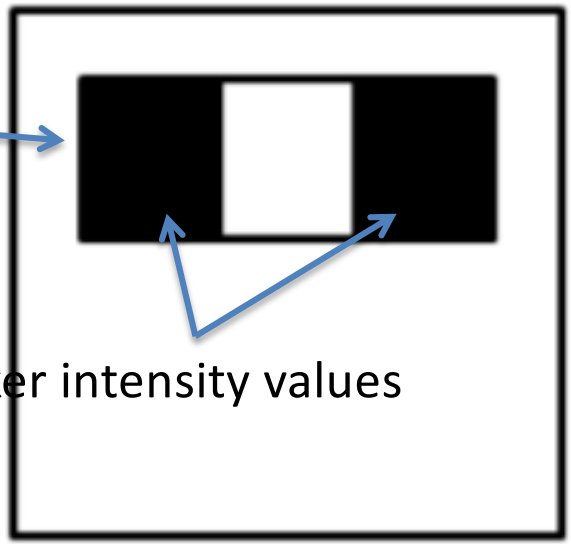


Why light and dark rectangular regions? =>

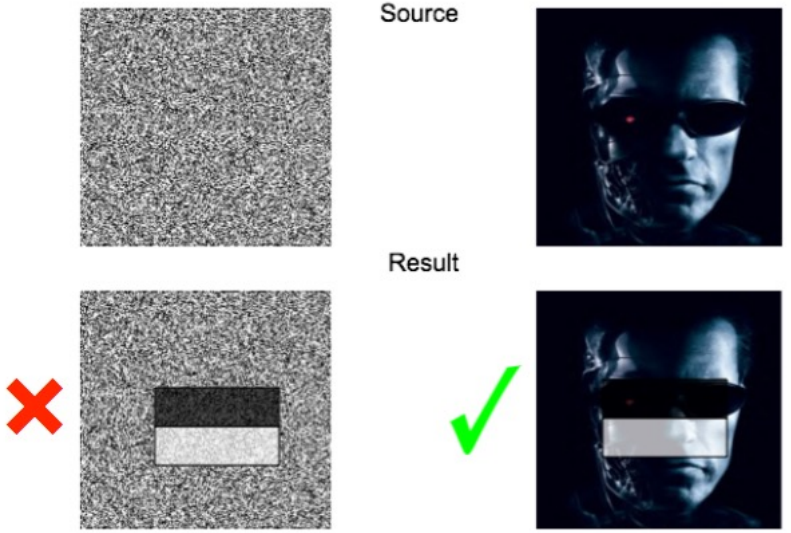
Viola-Jones Face Detector & **Feature Values**

Feature values are obtained by computing the **difference between the sum of the pixel intensities** in the light and dark rectangular regions

filter can be used to detect
the two eyes and
the bridge of the nose



the eyes typically have darker intensity values

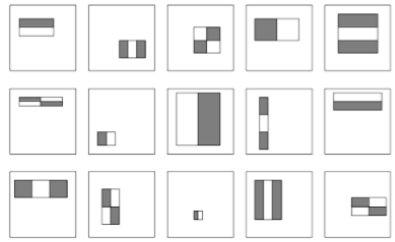


Viola-Jones Face Detector & **Integral Image**

Set of Haar-like filters of **different** sizes need to be applied to each window and the filter responses must be **combined** in an appropriate way in order to detect a face

=> more than **180,000** different Haar-like features can be derived from a detection window of size 24 × 24 pixels

=> to reduce computational burden calculate **integral image**



Input image

1	2	2
2	3	0
1	0	3

→

1	3	5
3	8	10
4	9	14

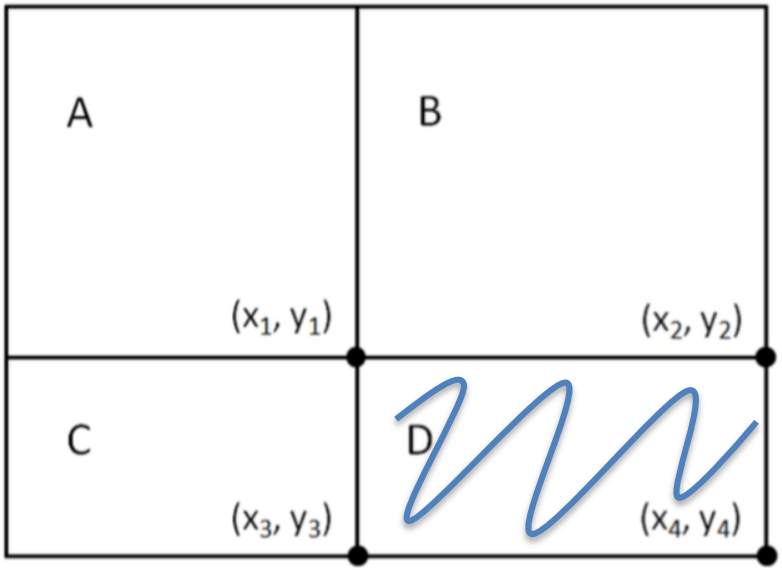
Integral image

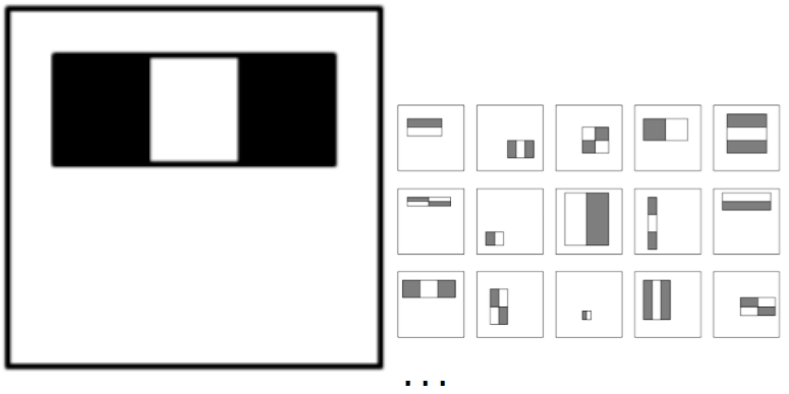
$$S(x,y) = \sum_{1 \le x' \le x, 1 \le y' \le y} I(x',y')$$

=> the sum of all pixel intensities **above** and to the **left** of (*x*,*y*) in the original image

Viola-Jones Face Detector & Integral Image

Sum of pixel values within any **arbitrary rectangular** region in the original image can be computed based on just **four** array accesses in the integral image =>





$$S(x_4, y_4) - S(x_2, y_2) - S(x_3, y_3) + S(x_1, y_1)$$

Viola-Jones Face Detector & **Adaboost**

Achieving real-time => it is essential to determine a small **subset of** discriminative **features** from the complete set of features available within each window

=> Variant of the **Adaboost** algorithm can be used to **select the discriminative features** as well as to train the **classifier** function (integrated)

=> Classifier function combines the feature values using appropriate **weights** … and the face is found if the combined value is greater than a threshold

VJ FD & Adaboost

Construct a **weak classifier** that predicts whether the window *w* is a face image

sign +/-

feature value obtained using filter *j* in window *w*

$$h_j(w) = \begin{cases} 1 & \text{if } p_j f_j(w) \leq p_j \theta_j \\ 0 & \text{otherwise} \end{cases}$$

threshold

L is the total number of filters

Given a training set of *n* example images that includes both faces and non-faces, the classifier can be **trained** =>

VJ FD & Adaboost & **Selection of Features**

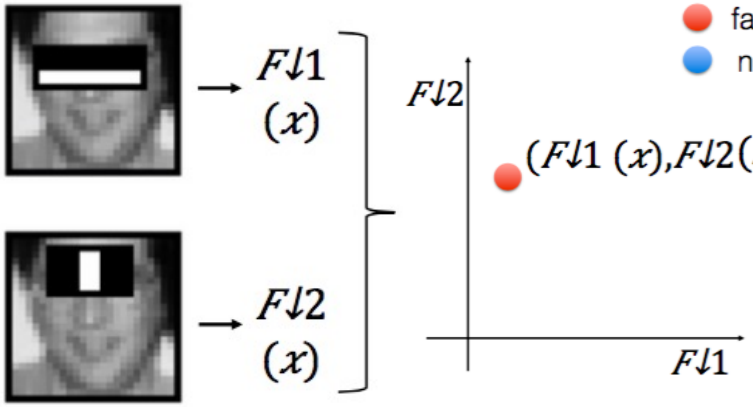
- Boosting is a **supervised learning method** that combines *weak learners* into a more accurate (*strong*) classifier
- A weak learner is a classifier that only needs to exceed **chance** performance
- Training consists of multiple boosting rounds
  - During each round, select a weak learner that does well on examples that were **hard** for the weak learners in previous rounds
  - "Hardness" is captured by **weights** attached to training examples

=>

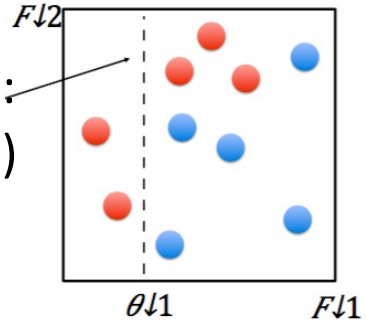
VJ FD & Adaboost & **Boosting Illustration**

vertical or horizontal half-planes

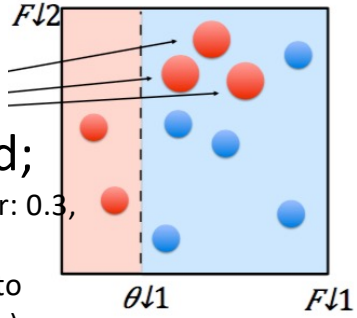
weak classifier: *h*(window *x*, feature value *F*, sign/polarity *p*, threshold θ)



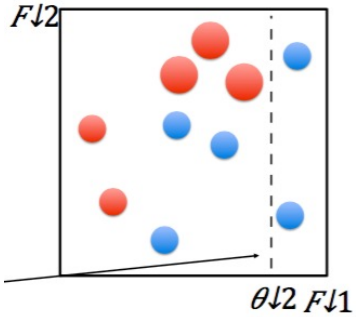
weak classifier 1:
h(x,$F\downarrow$1,1,$\theta\downarrow$1)



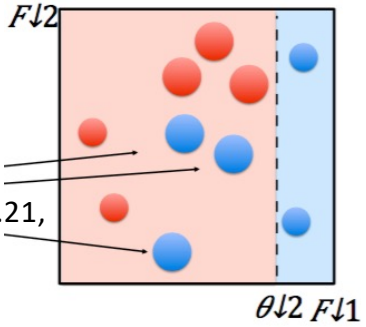
weights increased;
weighted error: 0.3,
α = 0.41
(proportional to weighted error)



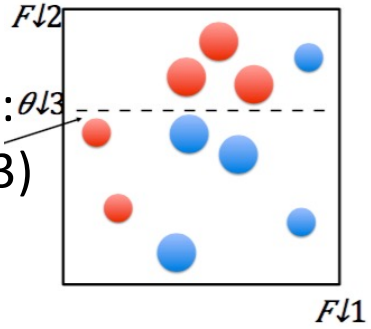
weak classifier 2:
h(x,$F\downarrow$1,1,$\theta\downarrow$2)



weights increased;
weighted error: 0.21,
α = 0.65
(proportional to weighted error)



weak classifier 3:
h(x,$F\downarrow$2,-1,$\theta\downarrow$3)
weighted error: 0.14,
α = 0.92
(proportional to weighted error)



Given n example image windows $(w_1, y_1), (w_2, y_2), \cdots, (w_n, y_n)$, where $y_i = 0, 1$ for non-face and face examples

training data

Let $l = \sum_{i=1}^{n} y_i$ be the number of face examples and $m = (n - l)$ be the number of non-face examples. Initialize weights $\tau_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$

weights

for $t = 1, 2, \cdots, T$ **do**

T most discriminative features among the *L* features (*T*<<*L*)

Normalize the weights, $\tau_{t,i} \leftarrow \frac{\tau_{t,i}}{\sum_{k=1}^{n} \tau_{t,k}}$

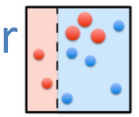
For each weak classifier $h_j(w)$, compute the weighted error $\varepsilon_j \leftarrow \sum_{i=1}^{n} \tau_{t,i} |h_j(w_i) - y_i|$

find weak classifier that achieves the lowest weighted training error

Select the best weak classifier with respect to the weighted error, $h_t(w) \leftarrow h_q(w)$ and $\varepsilon_t \leftarrow \varepsilon_q$, where $q \leftarrow \arg\min_j \varepsilon_j$

Update the weights: $\tau_{t+1,i} = \tau_{t,i} \beta_t^{1-e_i}$, where $e_i = 0$ if example w_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\varepsilon_t}{1-\varepsilon_t}$

raise the weights of training examples misclassified by current weak classifier



end

$\Rightarrow$ calculate final strong classifier

VJ FD & Adaboost & **Strong Classifier**

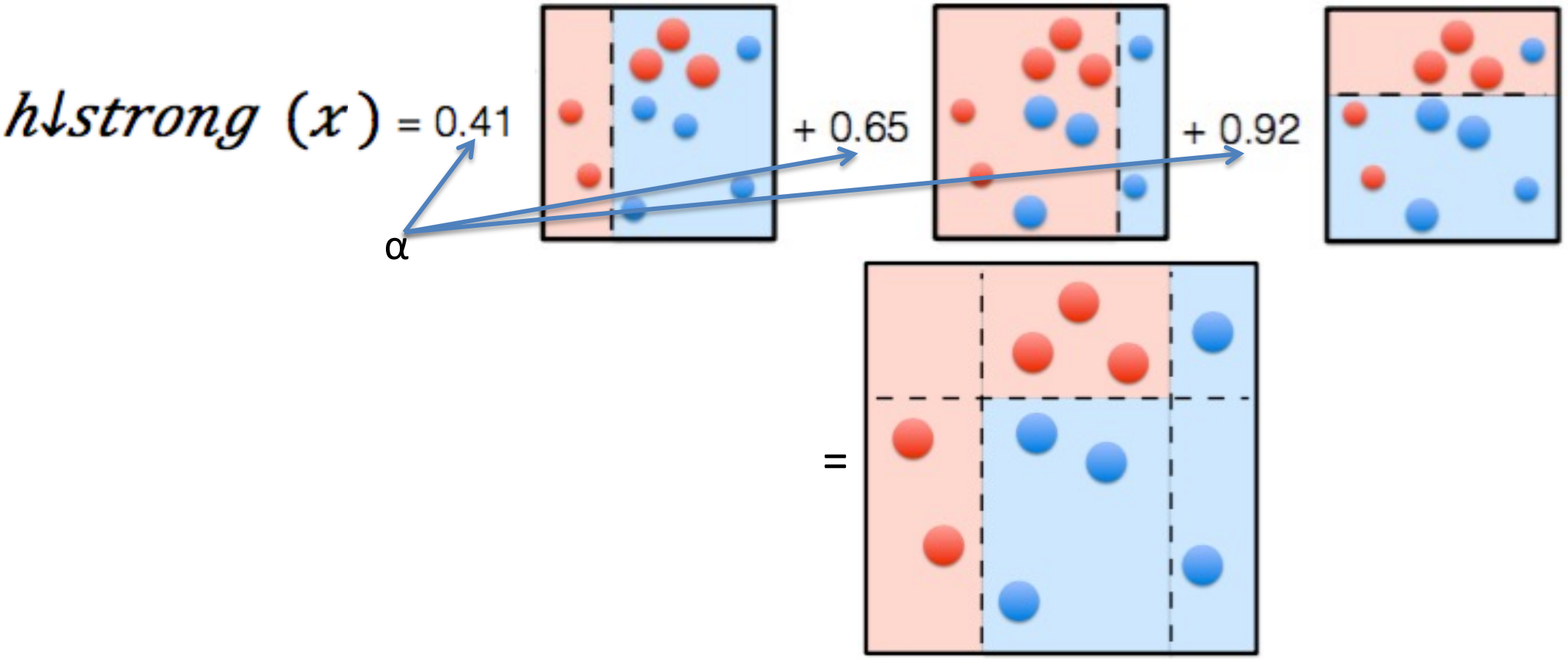
Final strong classifier is a **weighted linear combination** of the *T* best weak classifiers based on the selected features …

where the weights are proportional to the discriminative power of the features (inversely proportional to the training **error** rate)

$$H(w) = \begin{cases} 1 & \text{if } \sum_{t=1}^{T} \alpha_t h_t(w) \geq \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise,} \end{cases}$$

$$\alpha_t = \log \frac{1}{\beta_t}$$

VJ FD & Adaboost &
Strong Classifier Boosting Illustration



VJ FD & Adaboost **Detection Stage**

- During the actual face detection stage **only the *T* feature values** need to be computed for each window and the outputs of the weak classifiers based on these features

\+ Enables **rapid** categorization

– Original image must still be scanned with **windows of different sizes** to determine the location of the faces

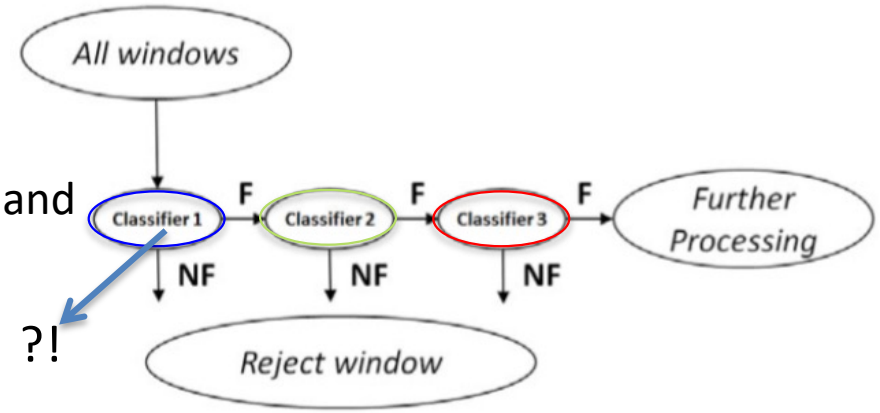
=> speed-up?

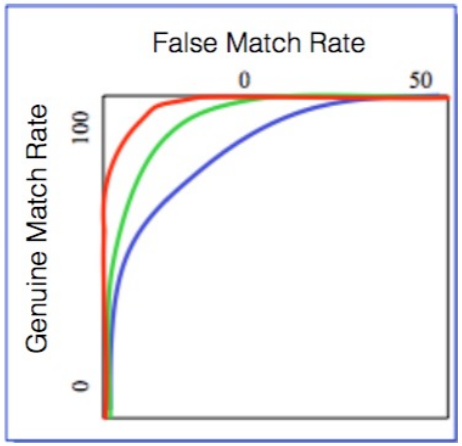
VJ FD & Adaboost & **Cascades**

Classifiers with a simple & smaller set of filters can be first used to screen out a **large** number of **non-face** images in the early stage ...
and then **more powerful classifiers** can be used in the later stage

What do we need to reach a GMR ≈ 0.9 and a FMR ≈ 10^{-6}?

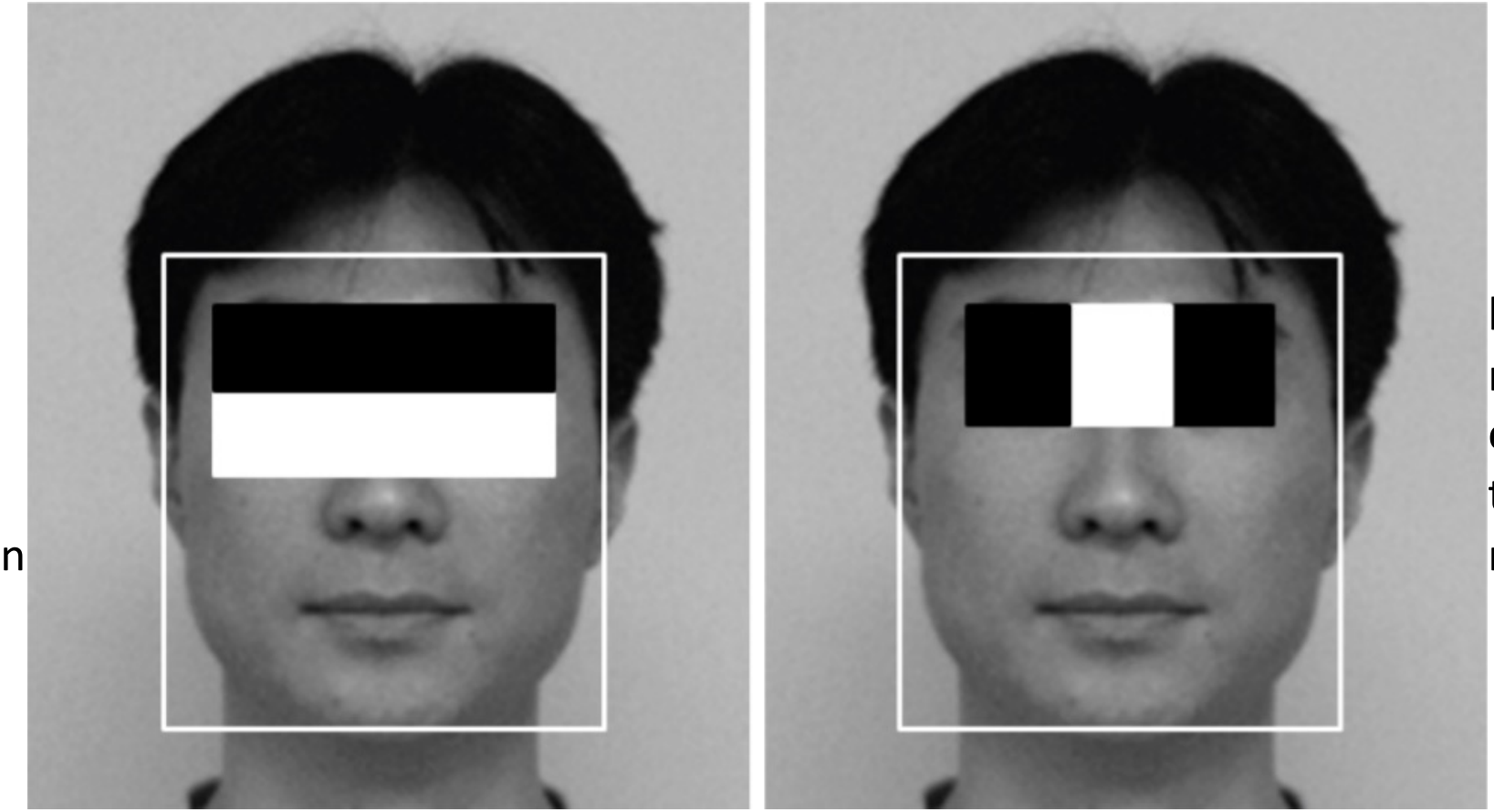
- Assume each stage has a GMR of 0.99 and a FMR of about 0.3
- Need 10 stages:
  ($0.99^{10} \approx 0.9$) and ($0.3^{10} \approx 6\times10^{-6}$)





VJ FD & **2 most effective Haar-like features**

eye region is typically darker than the cheek region

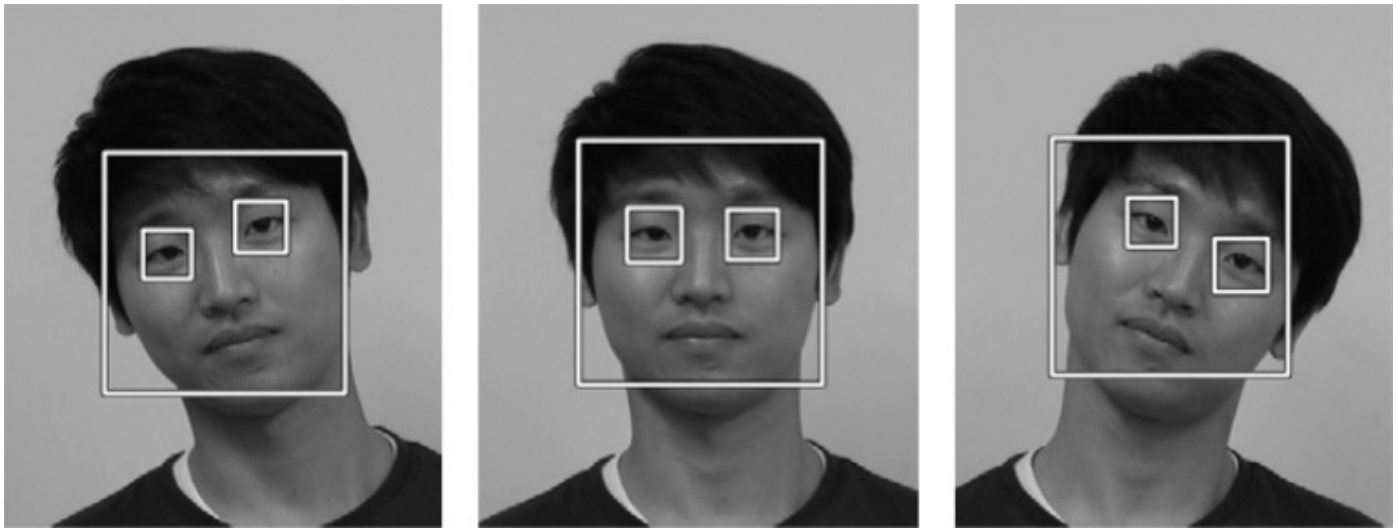


brighter nose bridge compared to the eye regions

able to reject about 60% of non-faces

Can We Use It for Detection of Eyes,…?

Eye detector can be constructed by simply replacing the training set with examples of eye images and non-eye images

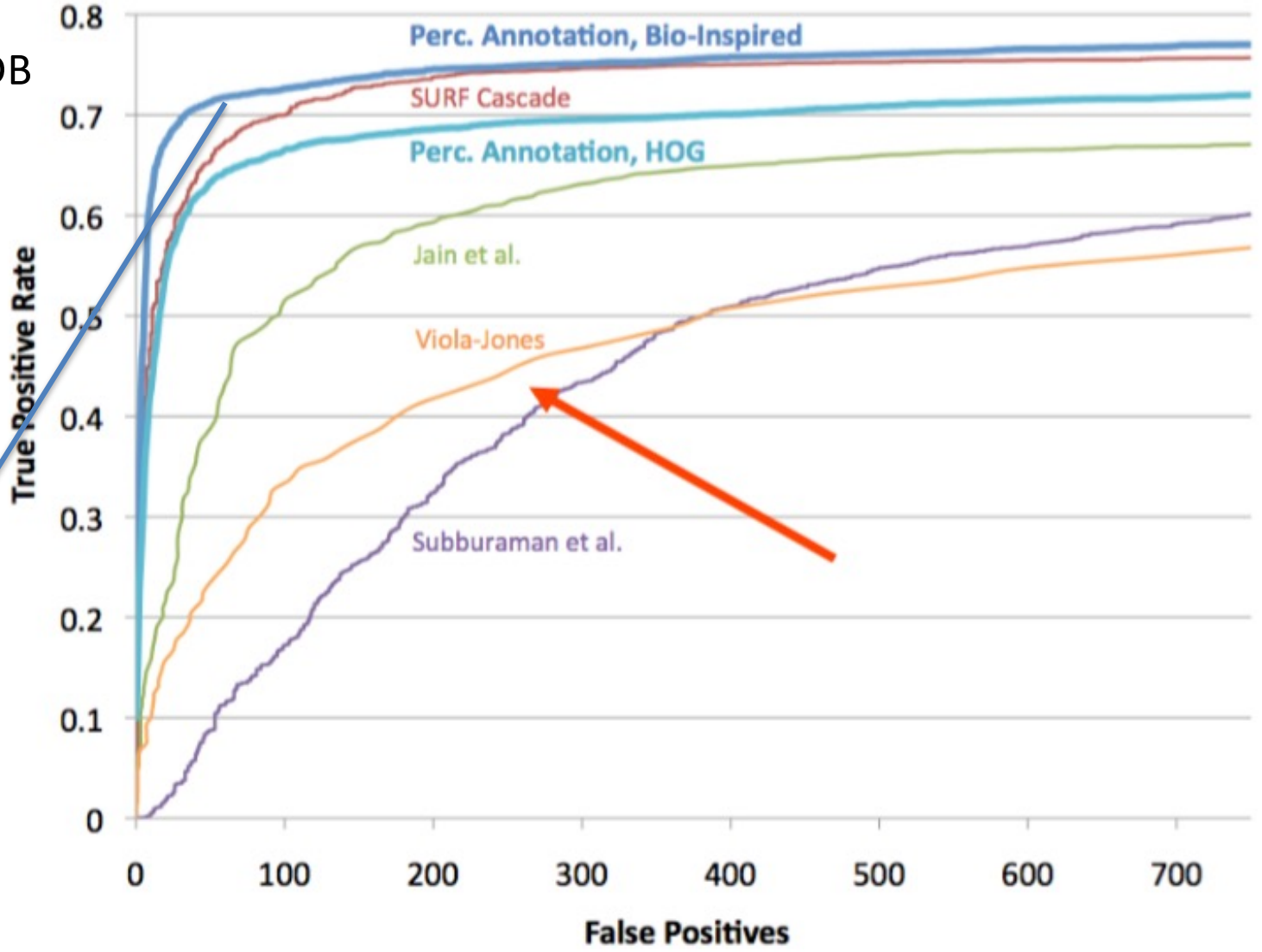


VJ FD & **Practical Considerations**

- VJ FD has a number of open **parameters**, which affect the speed and accuracy
- Factors to consider:
  - Number of overlapping detections
  - Scale factor
  - Minimum and maximum size of faces
  - Number of faces
  - Step size
  - …

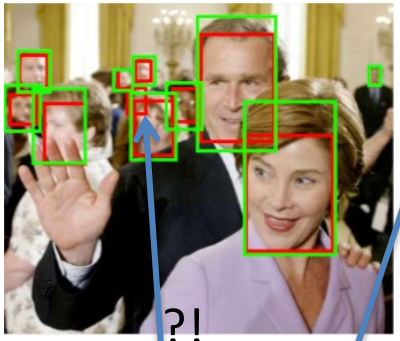
FD Evaluation

FDDB



FD with CNNs

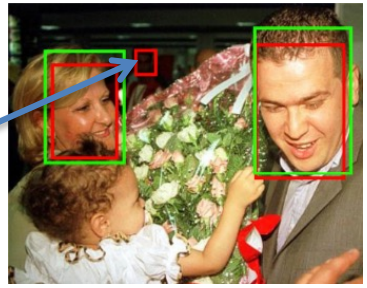
ground-truth annotations



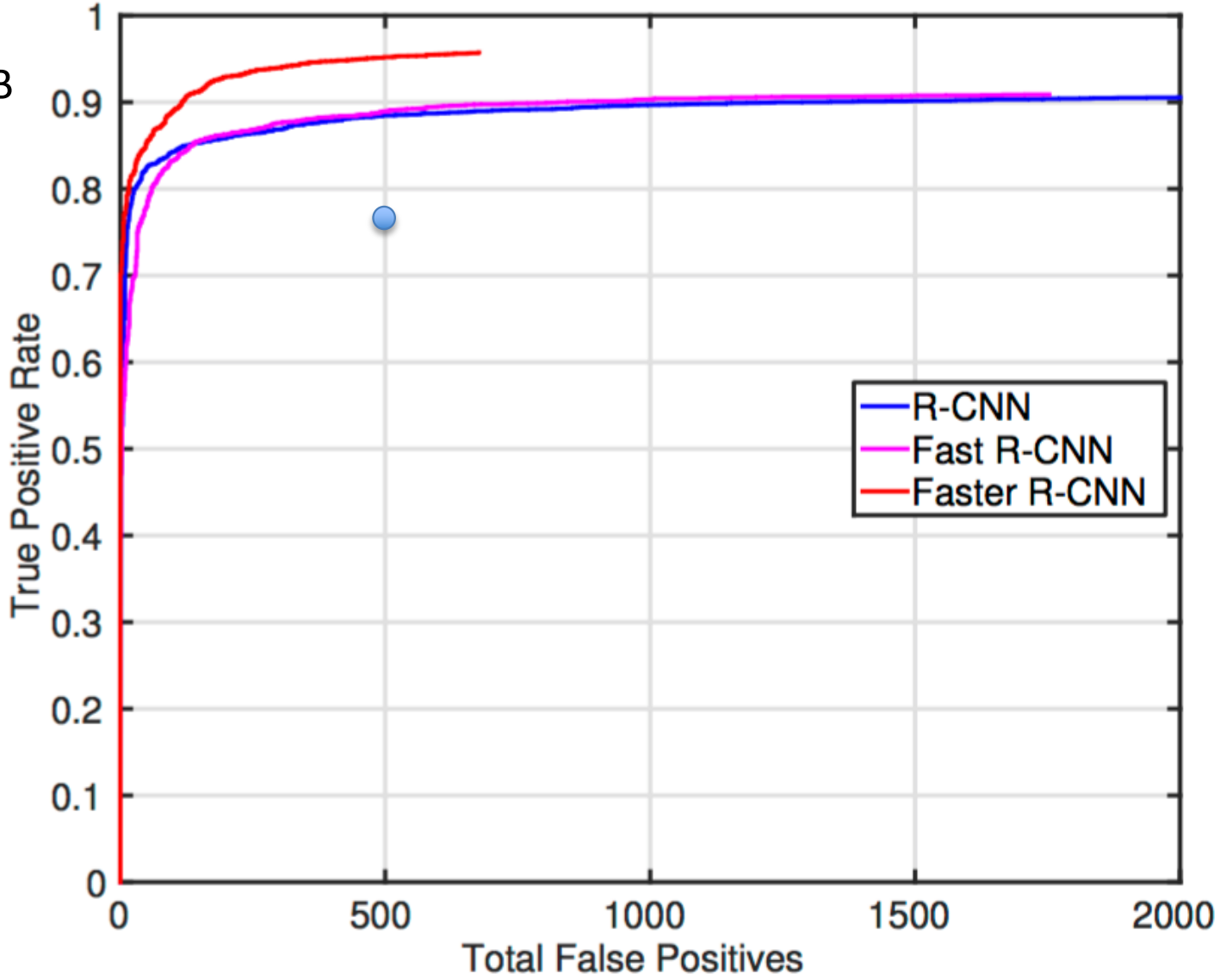
?!

Faster R-CNN detections

TRUE



FDDB



Open Research Questions?

Face detection is not (completely) solved

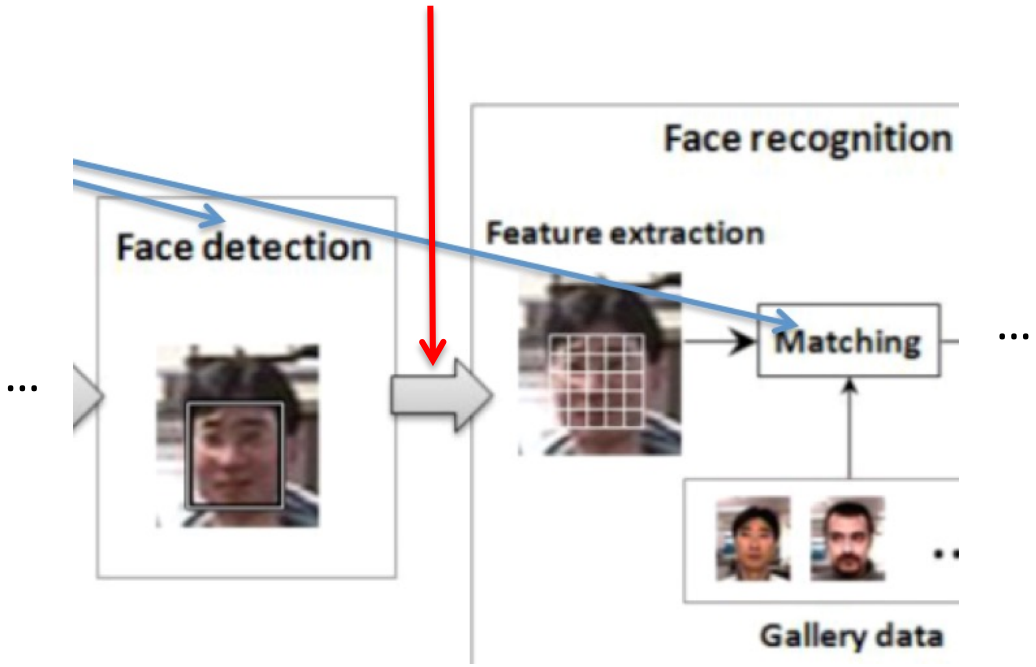
=> Occlusion

=> Pose

=> Low resolution faces

What Comes Next?

Is there something missing? =>



NORMALIZATION:
Pose (landmarking, **alignment**)
Illumination
...

Alignment by Geometric **Warping**

- Input: eyes positions

(252,364) (420,366)

& output parameters: horizontal / vertical offset, size

- Output:
  scaled, rotated and cropped image
- Problem:
  only *in-plane* rotation addressed

basic idea: rotate the face around one eye

0.1 (10%), 0.1 (10%), (200,200)

0.2 (20%), 0.2 (20%), (200,200)

0.3 (30%), 0.3 (30%), (200,200)

size of the output image

0.2 (20%), 0.2 (20%), (70,70)

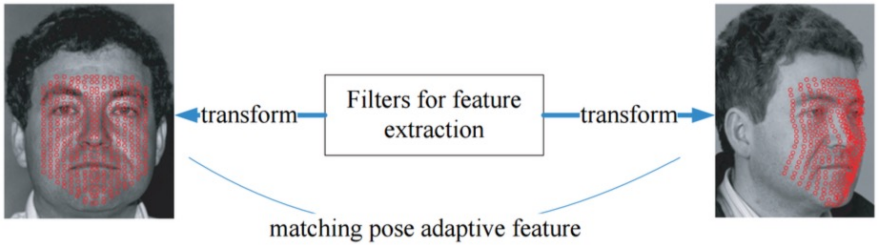
percent of the image you want to keep next to the eyes

Facial **Landmark Detection**

Find location of facial landmarks to >>>

Align / Frontalize

Use with pose adaptive feature extractors



Morph and synthesize

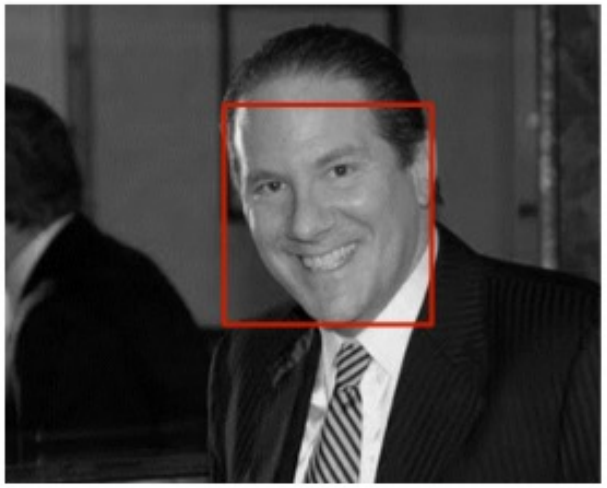
…

Landmark Detection

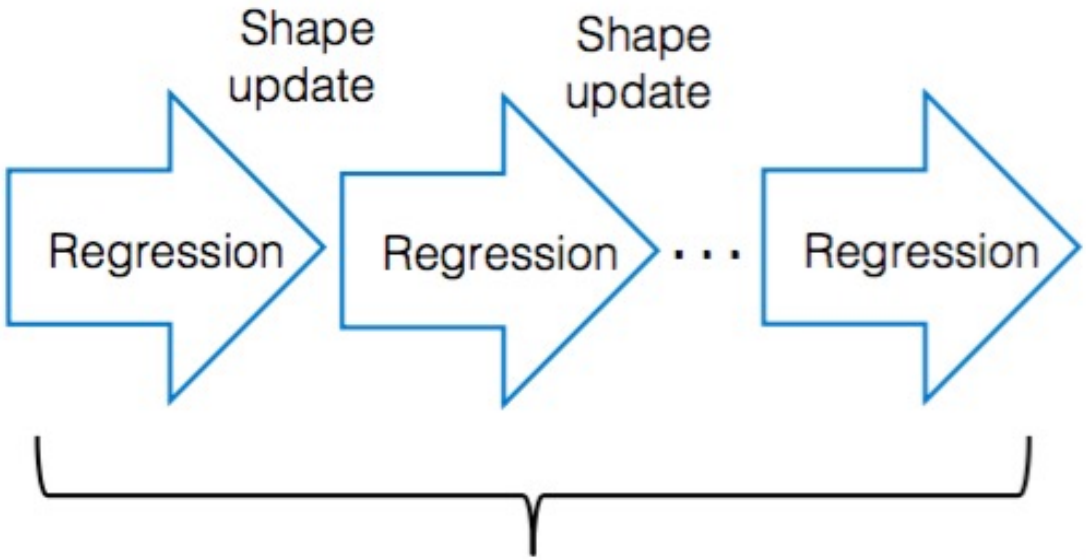
- Input: facial image & face position (detection)
- Output: landmarks (fiducial points)
- Approaches:
  - Active appearance models
  - Constrained local models
  - Parts-based methods
  - **Cascaded regression**
  - …

LD Using **Cascaded Regression**

Framework =>

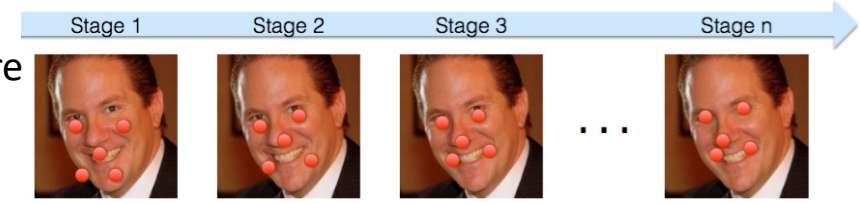


Input image
(face detection)



Landmarking results

Cascaded regression



At each stage calculate feature vector from current shape and perform regression

Key ideas:

Learn regression model**s**

Refine landmark positions by cascading regression models

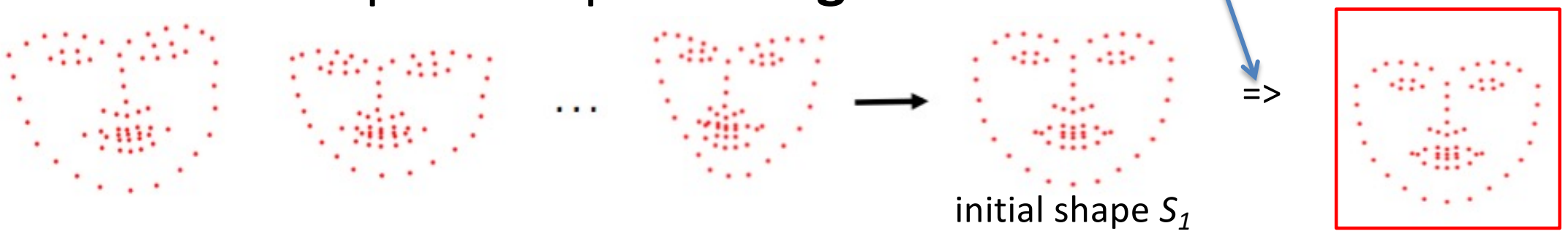
LD & **Training**

- Training data: sample images with annotated landmarks

- Output: regression model (and feature extractor) for **each** stage in cascade

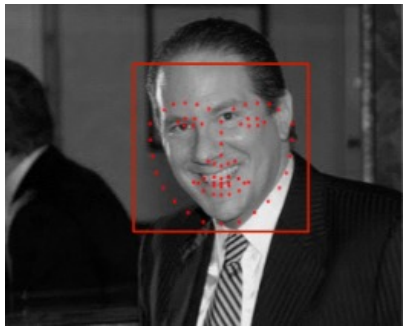
LD & **Training Procedure**

1. Compute initial shape and place it into the image (face detector output)
   – Initial shape is simple **average** of reference annotations



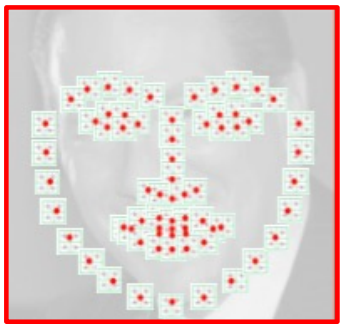
LD & Training Procedure

2. For **stage** t in cascade
   a) Compute feature vector x_i at current shape vertices S_t on image I_i



input image I_i

=>



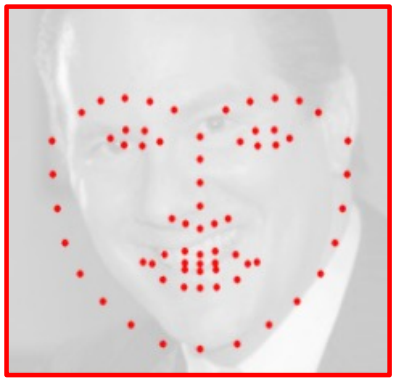
$x_i = F(I_i, S_t)$

Typically used features:
Pixel differences
Pixel comparisons
HoG
…

   b) Repeat for all n training input images and construct regressor matrix: $X_t=[x_1,\ldots,x_i,\ldots,x_n]$

LD & Training Procedure

2. (For stage t in cascade)
   c) Compute required displacement (shape increment) ΔS



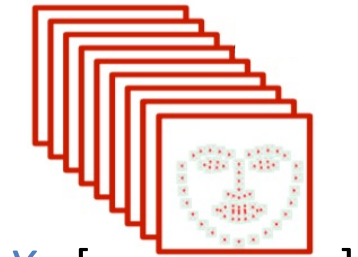
$\Delta S = S_r - S_t$

current shape S_t target/reference shape S_r

   d) Repeat for all n training images and construct regression output matrix: $\Delta_t=[\Delta S_1,\ldots, \Delta S_i,\ldots, \Delta S_n]$

LD & Training Procedure

2. (For **stage** t in cascade)
   e) Compute regression model R from features (regressor matrix X_t) to displacements (regression output matrix Δ_t)



$X_t=[x_1,\ldots,x_i,\ldots,x_n]$

R

=>

$\Delta_t=[\Delta S_1,\ldots,\Delta S_i,\ldots,\Delta S_n]$

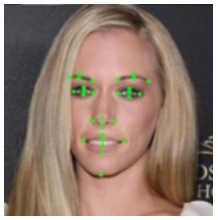
Typically used regression approaches:
Least Squares Regression
Regression Trees
Ridge Regression
Gaussian Process Regression
…

   f) Apply regression model R & update current shape estimate: $\Delta_t^* = R(X_t)$ => $S_{t+1} = S_t + \Delta_t^*$

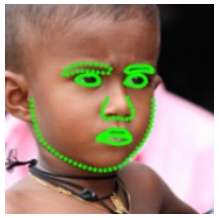
Illustration of the Regression

initial estimate & shape estimates at different stages

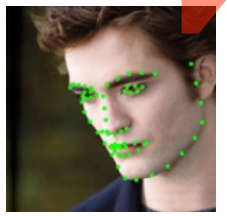
BTW:
different databases,
different number of landmarks



LFPW (29 landmarks)



HELEN (194 landmarks)

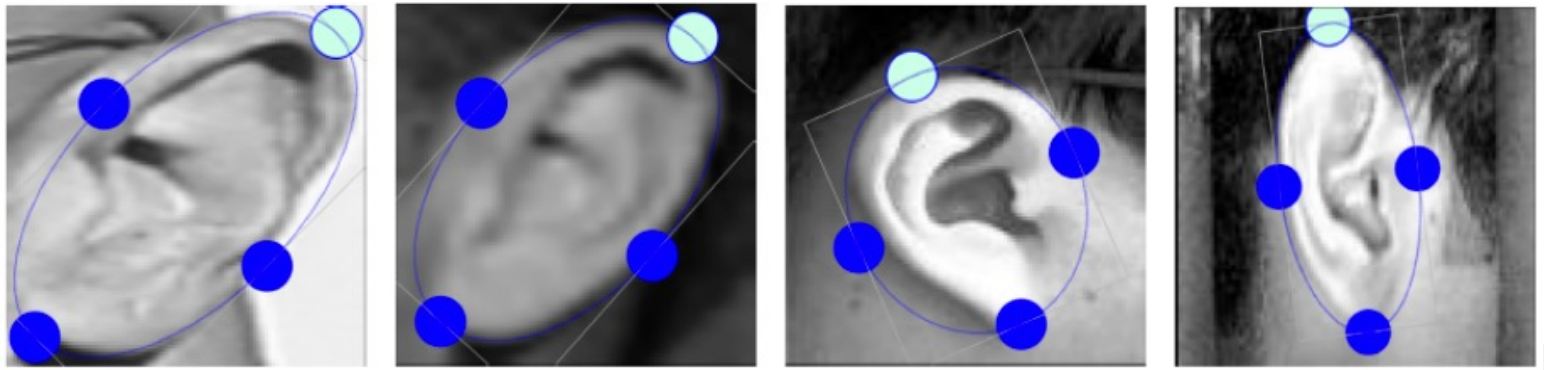


300-W (68 landmarks)

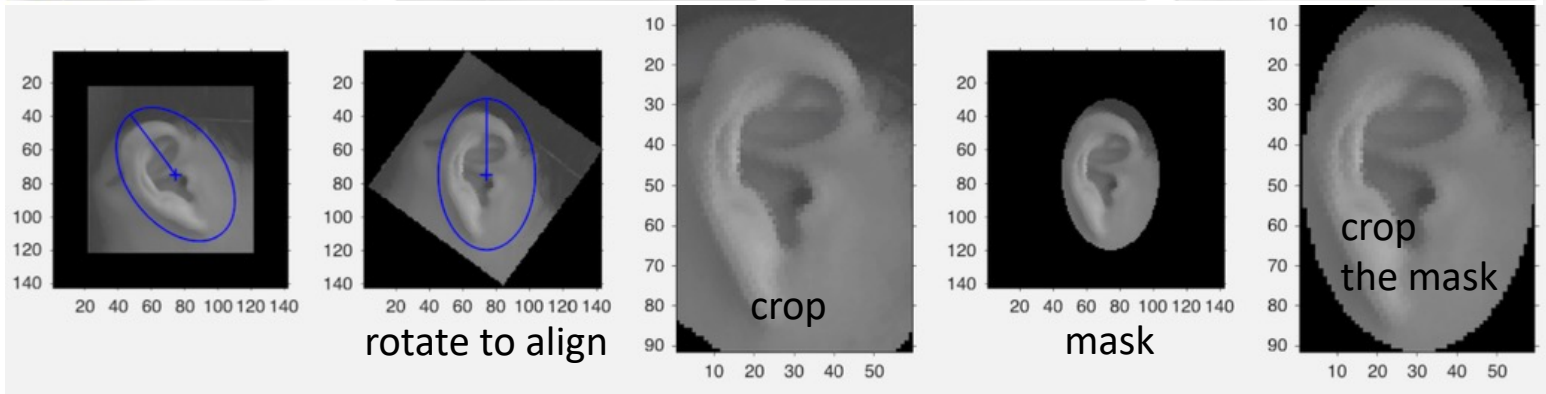
Can We Use CR on Other Modalities?

Cascade Pose Regression on ears:

elliptical model … on training data



evaluate position and direction of the model



see M. Ribič @ eprints/DKUL

Cascaded Pose Regression on Ears

descriptor	unaligned		aligned+cropped		+masked	
	Rank–1 (%)	EER (%)	Rank–1 (%)	EER (%)	Rank–1 (%)	EER (%)
LBP	83.1 ± 7.9	12.9 ± 4.3	94.4 ± 2.1	8.2 ± 2.8	91.9 ± 4.1	9.4 ± 4.3
BSIF	87.8 ± 5.4	11.9 ± 4.9	97.5 ± 1.6	6.0 ± 2.4	85.3 ± 5.4	14.1 ± 5.1
LPQ	84.7 ± 6.9	14.7 ± 4.6	95.0 ± 1.9	7.5 ± 2.8	61.5 ± 6.8	34.8 ± 7.2
RILPQ	86.2 ± 4.7	13.2 ± 4.1	94.7 ± 2.2	6.3 ± 1.9	90.0 ± 3.7	10.2 ± 5.0
POEM	85.0 ± 4.6	13.1 ± 4.1	94.4 ± 2.4	7.1 ± 2.4	92.2 ± 2.6	8.8 ± 3.9
HOG	70.3 ± 7.2	20.7 ± 3.8	84.6 ± 3.3	12.6 ± 5.0	84.7 ± 2.8	14.0 ± 5.6
DSIFT	65.5 ± 9.5	21.4 ± 6.0	83.1 ± 4.4	14.8 ± 5.9	79.1 ± 6.3	16.3 ± 4.8
Gabor	73.7 ± 8.3	19.1 ± 5.9	92.8 ± 3.8	10.7 ± 3.1	87.8 ± 2.9	11.6 ± 3.3

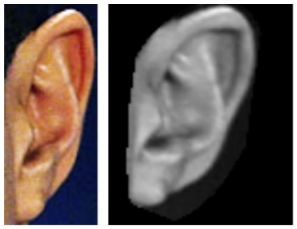
	Rank–1 (%)	EER (%)	Rank–1 (%)	EER (%)	Rank–1 (%)	EER (%)
LBP	52.2 ± 10.8	30.4 ± 7.7	46.9 ± 11.0	35.7 ± 6.5	49.9 ± 8.5	32.7 ± 6.6
BSIF	52.2 ± 13.8	30.2 ± 6.7	41.53 ± 8.2	37.9 ± 9.01	34.8 ± 6.1	40.2 ± 8.4
LPQ	48.6 ± 12.6	30.9 ± 6.1	18.4 ± 7.8	39.8 ± 6.2	23.2 ± 6.7	52.0 ± 6.7
RILPQ	51.3 ± 14.9	33.5 ± 9.3	43.3 ± 12.0	32.2 ± 4.2	35.7 ± 4.9	37.7 ± 5.3
POEM	56.6 ± 12.0	28.0 ± 5.9	49.5 ± 11.5	32.2 ± 7.6	48.6 ± 9.2	32.0 ± 4.9
HOG	50.9 ± 6.4	30.0 ± 9.7	48.6 ± 15.7	30.6 ± 8.1	43.3 ± 4.2	32.5 ± 7.1
DSIFT	52.2 ± 10.2	29.9 ± 7.2	49.0 ± 13.3	37.1 ± 11.6	45.9 ± 9.2	32.6 ± 7.7
Gabor	46.4 ± 6.4	30.3 ± 5.1	33.6 ± 7.7	36.5 ± 8.8	37.0 ± 9.4	36.8 ± 9.8

IITD I db –
in-plain images!

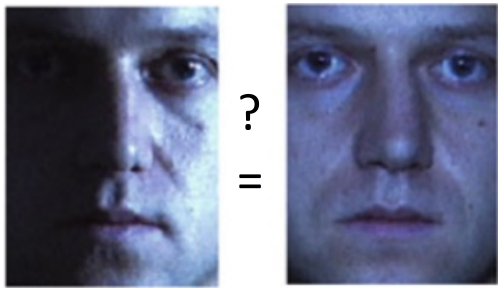
AWE db –
out-of-plain images!!!

↓

Producing info where there is non-given … is just not the way to go (for recognition purposes)!

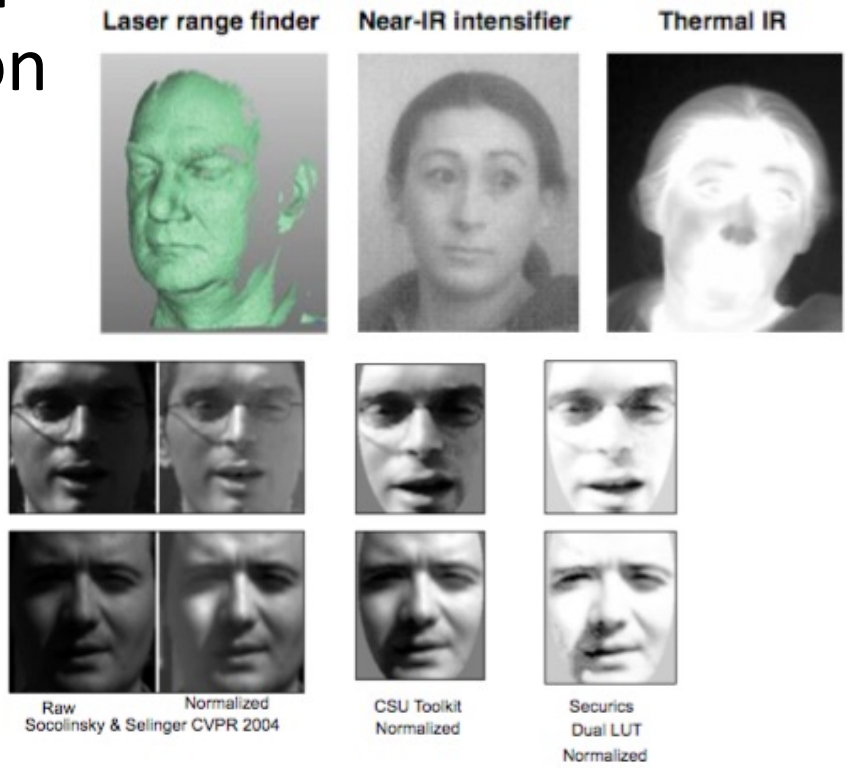


Lighting Normalization



How well will these faces match?

- **Active methods:** solve the problem at acquisition
  - 3D information acquisition
  - Near-infrared images
  - Thermal infrared images
- **Passive methods:** analysis of the acquired images =>



Passive Illumination Invariance Methods

- **Photometric normalization:**

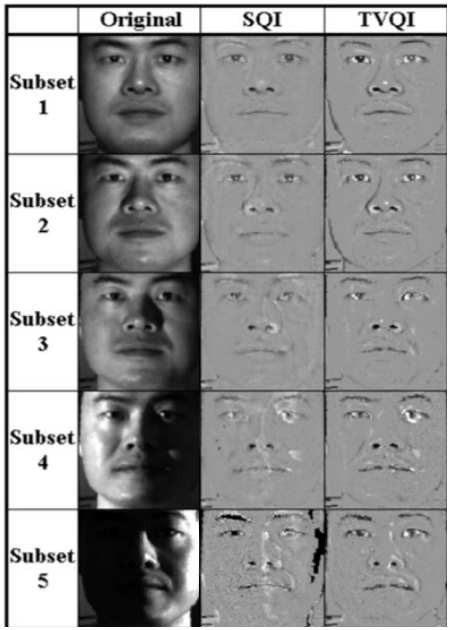
  Histogram normalization, Gamma intensity correction, Local normalization,...

- **Illumination variation modeling:**

  Linear subspaces, Illumination cone, Generalized photometric stereo,...

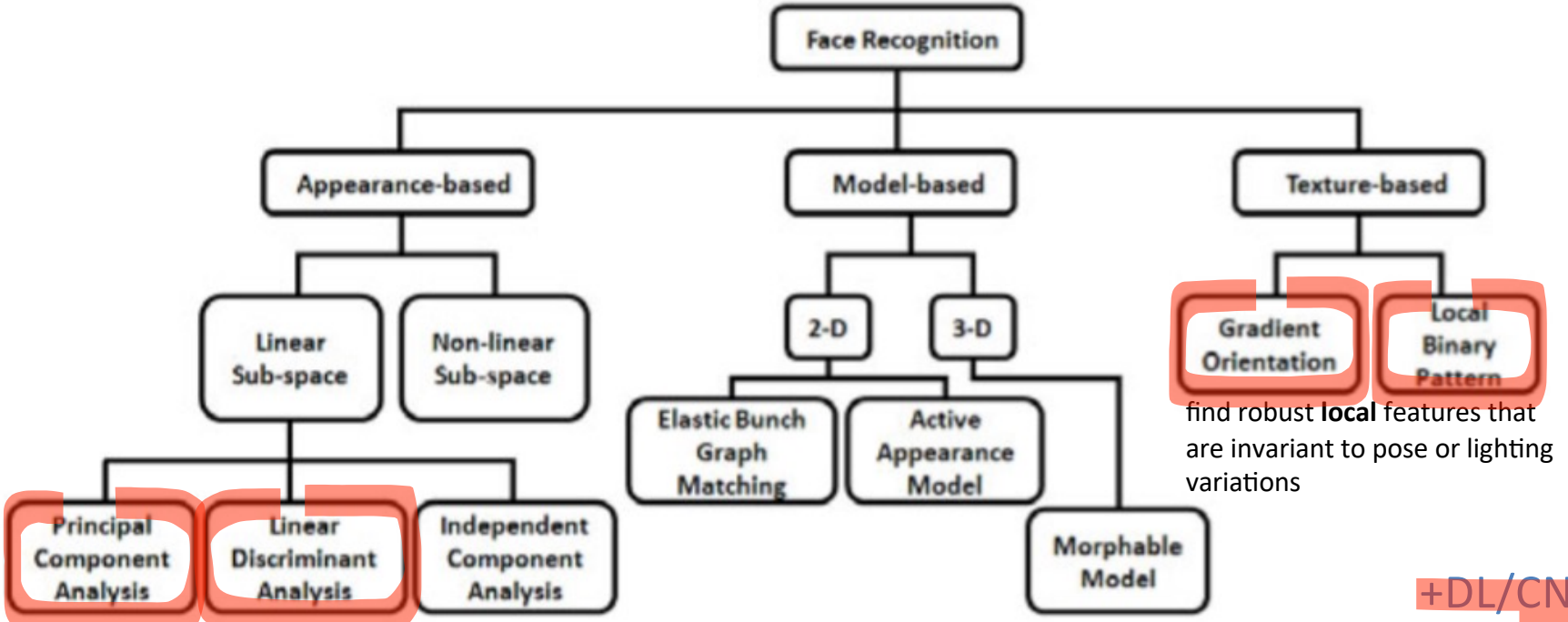
- **Illumination invariant features:**

  Direction of gradient, Shape from shading, **Q**uotient **i**mage, Eigen-phase, Local bin. pattern,...



Feature Extraction and Matching

Main approaches to match the detected face images / taxonomy:



- generate a compact representation of the **entire** face region
- sub-space is defined by a set of representative basis vectors, which are learned using a training set of images

facilitate matching of face images in the presence of **pose** variations

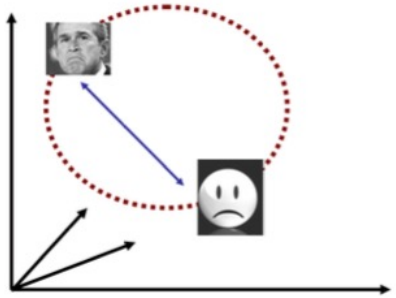
find robust **local** features that are invariant to pose or lighting variations

+DL/CNN-based

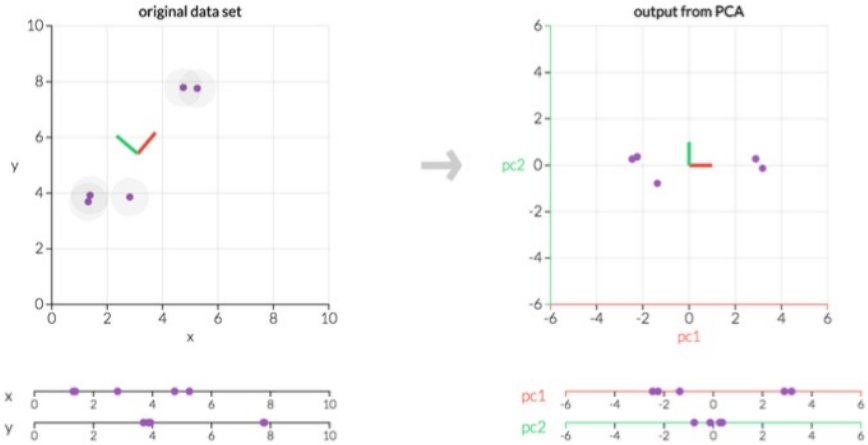
Appearance-based Face Recognition

- **Idea:** the pixel value at location (*x*,*y*) in a face image can be expressed as a weighted sum of pixel values in **all** the training images at (*x*,*y*)
- **Goal: goal** in linear subspace analysis is to find a **small** set of most representative basis faces
- **Matching:** Images can be **matched** by directly comparing their **vector of weights**

Subspace Methods



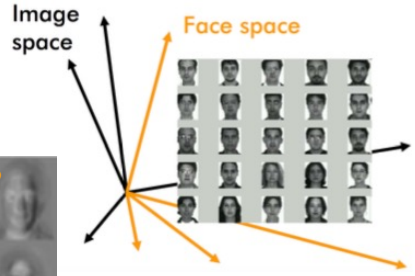
- An image is a point in high dimensional space
  => an $A \times B$ image is a point in $\mathbf{R}^{A \times B}$
- **Problem:** faces in the set of all possible images are highly correlated
- **Solution:** compress them to a low-dimensional subspace that captures key appearance features
- How do we do this? >>>

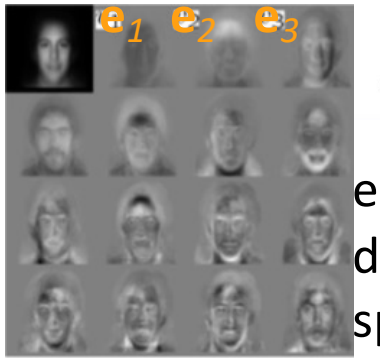


Principal Component Analysis

- Uses the training data to learn a subspace that accounts for as much **variability** in the training data as possible
- By performing an **Eigen** value decomposition of the **covariance** matrix of the data
- Algorithm >>>

1. $\mathbf{x}_1,\mathbf{x}_2,\cdots,\mathbf{x}_N$ is the training set & $\mathbf{x}_i$ represents a d-dimensional column vector;
   average of the training is: $\boldsymbol{\mu} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i$





each vector is a direction in face space

2. Define the data matrix $\mathbf{X} = [(\mathbf{x}_1 - \boldsymbol{\mu})\ (\mathbf{x}_2 - \boldsymbol{\mu})\ \cdots\ (\mathbf{x}_N - \boldsymbol{\mu})]$

3. Calculate the data covariance matrix $\mathbf{C} = \mathbf{X}\mathbf{X}^T$

4. Compute the Eigen vectors in $\mathbf{E}$ of the covariance matrix $\mathbf{C}$: $\mathbf{CE} = \lambda\mathbf{E}$, $\mathbf{E} = [\mathbf{e}_1,\mathbf{e}_2,\cdots,\mathbf{e}_d]$

5. Any data vector $\mathbf{x}$ can be represented as a weighted sum of the Eigen vectors
   Weights / Eigen coefficients can be computed as $\boldsymbol{\omega}=\mathbf{E}^T\mathbf{x}$
   ($\boldsymbol{\omega} = [\omega_1,\omega_2,\cdots,\omega_d]^T$ is a d-dimensional column vector, where ω_j is the weight associated with Eigen vector $\mathbf{e}_j$, for $j = 1,2,\cdots,d$)

Matching:
compute Eigen coefficients $\boldsymbol{\omega}^G$ and $\boldsymbol{\omega}^P$ corresponding to the gallery and probe face images >>> Euclidean distance between the two Eigen coefficients can be considered as a measure of (dis)similarity between the two face images

Any problems?

Projecting into the face space (using Eigenfaces):

$\mathbf{x}$ $\omega_1\mathbf{e}_1$ $\mathbf{x}\approx\boldsymbol{\mu}+\omega_1\mathbf{e}_1+\ldots$

Dimensionality

- Dimensionality of the Eigen coefficients ($\boldsymbol{\omega}$) is the same as the dimensionality of the original data $\mathbf{x}$, which is d
- Goal of linear subspace analysis is to reduce the dimensionality
- **Solution:** consider a lower dimensional subspace $\mathbf{E}'$, which is spanned by only d' ($d' << d$) Eigen vectors from $\mathbf{E}$ corresponding to the d' **largest** Eigen values
  - Eigen vector corresponding to the **largest** Eigen value in PCA accounts for **maximum** variability in the data and is called the **principal axis**
  - It is possible to account for most of the variability in the data by selecting **only a few** Eigen vectors corresponding to the largest Eigen values in the descending order

ORL db & seven largest Eigen values (Eigenfaces) >

Reconstruction and Errors

- Only selecting the **top** *P* Eigenfaces **reduces** the dimensionality
- **Fewer** eigenfaces = more information loss and less discrimination

Let's Optimize Something Else …

- **Eigenfaces** attempt to maximize the scatter of the training images in faces space
- Why not try to …
- **Fisherfaces** attempt to maximize the between class scatter, while minimizing the within class scatter
- Method name? >>>

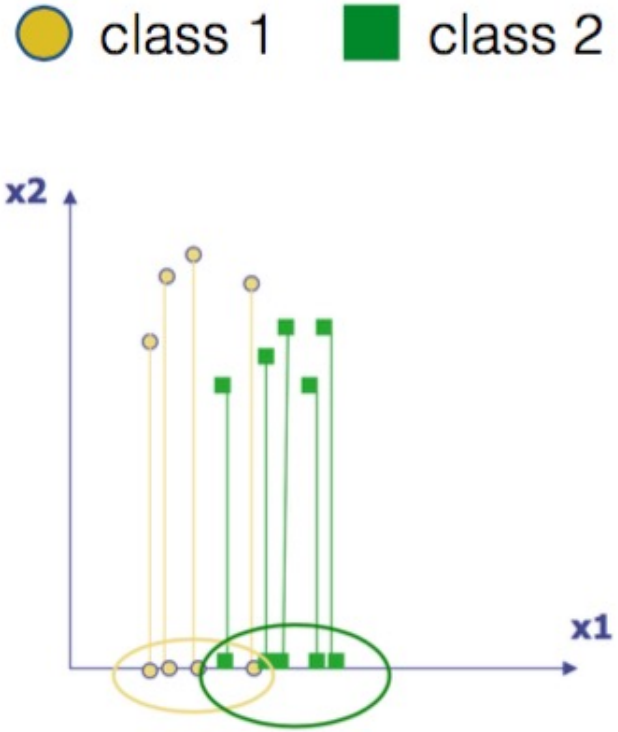
Linear Discriminant Analysis

- PCA is unsupervised learning method – class label (user ID information) is never used during the learning of the basis faces
- Accuracy based on PCA cannot be expected to be very high

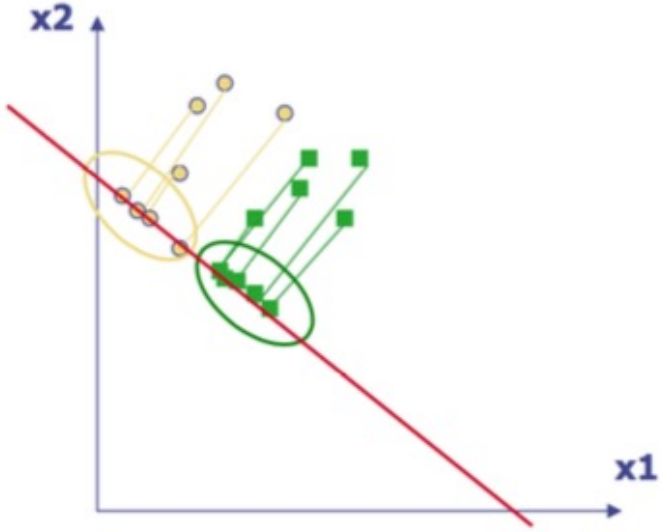
- LDA explicitly **uses the class** label of the training data and conducts subspace analysis with the objective of **minimizing** intra-class variations and **maximizing** inter-class variations

Illustration of the Projection

Separability?



Bad Projection



Good Projection

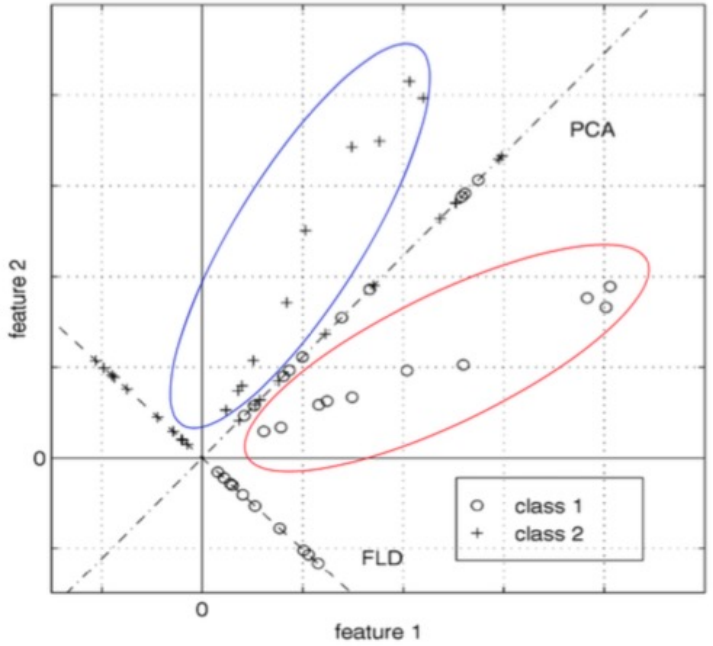
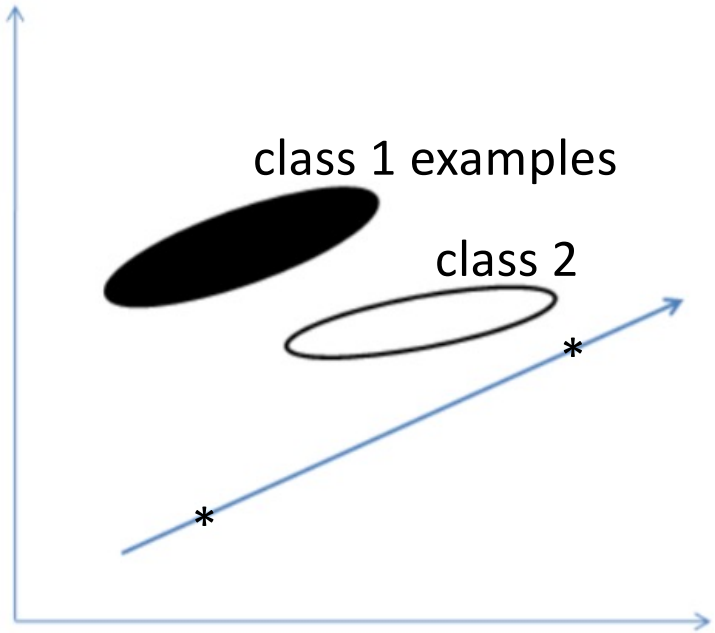
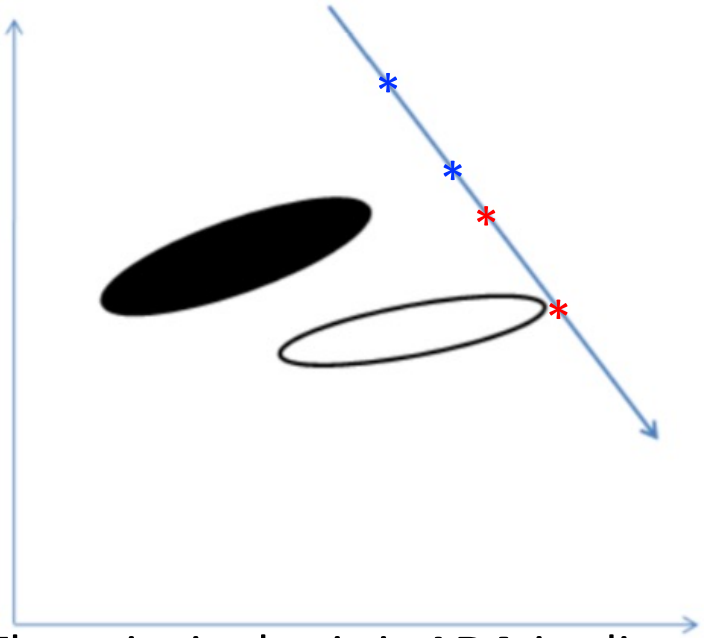


Illustration of the Projection



The principal axis in **PCA** is aligned such that when the **data** is projected onto this axis, the **variance is maximized**



The principal axis in **LDA** is aligned such that when the data is projected onto this axis, the **variance** within each **class** is **minimized** and the **separability** between the two **classes** is **maximized**

Algorithm >>>

1. $(\mathbf{x}_1,y_1),(\mathbf{x}_2,y_2),\ldots,(\mathbf{x}_N,y_N)$ be the training set,
   $\mathbf{x}_i$ represents a d-dimensional column vector,
   $y_i \in \{1,2,\cdots,c\}$ $(=j)$ is the corresponding class label, c is the number of classes
   Compute the average of each class j: $\boldsymbol{\mu}_j = \frac{1}{N_j}\sum_{y_i=j}\mathbf{x}_i$ $\quad \boldsymbol{\mu} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i$

2. Define the within and between class scatter matrixes: $\mathbf{S}_w = \sum_{i=1}^{c}\sum_{y_i=j}(\mathbf{x}_i-\boldsymbol{\mu}_j)(\mathbf{x}_i-\boldsymbol{\mu}_j)^T$

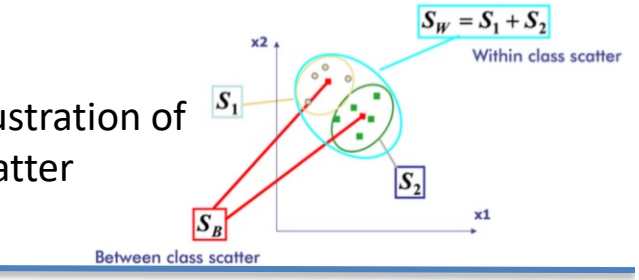
   Scatter of class j: S_j

$$\mathbf{S}_b = \sum_{j=1}^{c} N_j(\boldsymbol{\mu}_j-\boldsymbol{\mu})(\boldsymbol{\mu}_i-\boldsymbol{\mu})^T$$

3. Minimize $\mathbf{S}_w$ and maximize $\mathbf{S}_b$ simultaneously
   by maximizing $\mathbf{S}_w^{-1}\mathbf{S}_b$ >>> solve Eigen system:
   $\mathbf{S}_w^{-1}\mathbf{S}_b\,\mathbf{E} = \lambda\mathbf{E}$

4. Any data vector $\mathbf{x}$ can be represented as a weighted sum of the Eigen vectors
   Weights / Eigen coefficients are computed as $\boldsymbol{\omega}=\mathbf{E}^T\mathbf{x}$

illustration of scatter



Matching:
See PCA ;-)

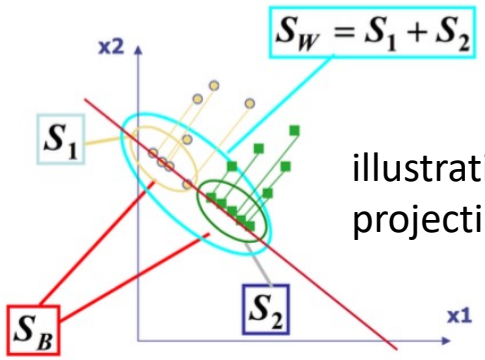


illustration of projection

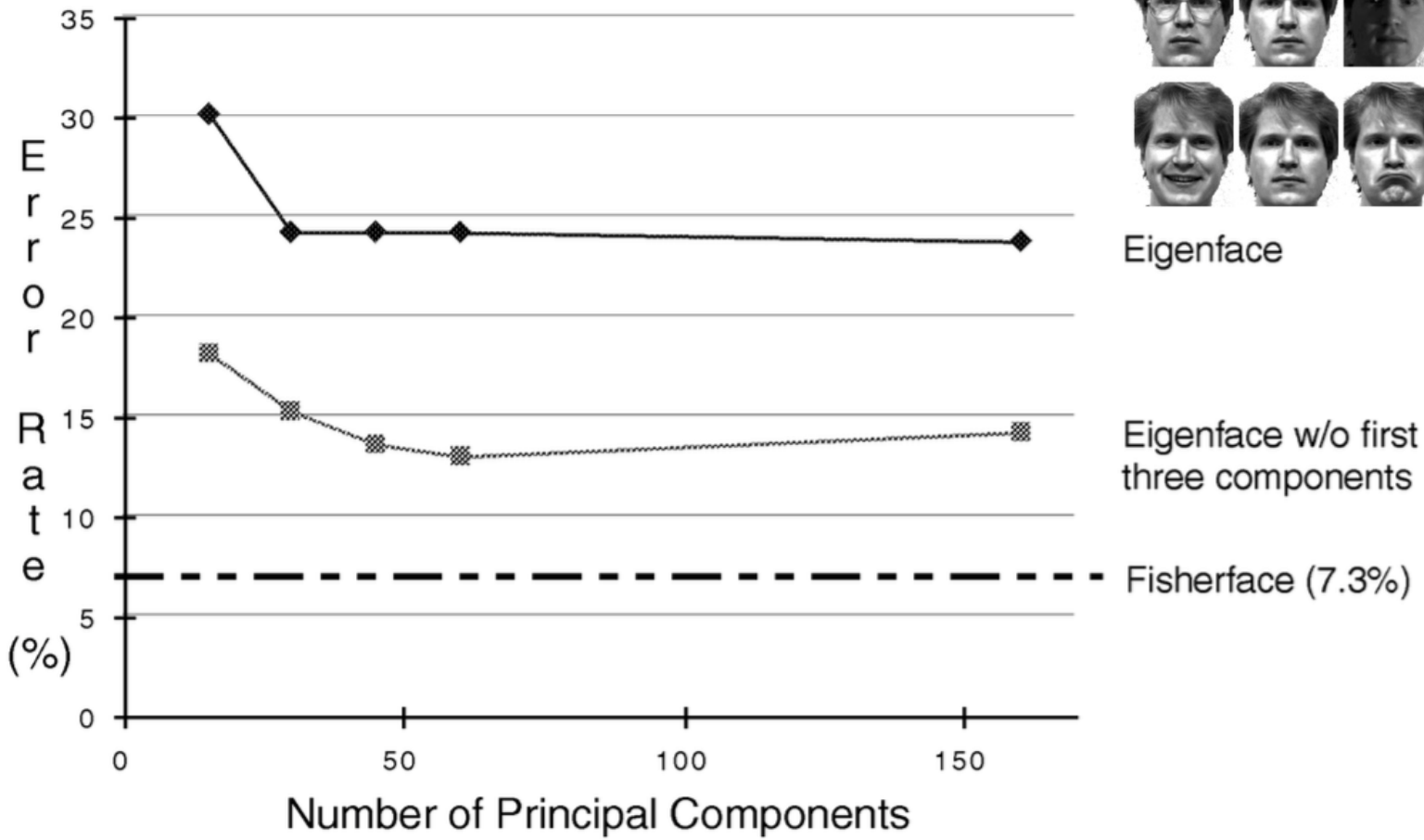
What if $N < d$?

- $\mathbf{S}_w$ often becomes singular
- Solution? >>>
- First apply **PCA** to the training samples to reduce the data dimensionality and then apply **LDA** to the transformed data having lower dimensionality

PCA & ICA

ORL db & seven largest Eigen values (Fisherfaces) >

Eigenfaces vs. Fisherfaces

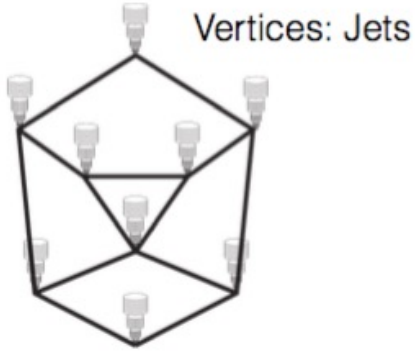


Model-based face recognition

- Try to derive a **pose**-independent representation
- Requires the detection of several fiducial/landmark points
- Increased complexity compared to appearance-based techniques

Elastic Bunch Graph Matching – Face Bunch Graph

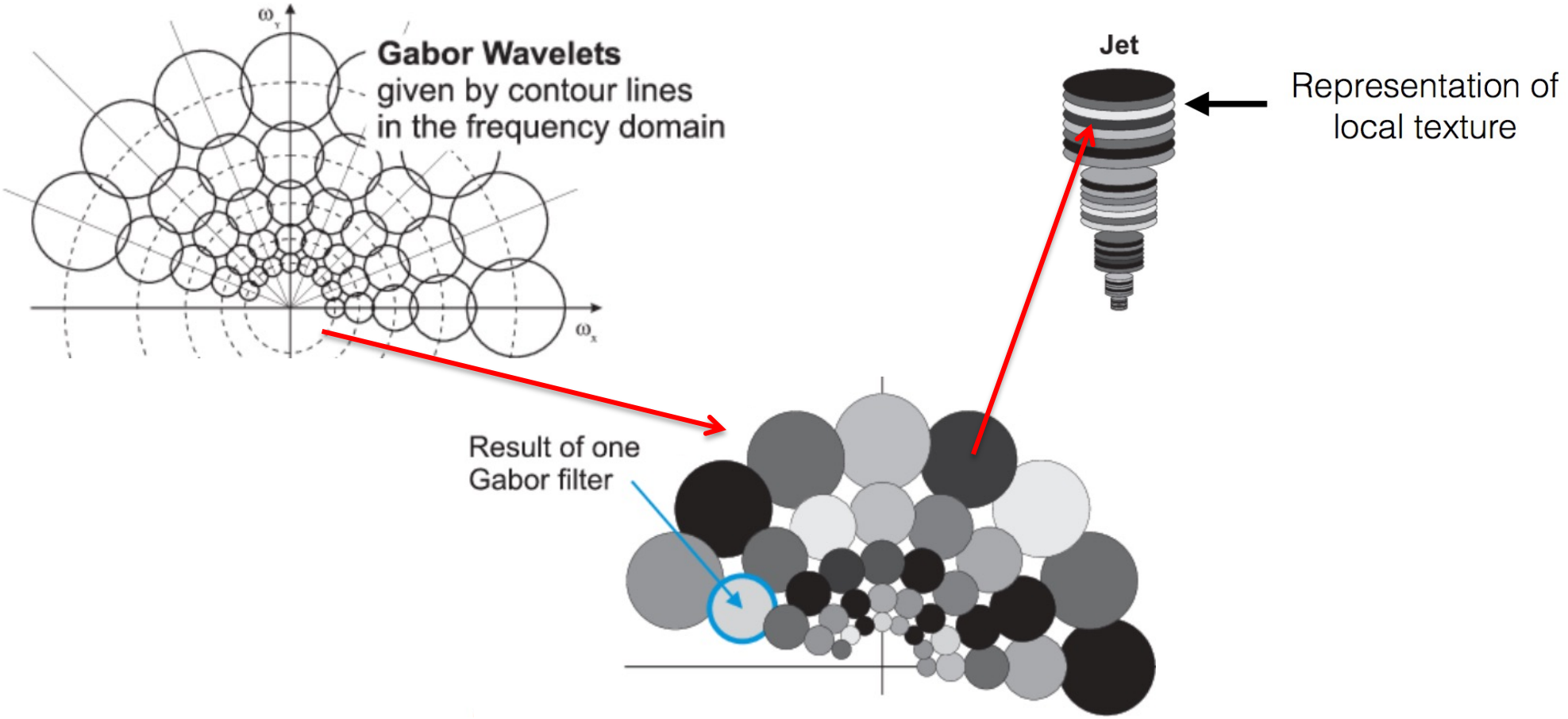
- It represents a face as a **labeled** image **graph** with each **node** being a fiducial point
- Node of the graph is labeled with a set of **Gabor** coefficients (**jet**)
- Jet **characterizes** the local texture information
  - The Gabor coefficient(!) at a location in the image is obtained by **convolving** the image with a complex 2D Gabor filter centered at that location
  - By **varying** the orientation and frequency of the Gabor filter, a set of coefficients or a Gabor **jet** is obtained



- **Connection** of any two nodes of the graph is labeled based on the average distance between the corresponding fiducial points

Jet

Transformation at **one** image location >>>



Construction of **FBG Model** in 2 Stages

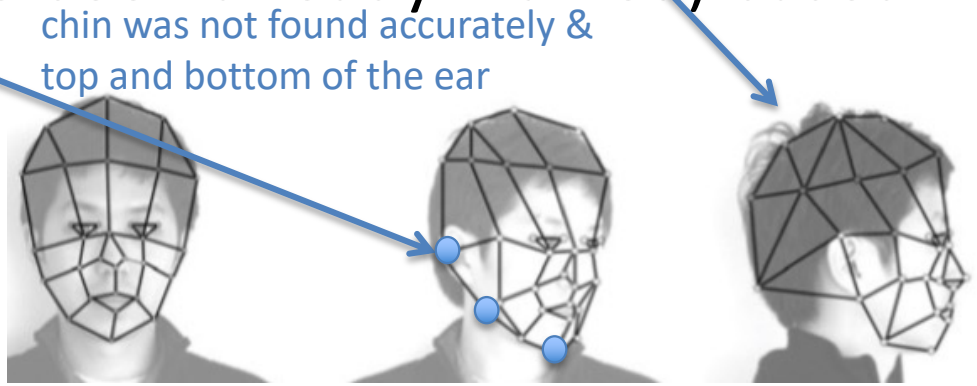
From a training set of face images with a **specific** pose (*n* poses)

Stage 1:

a) **Manually** mark the desired fiducial points and define the geometric structure of the **image graph** for one (or a few) initial image(s)

b) Obtain the rest semi-automatically, by comparing the new images to **model graphs** (images that have been already marked) based on the extracted Gabor jets

Correspondence between the nodes of bunch graphs belonging to different poses is specified manually

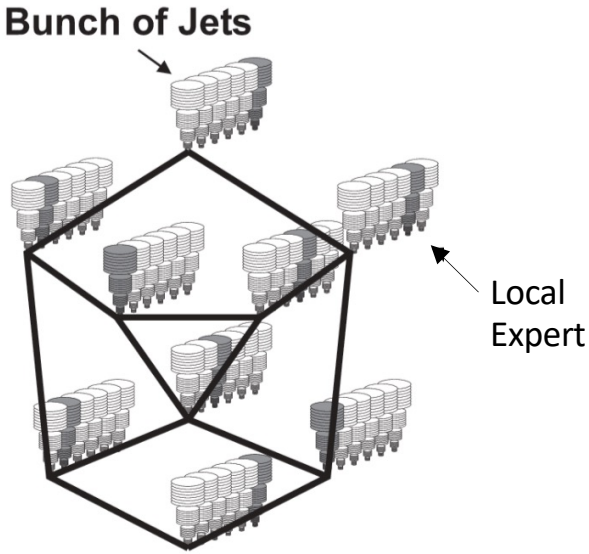


Stage 2

a) **Combine** a representative set of individual graphs in a stack-like structure

b) Set of jets corresponding to the same fiducial point is called a **bunch**

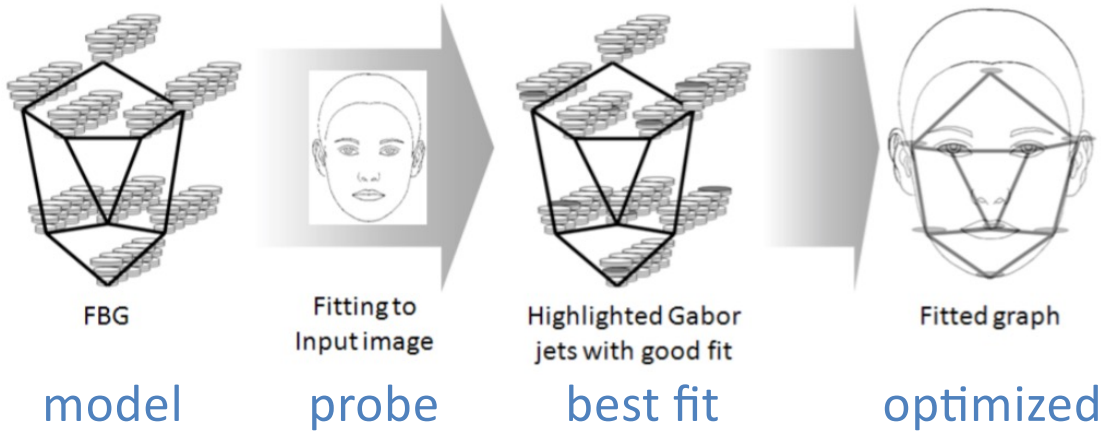
c) Bunch represents **local variations** in the associated fiducial point **among the population**



– Eye bunch includes jets from open, closed, male, female etc. eyes

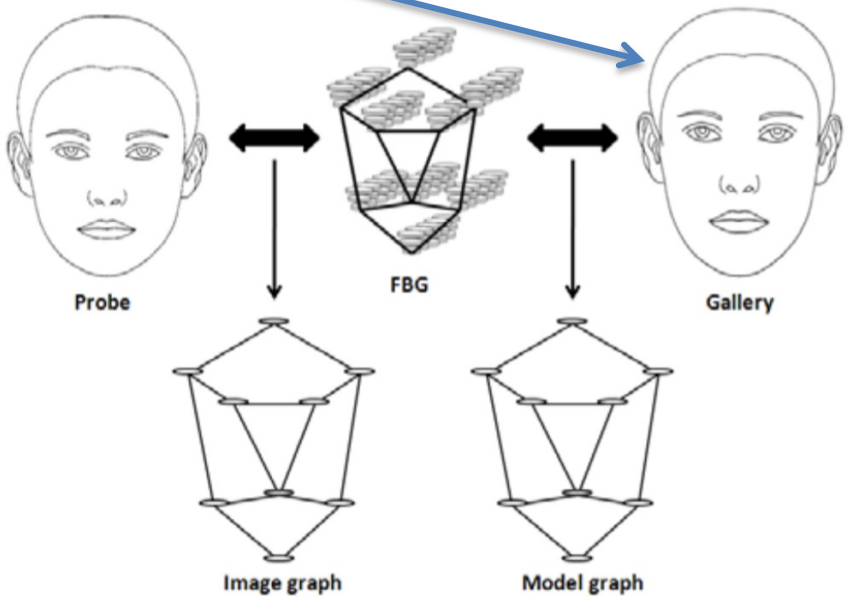
Recognition with FBG Model

1. Execute Elastic Bunch Graph Matching on **probe** image:
   a) Take the **mean** model Gabor jets at each bunch and find the approximate face position at some discrete locations
   b) Refine the position of the face by similarity between a Gabor jet in the given image and a **bunch** of jets in the FBG model
   c) Precisely locate the fiducial points by moving all the nodes locally and relative to each other to **optimize** the graph similarity further



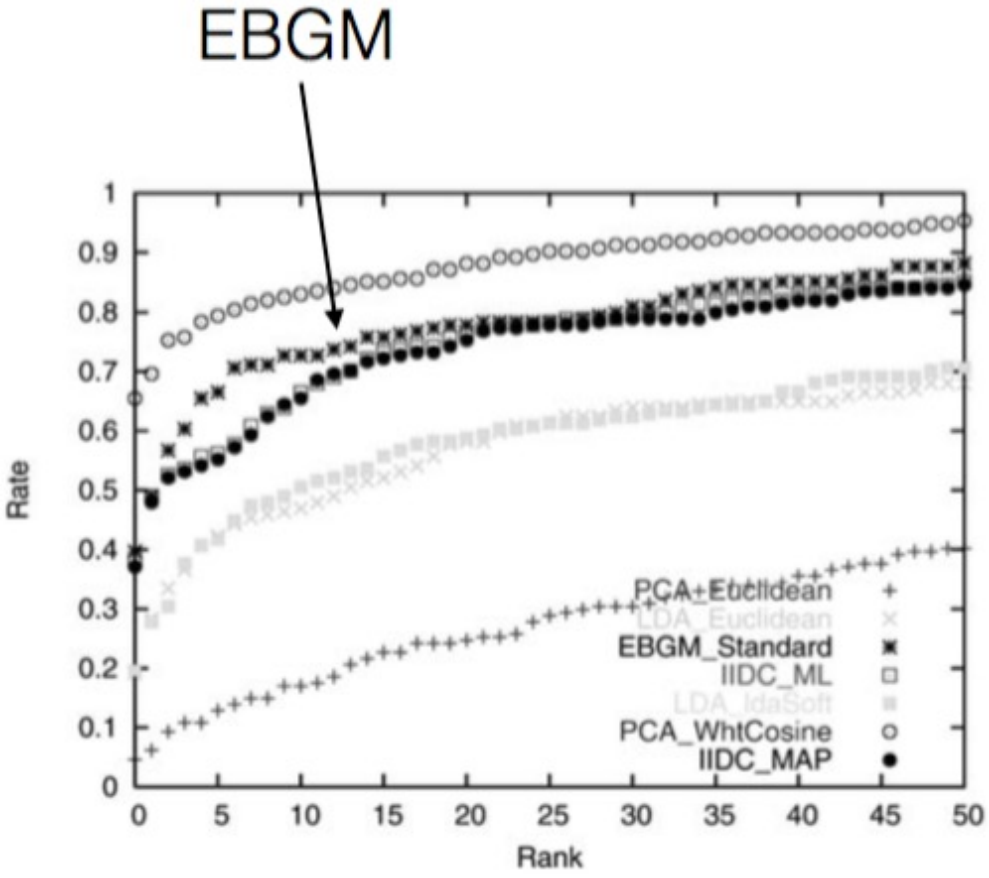
Recognition with FBG Model

2. Comparison with **all** model graphs
   - Similarity between **image graph** from the probe image and the **model graph** is computed as the **average** similarity between the jets at the corresponding fiducial points
   - Can be matched successfully even with some missing nodes
3. Choose **best similarity score** and apply a **threshold** to determine if it's a valid match

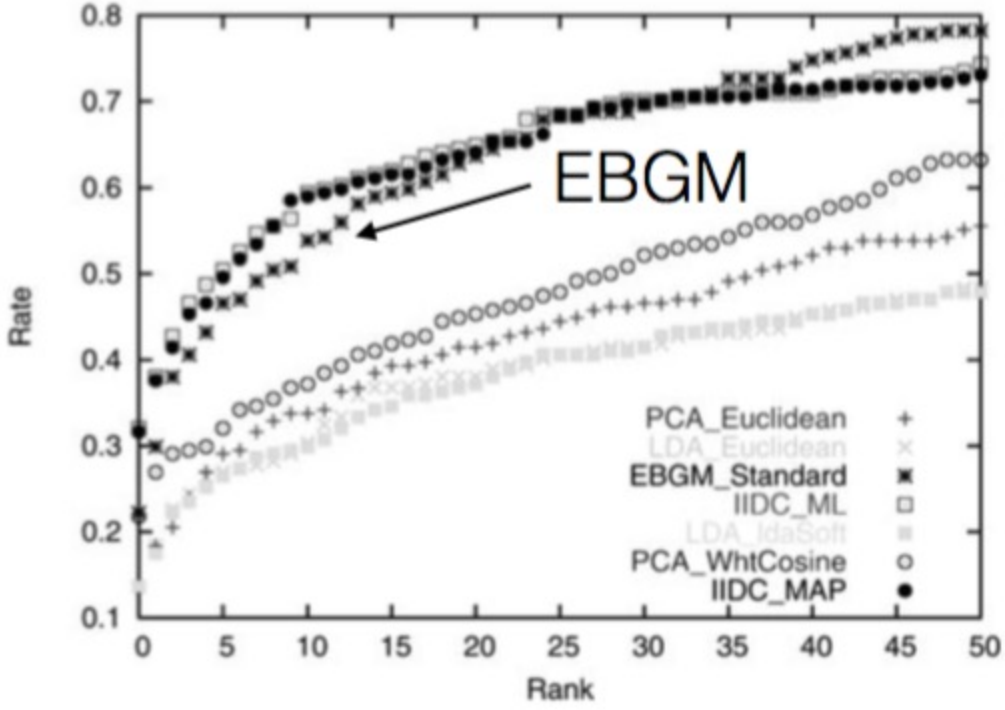


Results

DUP1 DUP2



FERET DUP1



FERET DUP2

Texture-based Face Recognition

- Raw pixel intensity values are quite sensitive to changes in ambient lighting and facial expressions
- Alternative: use more robust feature representation schemes
- Characterize the texture of an image using the **distribution** of **local** pixel values

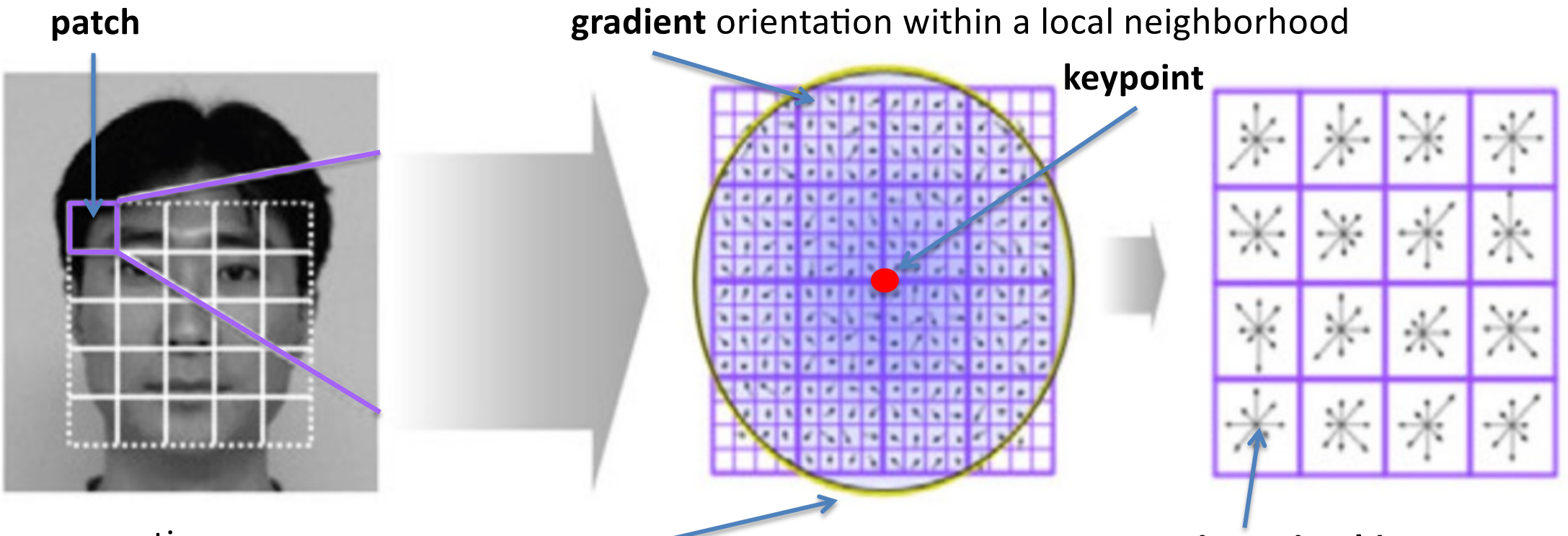
Scale Invariant Feature Transform

- **2 major steps:**
  - key point extraction
  - descriptor calculation in a local neighborhood at each key point
- Achieves tolerance against pose variations
- **Problem:** number of key points in SIFT could be large (in the order of hundreds)
  - Finding the correspondences between the key points from two different images is a challenging task
- **Assumption:** the face images are roughly pre-aligned (e.g. using the eyes)
  - Key point detection process can be bypassed and the descriptor can be constructed directly from the entire face image (Dense SIFT)

Scale Invariant Feature Transform

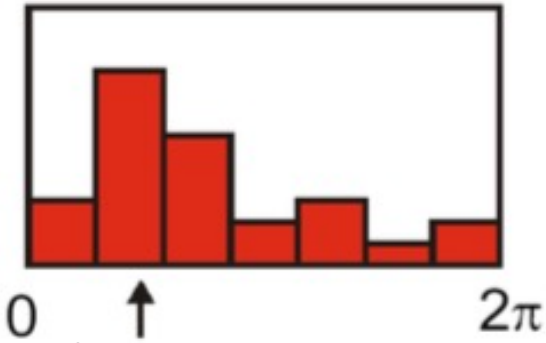
- **Descriptor:** usually a histogram of gradient orientations within a local neighborhood
- Face image is typically divided with multiple **patches** and the SIFT descriptor is constructed from each patch
- Final descriptor is obtained by **concatenating** all the descriptors from all the patches

SIFT Illustration



Orientation Assignment

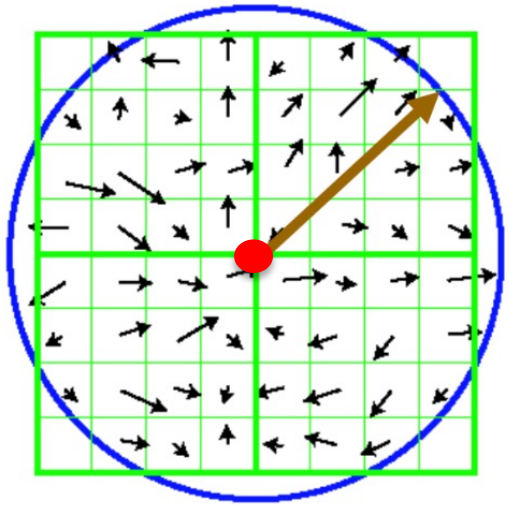
1. Create histogram (36 bins) of local gradients at selected scale
2. Assign canonical orientation (and magnitude) at peak of smoothed histogram



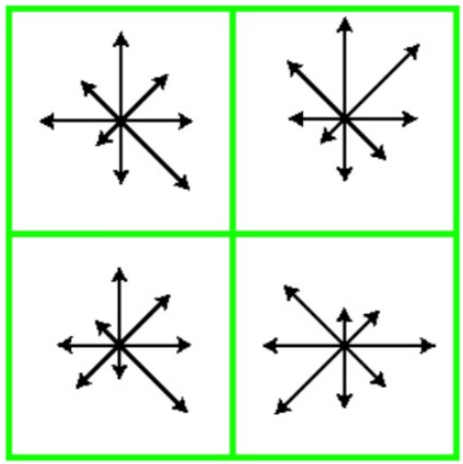
chosen orientation
(if two major orientations, choose both)

Patch/Keypoint Descriptor

Example: 1 patch = 1 keypoint, 2×2 windows, each sized 4×4 windows, weights



gradients on the image within the patch

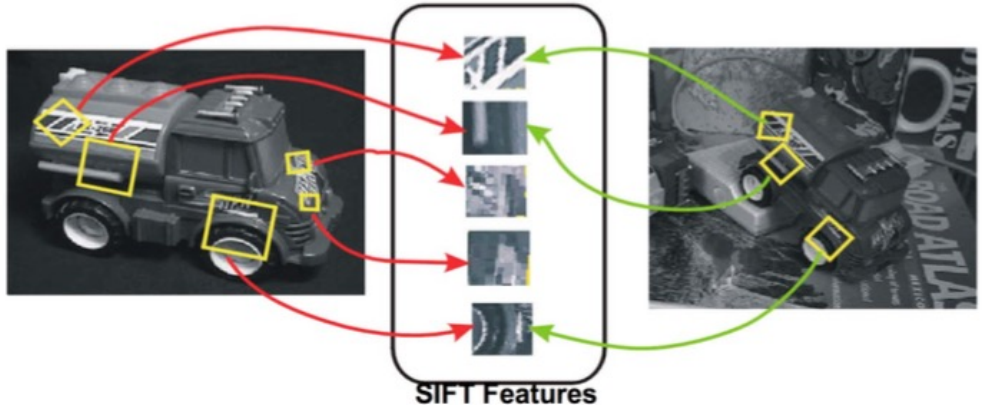


orientation histograms for 4 windows (1 patch)

- Orientation histograms have 8 bins (orientations)
- Each patch here is described with 2*2*8=32 features (32D vector)

Normally in practice: 1 patch, **4×4** windows, each sized 4×4 windows, weights

Advantages of SIFT



- **Locality:** features are local
- **Distinctiveness:**
  features can be matched to a large database
- **Quantity:**
  many features can be generated for small faces
- **Efficiency: (**close to) real time performance
- **Extensibility:**
  can easily be extended to a wide range of feature types

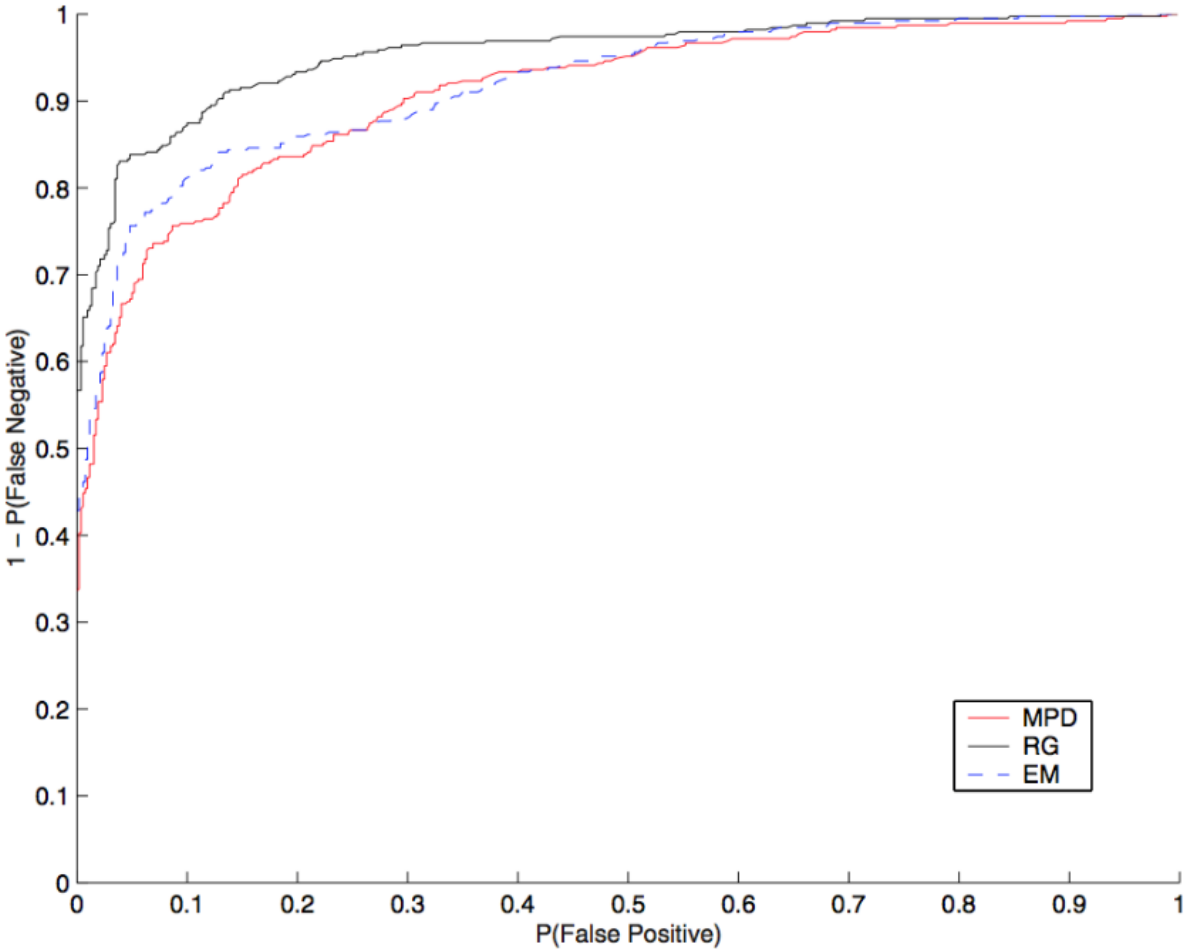
SIFT for Face Recognition

Three SIFT strategies:

1. **Minimum pair distance:** compute distance between all descriptors in each face image
2. **Match eyes and mouth:** compute distance only between descriptors in these regions
3. **Match on a regular grid:** divide face up into a regular grid, match descriptors in subimages

SIFT Results

BANCA db



MPD = Min. Pair Distance

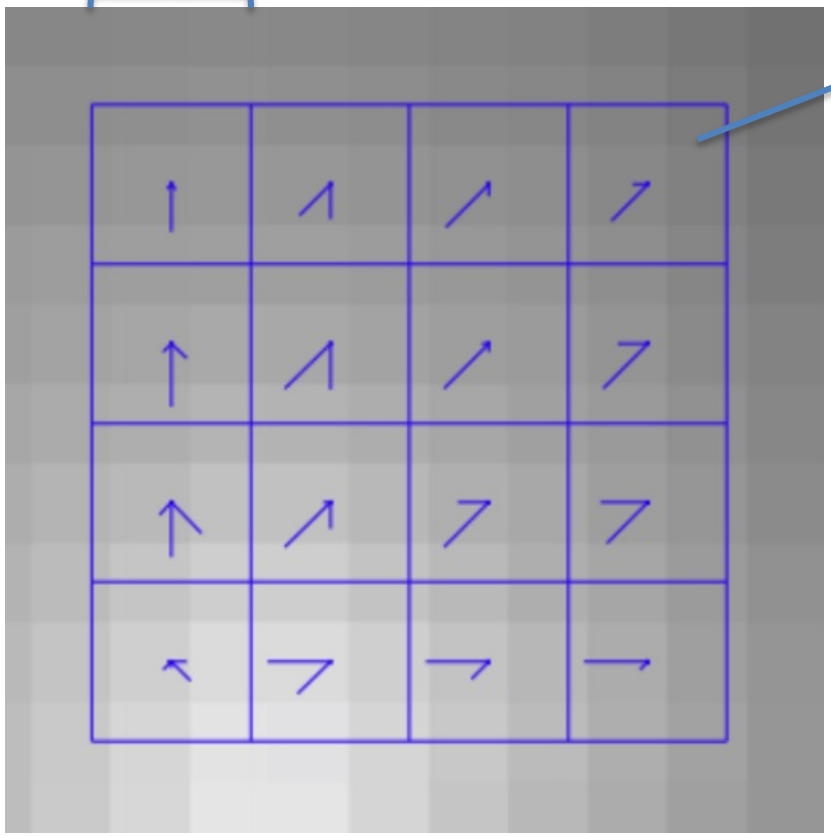
EM = Eyes and Mouth

RG = Regular Grid

Histogram of Oriented Gradients

>>> image descriptors invariant to 2D rotation

8 pixels => 8×8 patch => 8×8 HOG descriptor



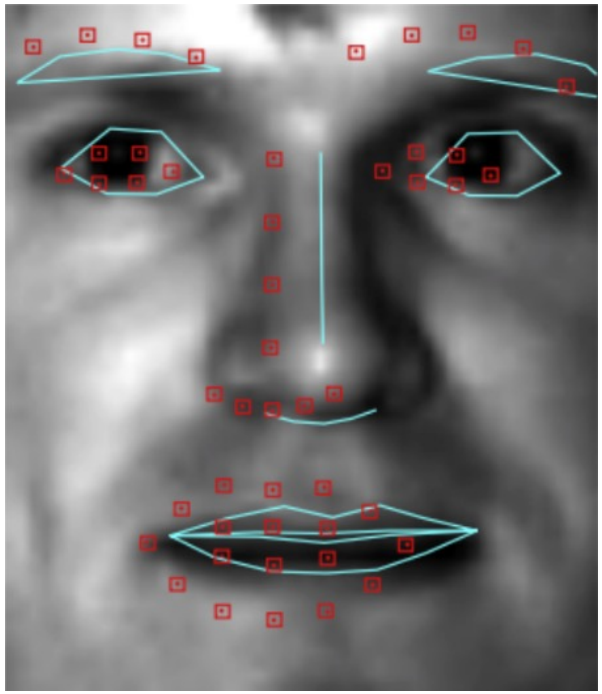
Each cell shows the orientation of the gradients

HOG counts occurrences of edge orientations

Arrange many of these:
- in local neighborhoods
- or across entire image (sparse -> dense)

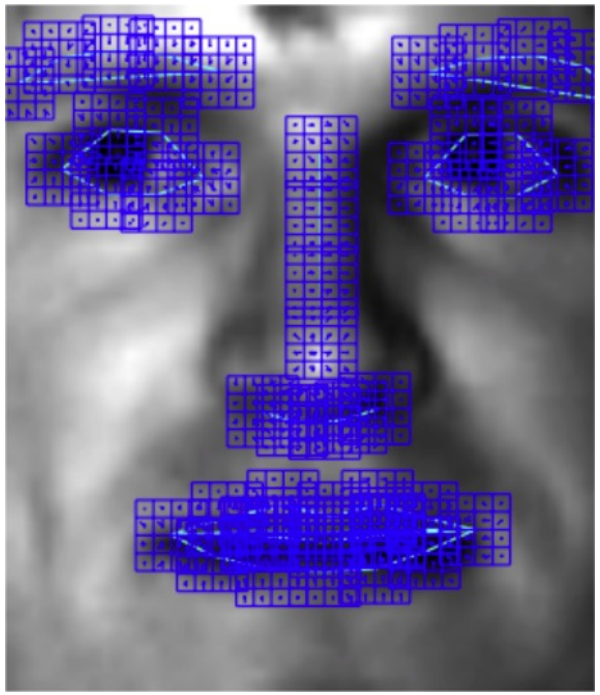
HOG for Face Recognition

Use of informative local neighborhoods …

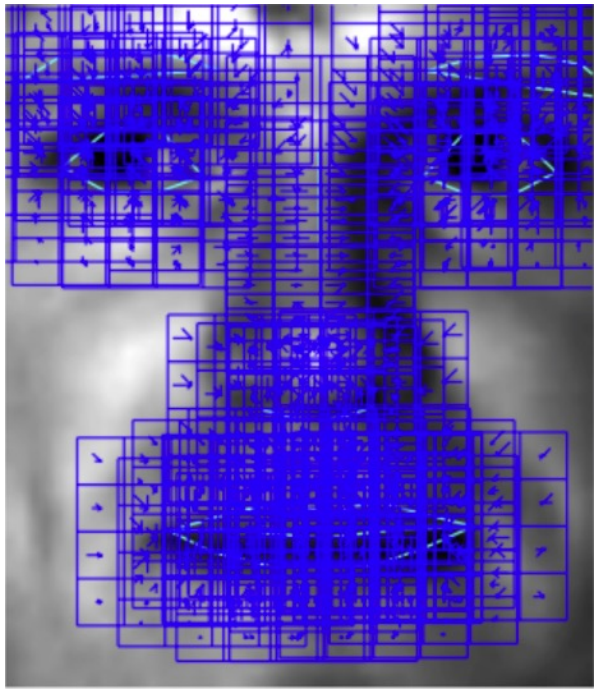


initialized landmarks (red boxes) and result for the Active Appearance Model fitting

>>>



24×24 HOG descriptors

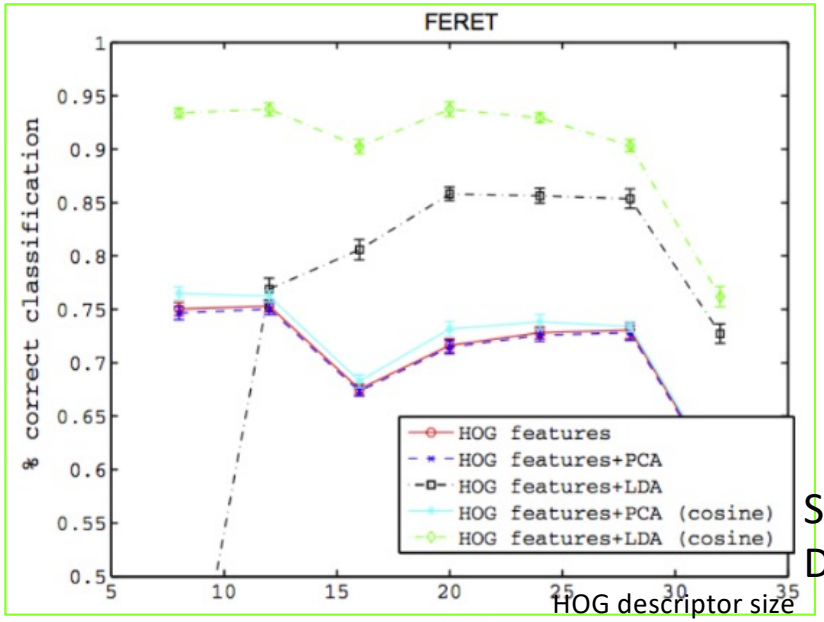


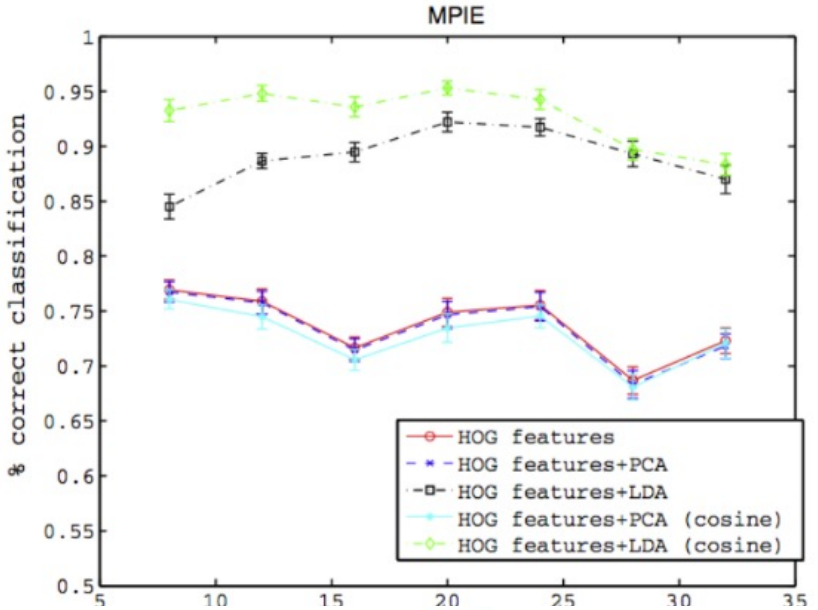
64×64 HOG descriptors

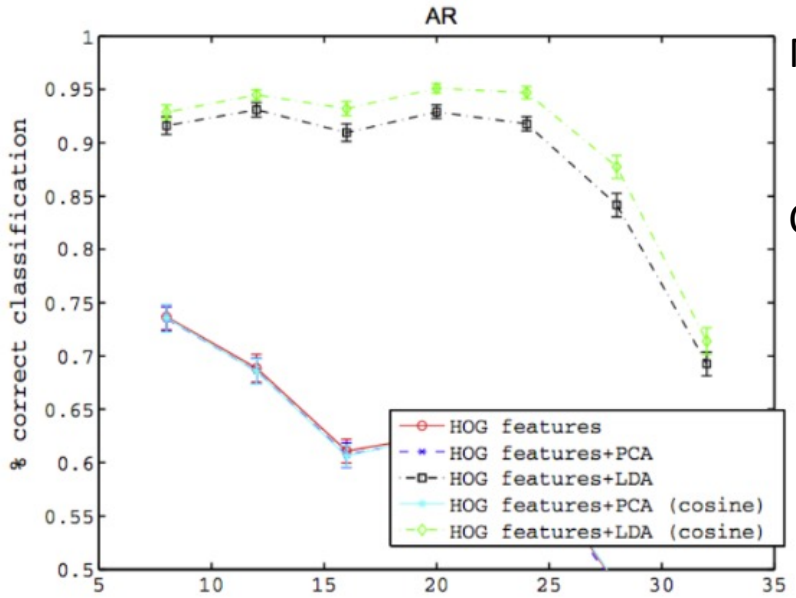
4 databases: FERET, MPIE, AR, Yale =>

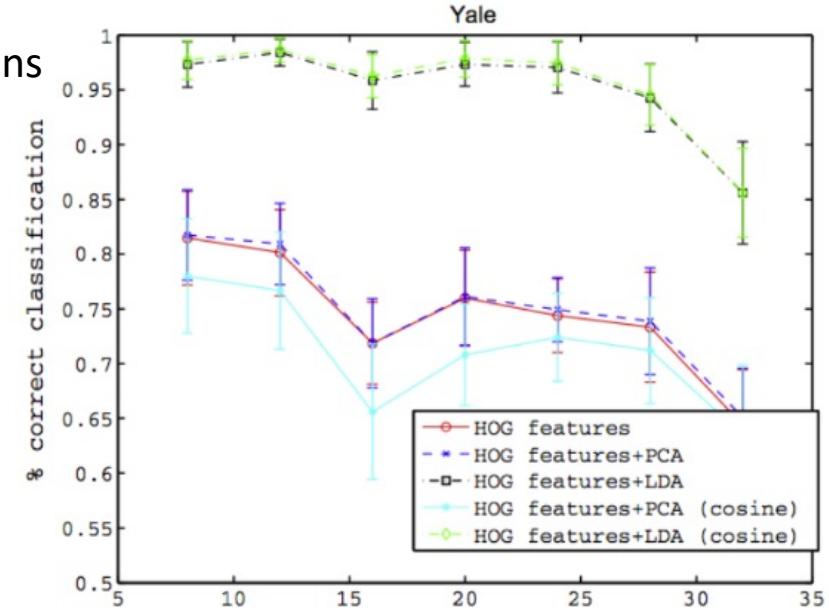
HOG Results

across entire image









- Smaller descriptor sizes better
- Dimensionality reduction doesn't help
- Minimizing intra-class variations and maximizing inter-class variations helps a lot
- Cosine distance better

FERET

	fb (%)	fc (%)	dup1 (%)	dup2 (%)
PCA Euclidean	74.3	5.6	33.8	14.1
PCA Mahal. cosine	85.3	65.5	44.3	21.8
LDA	72.1	41.8	41.3	15.4
Bayesian	81.7	35.0	50.8	29.9
Bayesian map	81.7	34.5	51.5	31.2
Gabor ML	87.3	38.7	42.8	22.7
Hybrid Approach → HOG-EBGM	**95.5**	81.9	60.1	55.6
8 × 8 patch	91.4	83.0	70.2	62.0
12 × 12 patch	93.0	82.0	70.8	63.3
16 × 16 patch	88.4	68.0	68.7	60.7
20 × 20 patch	93.7	75.3	70.2	60.3
24 × 24 patch	94.2	70.1	66.8	56.8
28 × 28 patch	91.6	42.8	60.0	56.0
Best Approach → Combination 8 × 8–28 × 28 (product rule)	95.4	**84.0**	**74.6**	**69.2**

decision-level combination of results

Computational Cost

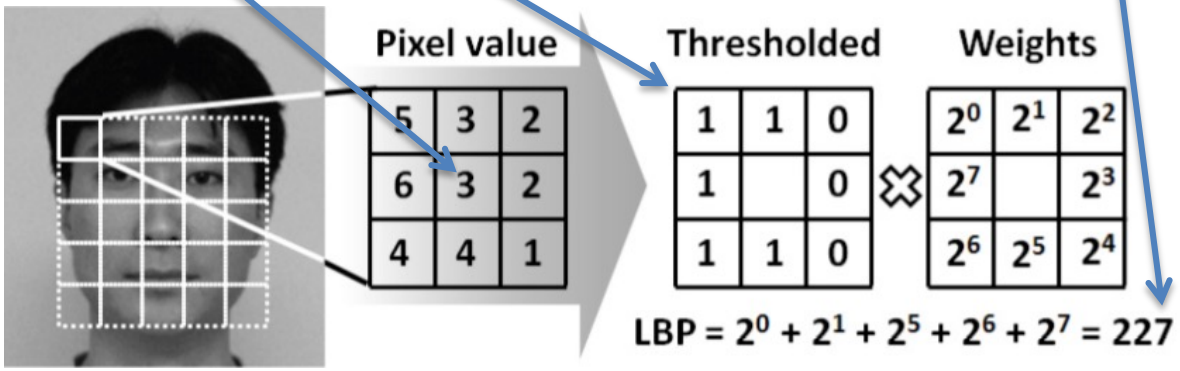
Times in seconds given for **all** 3540 FERET images

Patch size	From landmarks	From regular grid
8 × 8	203.8	388.3
12 × 12	210.2	200.7
16 × 16	205.6	73.7
20 × 20	211.2	75.1
24 × 24	212.9	75.9
28 × 28	218.2	27.7
Total	1261.9	841.4

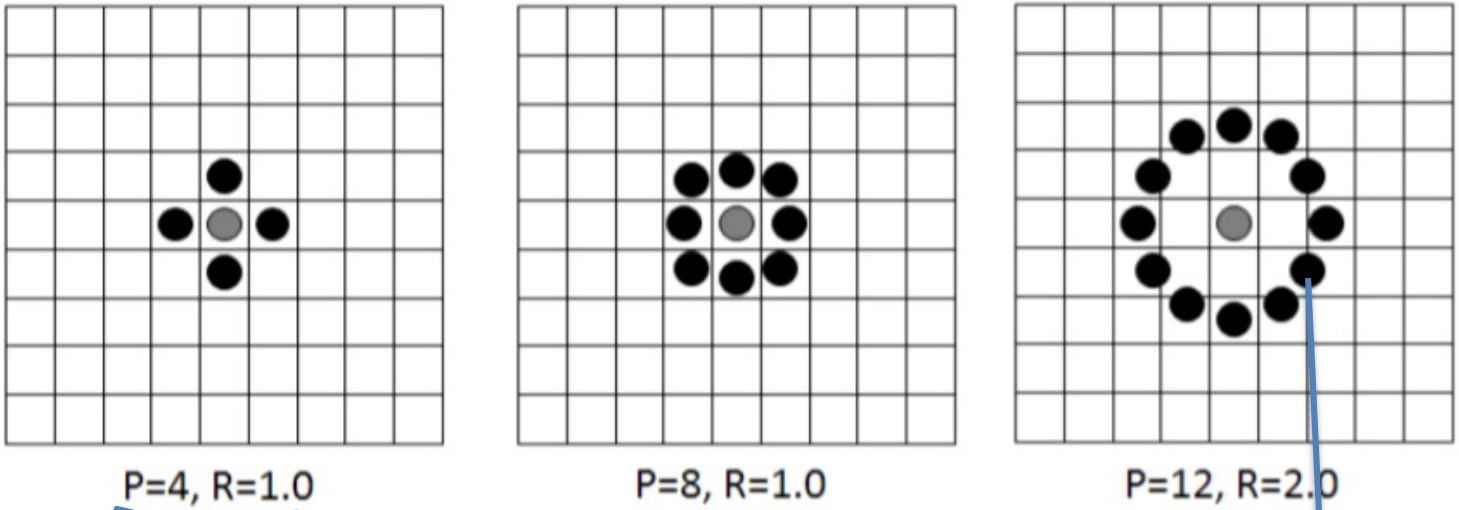
For locating landmarks add about 2.5 s per image

Local Binary Patterns

- LBP features are usually obtained from image pixels of a **3×3** neighborhood region
- Compare the 8 neighboring pixel intensity values to the intensity value of the **central** pixel in the region and represent the result as an 8-bit **binary code**
- Binary code can be further converted into a **decimal number** by applying weights to each bit and computing the sum



Multiscale LBP



number of sampling points

radius – the distance of the sampling points from the center pixel

If a sampling point is off the pixel grid, bilinear interpolation of pixel values can be applied to obtain the intensity value of the sampling point

MLBP images encoded at different scales

$LBP_{(P=)8,(R=)1}$

$LBP_{8,3}$

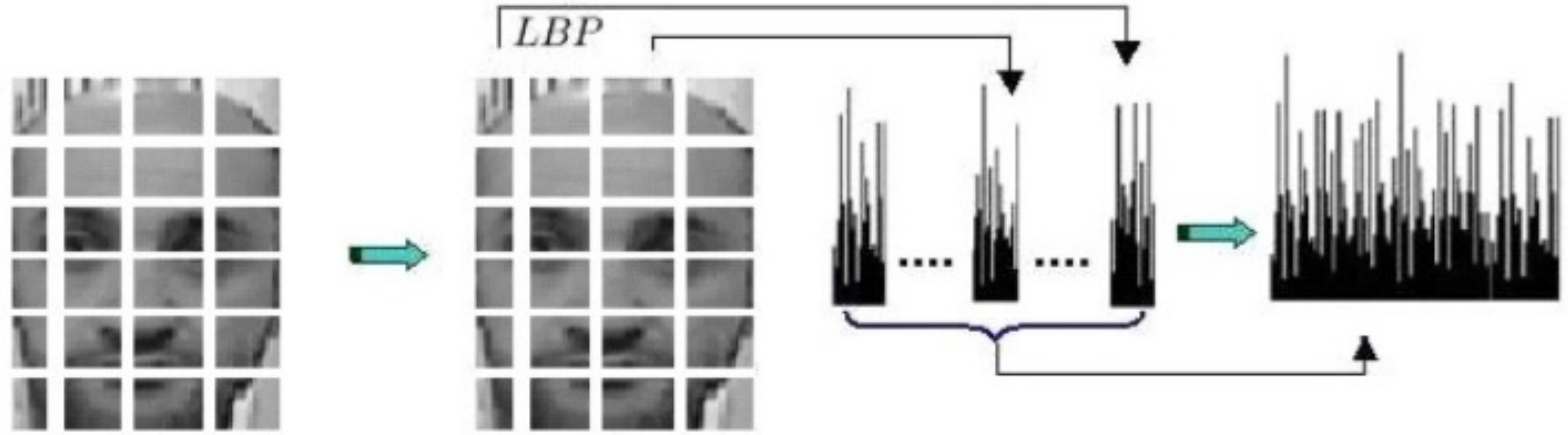
$LBP_{8,5}$

detection of **micro** details

highlight the **macro** features

Histogram of LBPs and Matching

- After LBP encoding of each pixel, the face image is divided into several smaller windows and the **histogram** of LBPs **in each window** is computed
- Number of **bins** in the histogram is 8 (LBP) and 2^P (MLBP)
- Global feature vector is generated by **concatenating** histograms of all the individual windows and normalizing the final vector
- 2 face images can be matched by computing the **similarity** (or distance) between their feature vectors

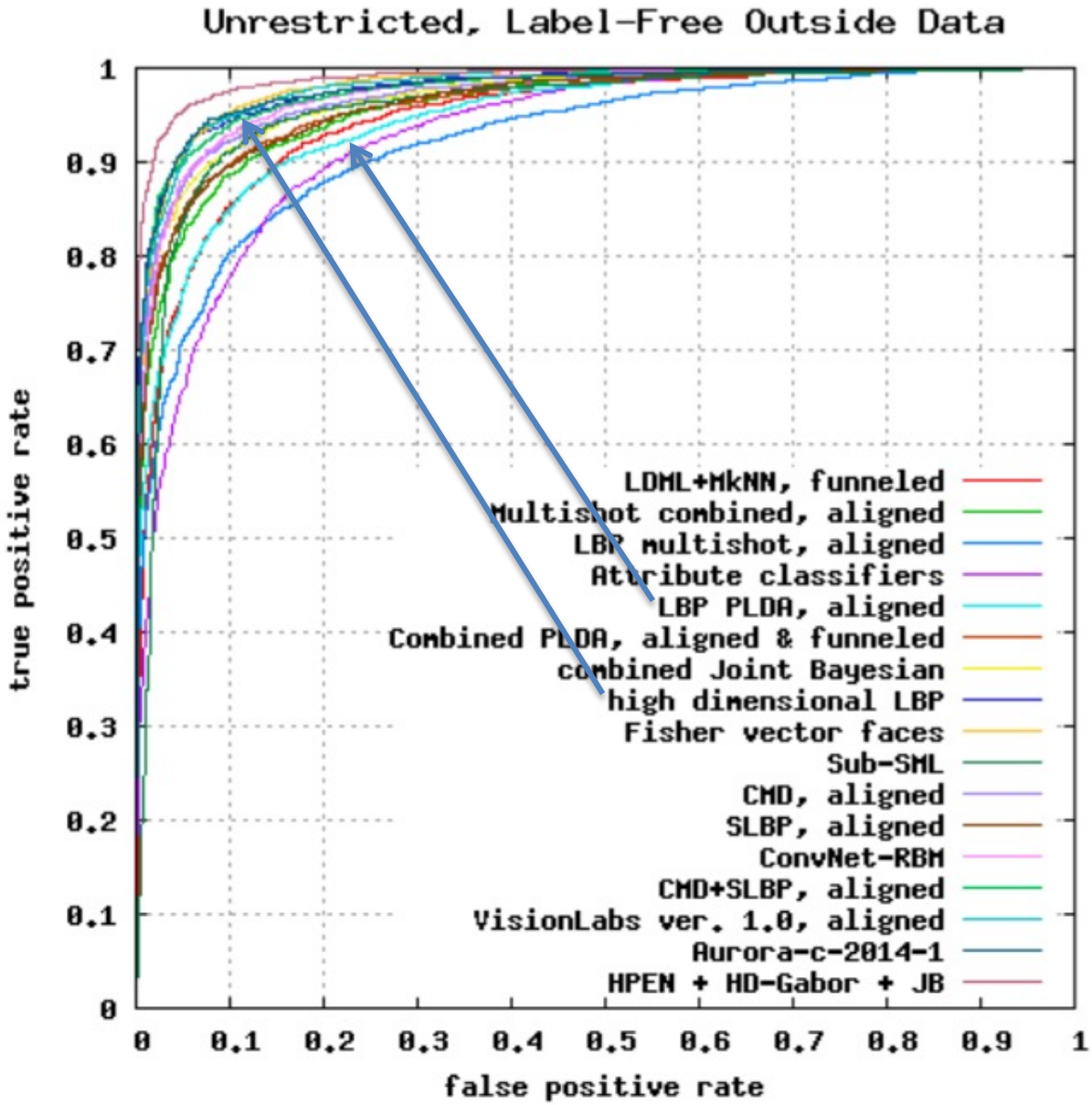


+region weighting:

black squares = 0
dark gray = 1
light gray = 2
white = 4

LBP Results

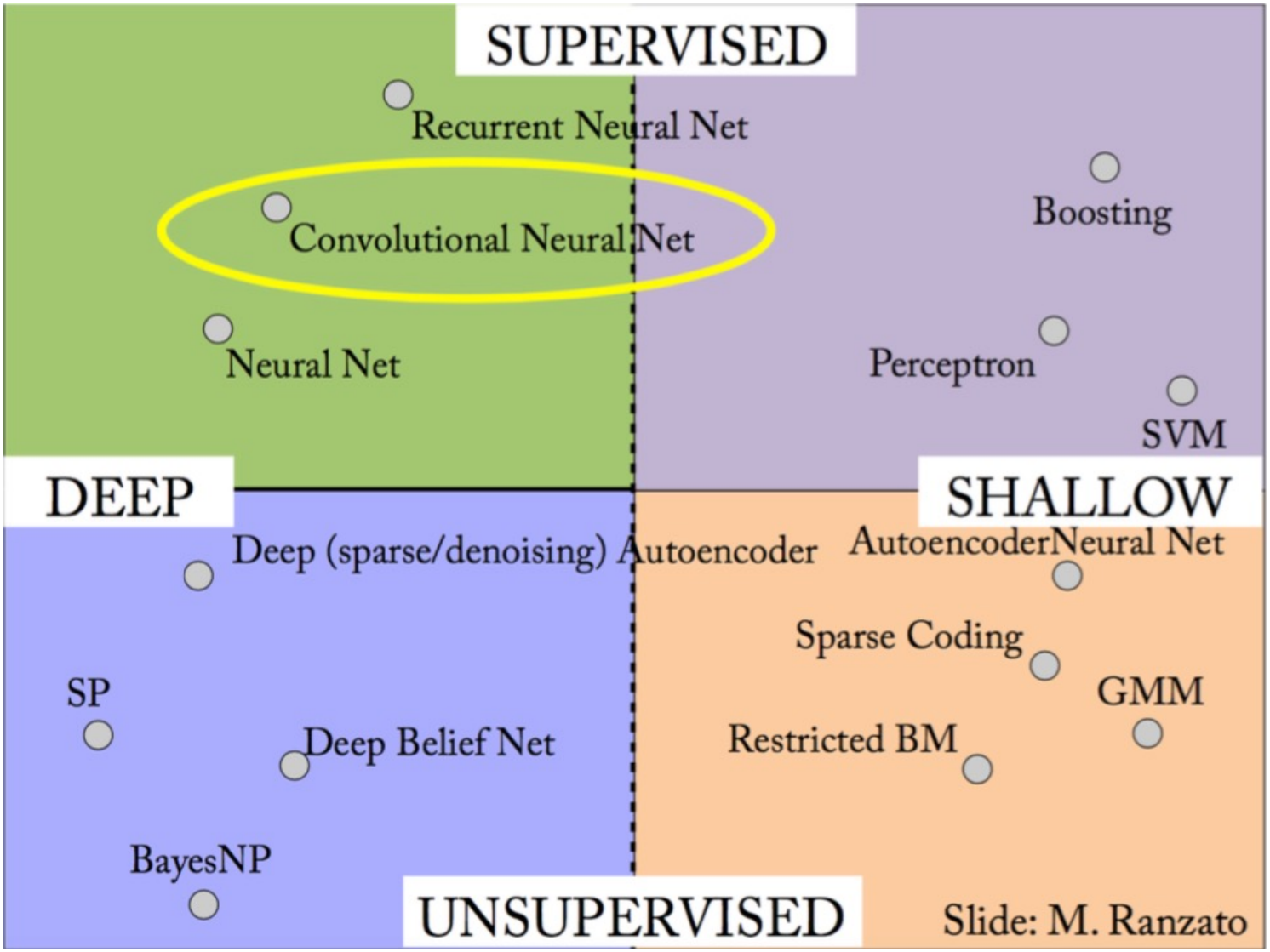
LFW db



Deep Learning Approach

- Representations we have discussed are not strongly invariant!

- **Solution:** multi-layer neural network
- Layers: hierarchy of feature extractors (from simple to complex)

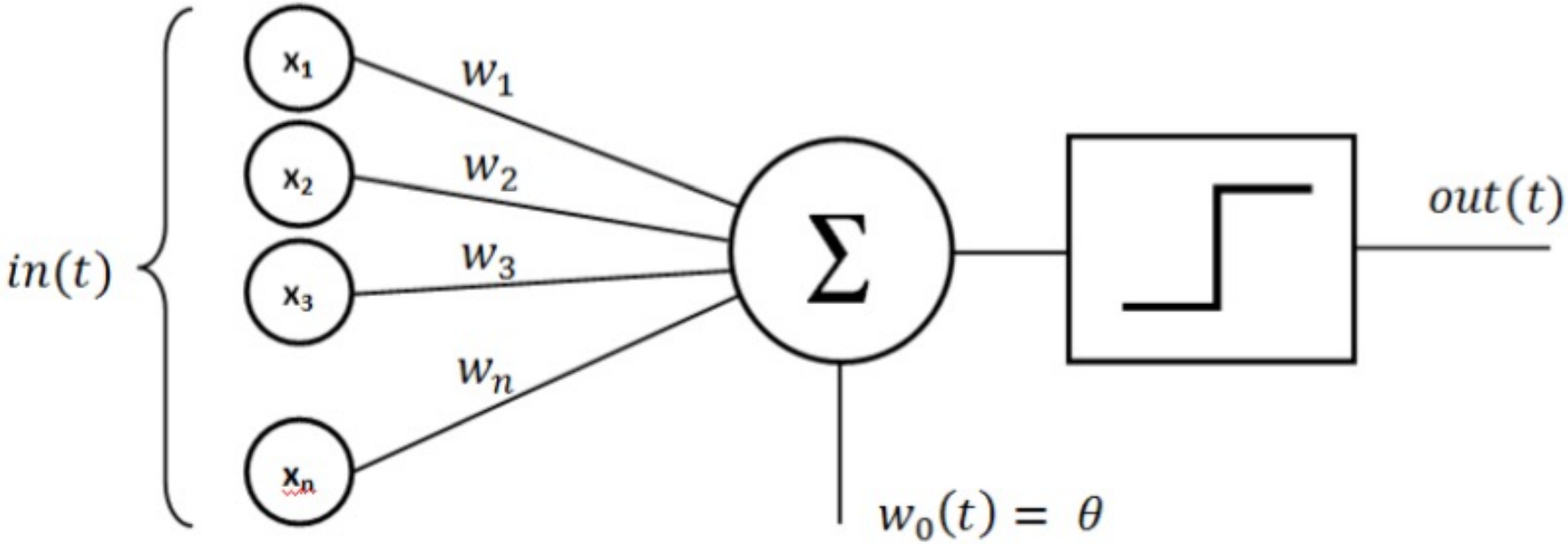

SUPERVISED
Recurrent Neural Net
Convolutional Neural Net
Neural Net
Boosting
Perceptron
SVM
DEEP
SHALLOW
Deep (sparse/denoising) Autoencoder
AutoencoderNeural Net
Sparse Coding
GMM
SP
Deep Belief Net
Restricted BM
BayesNP
UNSUPERVISED
Slide: M. Ranzato

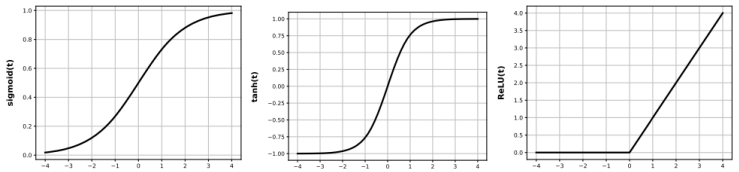
Introduction to Neural Networks

Artificial Neural Networks (ANNs):

- Inspired by the human brain
- Composed of **artificial neurons** connected by weighted links
- Weights represent the strength of connections
- Learning occurs by **adjusting** these weights

But it All Started with the … **Perceptron**

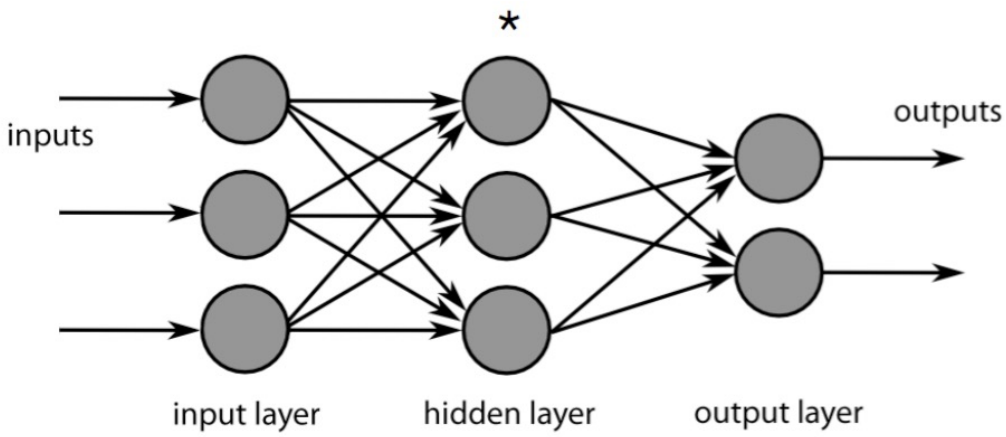




(a) sigmoid (b) tanh (c) ReLU

From Perceptron…to Layer...to Artificial NN

- Single **layer** perceptrons cannot model non-linear functions
- Adding a **hidden layer** with a non-linear activation function addresses this
- ANN contains sets of adaptive weights
  - Learned during training



Feedforward Neural Network (FNN)

- **Key characteristics:**

  Unidirectional data flow: i**nput → hidden layers → output**

  (layers: **input layer**, **hidden layers**, **output layer)**
- **Role of layers:**
  - **Input layer:** represents the input features
  - **Hidden layers:** apply transformations using activation functions
  - **Output layer:** produces the final output

Neural Network Training Process

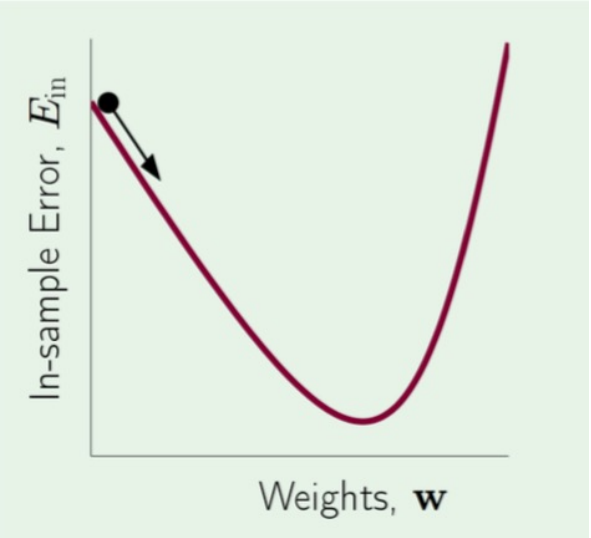
- **Goal of training:** minimize error using a **loss function**
- **Training steps:**
  1. **Forward propagation:**
     Compute output based on input and current weights
  2. **Loss calculation:**
     Measure the difference between predicted and actual output
  3. **Backpropagation:** calculate gradients for weight adjustments
  4. **Optimization:** update weights using algorithms (e.g., SGD)

Set the Weights

- We need a **strategy** to set the weights with respect to observed error on the training set >>>

1. Initialize $\mathbf{w}(0)$
2. For $t = 0, 1, 2, \ldots$ [to termination]
3. $\mathbf{w}(t+1) = \mathbf{w}(t) - \alpha \nabla E_{\text{in}}(\mathbf{w}(t))$
4. Return final $\mathbf{w}$

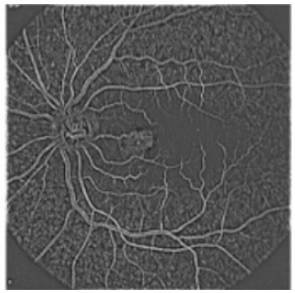
α = learning rate

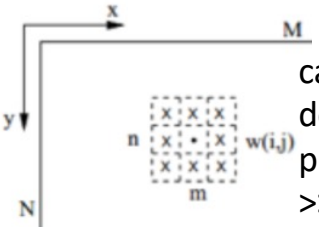


- Use: Stochastic Gradient Descent
- What's an efficient way to calculate the gradients? Backpropagation!

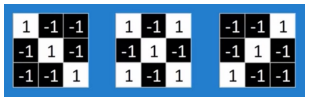
From Deep Learning … to **CNN**

- What is deep learning?
  - The simple answer: **Just multilayer artificial neural network!**
- What is convolution?
  - The simple answer: **Image filtering!**





calculate dot product >>>



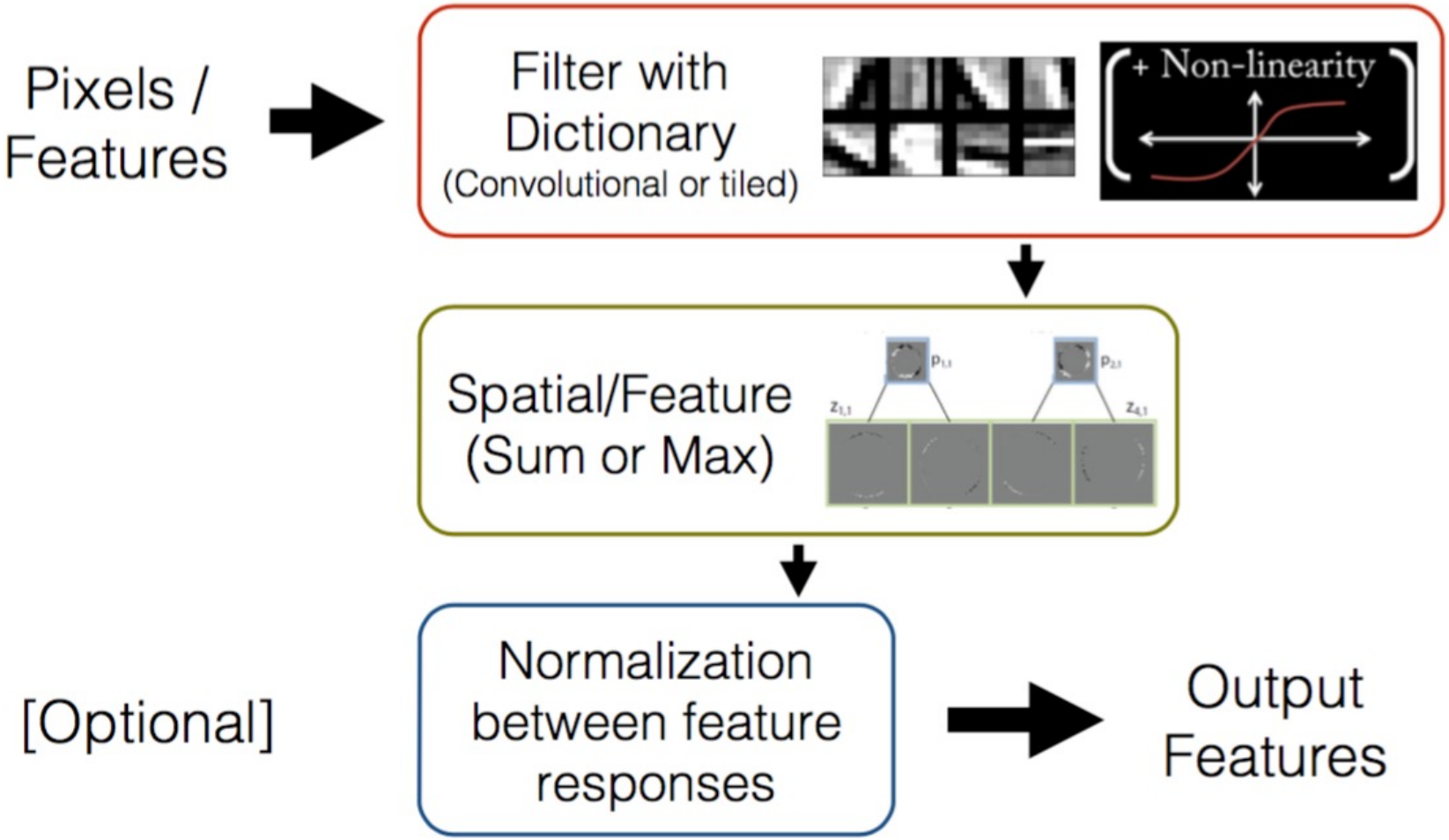
$$\mathbf{w}(x,y) * \mathbf{f}(x,y) = \sum_{i=\lfloor -m/2 \rfloor}^{\lfloor m/2 \rfloor} \sum_{j=\lfloor -n/2 \rfloor}^{\lfloor n/2 \rfloor} w(i,j)\, f(x-i, y-j)$$

Convolutional Neural Networks (CNN)

- **Best suited for:** image and spatial data processing
- **Core components of CNNs:**
  - **Convolutional layers:** extract spatial features using filters
  - **Pooling layers:** reduce feature map size while retaining key information
  - **Fully-connected layers:** flatten and classify features
- **Hyperparameters in convolutional layers:**
  - Filter/kernel size (F)
  - Depth (K)
  - Stride (S)
  - Padding (P)

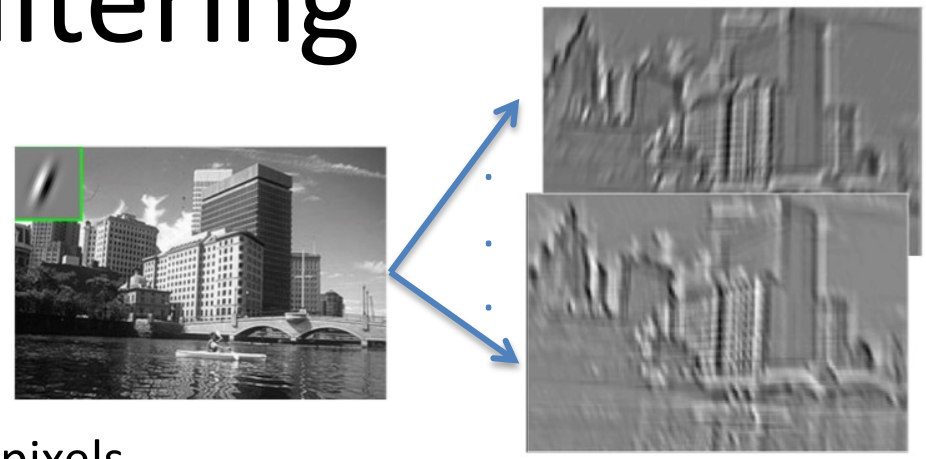
Hyperparameter	Definition	Impact
Filter Size (F)	Dimensions of the filter	Controls the size of the receptive field
Depth (K)	Number of filters	Determines the number of feature maps
Stride (S)	Step size for filter movement	Affects output size and computational cost
Padding (P)	Extra pixels added to the edges	Preserves edge information and controls output size

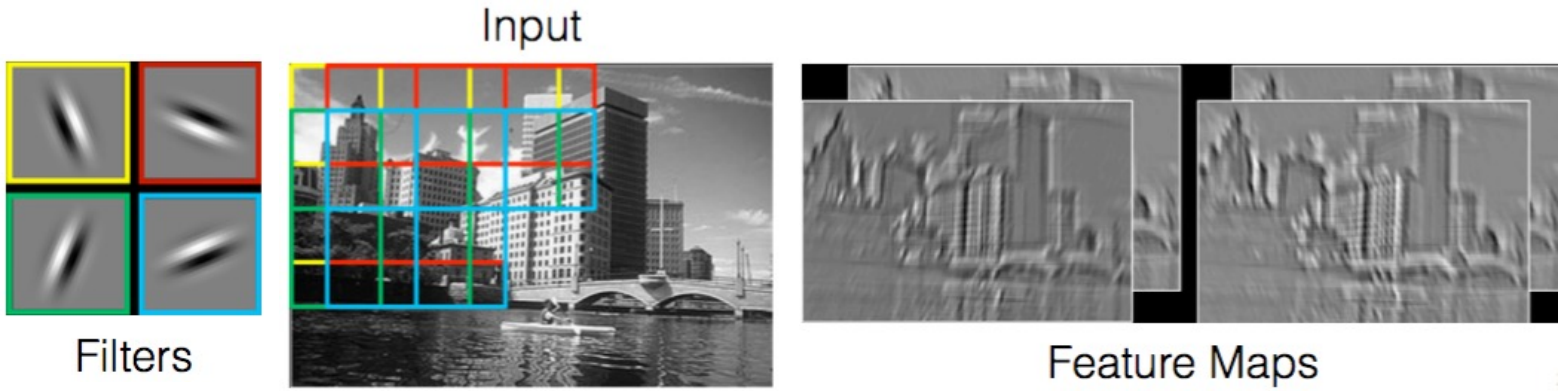
Layer(!) Components



Filtering

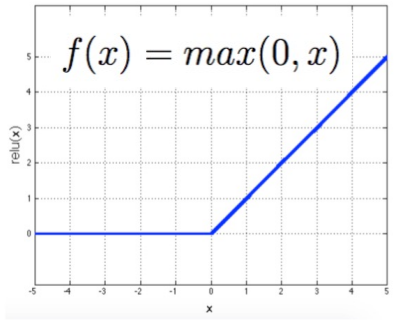
- Convolutional
  - Dependencies are local
  - Stride 1, 2,... (faster, less mem.)
- Tiled
  - Filters repeat every *n* (filter size) pixels
  - More filters than convolutions for given number of features





- Non-linearity
  - Use Rectified Linear Units (ReLU) to simplify backpropagation, make learning faster, avoid saturation issues

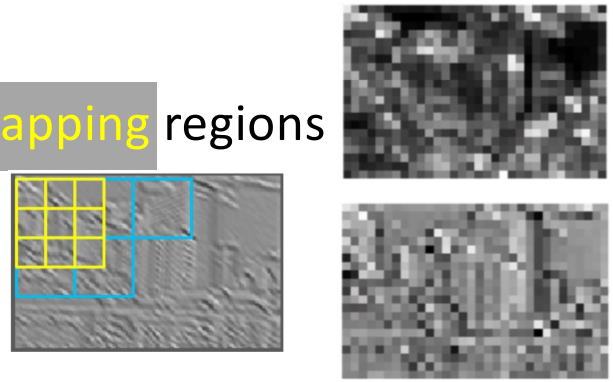
Property	ReLU
Formula	$f(x) = \max(0, x)$
Range of Output	[0, ∞)
Gradient	1 for $x > 0$; 0 for $x \leq 0$
Non-linearity	Yes
Vanishing Gradient Issue	No (for positive inputs)
Dying Neuron Issue	Yes (for negative inputs)
Computational Cost	Very Low

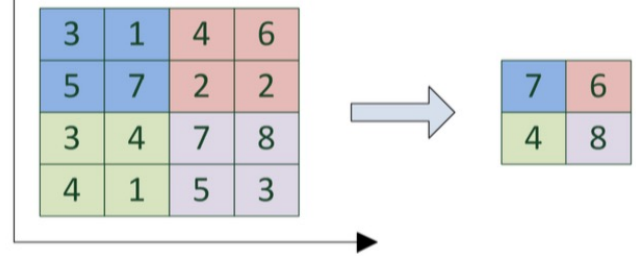


Preferred Option

Pooling

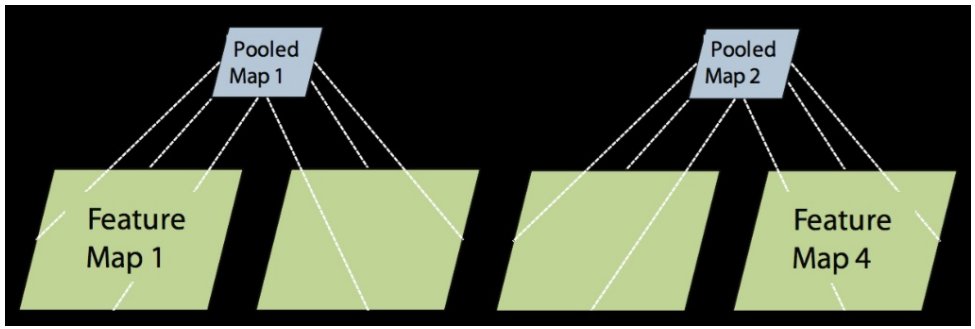
- **Spatial** pooling
  - Non-overlapping / overlapping regions
  - Sum / Max



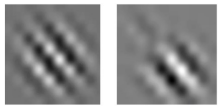


Maximal pooling with a 2×2 filter and stride=2

- Pooling across **feature groups**
  - Additional form of inter-feature pooling
  - MaxOut networks

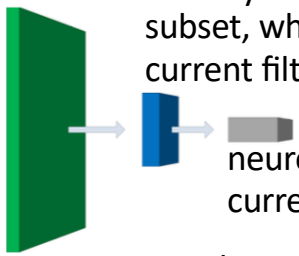


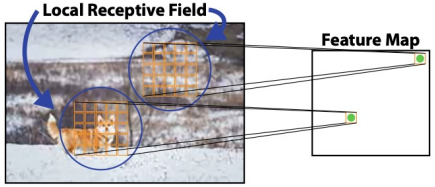
- Role of (spatial) pooling
  - Invariance to small transformations
  - Larger receptive fields (see more of the input)



Convolutional layer:

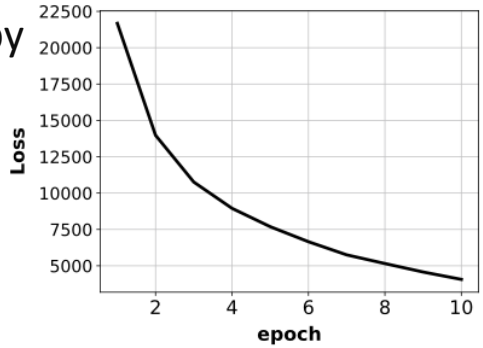
subset, where only the part that influences current filter computation is used – receptive field

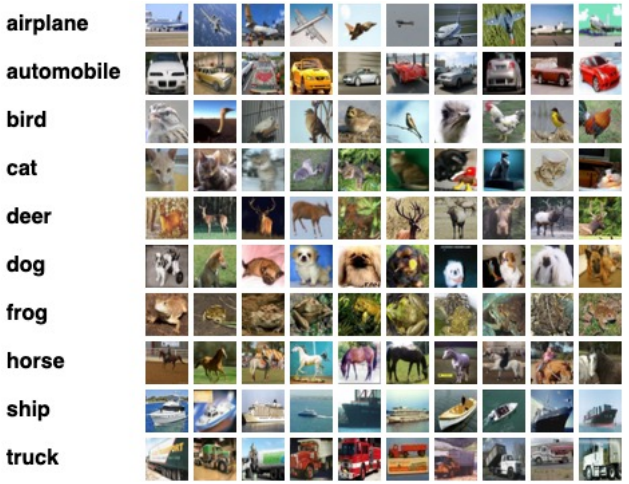




Training CNNs

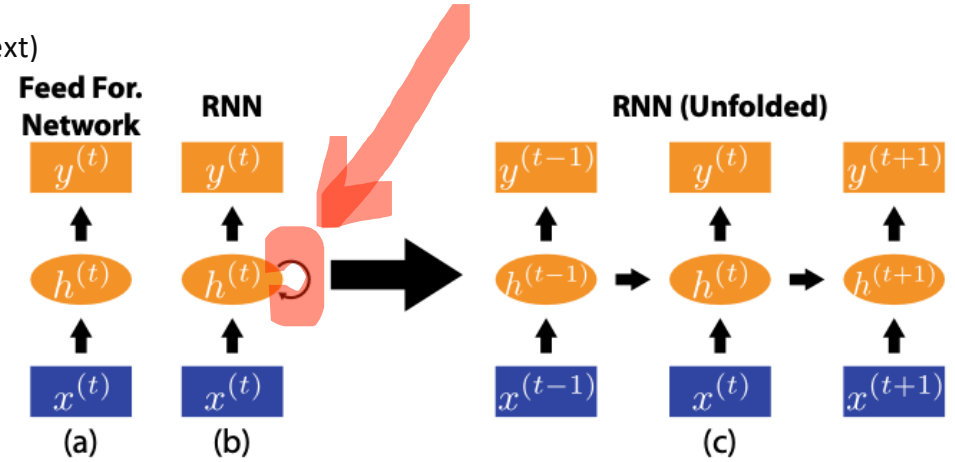
- **Dataset example:** CIFAR-10 (10 object classes)
- **Training steps:**
  1. Initialize weights
  2. Perform forward propagation
  3. Calculate loss using cross-entropy
  4. Perform backpropagation
  5. Optimize weights using SGD
- **Key training parameters:**
  - Learning rate
  - Batch size
  - Momentum





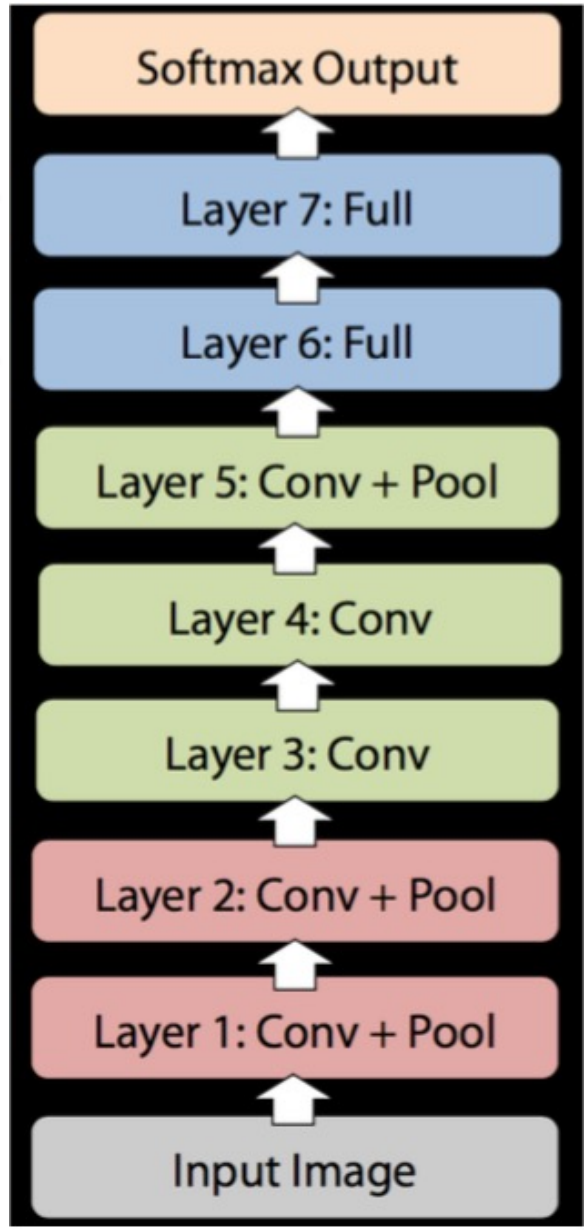
Key Differences Between FNN, CNN, and RNN

- **Feedforward Neural Network (FNN):**
  - **Unidirectional data flow** from input to output
  - Best suited for **static input data** (e.g., tabular data, pre-processed features)
- **Convolutional Neural Network (CNN):**
  - **Local connectivity** via filters for analysing regions of input data
  - Designed for **spatial feature extraction** from grid-like data (e.g., images)
- **Recurrent Neural Network (RNN):**
  - **Temporal dependence** where current outputs depend on previous states
  - Suitable for **sequential data processing** (e.g., time-series, speech, text)
- **Applications:**
  - **FNN:** general-purpose predictions and classification tasks
  - **CNN:** image analysis and spatial pattern recognition
  - **RNN:** sequence modelling and temporal pattern recognition



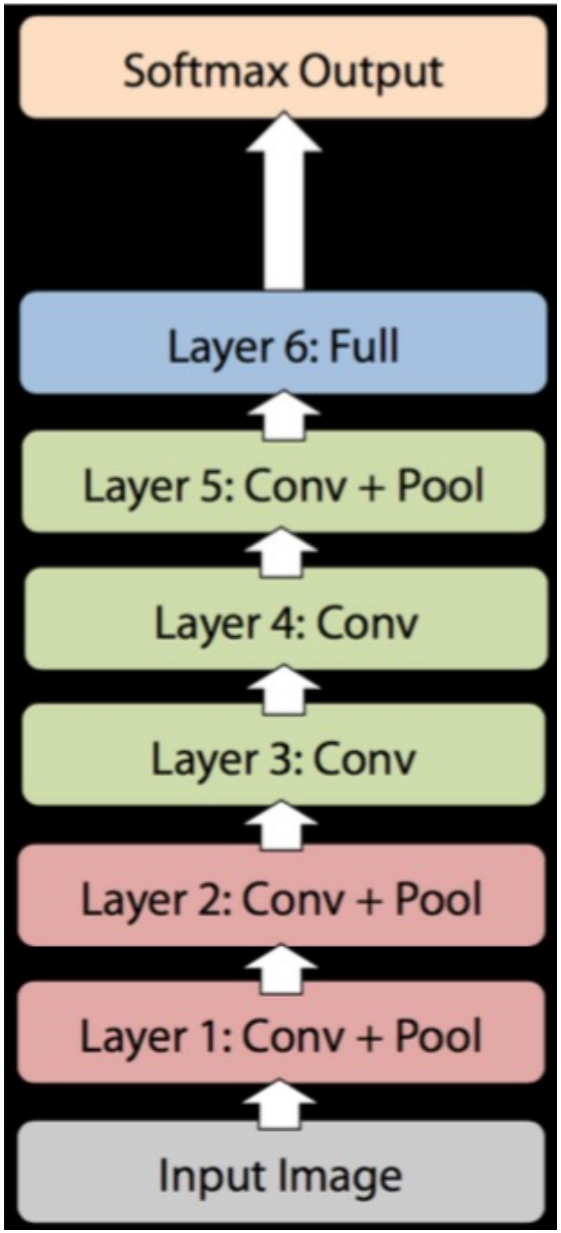
AlexNet Architecture

- 8 layers
- Trained on ImageNet
- 18.2% top/rank-5 error (2009?!)



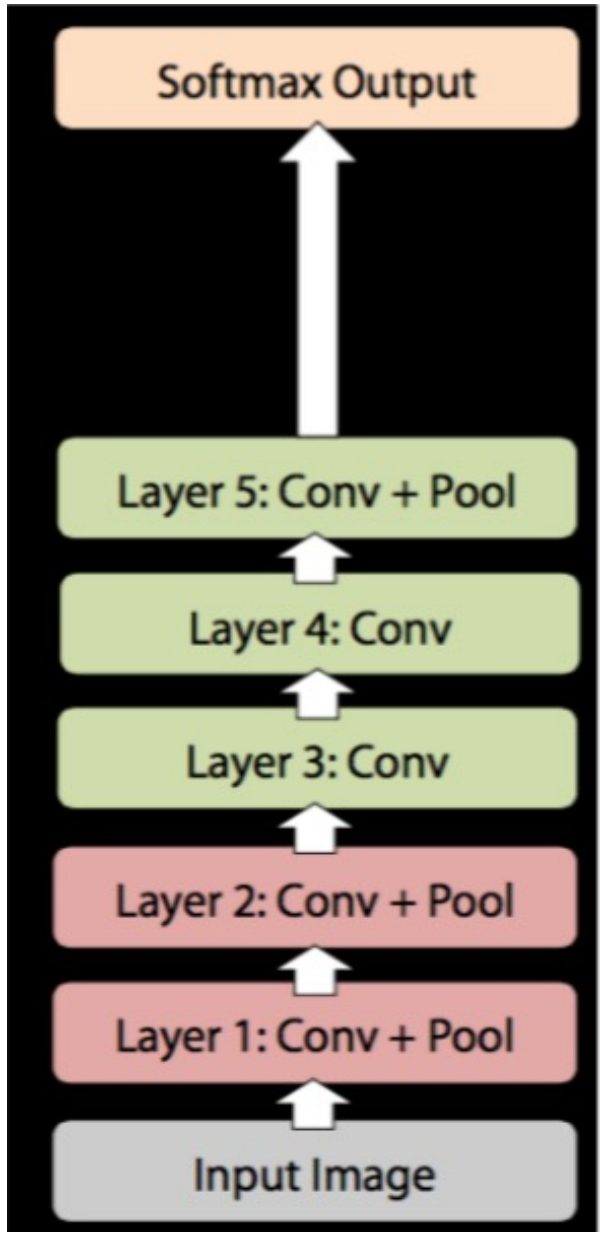
Test by Removal of Layers

- Remove top fully connected layer (Layer 7)
- Drop **16 million** parameters
- **Only** 1.1% drop in performance



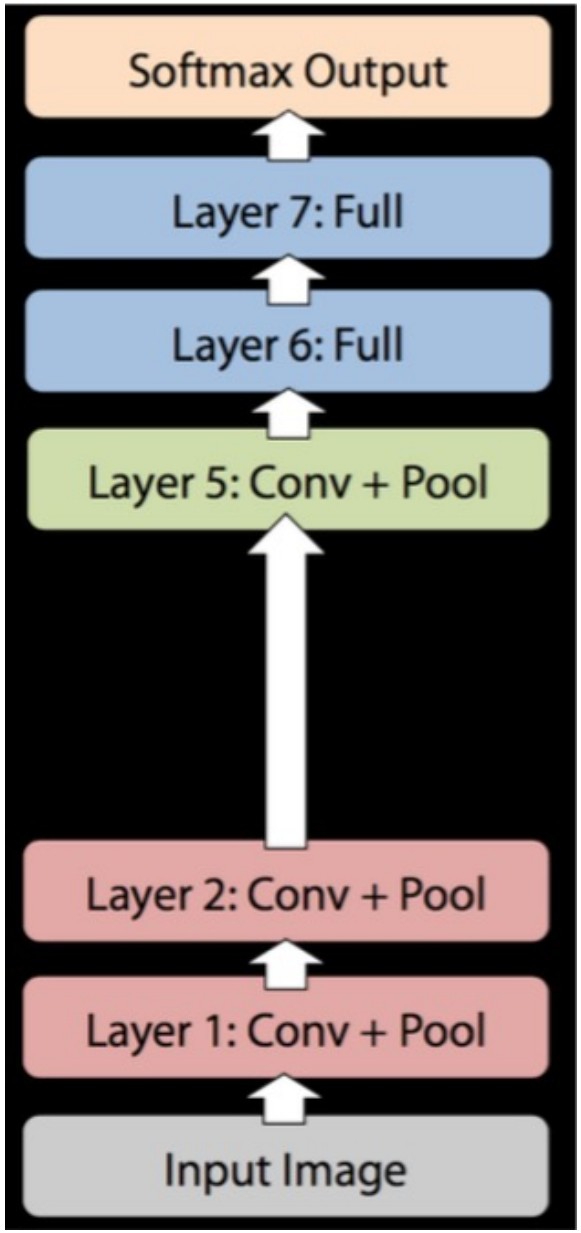
Test by Removal of Layers

- Remove both fully connected layers (Layers 6 & 7)
- Drop ~50 million parameters
- 5.7% drop in performance



Test by Removal of Layers

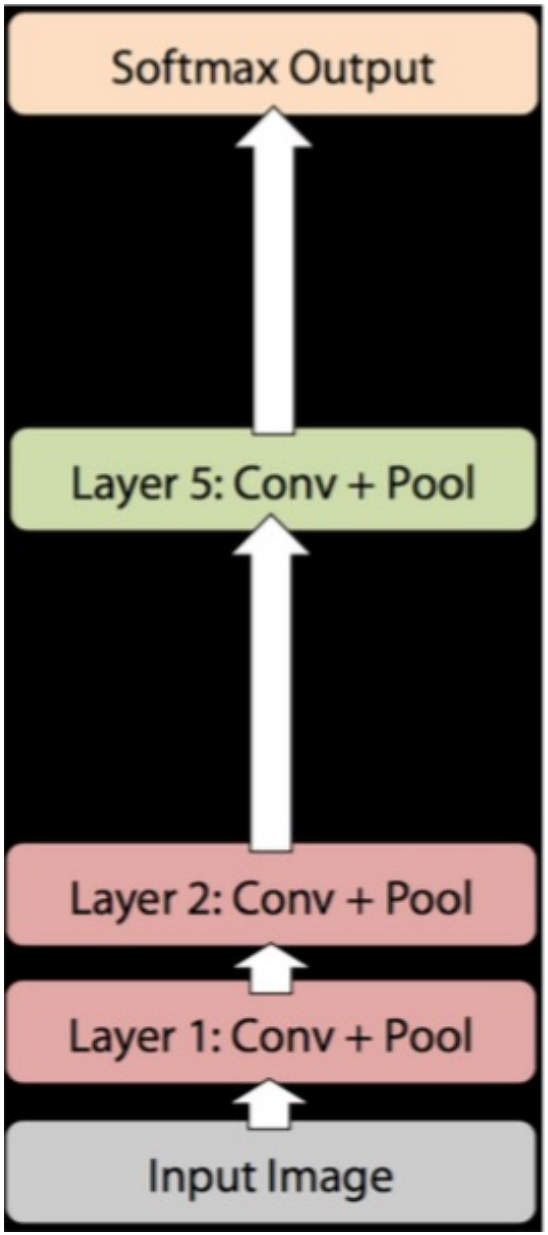
- Now remove the upper feature extractor layers (Layers 3 & 4)
- Drop ~1 million parameters
- 3.0% drop in performance



Test by Removal of Layers

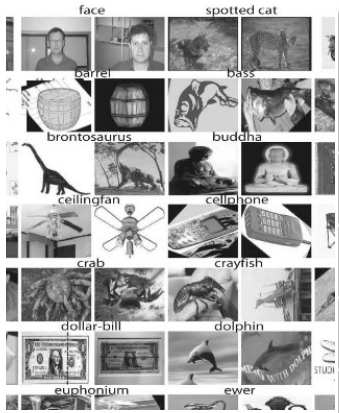
- Now remove the upper feature extractor layers and fully connected layers (Layers 3, 4, 6, 7)
- Now only 4 layers
- 33.5% drop in performance

- **CONCLUSION: Depth of network is key**



Analysis of the discriminative information contained in each layer

Take features **from each layer** (ImageNet-pretrained) and input them into linear SVM or Softmax



	Cal-101 (30/class)	Cal-256 (60/class)
SVM (1)	44.8 ± 0.7	24.6 ± 0.4
SVM (2)	66.2 ± 0.5	39.6 ± 0.3
SVM (3)	72.3 ± 0.4	46.0 ± 0.3
SVM (4)	76.6 ± 0.4	51.3 ± 0.1
SVM (5)	$\mathbf{86.2 \pm 0.8}$	65.6 ± 0.3
SVM (7)	$\mathbf{85.5 \pm 0.4}$	$\mathbf{71.7 \pm 0.2}$
Softmax (5)	82.9 ± 0.4	65.7 ± 0.5
Softmax (7)	$\mathbf{85.4 \pm 0.4}$	$\mathbf{72.6 \pm 0.1}$

DBs

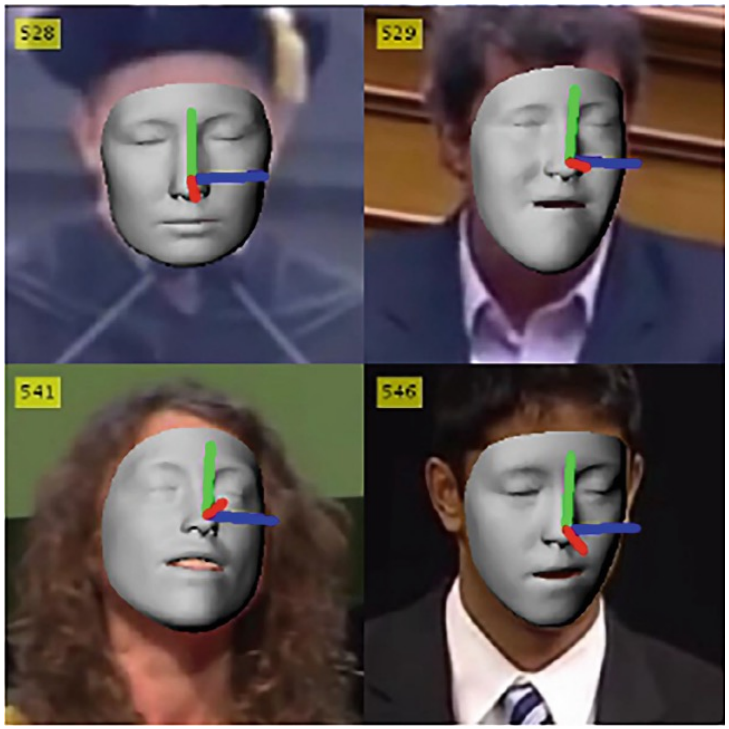
#training images per class

Higher layers generally produce more discriminative features,
i.e.
as the feature hierarchies become deeper, they learn increasingly powerful features

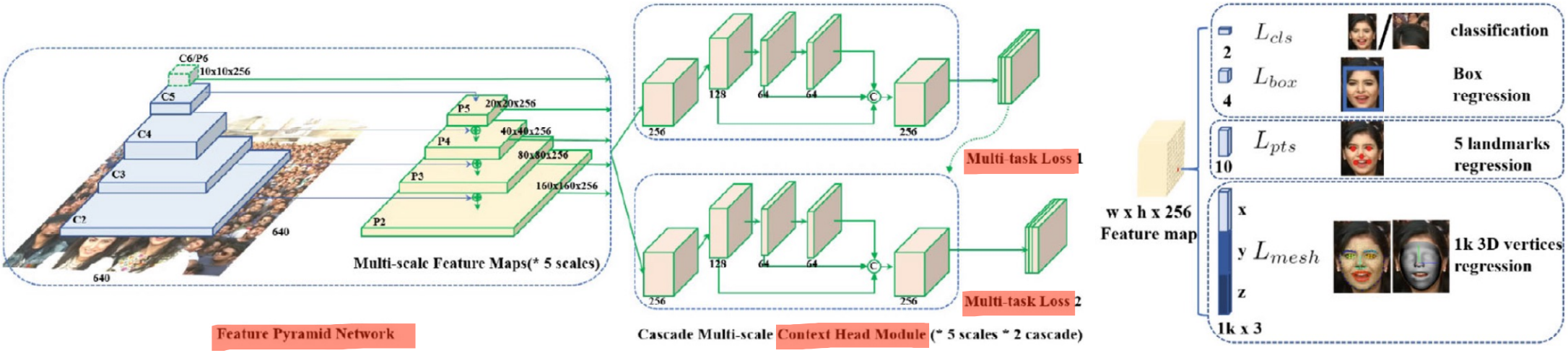
Why does this work so well?
Most probable explanation:
lots of data!

RetinaFace – Convolutional Neural Network-Based Face **Detector**

- **RetinaFace predictions – multi-task network:**
  1. **Face classification:** region contains a face or not (probability)
  2. **Bounding box:** position and size of the face
  3. **Face landmarks:** five key facial points (e.g., eyes, nose, mouth)
  4. **3D face mesh:** 3D structural parameters for face alignment
- **Key strength:** simultaneous prediction of **multiple facial attributes** in a single forward pass

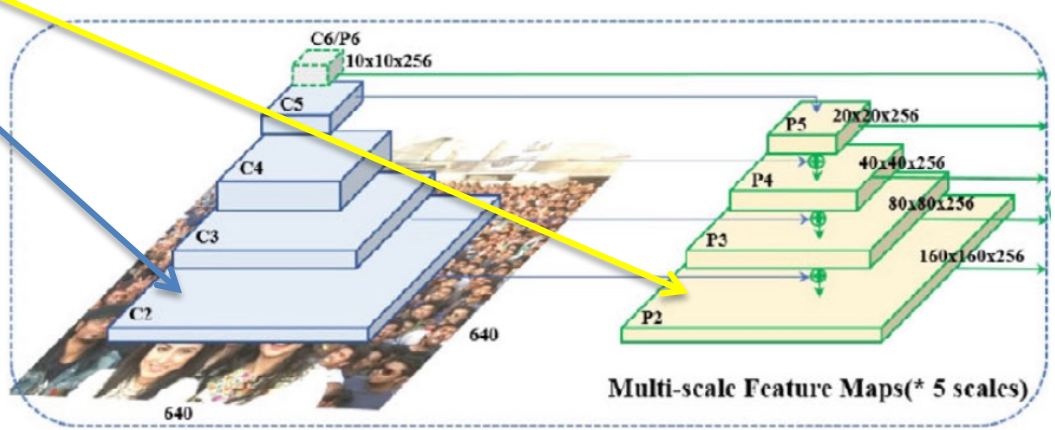


RetinaFace Network Components



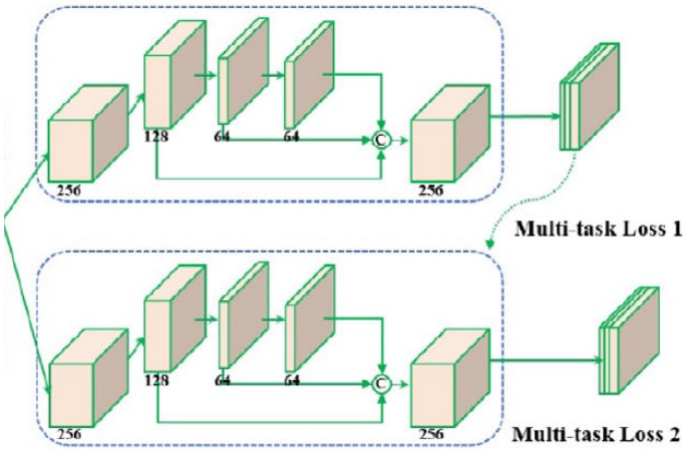
RetinaFace – Feature Pyramid

- **Feature Pyramid Network (FPN):**
  - **Bottom-up pathway:** extracts **high-level semantic features**
  - **Top-down pathway:** enhances **low-resolution features** with semantic information
  - Ensures effective face detection across **multiple scales**
- **Purpose:**
  - Small face detection in low-resolution regions
  - Consistent accuracy across varying face sizes



RetinaFace – Context Head Modules

- **Anchor-based detection:**
  - Predictions are made at specific **anchor points** in the feature map
  - Each anchor box predicts **face classification**, **bounding box**, **landmarks**, and **3D face mesh**
- **Sequential refinement:**
  - First head predicts coarse bounding boxes and landmarks
  - Second head refines these predictions for improved accuracy

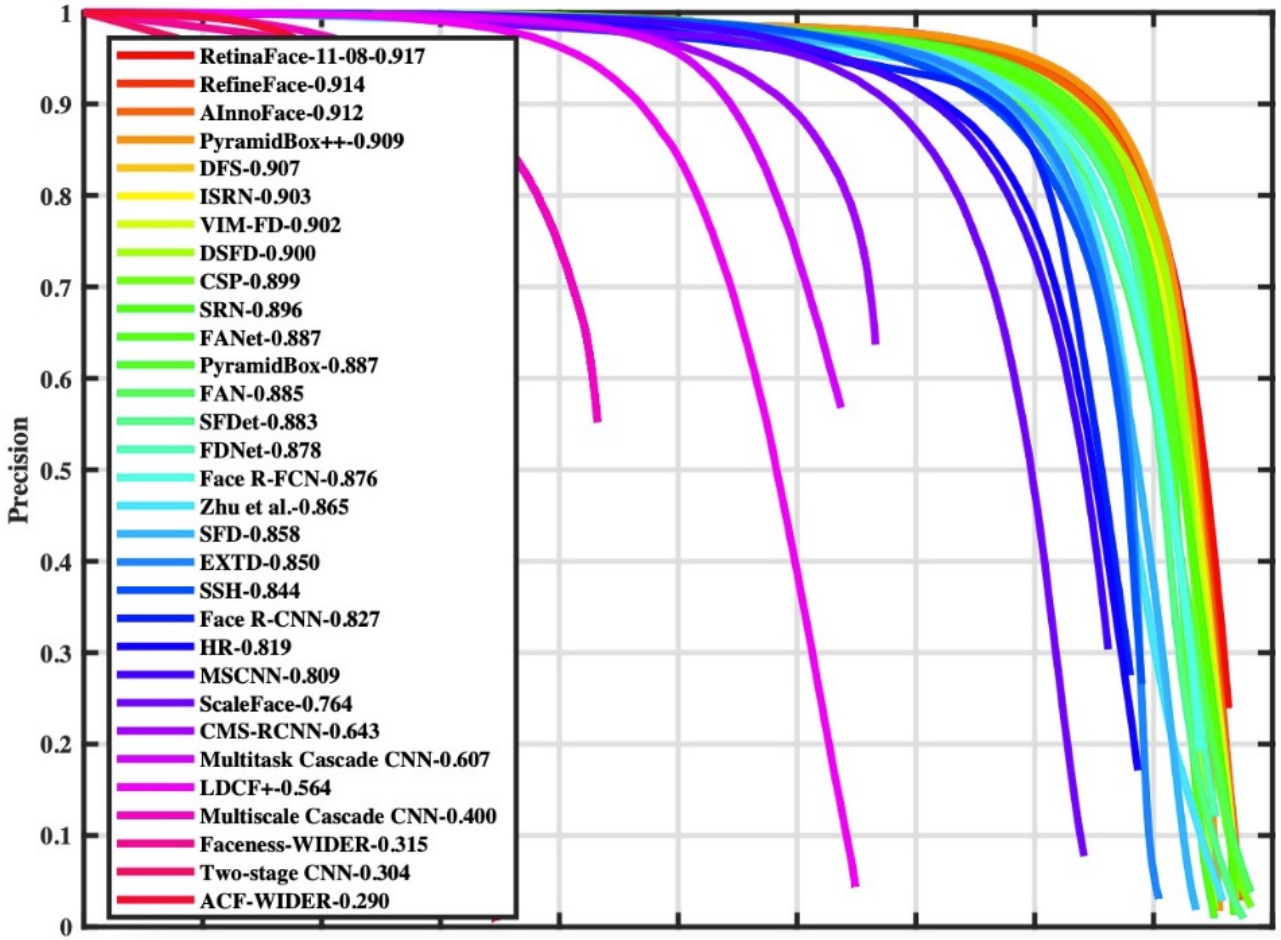


RetinaFace – Multi-Task Loss Function

Loss function components:

- **Classification loss (*L_cls*):** determines if a region contains a face
- **Bounding box loss (*L_box*):** adjusts predicted face location and size
- **Landmark loss (*L_pts*):** refines key landmark predictions
- **3D mesh loss (*L_mesh*):** aligns predicted 3D mesh with ground-truth
- Each component is weighted equally to ensure balanced training across all tasks

RetinaFace – Results



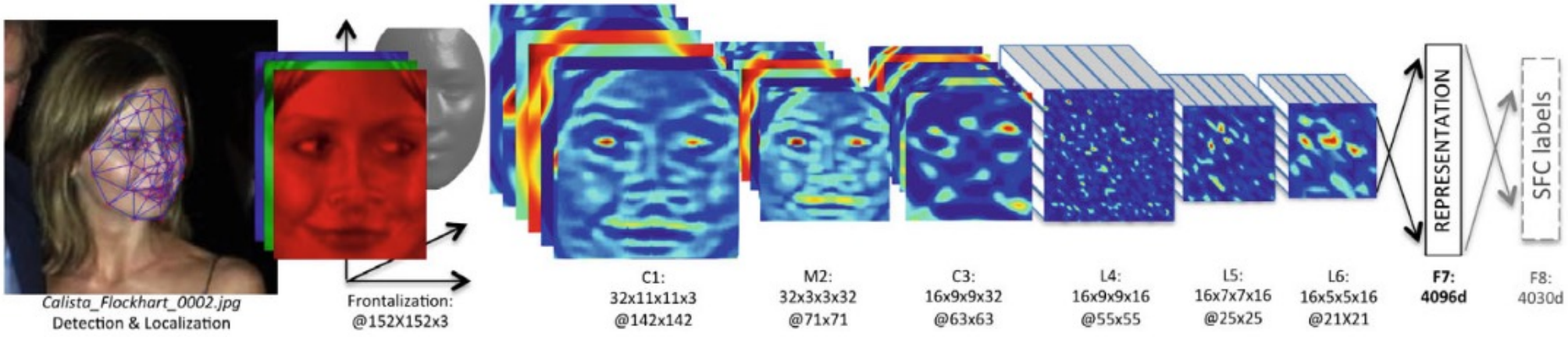
Network Architectures for Deep Face **Recognition**

- **Convolutional Neural Networks (CNNs):**
  - Stack of **convolutional layers** extracting spatial features
  - Non-linear **activation functions** enhance feature representation
  - **Weight Sharing:** reduces complexity and computational cost
- **DeepFace model (2014):** early model using **locally connected layers** for region-specific features
- **Modern approaches:**
  - Variants of **ResNet** dominate state-of-the-art models
  - **Residual learning:** prevents gradient vanishing in deep networks
  - **Depth matters:** deeper architectures generally yield better results

Loss Functions for Training Face Networks

- **Objective of loss functions:**
  - Minimize intra-class variance
  - Maximize inter-class separation
- **Common loss functions:**
  - **Cross-entropy loss:** standard **classification loss** using **softmax activation**
  - **Triplet loss:** enforces a **margin** between genuine and imposter pairs
  - **Angular margin loss:** adds geometric constraints to the classification objective

DeepFace


Calista_Flockhart_0002.jpg
Detection & Localization
Frontalization:
@152X152x3
C1:
32x11x11x3
@142x142
M2:
32x3x3x32
@71x71
C3:
16x9x9x32
@63x63
L4:
16x9x9x16
@55x55
L5:
16x7x7x16
@25x25
L6:
16x5x5x16
@21X21
F7:
4096d
REPRESENTATION
F8:
4030d
SFC labels

Results on **L**abeled **F**aces in the **W**ild Db

- Google FaceNet (Schroff et al., CVPR 2015): recognition accuracy **99.63%±0.09%** (22 layers deep model)
- DeepID2+: 99.47%±0.12%
- DeepFace: 97.35%±0.25% (2014, first significant success using CNNs for face recognition)

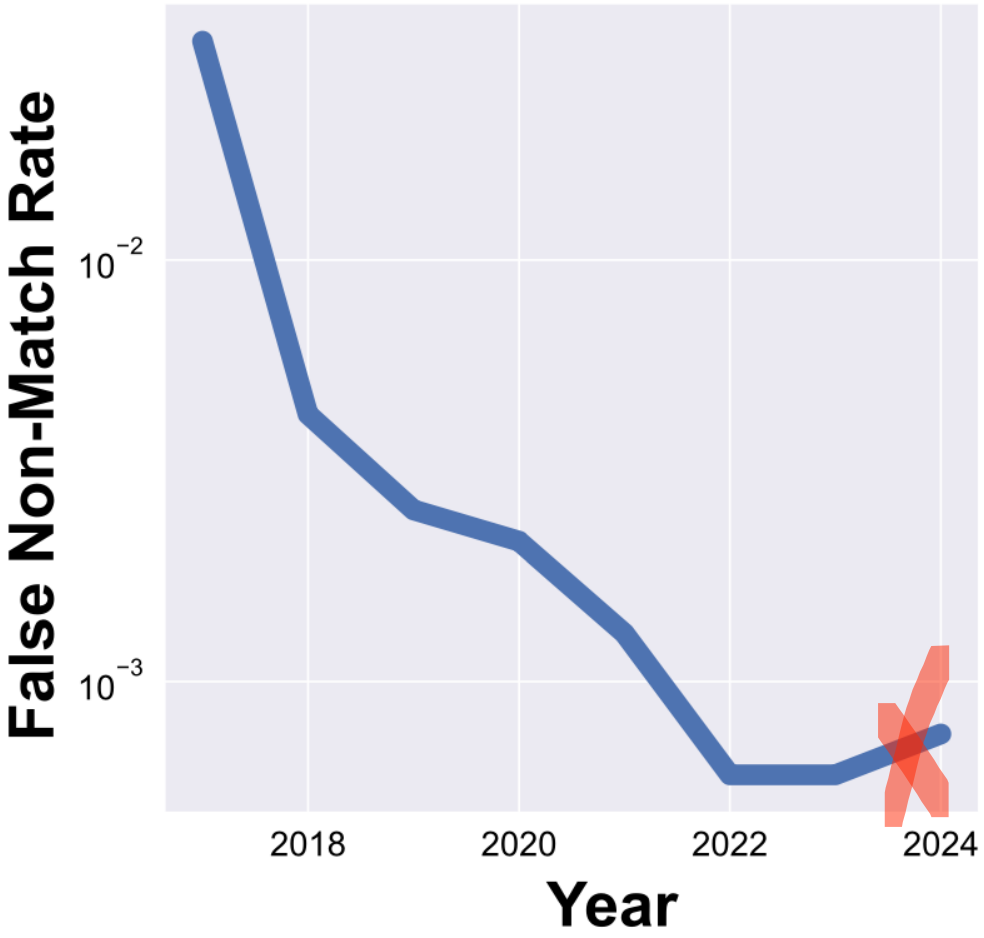
- This reduces the error reported for DeepFace **by more than a factor of 7** and improves the previous state-of-the-art reported for DeepId2+ **by 30%**

- Jan. 2017: 7 with even better results (see LFW web page), best being Dahua-FaceImage: 99.78%±0.07%, but all commercial recognition systems which algorithms have not been published and peer-reviewed
- Jan. 2025: Innovative Technology (commercial): 99.88% ± 0.04%

Results on **L**abeled **F**aces in the **W**ild Db

All pairs of images that were incorrectly classified

FRTE Verification Results (6/2017–3/2024)



Multibiometrics

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Volume 2014, Article ID 585139, 15 pages
http://dx.doi.org/10.1155/2014/585139

Hindawi

Research Article

Strategies for Exploiting Independent Cloud Implementations of Biometric Experts in Multibiometric Scenarios

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[illegible] a new computing model for various [illegible] recognition, where scalability, processing [illegible] generation of recognition technology. Next to the availability of computing resources, another important aspect of cloud computing with respect to biometrics is accessibility. Since biometric cloud services are easily accessible, it is possible to combine different existing implementations and design new multibiometric services that next to almost unlimited resources also offer superior recognition performance and, consequently, [illegible] to its [illegible] implementations. Unfortunately, the literature on the best strategies of how to combine existing implementations of cloud-based biometric experts into a multibiometric service is virtually nonexistent. In this paper, we try to close this gap and evaluate different strategies for combining existing biometric experts into a multibiometric cloud service. We analyze fusion strategies from different perspectives such as performance gains, training complexity, or resource consumption and present results and findings important to software developers and other researchers working in the areas of biometrics and cloud computing. The analysis is conducted based on two biometric cloud services, which are also presented in the paper.

Why?

- Different biometric sources usually **compensate** for the inherent limitations of one another
- Provide a wider population **coverage** (reduce the failure to enroll rate)

How?

- By consolidating the information given by multiple biometric sources – information fusion

Why the Accuracy Improves?

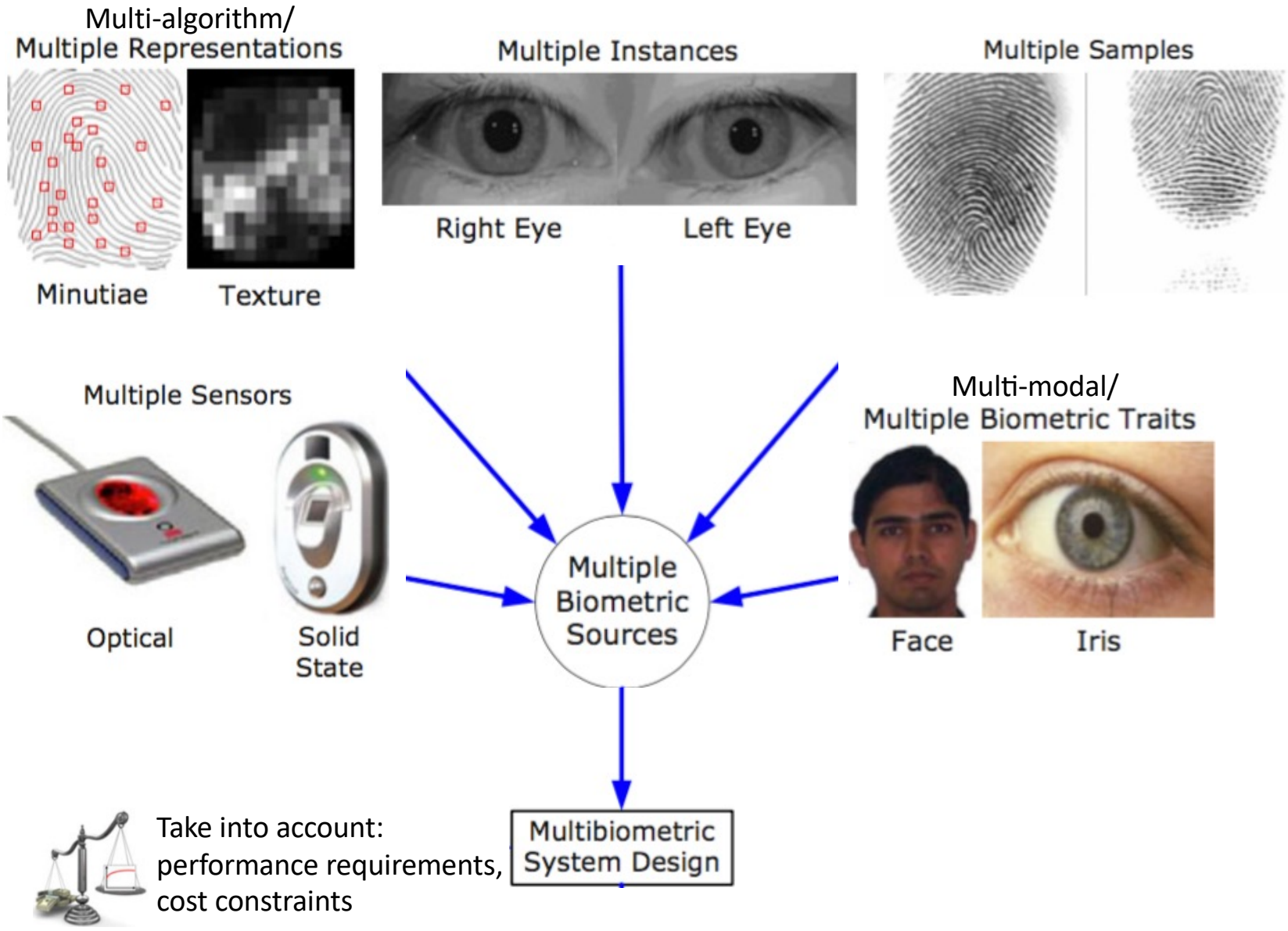
1. Fusion **increases** the dimensionality of the feature space and **reduces** the overlap between the feature distributions of different individuals

   $\Rightarrow$ combination of multiple biometric sources is more unique

2. Noise, imprecision, inherent drift (caused by factors like aging) in a subset of the biometric sources can be compensated by the **discriminatory** information provided by the **remaining** sources

   $\Rightarrow$ multiple biometric sources provide redundancy and fault-tolerance

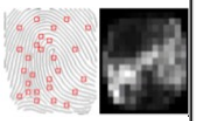
Five Classes of Sources of Multiple Evidence =>



Dependencies

Between the Design Choices in a Multibiometric System

Multibiometric sources	Type of information fused				Acquisition architecture		Processing architecture	
	Raw data	Features	Scores	Decisions	Serial	Parallel	Serial	Parallel
Multiple sensors	✓	✓	✓	✓	✓	✓	✓	✓
Multiple representations	×	✓	✓	✓	×	✓	✓	✓
Multiple matchers	×	×	✓	✓	×	✓	✓	✓
Multiple instances	×	✓	✓	✓	✓	✓	✓	✓
Multiple samples	✓	✓	✓	✓	✓	×	✓	✓
Multiple traits	×	✓	✓	✓	✓	✓	✓	✓



✓ Compatible

× Not compatible

Evidences derived from **a single** biometric trait

Multi-sensors Systems

(Additional Properties)

(built in registration)

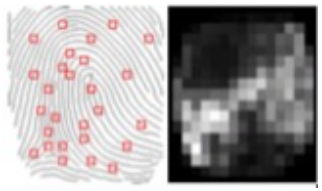
+ also **assist** in the intermediate processing stages of the individual component systems like image segmentation and registration

– cost

+ two fingerprint images can be processed independently without spatially registering them

– larger enrollment and verification times

Multi-algorithm Systems
(Additional Properties)

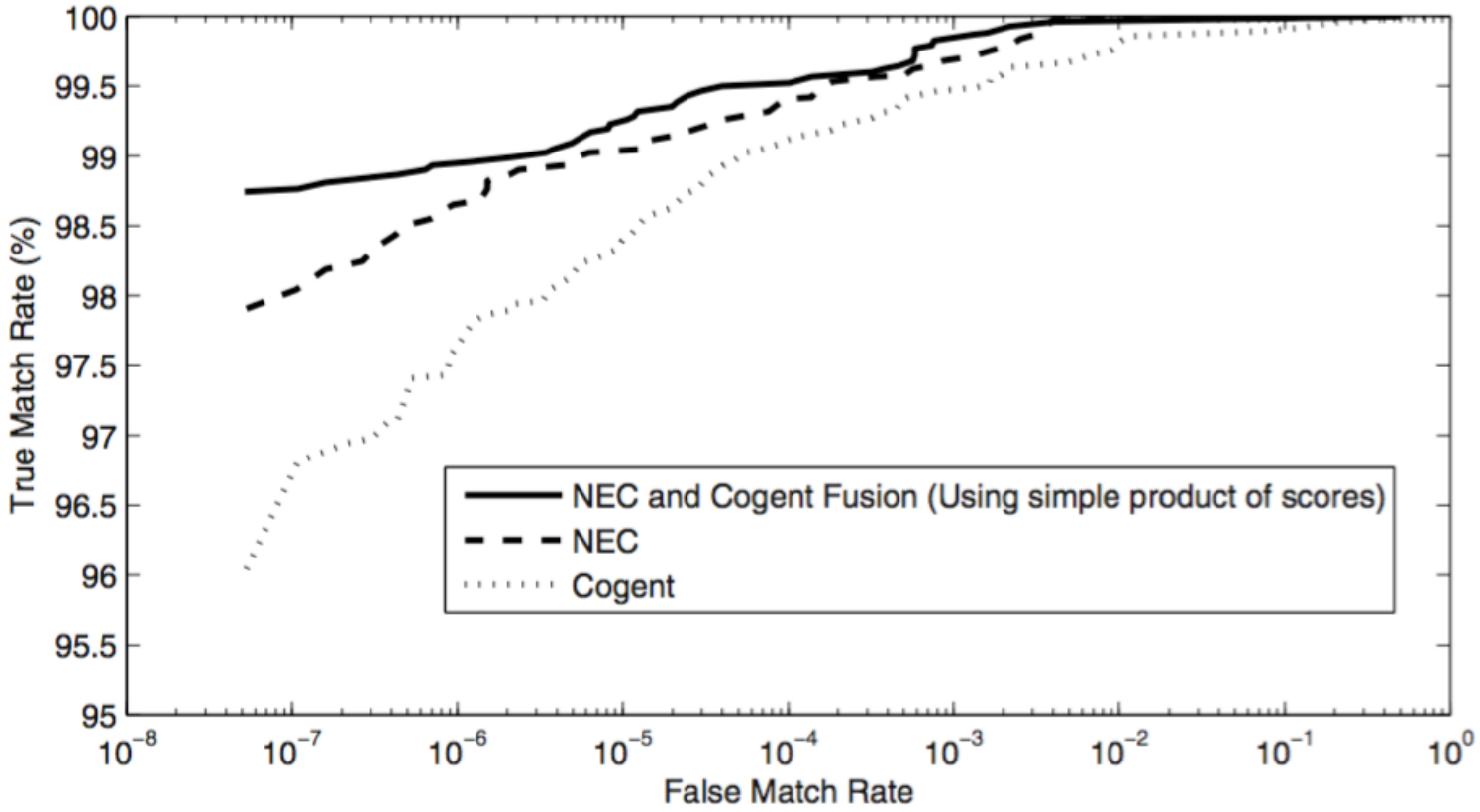


+ cost-effective => does not require the use of new sensors

+ no additional inconvenience for the user

– multiple sources tend to be **correlated** (noisy input will affect both the texture-based and minutiae-based algorithms but to different extents) => limits the possible improvement in the matching accuracy

Accuracy Improvement in a Multi-algorithm Fingerprint Verification System



NIST Fingerprint Vendor Technology Evaluation 2003, two most accurate matchers

Multi-instance Systems

(Additional Properties)

- Also known as multi-unit systems

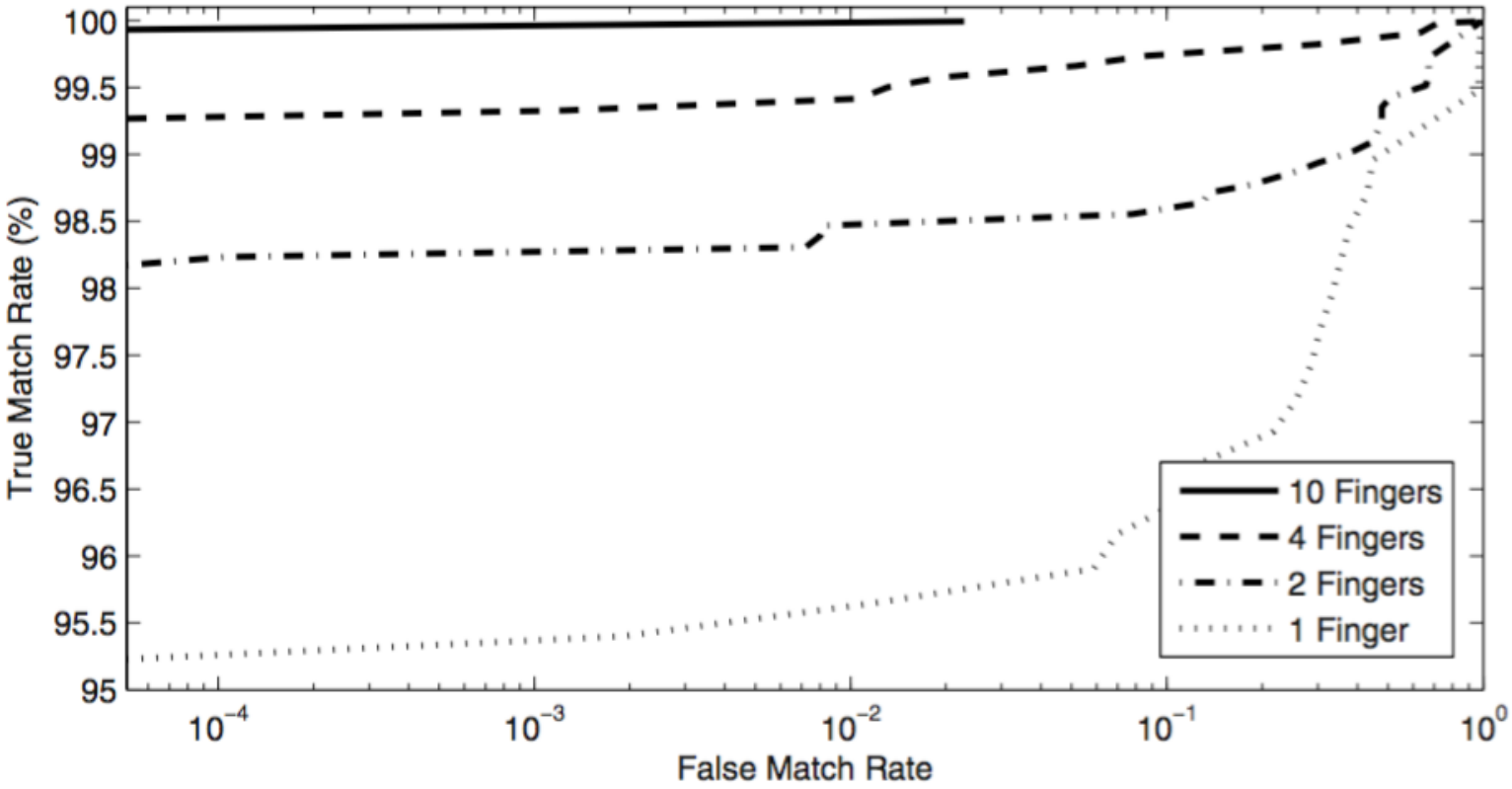
+ Do **not** necessitate the introduction of new sensors nor do they entail the development of new feature extraction or matching algorithms

+ Especially beneficial for users whose biometric traits cannot be reliably captured due to inherent problems (dry finger skin, drooping eyelids, etc.)

– In some cases a new sensor arrangement might be necessary in order to facilitate the simultaneous capture of the various units/instances

rapid acquisition of all ten fingers in three steps (by Identix)

Accuracy Improvement in a Multi-instance Fingerprint Verification System



FpVTE 2003

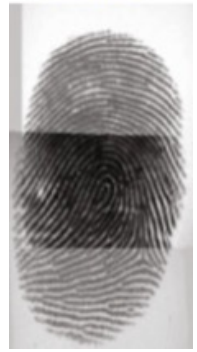
Multi-sample Systems

(Additional Properties)

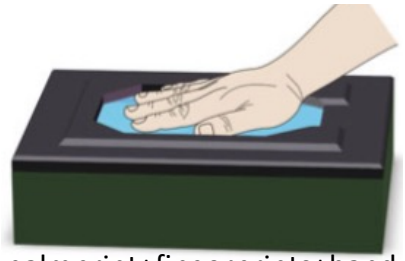
Note: if captured simultaneously it would be considered as a multi-sensor system

- To account for the **variations** that can occur in the trait /or/ to obtain a more **complete** representation of the underlying trait
  - Variations in the facial pose
  - To form a complete fingerprint

Mosaiced Fingerprint Image



- Determine the number of samples that need to be acquired (variability, typicality)
  - Determine sample collection protocol
  - Automatically select the “optimal” subset



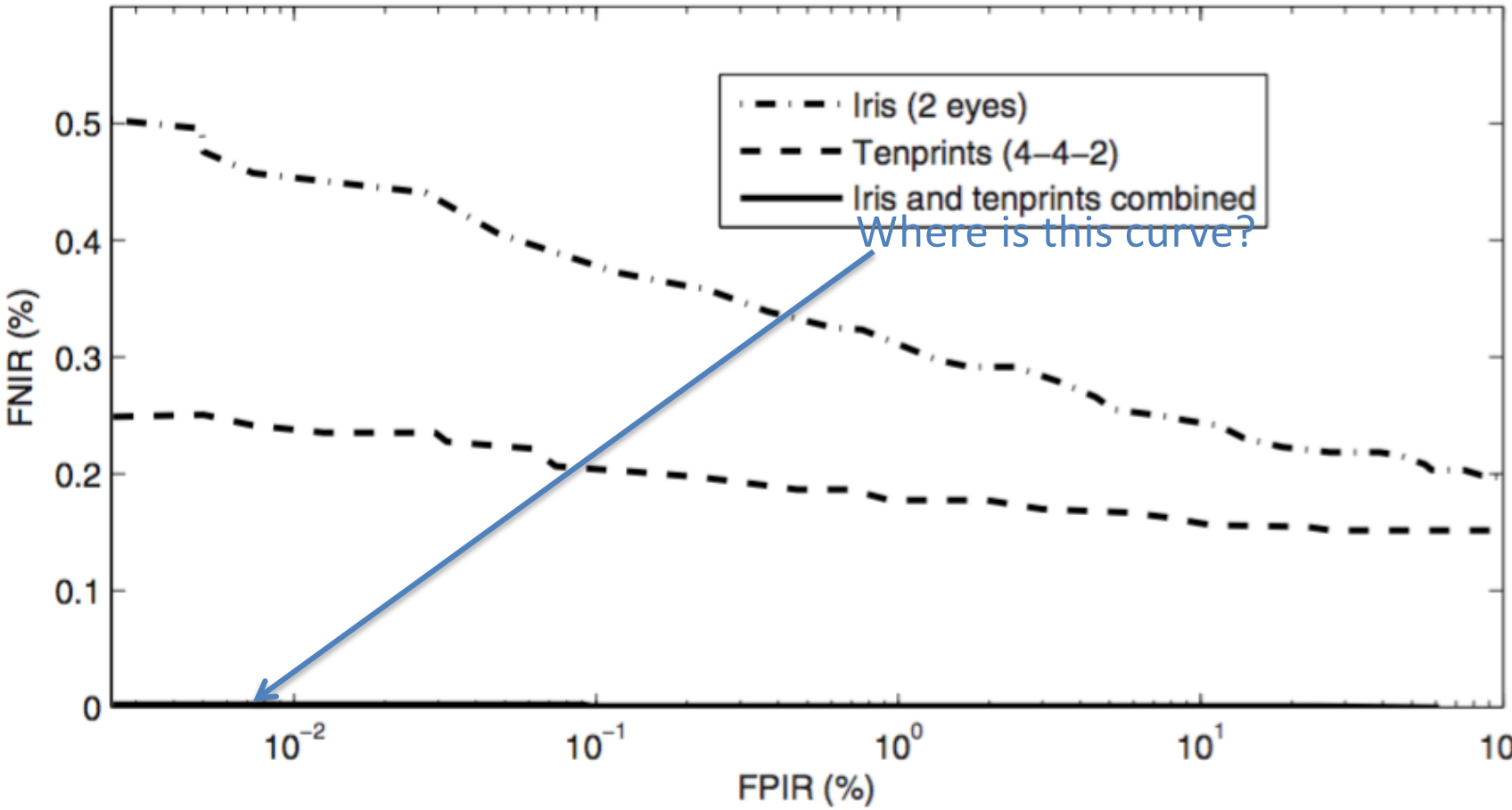
palmprint+fingerprints+hand-shape

Multi-modal Systems

(Additional Properties)

+ Traits are expected to be uncorrelated
  + Leads to **greater improvement** in performance compared to other types of multibiometric systems
  – Voice and lip movement could exhibit significant correlation
– Higher/highest cost
– There is a diminishing return after strong biometric traits (e.g. fingerprint and iris) have already been fused
  – Depending on the fusion scheme, there may even be a performance **degradation** if many biometric traits with lower accuracy are added

Accuracy Improvement in a Multi-instance and Multi-modal Biometric Identification System



UID India,
subset of 20k subjects

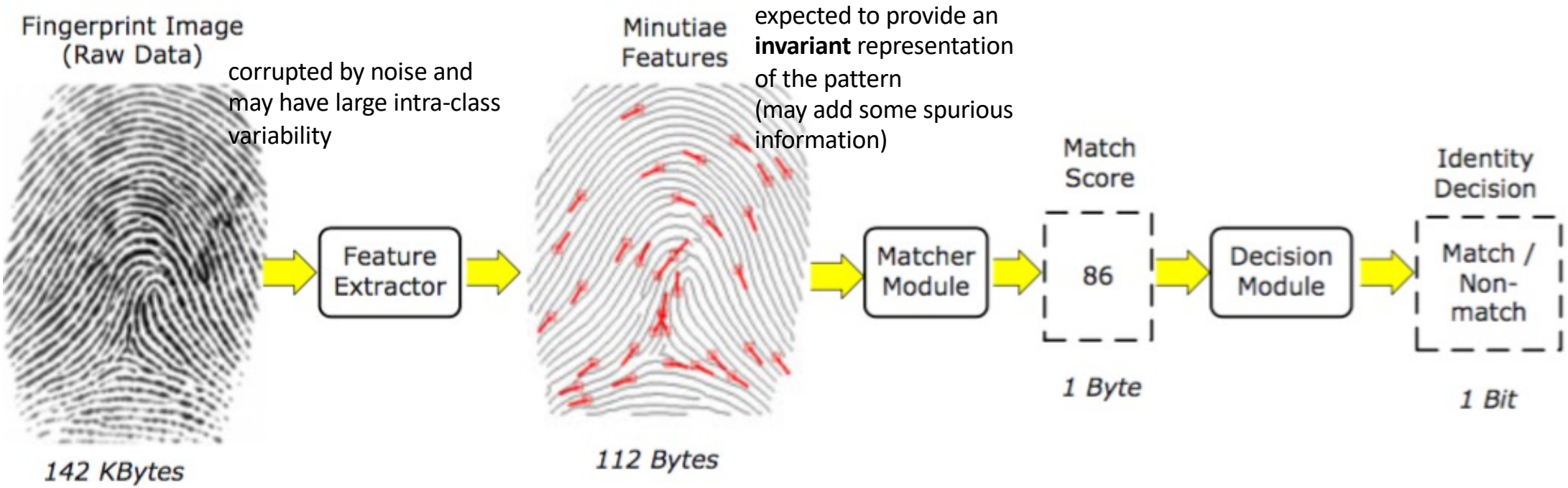
Multiple Biometric Sources: Final Remarks

- *Multi-instance* and *multi-modal* biometric systems are more **popular** than the others =>
- Use multiple instances or multiple traits or a **combination** of both

- Define: **Hybrid multibiometric systems**
  - Since the UID system in India uses all ten fingers and two irises, it is both multi-instance and multimodal; such systems are called hybrid multibiometric systems
- Define: **Multi-factor authentication**
  - Use biometric traits in conjunction with non-biometric authenticators like tokens or password

Fusion

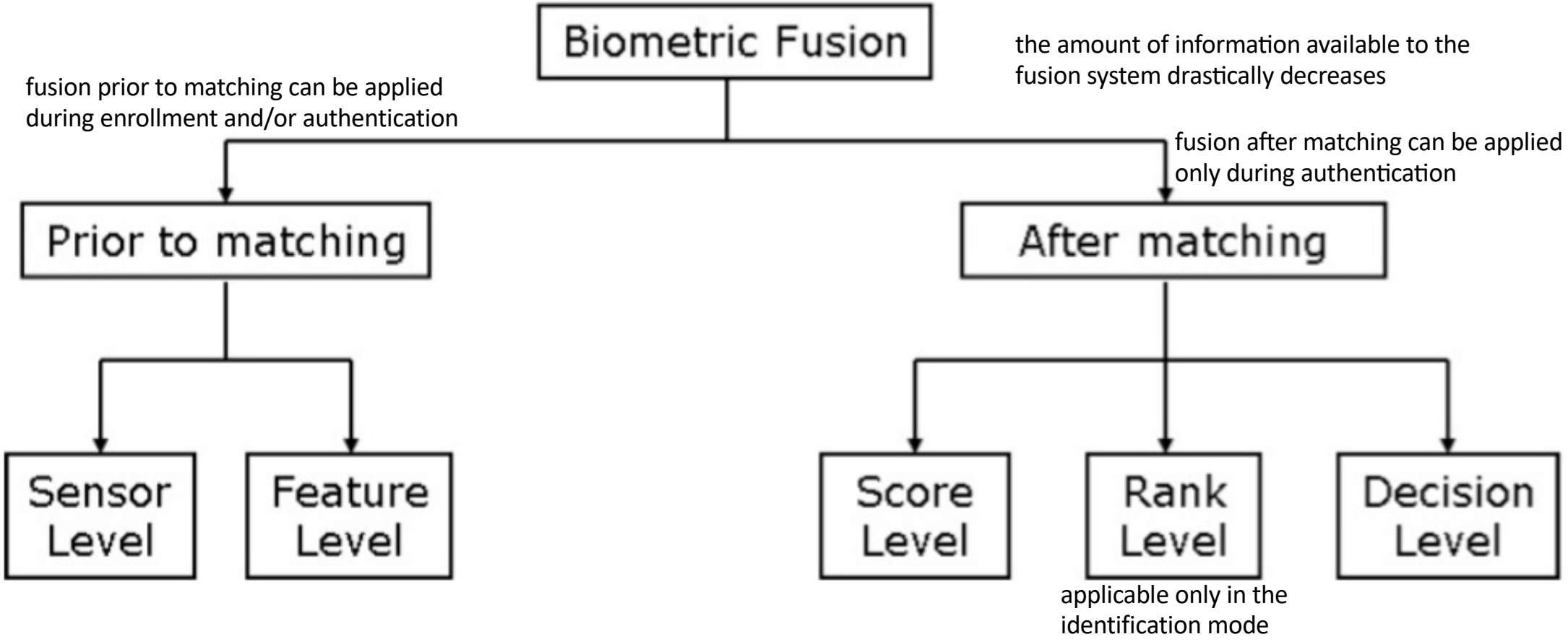
A fundamental issue in the design of a multibiometric system is to determine the **type** of information that should be **consolidated** by the fusion module:



The amount of information available for fusion **gets reduced** as one progresses along the various processing modules of a biometric system &

Fusion can be accomplished by utilizing the information available **in any** of the biometric modules

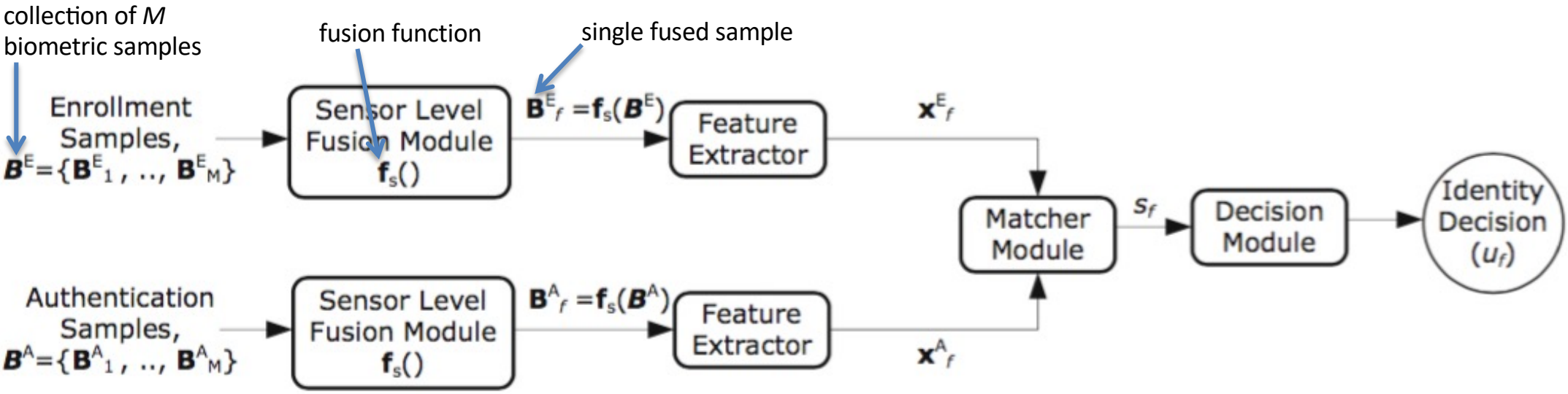
Fusion Levels



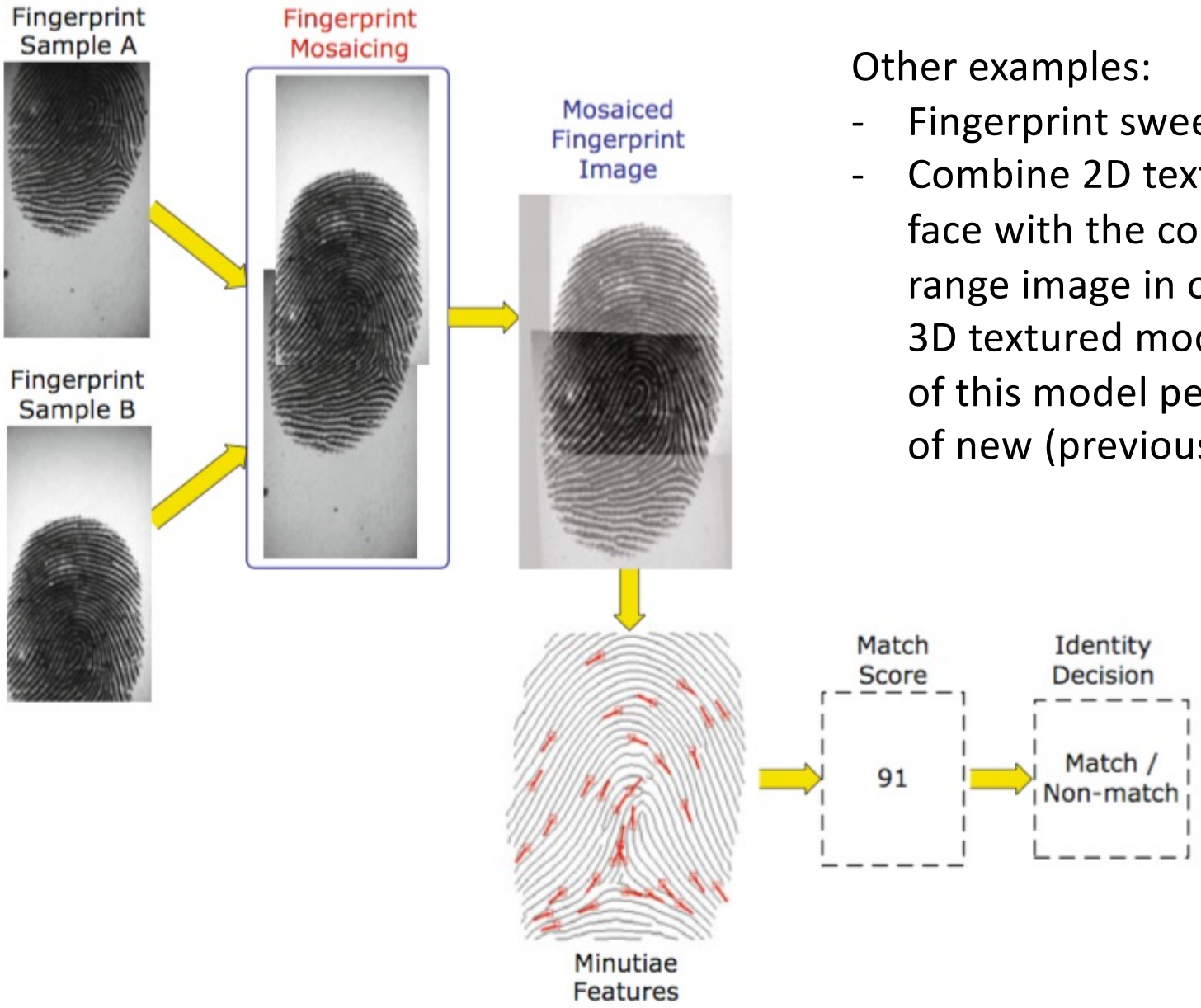
Sensor-level Fusion

Also known as image-level or pixel-level fusion

Consolidates evidence presented by multiple sources of raw data **before** they are subjected to feature extraction



Applicable only for multi-sensor and multi-sample systems >>>

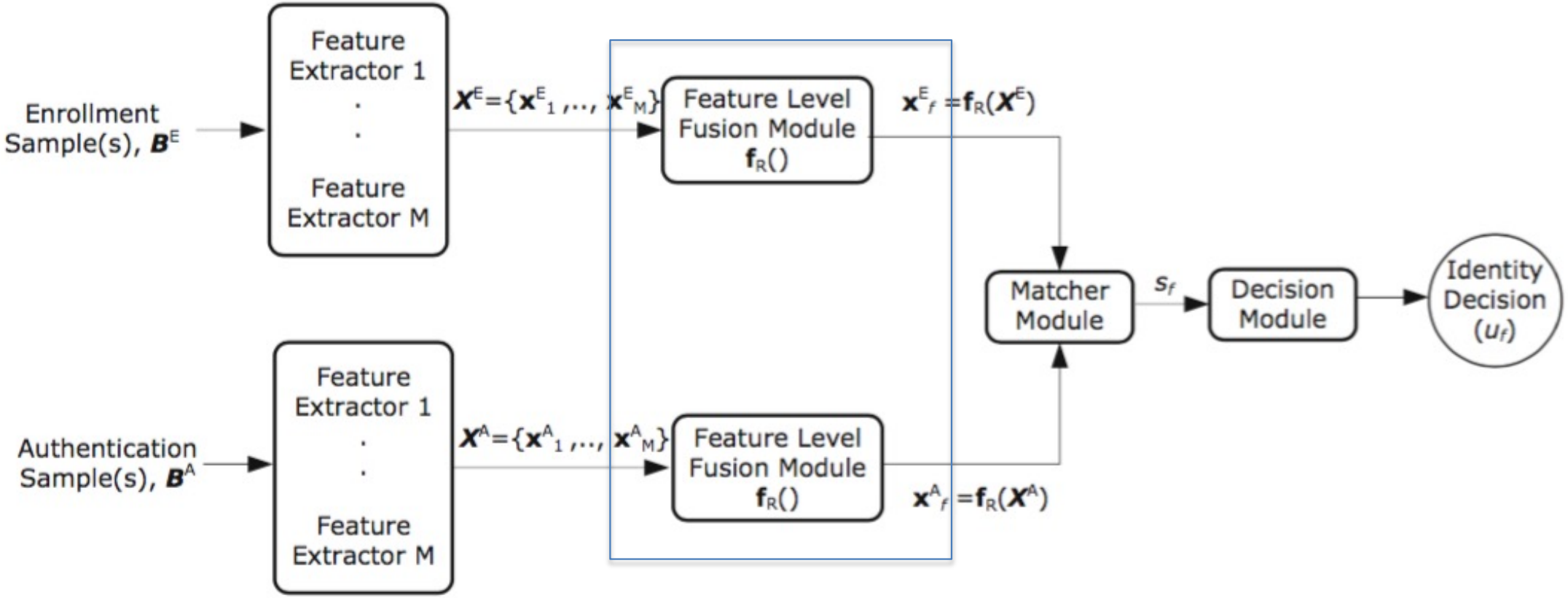


Other examples:

- Fingerprint sweep-sensors
- Combine 2D texture of a person's face with the corresponding 3D range image in order to create a 3D textured model; the availability of this model permits the generation of new (previously unseen) 2D images

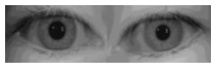
Feature-level Fusion

Feature/representation-level fusion **consolidates** evidence presented by two+ **different** biometric feature sets of the same individual



Categorization of Feature-level Fusion Schemes

- Homogeneous class
  - Feature sets to be combined are obtained by applying **the same** feature extraction algorithm to **multiple** samples of **the same** biometric trait
  - Applicable to multi-sensor and multi-sample systems
- Heterogeneous class
  - Feature sets originate from **different** feature extraction algorithms or from samples of **different** biometric traits (or different instances of the same trait)



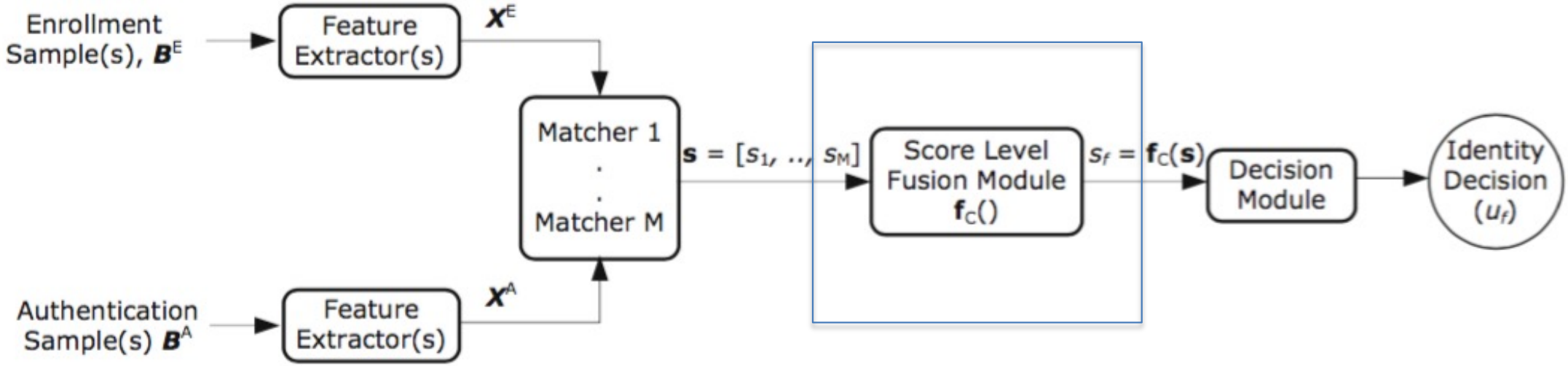
Score-level Fusion

Also known as fusion at the measurement or confidence level

Match scores output by different biometric matchers are **consolidated**

Relatively easy to access and combine the scores

Most commonly used approach

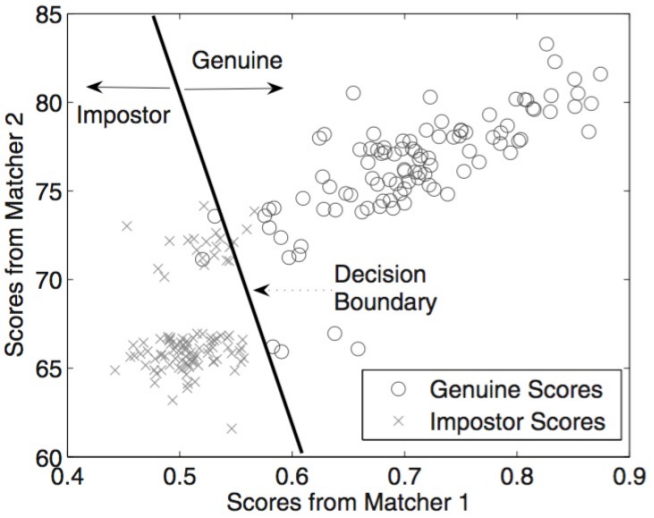


Challenges of non-homogenous match score**s**: - distance or similarity measure output
- different numerical ranges of outputs
- different probability distributions of the matchers

Fusion methodologies will vary depending on whether the system operates in the ver./id. mode ->

Score-level Fusion in Verification Mode

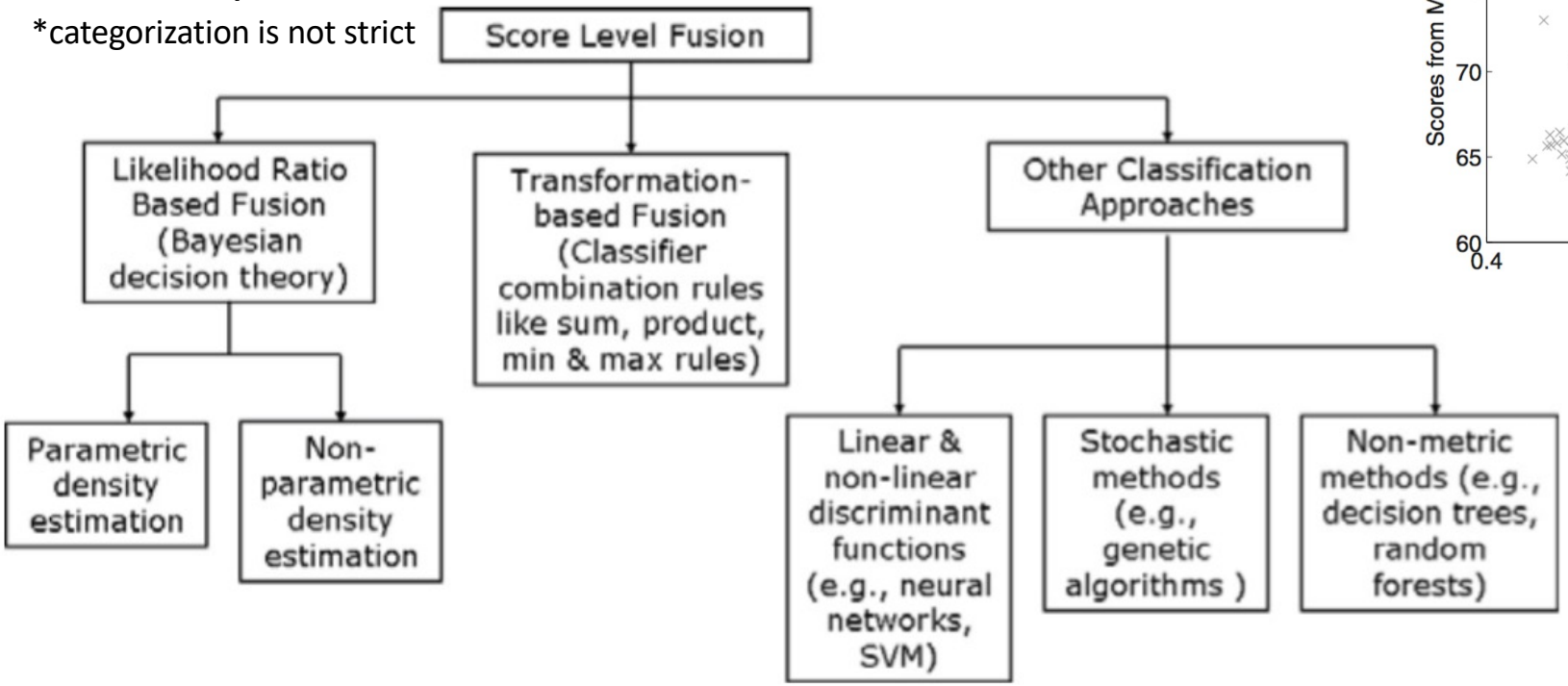
Based on the training set of match scores from the genuine and impostor classes, a classifier learns a decision boundary between the two classes



*boundary can be quite complex depending on the nature of the classifier

Taxonomy:

*categorization is not strict



Likelihood-ratio-based Score Fusion

Based on the **Bayesian** decision theory &
the **Neyman-Pearson** approach to statistical hypothesis testing

Bayes formula:

prior probabilities of observing the impostor and genuine classes $P(\omega_0)$ $P(\omega_1)$

conditional density function

$$P(\omega_j|\mathbf{s}) = \frac{p(\mathbf{s}|\omega_j)P(\omega_j)}{p(\mathbf{s})}$$

posterior probability

$$p(\mathbf{s}) = \sum_{j=0}^{1} p(\mathbf{s}|\omega_j)P(\omega_j)$$

Decide between the genuine and impostor classes based on the observed match score vector $\mathbf{s}$ >>>>> define Bayesian rule through density estimation

Transformation-based Score Fusion

- Estimating *M*-dimensional densities is a challenging problem
- Approximate the general minimum error rate of the Bayes decision rule
- Approximations of the posterior probabilities >>>>>

Approximation Achieved Using Different Classifier Combination Rules

- Assumptions:
  - Match scores of different matchers are statistically **independent**
  - Prior probabilities of the genuine and impostor classes are **equal**
- Decision rule: Decide ω_1 if

$$h(P(\omega_1|s_1), P(\omega_1|s_2), \cdots, P(\omega_1|s_M)) > h(P(\omega_0|s_1), P(\omega_0|s_2), \cdots, P(\omega_0|s_M))$$

classifier combination rules function >>>>>

Combination Rule	$h(P(\omega_j\|s_1), P(\omega_j\|s_2), \cdots, P(\omega_j\|s_M)) =$
Product	$\prod_{m=1}^{M} P(\omega_j\|s_m)$
Sum	$\sum_{m=1}^{M} P(\omega_j\|s_m)$
Max	$\max_{m=1}^{M} P(\omega_j\|s_m)$
Min	$\min_{m=1}^{M} P(\omega_j\|s_m)$
Median	$median_{m=1}^{M} P(\omega_j\|s_m)$

Product rule is a direct implication of the assumptions

Limitation: it is sensitive to errors in the estimation of the posterior probabilities

Sum rule is generally more effective

Especially when the estimates of the marginal posterior probabilities are unreliable

The sum rule can be derived from the product rule by assuming that the posterior probabilities do not deviate dramatically from the prior probabilities

The max, min, and median rules can also be obtained by various approximations of the product and sum rules

Approximation Achieved By Transforming the Match Scores into a Common Domain

- Perform **score normalization** (with a monotonic function): min-max, z-score, median, tanh (not sensitive to outliers)
- Once the match scores from different matchers are normalized, classifier combination rules such as the sum, max, and min rules can be applied to obtain the fused match scores
- The corresponding combination rules are referred to as **sum of scores**, **max score**, and **min score** fusion rules, because the normalized match scores may **not** have any direct probabilistic interpretation (normalized scores may not even lie in the interval [0,1]) – product rule is generally not applicable
- Decision rule: $\text{Decide } \omega_1 \text{ if } h(g_1(s_1), g_2(s_2), \cdots, g_M(s_M)) > \tau$

Score-level Fusion in Identification Mode

- Key difference:
  - In a verification system the score data is in the form of a **vector** $\mathbf{s} = [s_1, s_2, \cdots, s_M]$
  - Score data in an identification system is in the form of a $N \times M$ **matrix** $\mathbf{S} = [s_{n,m}]$, where $s_{n,m}$ is the match score output by the *m-th* matcher corresponding to the *n-th* identity
- With minor modifications, many of the decision rules designed for the verification scenario can also be **extended** to the identification mode

Rank-level Fusion

- When a biometric system operates in the **identification** mode, the output of the system can be viewed as a ranking of the enrolled identities
- **Consolidate** all the ranks output by the individual biometric subsystems in order to derive a consensus rank for each identity
  - – Ranks reveal less information than match scores
  - + Unlike match scores, the rankings output by multiple biometric systems are comparable => no normalization is needed

- Let $\mathbf{R} = [r_{n,m}]$ be the rank matrix in a multibiometric system, where $r_{n,m}$ is the rank assigned to identity I_n by the m-th matcher
- Let $\hat{r}_n$ be a statistic computed for user I_n such that the user with the lowest value of $\hat{r}$ is assigned the highest consensus (or reordered) rank >>>>>

Highest Rank Method

- Each user is assigned the highest rank (**minimum** r value) as computed by different matchers
- Statistic for user I_n is: $\hat{r}_n = \min_{m=1}^{M} r_{n,m}$
- Useful only when the number of users is large compared to the number of matchers
- It can utilize the strength of each matcher effectively

Borda Count Method

- Uses the **sum** of the ranks assigned by the individual matchers
- Statistic for user I_n is: $\hat{r}_n = \sum_{m=1}^{M} r_{n,m}$
- Measure of the degree of agreement among the different matchers
- Assumptions:
  - ranks assigned to the users by the matchers are statistically independent
  - all the matchers perform equally well

Logistic Regression Method

- Generalization of the Borda count method where a **weighted sum** of the individual ranks is calculated
- Statistic for user I_n is:

$$\hat{r}_n = \sum_{m=1}^{M} w_m r_{n,m}$$

- Weights are determined by logistic regression
- Useful when the different biometric matchers have significant differences in their accuracies
- Requires a training phase

Illustration
(two matchers,
four users):

Face Matcher 1

User Identity	Rank
Alice	3
Bob	1
Charlie	4
Dick	2

Face Matcher 2

User Identity	Rank
Alice	1
Bob	2
Charlie	3
Dick	4

Rank Fusion Module

	Highest Rank		Borda Count		Logistic Regression ($w_1 = 0.8$, $w_2 = 0.2$)	
User Identity	Fused Score	Reordered Rank	Fused Score	Reordered Rank	Fused Score	Reordered Rank
Alice	1	1	4	2	2.6	3
Bob	1	2	3	1	1.2	1
Charlie	3	4	7	4	3.8	4
Dick	2	3	6	3	2.4	2

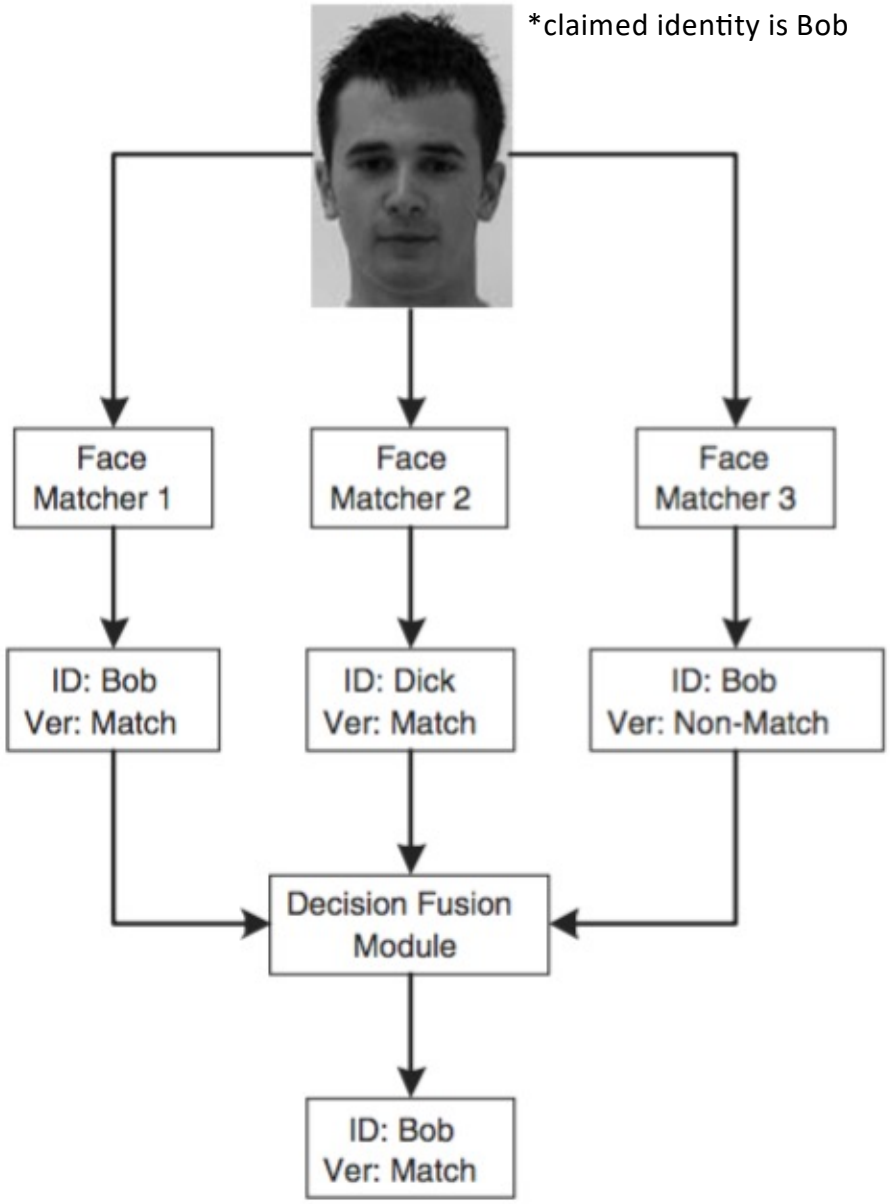
breaking the ties randomly ;-)

Decision-level Fusion

- Abstract/decision-level
- Commercial off-the-shelf (COTS) biometric matchers provide access only to the final recognition decision
- Methods: AND/OR rules (rarely used in practice), majority voting (most common approach), weighted majority voting, Bayesian decision fusion, the Dempster-Shafer theory of evidence, behavior knowledge space,...

Majority Voting

- Input biometric sample is assigned to that class on which a **majority of the matchers agree (half)**
- Assumes that all the matchers perform equally well
- No apriori knowledge about the matchers is needed
- No training is required to come up with the final decision



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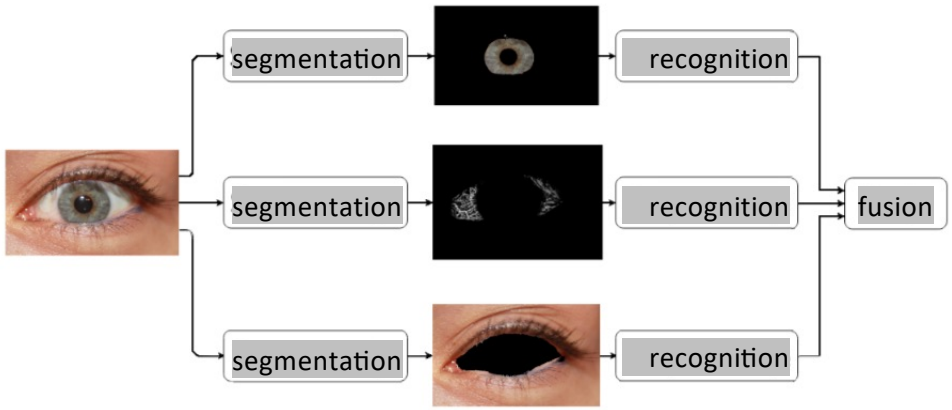

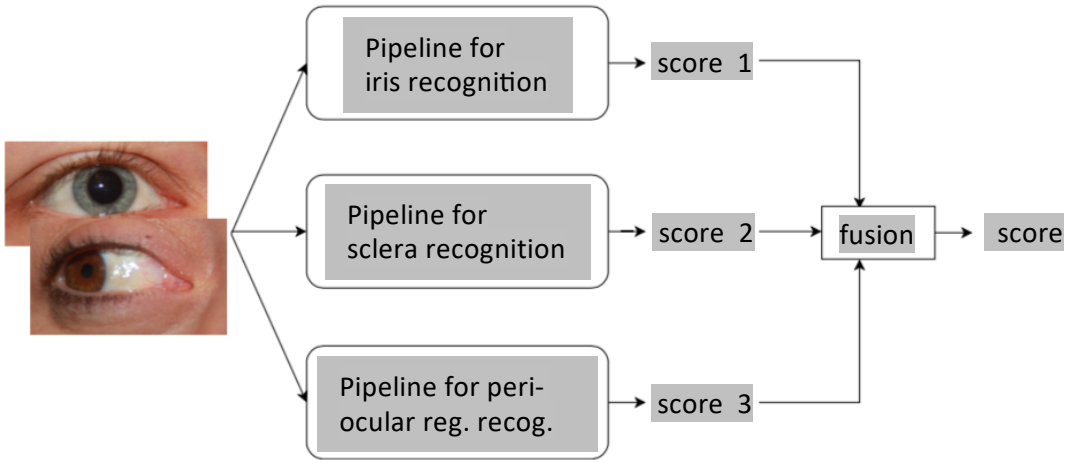

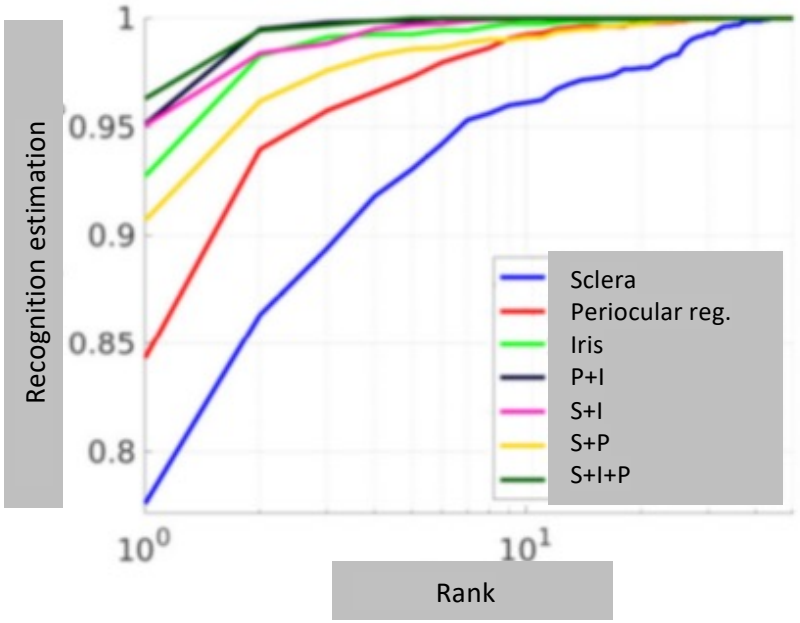

DEEP LEARNING METHODS FOR BIOMETRIC RECOGNITION BASED ON EYE INFORMATION

Peter Rot (2018) *DEEP LEARNING METHODS FOR BIOMETRIC RECOGNITION BASED ON EYE INFORMATION.* MSc thesis.

PDF
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Abstract

The accuracy of ocular biometric systems is critically dependent on the image acquisition conditions and segmentation methods. To minimize recognition error robust segmentation algorithms are required. Among all ocular traits, iris got the most attention due to high recognition accuracy. New modalities such as sclera blood vessels and periocular region were also proposed as autonomous (or iris-complementary) modalities. In this work we tackle ocular segmentation and recognition problems using deep learning methods, which represent state-of-the-art in many computer vision related tasks. We individually evaluate three recognition pipelines based on different ocular modalities (sclera blood vessels, periocular region, iris). The pipelines are then fused into a single biometric system and its performance is evaluated. The main focus is sclera recognition in the scope of which we i) create a new dataset named SBVPI, ii) propose and evaluate segmentation approaches, which won the first place on SS(ER)BC competitions, and iii) develop and evaluate the rest of the sclera-based recognition pipeline. The next contribution of this work is multi-class eye segmentation technique, which gives promising results. We also propose and evaluate deep learning pipeline for periocular recognition. For iris recognition we use an existing pipeline and evaluate it on our dataset. With deep learning we achieve promising recognition results for each individual modality. We further improve recognition accuracy with multi-modal fusion of all three modalities.

Item Type:	Thesis (MSc thesis)
Keywords:	deep learning, convolutional neural networks, sclera, iris, periocular information, segmentation, recognition
Number of Pages:	117
Language of Content:	Slovenian

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State-of-the-Art
in
Fingerprint and Iris Recognition
(using Deep Learning)

Deep Learning

Advantages:

- Faster on specialized HW
- Pattern recognition from noisy data
- Better generalization

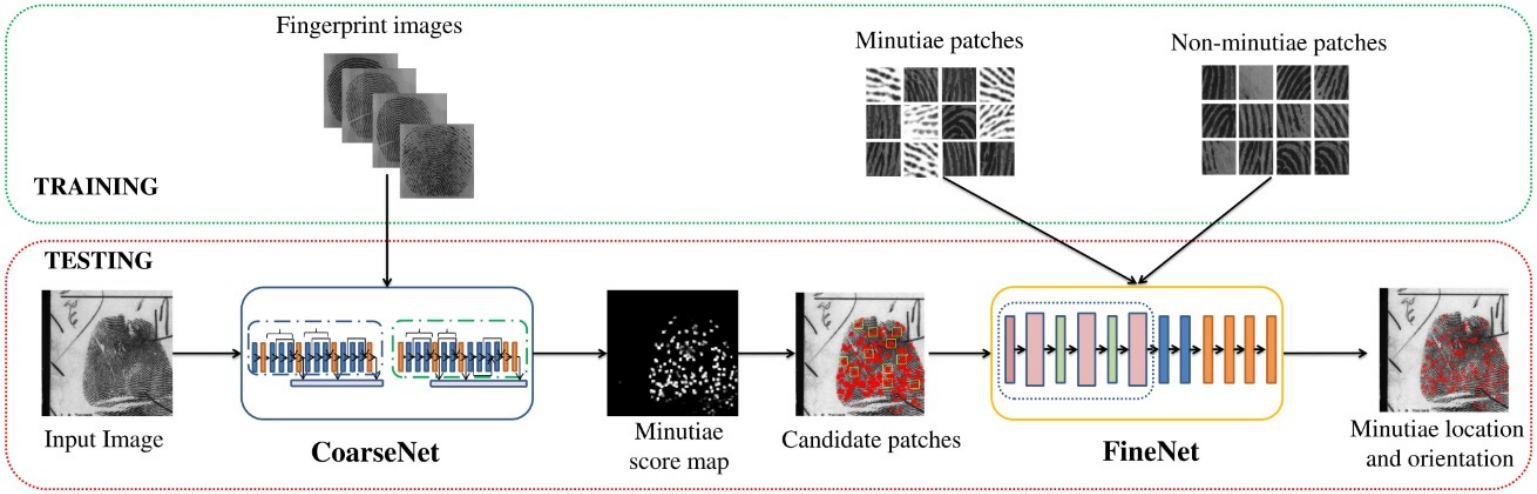
Disadvantages:

- **Black box**
- **Requires a lot of training data**

The biggest challenge in the context of fingerprint recognition

FPR Early Models

- Parts of traditional pipeline are substituted with DL counterparts
- Improving preprocessing, feature extraction
- Example:
  - MinutiaeNet (2018)



First Fully End-to-End Models

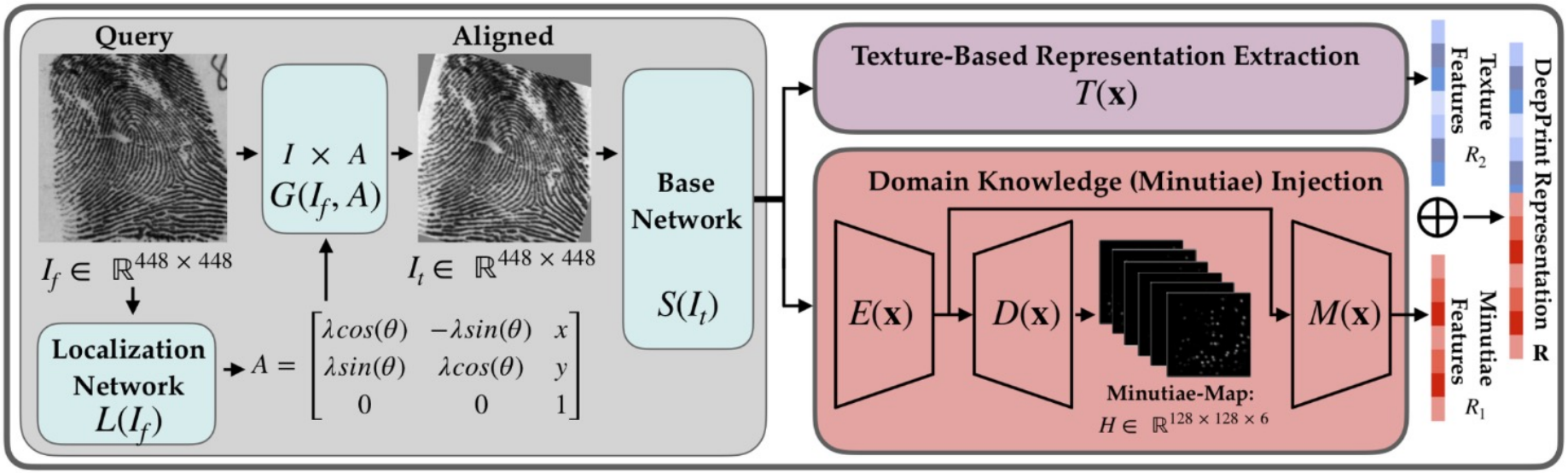
- Goal: Produce a feature vector as a fixed-length representation
- Matching = distance/similarity between feature vectors

- Properties:
  - Methods use CNN encoders
  - Domain knowledge added to guide learning (e.g., minutiae points)
  - Matching is now part of the learning process (affects loss)
  - Contrastive learning to learn relations between feature vectors

- **99.5%+** on NIST benchmark fingerprint datasets

DeepPrint (2019)

- Uses a Localisation network to first align the input image
- Predicts minutiae points as intermediate calculation
- Representation is a combination of minutiae and texture data

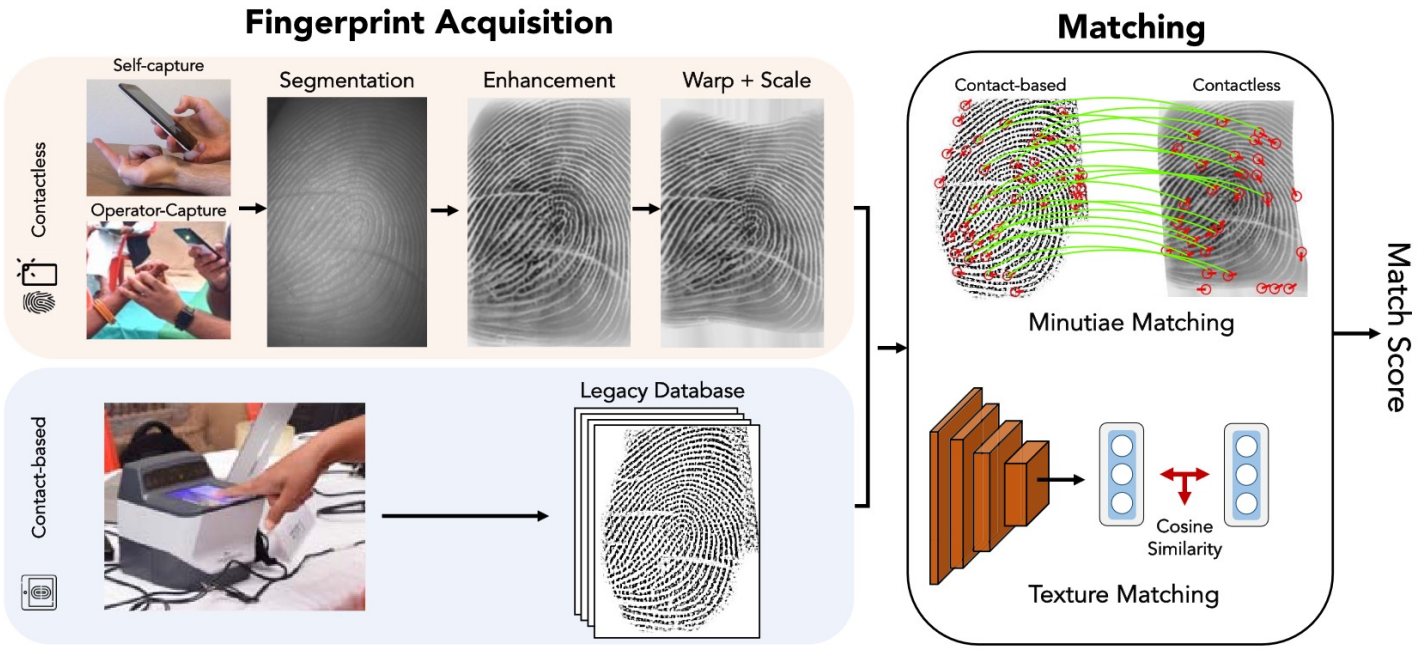


DeepPrint (2019)

Algorithm	Description	HR @ PR = 1.0%[1] (NIST SD4)[2]	HR @ PR = 1.0% (NIST SD14)[3]	Template Size (bytes)	Gallery Size[4]
Jain *et al.* [22], [23]	**Fingercode**: Global representation extracted using Gabor Filters	N.A.	N.A.	640	N.A.
Cappelli *et al.* [24]	**MCC**: Local descriptors via 3D cylindrical structures comprised of the minutiae-set representation	93.2%	91.0%	1,913	2,700
Cao and Jain [25]	**Inception v3:** Global deep representation extracted via Alignment and Inception v3	98.65%	98.93%	8,192	250,000
Song and Feng [26]	**PDC**: Deep representations extracted at different resolutions and aggregated into global representation	93.3%	N.A.	N.A.	2,000
Song *et al.* [27]	**MDC**: Deep representations extracted from minutiae and aggregated into global representation	99.2%	99.6%	1,200	2,700
Li *et al.* [28]	**Finger Patches:** Local deep representations aggregated into global representation via global average pooling	**99.83%**	99.89%	1,024	2,700
Proposed	**DeepPrint**: Global deep representation extracted via multi-task CNN with built-in fingerprint alignment	99.75%	**99.93%**	**200**†	**1,100,000**

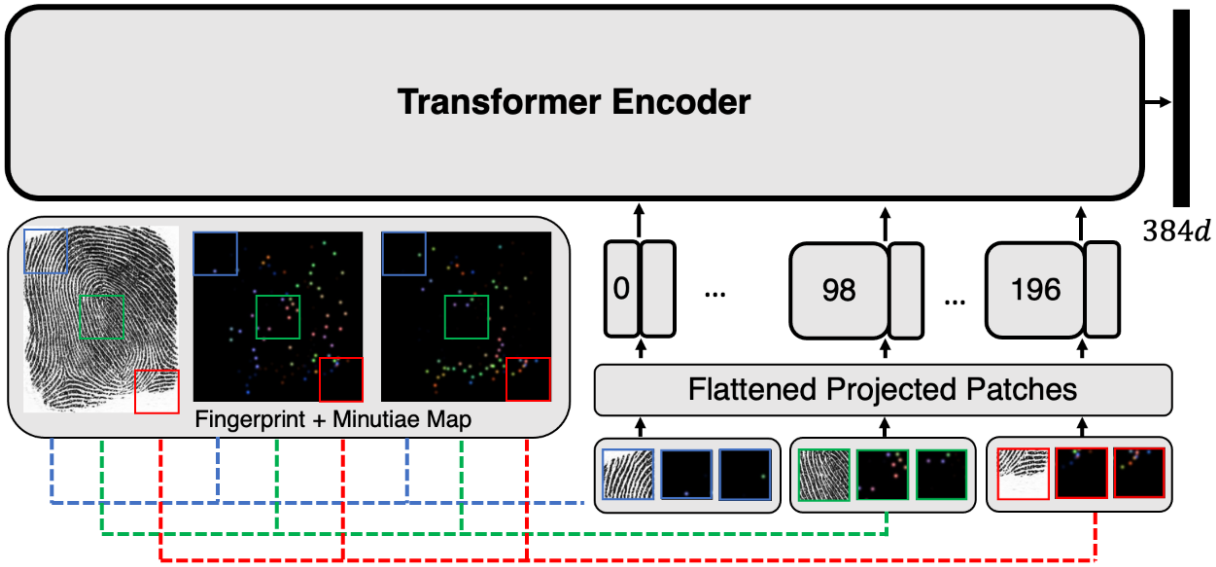
C2CL (2021)

- Extends DeepPrint to work on **contactless** fingerprints
- Query images can be captured using a smartphone camera



Contemporary State-of-the-Art Models

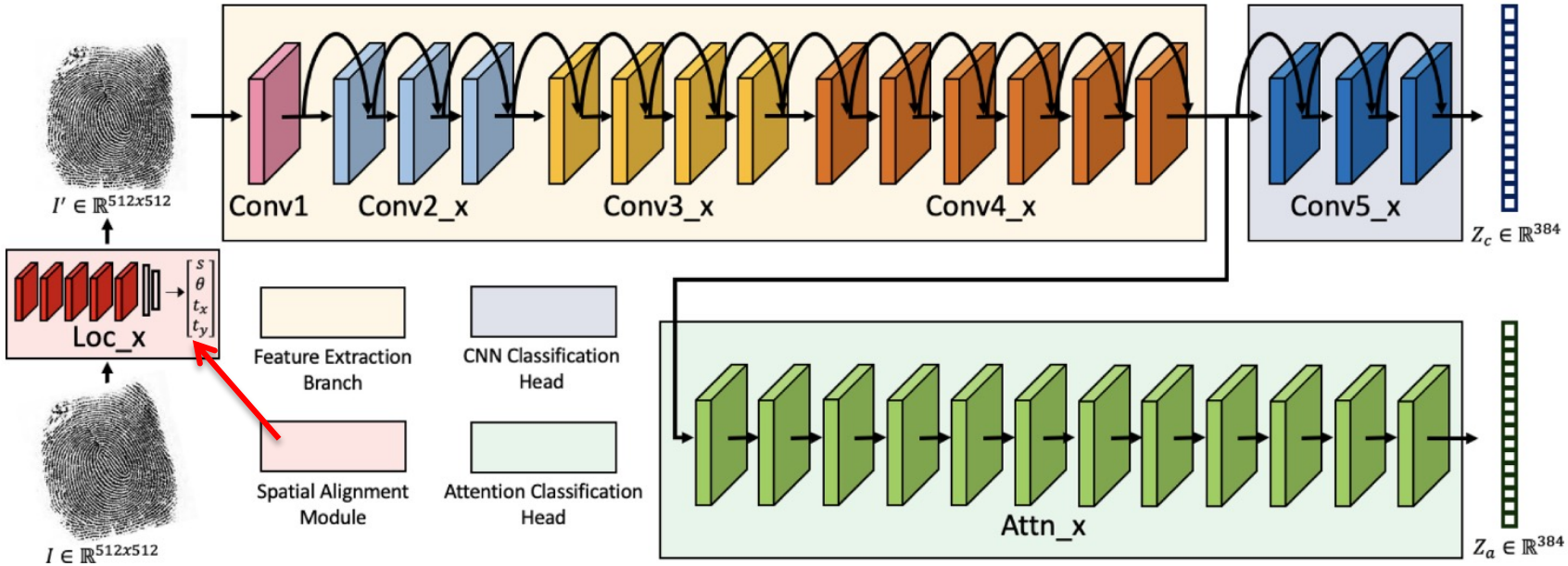
- Use of attention, vision transformers (ViTs) and other tranformer architectures
- Interpretability is important



Minutiae-Guided Fingerprint Embeddings via Vision Transformers (2022)

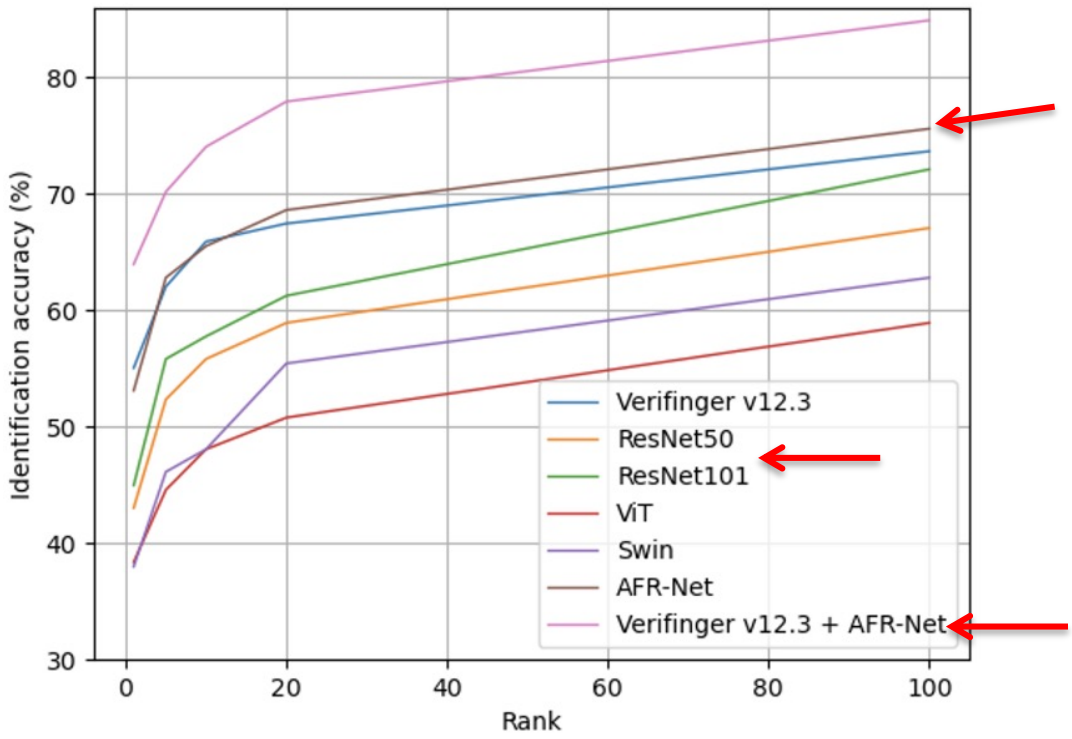
AFR-Net (2023)

- Combination of CNN and ViT
- **ArcFace loss** (state-of-the-art in face recognition!)

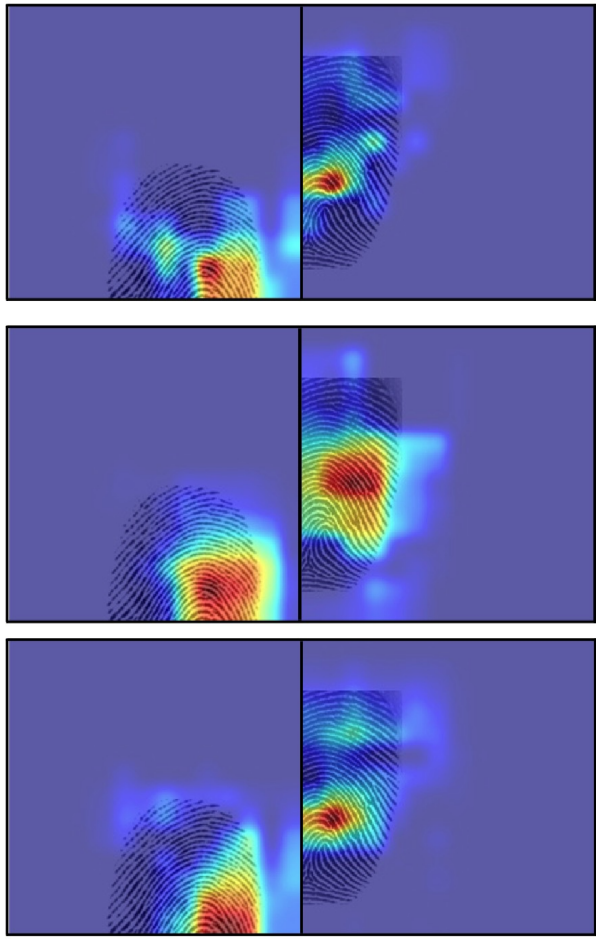


AFR-Net (2023)

Performance on a very hard fingermarks dataset (SD27)

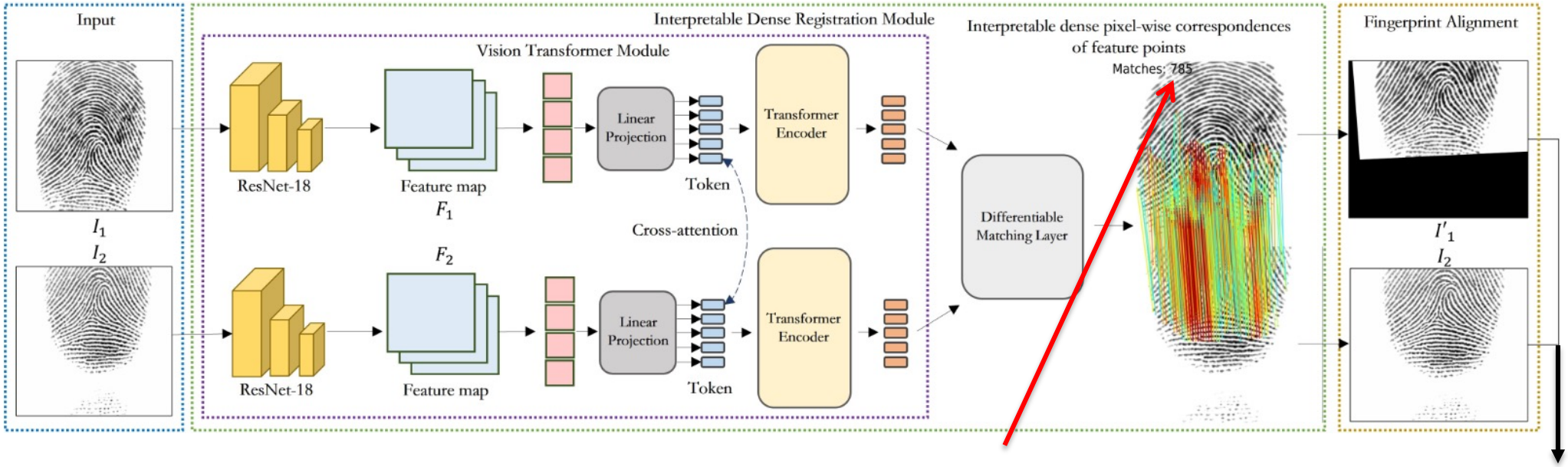


Saliency shows correspondence between misaligned samples



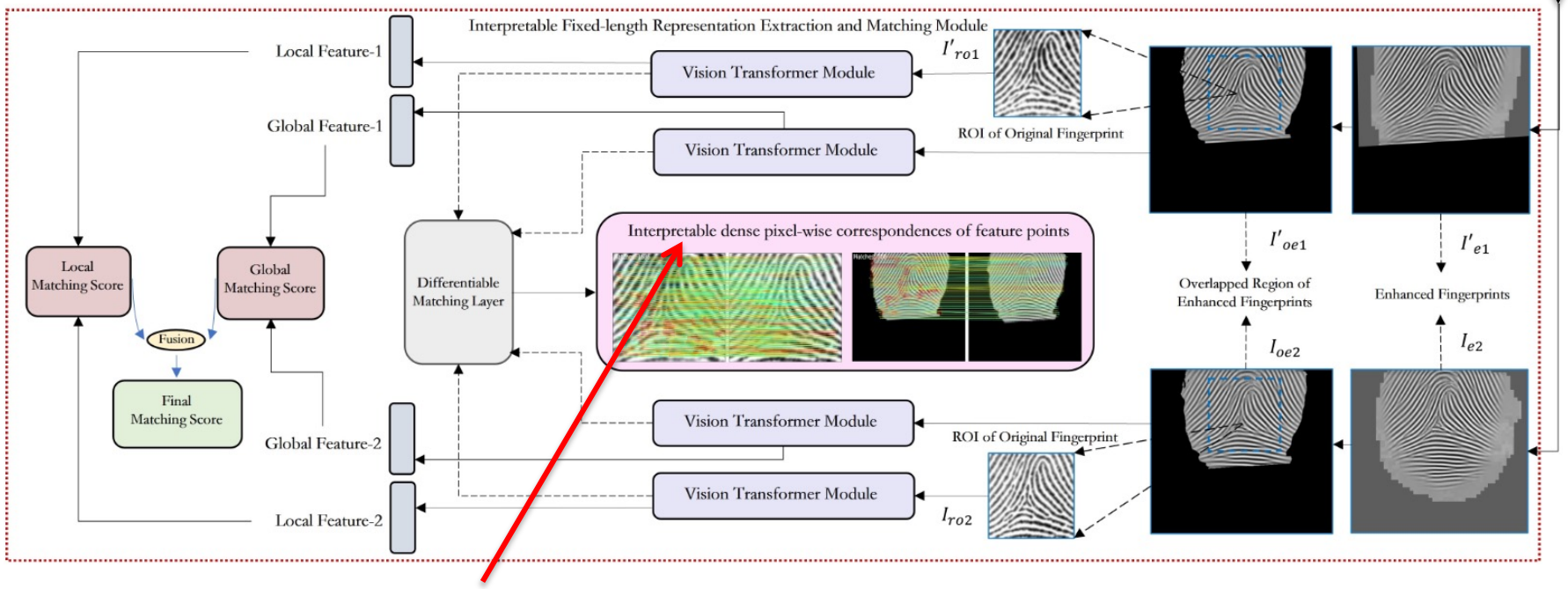
IFViT (2024)

- Does not use domain knowledge
- Step 1: Calculate the **alignment** between images using CNN



IFViT (2024)

- Step 2: Compute **matching score** using a (complex) ViT siamese network (ArcFace loss during training)

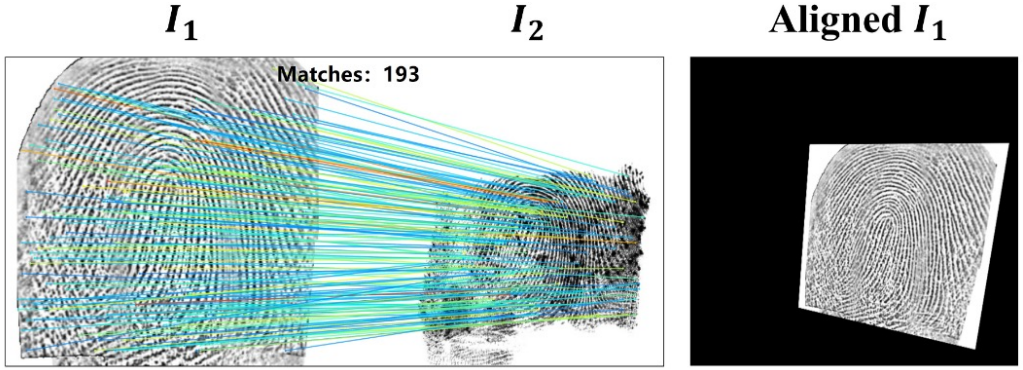


IFViT (2024)

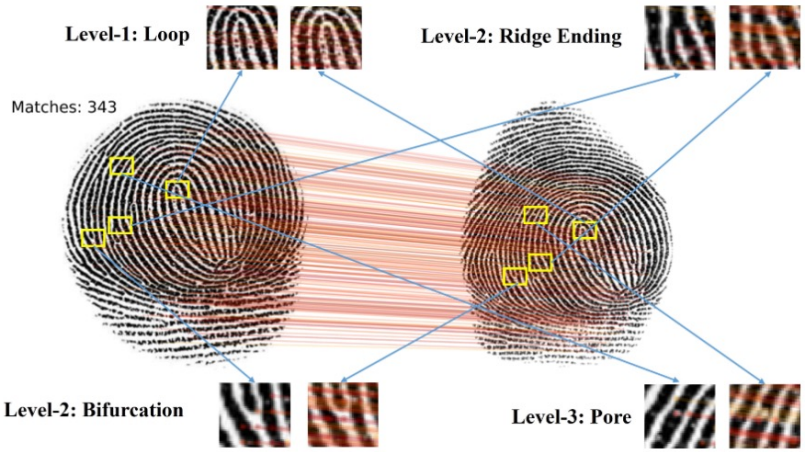
Superior performance on various types of datasets

Method	EER					
	FVC 2004 DB1	FVC 2004 DB2	FVC 2004 DB3	MOLF	NIST SD4	NIST SD302a
IFViT	**1.87%**	**1.83%**	**1.49%**	**2.33%**	**2.71%**	**2.31%**
DeepPrint	2.70%	3.39%	2.05%	5.18%	2.85%	5.03%
AFRNet	2.43%	2.95%	1.56%	5.33%	2.80%	6.29%

The resulting matches are much denser compared to minutiae points



Matches based on various levels of details



DL-FPR Key Takeaways

- Performance peaked on fingerprint datasets (99.9%+ accuracy)
- Focus now on:
  - Contactless, fingermarks, sensor interoperability
  - One-shot learning and ViTs (in combination with CNNs)
  - Explainability

- **Even cheapest commercial systems (e.g., Verfinger) still outperform published methods, but are typically slower**
  - Private vendors have more data & resources
  - Commonly compromise time in favour of better performance

The Shift to Deep Learning Iris Recognition

- **Why CNNs?**
  - Extract hierarchical features automatically
  - Robust to variations in lighting, occlusions, and noise
- **Milestones in CNN-based iris recognition:**
  - DeepIrisNet (2016)
  - IrisNet (2018)
  - Attention-based Models (2019+)
- **Introduction to end-to-end learning**:
  From raw iris image to classification without handcrafted feature extraction

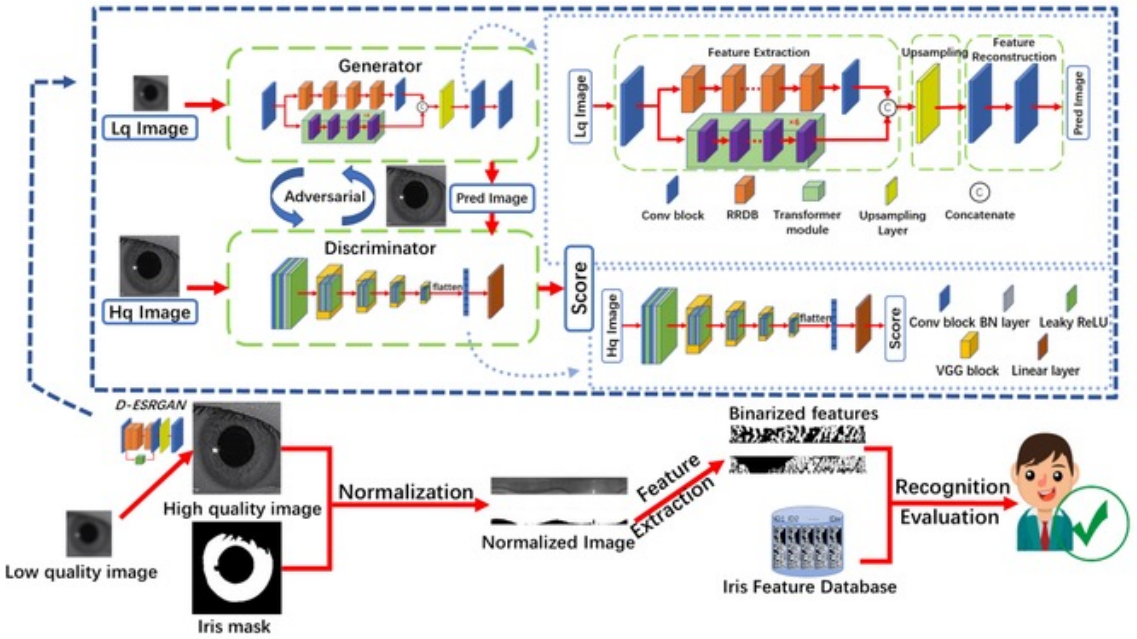
Modern CNN Architectures for Iris Recognition

- **(Still) popular CNN architectures adapted for iris recognition:**
  - VGG-16, ResNet, MobileNet
  - EfficientNet (for mobile applications)
- **Custom-built CNNs for iris recognition:**
  - DeepIrisNet (2016)
  - IrisCNN (2018)
  - Capsule Networks for Iris Recognition (2019)
- **Advantages over traditional methods:**
  - Robust against noise and occlusions
  - Higher accuracy in unconstrained environments

Transformer-Based Iris Recognition

- **ViTs for iris recognition:**
  - Self-attention mechanisms capture long-range dependencies
  - Outperforms CNNs in some cases for occluded iris images

- **Hybrid models (CNN + transformer):**
  - Combine local feature extraction (CNN) with global context learning (ViT)



SOTA Feature Extraction and Matching Techniques

- **Deep feature embedding techniques:**
  - Triplet loss for feature space learning
  - Contrastive loss for better discrimination
- **Attention mechanisms:**
  - Self-attention for iris segmentation and recognition
  - Spatial attention for occlusion handling
- **Matching strategies:**
  - Hamming distance vs. learned metric spaces
  - Siamese networks for iris verification

Datasets and Benchmarking

- **Popular datasets:**
  - CASIA-IrisV4
  - IITD Iris Database
  - ND-IRIS-0405
  - OpenEDS (for synthetic iris segmentation)
- **Benchmarking protocols:**
  - Equal Error Rate (EER)
  - False Acceptance Rate (FAR) / False Rejection Rate (FRR)
  - Rank-1 Accuracy (for identification tasks)

ConvNeXt – Overview

- A family of pure convolutional neural networks (ConvNets)
- Inspired by ViTs
- It retains convolutional inductive biases while improving upon ResNet designs

- Chosen for its small size and high efficiency

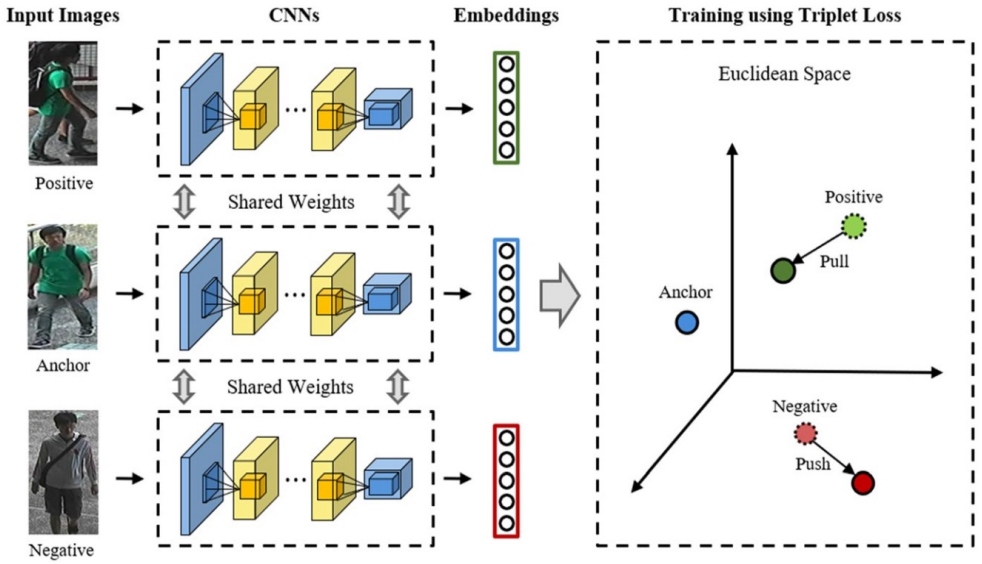
ConvNeXt – Batch-Hard Triplet Mining

- **Triplet** loss optimizes the feature space by minimizing intra-class distance and maximizing inter-class distance
- Batch-**hard** mining selects the hardest positive and hardest negative samples within each batch to maximize learning
- This approach **improves** generalization and discrimination in biometric recognition

Positive: image from the same class as the anchor

Anchor: image used as reference

Negative: image from a different class



ConvNeXt – Training Pipeline

1. **Preprocessing:**
   normalization, segmentation, and enhancement of iris images
2. **Model training:**
   ConvNeXt-Tiny trained with triplet loss
3. **Feature extraction:**
   deep feature embeddings are obtained
4. **Matching:**
   cosine similarity or Euclidean distance used for verification

ConvNeXt – Advantages

- ConvNeXt-Tiny achieves competitive accuracy on public iris datasets
- Improved generalization compared to traditional CNN-based models
- Reduced computational complexity, making it feasible for real-time applications

ConvNeXt – Challenges in SOTA Iris Recognition

- **Occlusions and illumination variability**
  - Solutions: GAN-based inpainting, attention-based occlusion handling
- **Cross-sensor generalization:**
  - Different iris cameras cause domain shifts
  - Domain adaptation techniques required
- **Privacy concerns and security:**
  - Adversarial attacks on iris recognition
  - Liveness detection methods against presentation attacks

ConvNeXt – Future Directions

- **Lightweight models for edge deployment:**
  - Knowledge distillation for smaller, faster iris models
  - MobileNetV3-based iris recognition
- **Self-supervised learning for low-data regimes:**
  - Reduces need for large labeled datasets
- **Explainability in deep iris recognition:**
  - Understanding CNN and transformer decision-making